CAREERS360
PREPARATION Series

Re-NEET 2026

Complete Answer Key and Solution PDF (All Codes)

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About This eBook

Re-NEET 2026 Complete Answer Key and Solution PDF (All Codes) is a comprehensive resource designed to help aspirants accurately evaluate their performance in the Re-NEET 2026 examination. This eBook includes the official answer keys and detailed solutions for all question paper codes, enabling students to verify answers, understand concepts, and identify areas for improvement.

Whether you're estimating your score, analysing mistakes, or revising important topics, this guide serves as a one-stop reference for post-exam evaluation.

What's Included?

- Complete Answer Keys for all Re-NEET 2026 paper codes

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- Detailed, step-by-step solutions for every question
- Subject-wise coverage of Physics, Chemistry, and Biology
- Explanations based on NCERT concepts and NEET exam patterns
- Easy-to-follow format for quick answer verification
- Performance analysis support for score estimation

Who Should Download This eBook?

- Re-NEET 2026 candidates seeking answer verification
- Students preparing for future NEET examinations
- Educators and mentors assisting NEET aspirants
- Parents looking to assess student performance accurately

Key Benefits

- Calculate your expected score with confidence
- Understand the logic behind every correct answer
- Identify strengths and weak areas across subjects
- Improve exam strategy for upcoming medical entrance tests
- Access all paper codes in a single, convenient PDF

Download the Re-NEET 2026 Complete Answer Key and Solution PDF (All Codes) to evaluate your performance, learn from detailed explanations, and stay ahead in your medical entrance preparation journey.

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Important Details About the Exam

The National Eligibility cum Entrance Test (NEET 2026) was conducted by the National Testing Agency (**NTA**) on **June 21, 2026**, from **2:00 PM to 5:15** PM across multiple exam centres in India and abroad.

The exam was held smoothly in offline mode, with lakhs of medical aspirants appearing for one of the most competitive entrance exams in the country. Students reported on time at their allotted centres and followed strict guidelines regarding entry, dress code, and examination protocols.

Particulars	Details
Mode of Exam	Offline (Pen & Paper)
Exam Duration	195 minutes (3 hours 15 minutes)
Question Type	MCQs (Multiple Choice Questions)
Official Website	https://neet.nta.nic.in/

NEET 2026 Marking Scheme

- ✅ Correct Answer: **+4 marks**
- ❌ Incorrect Answer: **-1 mark**
- ➖ Unattempted: **0 marks**

Subject-wise distribution of Questions and Marks:-

Subject	No of Questions	Marks
Physics	45	180
Chemistry	45	180
Biology	90	180
Total	180 Questions	720 Marks

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NEET 2026 Answer Key

Question Number (Set 70)	Answer Key of Set 70	Question Number (Set 50)	Answer of Set 50	Question Number (Set 60)	Answer of Set 60	Question Number (Set 80)	Answer of Set 80
1	4	19	3	18	2	40	2
2	1	10	4	27	4	44	3
3	4	31	3	25	4	45	2
4	2	5	1	40	1	37	4
5	4	24	3	42	4	39	2
6	2	36	1	16	4	32	4
7	2	8	1	8	3	14	4
8	3	16	2	26	2	29	1
9	2	12	1	11	2	22	4
10	4	13	3	35	2	15	2
11	2	30	1	1	4	12	4
12	1	3	4	22	Bonus	42	3
13	1	9	4	24	3	30	3
14	3	32	2	17	4	31	1
15	4	17	3	5	4	43	2
16	4	6	3	37	3	20	2
17	4	43	3	6	2	18	2
18	2	45	1	10	4	1	4
19	1	21	4	7	2	16	3
20	2	11	1	41	3	24	4
21	4	44	3	9	2	33	2
22	Bonus	38	Bonus	36	4	5	BONUS

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23	1	**1**	**4**	**28**	**4**	**25**	**3**
24	3	**22**	**2**	**44**	**3**	**8**	**1**
25	4	**34**	**3**	**15**	**4**	**13**	**2**
26	2	**41**	**1**	**31**	**1**	**19**	**4**
27	4	**23**	**3**	**33**	**1**	**6**	**2**
28	4	**42**	**3**	**23**	**1**	**3**	**2**
29	3	**29**	**2**	**34**	**4**	**28**	**1**
30	3	**27**	**2**	**13**	**1**	**27**	**1**
31	1	**40**	**4**	**30**	**3**	**41**	**3**
32	1	**2**	**4**	**2**	**1**	**35**	**3**
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37	3	**14**	**2**	**12**	**1**	**26**	**1**
38	2	**37**	**1**	**19**	**1**	**9**	**4**
39	2	**18**	**1**	**29**	**3**	**23**	**4**
40	1	**26**	**4**	**43**	**4**	**2**	**3**
41	3	**33**	**2**	**45**	**3**	**21**	**1**
42	4	**35**	**3**	**38**	**2**	**7**	**2**
43	4	**39**	**3**	**32**	**1**	**17**	**2**
44	3	**15**	**2**	**20**	**2**	**11**	**1**
45	3	**28**	**2**	**4**	**2**	**4**	**1**
46	**1**	**86**	**4**	**75**	**2**	**79**	**3**
47	**1**	**68**	**4**	**80**	**2**	**78**	**3**
48	**1**	**67**	**4**	**46**	**2**	**76**	**3**
49	**4**	**64**	**3**	**86**	**1**	**56**	**2**
50	**2**	**85**	**1**	**72**	**3**	**64**	**4**
51	**1**	**76**	**4**	**74**	**2**	**61**	**3**
52	**1**	**72**	**4**	**67**	**2**	**83**	**3**
53 *	**1**	**65**	**3**	**48**	**2**	**60**	**3**
54	**4**	**53**	**3**	**90**	**1**	**85**	**2**
55	**4**	**47**	**3**	**55**	**1**	**54**	**2**
56	**4**	**50**	**3**	**66**	**1**	**66**	**2**

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57	2	84	1	61	3	48	4
58	3	79	2	82	4	70	1
59	1	59	4	85	2	84	3
60	2	82	1	78	3	50	4
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69	2	71	1	87	3	59	4
70	2	62	1	89	3	51	4
71	1	89	4	70	2	57	3
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74	2	57	1	68	3	75	4
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76	2	83	1	50	3	82	4
77	3	60	2	54	4	67	1
78	1	52	4	64	2	86	3
79	3	81	2	83	4	77	1
80	2	69	1	71	3	74	4
81	4	54	3	81	1	80	2
82	2	66	1	57	3	71	4
83	2	74	1	53	3	58	4
84	2	61	1	56	3	68	4

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85	3*	70	2	60	4	52	1
86	2	77	1	69	3	87	4
87	1	90	4	88	2	62	3
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141	2	150	1	178	3	154	4
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144	4	177	3	142	1	172	2
145	1	155	4	140	2	144	2
146	2	172	1	137	3	162	4
147	4	173	3	161	1	158	2

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148	2	136	1	166	3	146	4
149	1	143	4	139	2	160	3
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176	3	178	2	162	4	178	1
177	3	148	2	179	4	157	1
178	3	165	2	172	4	151	1
179	4	162	3	158	1	143	2
180	3	138	2	146	4	173	1

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Test Booklet Code

70

SUSHRUT

No. : 264844067

This Booklet contains 32 pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**.
3. Wherever the symbols/constants are not mentioned, they are to be considered as per their standard meaning/ value.
4. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate **must handover the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
8. **The CODE for this Booklet is 70. Make sure to enter this code in the OMR answer sheet.**
9. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
10. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
11. Each candidate must show on-demand his/her Admit Card to the Invigilator.
12. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
13. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
14. Use of Electronic/Manual Calculator is prohibited.
15. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination along with Public Examinations (Prevention of unfair means act 2024).
16. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
17. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
18. If a candidate marks more than one answers for a question in the OMR sheet, it will be treated as incorrect and negative marking will be applicable.

Name of the Candidate (in Capitals) : ANSAR AKHTER LONE

Roll Number : in figures 250221139

: in words Two five zero two two one four one three nine

Centre of Examination (in Capitals) : VISHWA BHARTI WOMENS COLLEGE

Candidate's Signature : _____ **Invigilator's Signature :** Sujata

Facsimile signature stamp of Centre Superintendent : _____

SEAL

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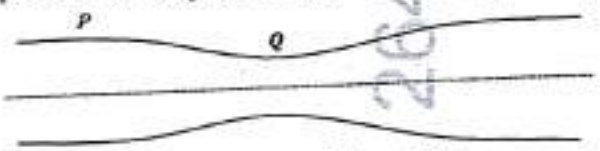
1. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is :
(Take speed of light $=3\times10^8$ ms^{-1}, charge of electron $=-1.6\times10^{-19}$ C and mass of electron $=9\times10^{-31}$ kg)

(1) 275 (2) $\frac{2}{450}$

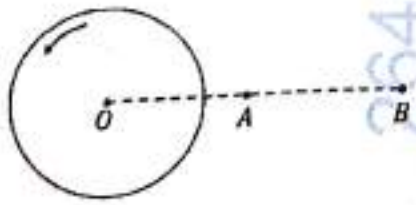
(3) $\frac{1}{250}$ (4) 225

2. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is 15 Nm^{-2}. The area of cross-section at P and Q are 40 cm^2 and 20 cm^2, respectively. The rate of flow of water through the pipe, in cm^{3}s^{-1}, is :
[Take density of water = 1000 kg m^{-3}]



(1) 400 (2) 100
(3) 200 (4) 300

3. A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2\times OA$. The value of $\frac{L_A}{L_B}$ is :



(1) 2 (2) $\frac{1}{4}$

(3) $\frac{1}{2}$ (4) 1

4. Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

(1) $2\mu_0\pi n^2r^2l$

(2) $\mu_0\pi n^2r^2l$

(3) $\mu_0 n^2r^2l$

(4) $(\mu_0/2\pi)n^2r^2l$

5. The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is :

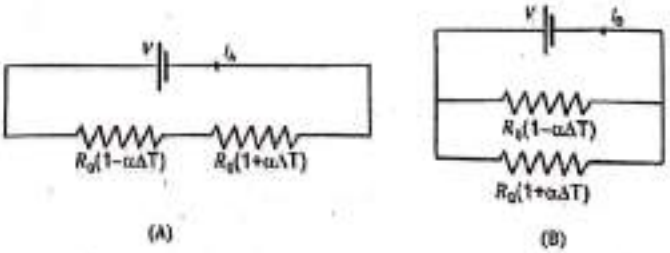
(1) $6\pi R^3\alpha\Delta T$

(2) $2\pi R^3\alpha\Delta T$

(3) $3\pi R^3\alpha\Delta T$

(4) $4\pi R^3\alpha\Delta T$

6. Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature of coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is :



(1) both I_A and I_B remain constant

(2) I_A remains constant while I_B increases

(3) I_A decreases while I_B increases

(4) I_A increases while I_B decreases

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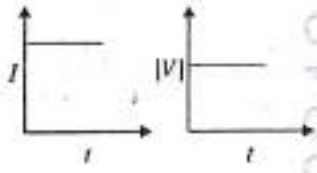
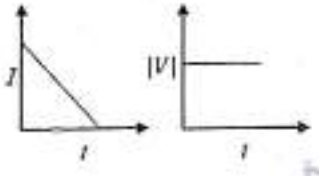
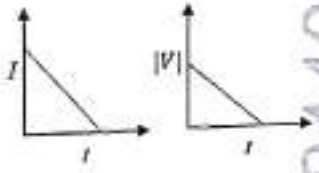
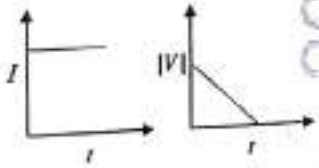
7. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by :

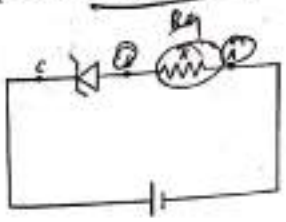
(1) 

(2) 

(3) 

(4) 

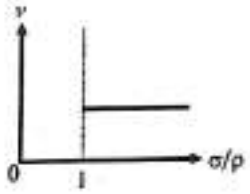
8. An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between points B and A is :

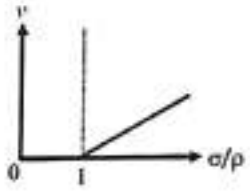


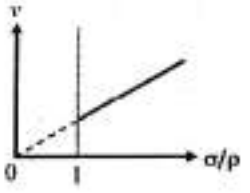
(1) 0 V (2) 3 V

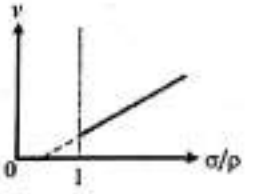
(3) 2 V (4) 1 V

9. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by :

(1) 

(2) 

(3) 

(4) 

10. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2 = \frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is :

(1) 2 (2) $\frac{1}{\sqrt{2}}$

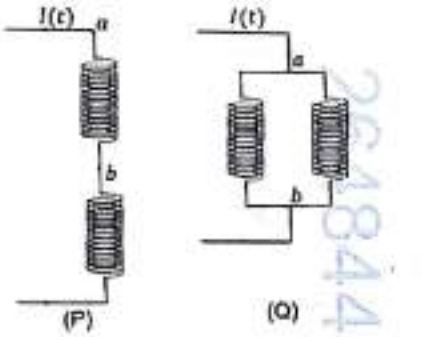
(3) 1 (4) $\sqrt{2}$

11. An ac voltage $V=220\sin(2\times10^3t)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :
(Given : L = 10 mH, C = 25μF, R = 100 Ω)

(1) 22.0 A (2) 2.2 A
(3) 5.5 A (4) 11.0 A

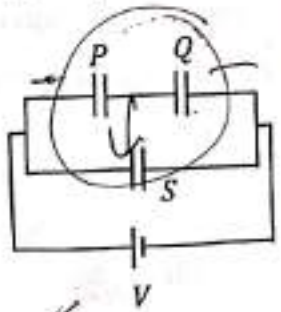
12. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is :

[Neglect the effect of mutual inductance.]



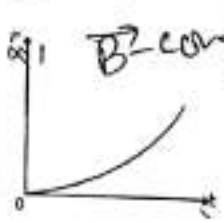
(1) 2 (2) 1/4
(3) 1/2 (4) 1

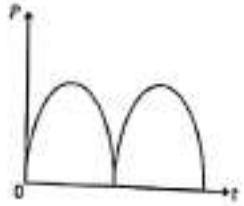
13. Three identical capacitors P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is :

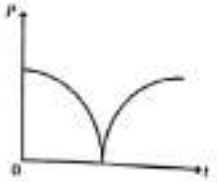


(1) 1/6 (2) 2/3
(3) 1/3 (4) 1/2

14. A conducting loop of finite resistance lies on the x - y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A=A_0(1+\sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :

(1)


(2)


(3)


(4)

15. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma=5/3$) decreases from 60K to 50K. The work done by the gas in the process is :

(Take the universal gas constant as $R=8.3\ J\ mol^{-1}\ K^{-1}$)

(1) 166 J
(2) 41.5 J
(3) 83 J
(4) 124.5 J

16. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1$ ms^{-2}, $\beta = 6$ ms^{-1} and $\gamma = 5$ m. The average speed of the particle, in ms^{-1}, from $t = 0$ to $t = 6$ s is :

(1) 0 (2) 12

(3) 6 (4) 3

17. The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the electromagnetic spectrum		Applications	
P	Microwave	I	For purifying the water
Q	UV rays	II	For warming the food
R	Gamma rays	III	For AM and FM communication systems
S	Radio wave	IV	For treating the Cancer cells

The **correct** option is :

(1) P-II, Q-IV, R-III, S-I

(2) P-I, Q-II, R-III, S-IV

(3) P-I, Q-IV, R-II, S-III

(4) P-II, Q-I, R-IV, S-III

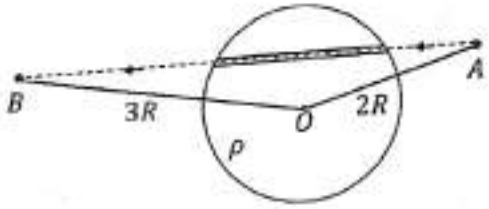
18. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_P/C_V of the gas is 8/7, then the value of f is :

(1) 1 (2) 4

(3) 3 (4) 2

19. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is :

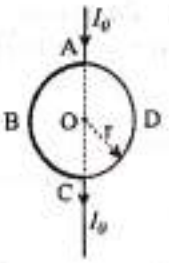
(ϵ_0 is the permittivity of vacuum)



(1) 18 (2) 2

(3) 6 (4) 9

20. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the center O of the loop is :



(1) $\frac{\mu_0 I_0}{2\pi r}$ (2) $\frac{\mu_0 I_0}{12r}$

(3) $\frac{\mu_0 I_0}{4r}$ (4) $\frac{\mu_0 I_0}{2r}$

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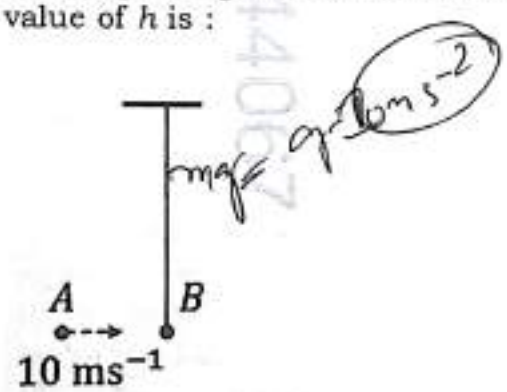
21. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed 10 ms^{-1} hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ ms^{-2}$ and neglecting the size of the bob, the value of h is :



(1) 2.5 (2) 8

(3) 7 (4) 5

22. An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\epsilon_r = 9$, has the electric field, $E_x = E_0 \sin(kz - 2\pi \times 10^6 t)\ Vm^{-1}$ where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is :

(1) The direction of propagation of the electromagnetic wave is along $+z$

(2) The speed of the electromagnetic wave inside the medium is $10^8\ ms^{-1}$

(3) The wavelength of the electromagnetic wave inside the medium is 300 m

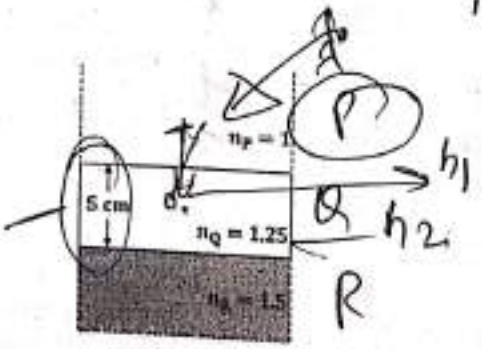
(4) The magnetic field is given by the relation $B_y = \frac{B_0}{v} \sin(kz - 2\pi \times 10^6 t)$ where v is the speed of the electromagnetic wave inside the medium

23. A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is :

(1) $\frac{1}{2}$ (2) 3

(3) 1 (4) $\frac{1}{3}$

24. Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5, respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm, is :



(1) 3 (2) 0

(3) 1 (4) 2

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25. Consider a fixed uniformly charged insulating sphere with radius R and total charge +Q. A point charge −q (q<<Q) with mass m is released from rest at a distance of 3R from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is :

(ϵ_0 is the permittivity of vacuum, neglect gravitational forces).

(1) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$

(2) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$

(3) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$

(4) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$

26. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed 10 ms^{-1}, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be 10 ms^{-2}, the value of θ is :

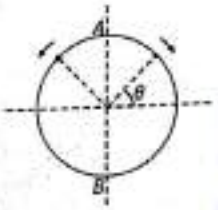
(1) $\tan^{-1}\left(2\sqrt{3}\right)$

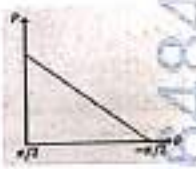
(2) $\tan^{-1}\left(\frac{1}{5}\right)$

(3) $\tan^{-1}\left(\frac{2}{5}\right)$

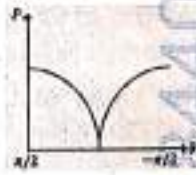
(4) $\tan^{-1}\left(\sqrt{3}/2\right)$

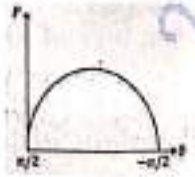
27. A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point $A\left(\theta=\frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point $B\left(\theta=-\frac{\pi}{2}\right)$. During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ ?



(1) 

(2)

(3) 

(4) 

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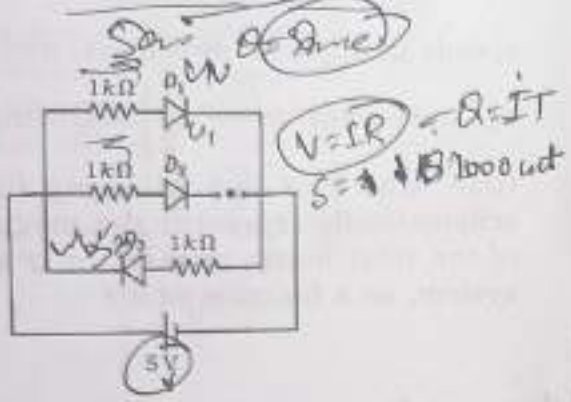
28. Three identical p-n junction diodes D_1, D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is :



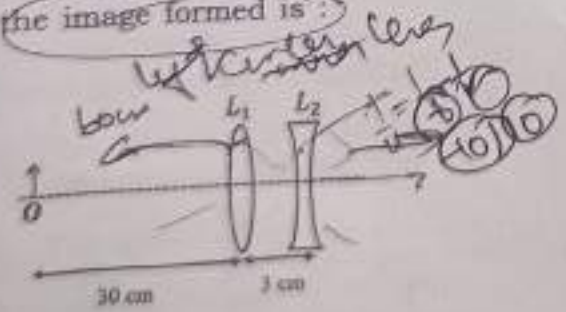
(1) $W_2 > W_1 = W_3$

(2) $W_1 > W_2 > W_3$

(3) $W_3 = W_1 > W_2$

(4) $W_3 > W_2 > W_1$

29. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and −10 cm, respectively. The position of the image formed is :



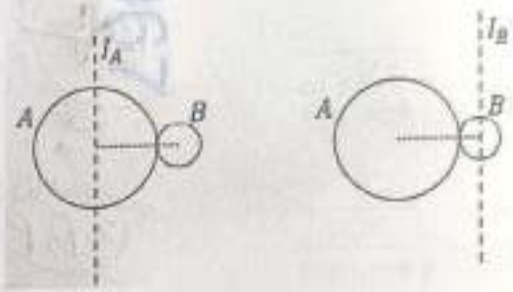
(1) 60 cm to the right of the concave lens

(2) 20 cm to the left of the concave lens

(3) 60 cm to the left of the concave lens

(4) 30 cm to the right of the concave lens

30. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius r < R and mass m < M. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is :



(1) 0

(2) $(M-m)(R+r)^2$

(3) $(m-M)(R+r)^2$

(4) $(m-M)(R-r)^2$

31. Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6} \times 10^5$ ms^{-1}. The radius of the orbit is $x \times 10^{-9}$ m. The value of x is :
[Take the mass of electron to be 9×10^{-31} kg, charge of electron $= -1.6 \times 10^{-19}$ C and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ N m^2 C^{-2}]

(1) 1 (2) 4

(3) 3 (4) 2

32. The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, then the correct option is :

(1) $n_A = \frac{1}{2} n_B$ (2) $n_A = n_B$

(3) $n_A = 2n_B$ (4) $n_A = \frac{1}{4} n_B$

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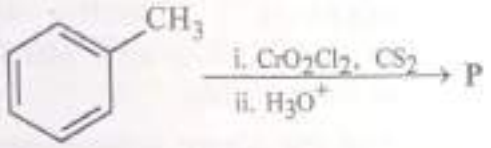
46. Consider the following reaction, and choose the correct option.



(1) Compound **P** is obtained by the hydrogenation of benzoyl chloride with Pd on $BaSO_4$.

(2) On treating compound **P** with saturated $NaHCO_3$ solution, brisk effervescence is observed.

(3) Compound **P** can be prepared by treating benzene with anhydrous $AlCl_3$ and CH_3COCl.

(4) On treatment with bromine water, compound **P** gives a white precipitate.

47. The formula of tetraammineaquachloridocobalt(III) chloride is

(1) $[Co(NH_3)_4(H_2O)Cl]Cl_2$

(2) $[Co(NH_3)_4Cl_2] \times H_2O$

(3) $[Co(NH_3)_4]Cl_3 \times H_2O$

(4) $[Co(NH_3)_4(H_2O)Cl]Cl$

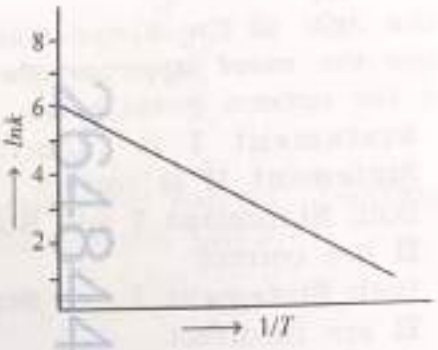
48. The lanthanide ion having four unpaired electrons is

(Given : Atomic numbers of Ce = 58, Nd = 60, Tb = 65 and Ho = 67)

(1) Ho^{3+} (2) Nd^{3+}

(3) Ce^{3+} (4) Tb^{3+}

49. For an elementary chemical reaction, the Arrhenius plot is given below.



If the energy of activation is 6.64 kJ mol^{-1} and R = 8.3 $J\,K^{-1}\,mol^{-1}$, the temperature at which the rate constant becomes e^2 min^{-1}, is

(1) 250 K (2) 125 K

(3) 150 K (4) 200 K

50. The green paramagnetic species formed by heating $KMnO_4$ at 513 K is

(1) KO_2 (2) K_2MnO_4

(3) Mn_3O_4 (4) MnO

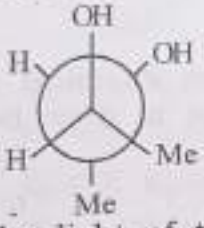
51. Given below are two statements:

Statement I:

trans-But-2-ene upon treatment with Br_2 in CCl_4 gives the following product

Statement II:

cis-But-2-ene upon treatment with alkaline $KMnO_4$ gives the following product



In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

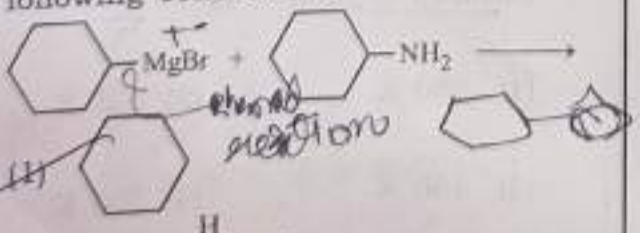
(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are correct

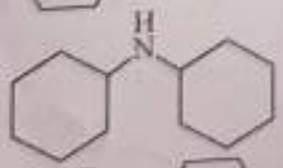
(3) Both **Statement I** and **Statement II** are incorrect

(4) **Statement I** is correct but **Statement II** is incorrect

52. One of the products formed in the following reaction is



(1)

(2) 

(3)

(4)

53. Given below are two statements:

Statement-I : Heating NaCl with concentrated H_2SO_4 and MnO_2 results in oxidation of Mn.

Statement-II : Heating NaI with concentrated H_2SO_4 and MnO_2 results in reduction of Mn.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **Statement-I** is incorrect but **Statement-II** is correct.

(2) Both **Statement-I** and **Statement-II** are correct.

(3) Both **Statement-I** and **Statement-II** are incorrect.

(4) **Statement-I** is correct but **Statement-II** is incorrect.

54. Among the following options, the correct trend in the electron gain enthalpy is

(1) I > Br > Cl > F

(2) F > Cl > Br > I

(3) Br > Cl > F > I

(4) Cl > F > Br > I

55. Given below are two statements:

Statement-I : $[Fe(ox)_3]^{3-}$ is chiral.

Statement-II : *trans* – $[Cr(H_2O)_2(ox)_2]^-$ is chiral.

(Given : oxH_2 = HOOC – COOH)

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **Statement-I** is incorrect but **Statement-II** is correct.

(2) Both **Statement-I** and **Statement-II** are correct.

(3) Both **Statement-I** and **Statement-II** are incorrect.

(4) **Statement-I** is correct but **Statement-II** is incorrect.

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56. The correct statement about peptides and proteins is

(1) In α-helices, the polypeptide chain is twisted into a left-handed screw (helix) through intramolecular hydrogen bonds.

(2) Tertiary structure of proteins has two or more polypeptide subunits.

(3) Only the proteins having a quaternary structure are biologically active.

(4) In β-pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

57. Given below are two statements:

Statement-I : Oxidation of *p*-nitrotoluene with acidic $KMnO_4$ gives an acid that is stronger than benzoic acid.

Statement-II : Reduction of *p*-nitrotoluene with Sn/HCl followed by neutralization gives an amine that is more basic than aniline.

In light of the above statements, choose the ***most appropriate*** answer from the options given below.

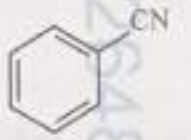
(1) **Statement-I** is incorrect but **Statement-II** is correct.

(2) Both **Statement-I** and Statement-II are correct.

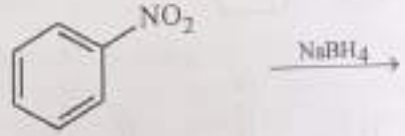
(3) Both **Statement-I** and **Statement-II** are incorrect.

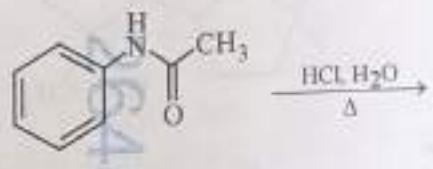
(4) **Statement-I** is correct but **Statement-II** is incorrect.

58. Identify the reactions which give aniline as the major product.

A. 

B.

C. 

D. 

Choose the correct answer from the options given below.

(1) C and D only

(2) A and B only

(3) B and D only

(4) A and C only

59. Two moles of an ideal gas undergo free expansion from 10 L to 100 L at 300 K. The values of ΔS_{system} and $\Delta S_{surroundings}$ are

(R is universal gas constant)

(1) $\Delta S_{system} = 4.606\ R$; $\Delta S_{surroundings} = 0$

(2) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 0$

(3) $\Delta S_{system} = 4.606\ R$; $\Delta S_{surroundings} = -4.606\ R$

(4) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 4.606\ R$

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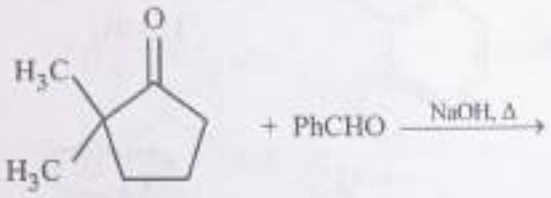
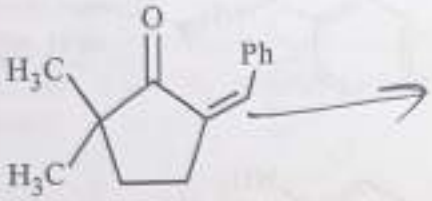
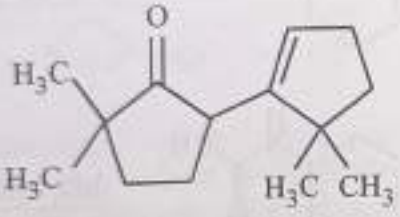
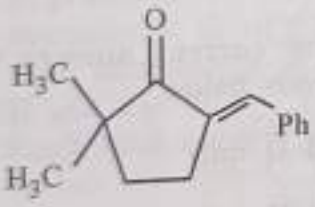
60. The compound that **CANNOT** be obtained from the aldol condensation reaction shown below, is



(1)


(2)


(3)


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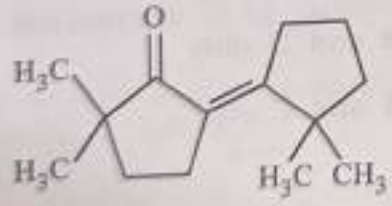


61. The complex which has *facial* and *meridional* isomers is

(Given : py = pyridine and en = H_2N–CH_2–CH_2–NH_2)

(1) $[Ni(en)_2(H_2O)_2]^{2+}$

(2) $[Cr(py)_3(Cl)_3]$

(3) $[Cr(H_2O)_6]^{3+}$

(4) $[Co(NH_3)_4(H_2O)_2]^{3+}$

62. The numbers 17.0145 and 21.0235 were rounded to three figures after the decimal point. The resulting numbers, respectively, are

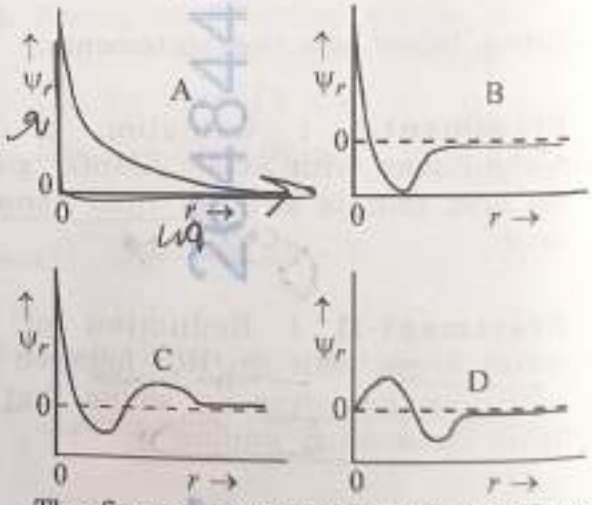
(1) 17.015 and 21.024
(2) 17.014 and 21.023
(3) 17.015 and 21.023
(4) 17.014 and 21.024

63. The amount of carbon dioxide evolved upon complete combustion of 116 g of *n*-butane is

(Given: atomic mass in amu H = 1, C = 12 and O = 16)

(1) 362 g
(2) 352 g
(3) 322 g
(4) 176 g

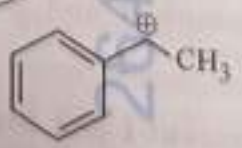
64. Consider the following schematic plots of orbital wavefunction (ψ_r) against distance (r) from the nucleus.



The figure representing two radial nodes in the orbital is

(1) D
(2) A
(3) B
(4) C

65. The following carbocation is stabilized by the interaction of the empty *p* orbital with



(1) empty σ^* and empty π^* orbitals
(2) filled σ and filled π orbitals
(3) empty σ and empty π^* orbitals
(4) empty σ^* and filled π orbitals

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66. A 1:3 electrolyte in an aqueous solution is

(1) $[Co(NH_3)_3(NO_2)_3]$

(2) $[CoCl_2(NH_3)_4]Cl$

(3) $[CoCl(NH_3)_5]Cl_2$

(4) $[Co(NH_3)_6]Cl_3$

67. The standard electrode potential (E°) for the half-cell reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$ at 298 K is
(Given: $E^\circ(Fe^{3+}/Fe) = -0.04$ V and $E^\circ(Fe^{2+}/Fe) = -0.44$ V at 298 K)

(1) +0.92 V (2) +0.40 V
(3) +0.76 V (4) −0.48 V

68. In potash alum, the ratio of K^+ and SO_4^{2-} ions is

(1) 3:2 (2) 1:2
(3) 2:1 (4) 2:3

69. Consider the following statements about the solutions formed by mixing two liquids.

A. An ideal solution thus formed obeys Raoult's law throughout the composition range.

B. Mixture of chloroform and acetone shows negative deviation from Raoult's law.

C. Mixture of aniline and phenol shows positive deviation from Raoult's law.

(1) A and C only
(2) A and B only
(3) B and C only
(4) A only

70. For a salt **XY**, which is a strong electrolyte, the plot of Λ_m versus $\sqrt{c}$ has a slope of −90.0 S cm^2 mol$^{-3/2}$ L$^{1/2}$ at 298 K. At 0.01 M concentration of **XY**, the value of Λ_m is 145.0 S cm^2 mol^{-1}. The limiting molar conductivity of Y^- ion ($\lambda^0_{Y^-}$, in S cm^2 mol^{-1}) at 298 K will be

(Given: $\lambda^0_{X^+} = 74.0$ S cm^2 mol^{-1})

(1) 76.0 (2) 80.0
(3) 100.0 (4) 90.0

71. Arrange the following compounds in the increasing order of polarity

A. $CH_3CH_2OCH_2CH_3$
B. CH_3CH_2OH
C. CH_3COCH_3
D. CH_3COOH

Choose the correct answer from the options given below.

(1) A < C < B < D
(2) A < B < C < D
(3) C < A < D < B
(4) C < A < B < D

72. According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is

(1) $Cl^- > NH_3 > H_2O > CO$
(2) $CO > NH_3 > H_2O > Cl^-$
(3) $CO > H_2O > NH_3 > Cl^-$
(4) $Cl^- > H_2O > NH_3 > CO$

73. The amino acid that gives a red-blood colour on treating its sodium fusion extract with sodium nitroprusside is

(1) serine (2) leucine
(3) threonine (4) methionine

74. In an acidic medium, 10 mL of 0.25 M oxalic acid is titrated with $KMnO_4$ solution. If the volume of $KMnO_4$ solution required to reach end point is 10 mL, the strength of the $KMnO_4$ solution is

(1) 0.15 M (2) 0.10 M
(3) 0.20 M (4) 0.25 M

75. The correct statement is

(1) Aluminium has five valence orbitals.
(2) Boron has a maximum covalency of four.
(3) Beryllium has three valence orbitals.
(4) Magnesium has a maximum covalency of four.

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76. Among the following, the compound having conjugated double bonds is

(1) hepta-1,6-diene

(2) hepta-1,3-diene

(3) hepta-1,4-diene

(4) hepta-1,5-diene

77. $2A \xrightarrow{k} B$ is a zero-order reaction, where $k = 1.0\ mol\ L^{-1}\ min^{-1}$. If the initial concentration of A is 2 M, then the time taken to complete 75% of the reaction will be

(1) 2.0 min (2) 1.5 min

(3) 0.75 min (4) 1.0 min

78. The correct order of solubility of the given salts in water at 298 K is

Salt	K_{sp} at 298 K
AgBr	5.0×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

(1) $Zn(OH)_2 > AgBr > Hg_2Cl_2$

(2) $Hg_2Cl_2 > Zn(OH)_2 > AgBr$

(3) $AgBr > Zn(OH)_2 > Hg_2Cl_2$

(4) $Hg_2Cl_2 > AgBr > Zn(OH)_2$

79. The correct decreasing order of oxidation state of the underlined atom in each molecule is

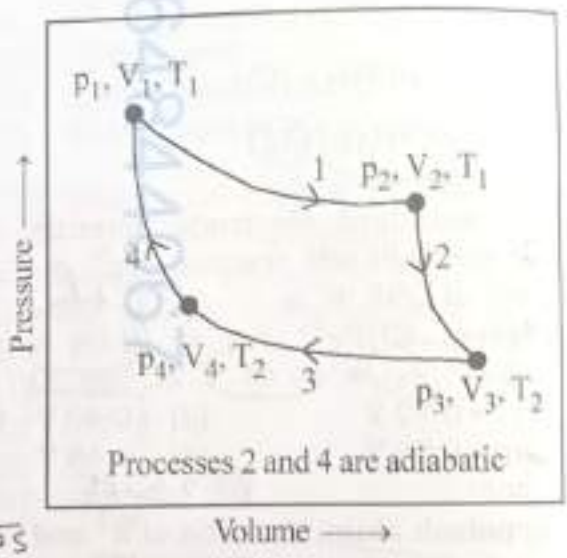
(1) $\underline{P}_4O_6 > \underline{Cl}_2O_7 > \underline{Al}H_3$

(2) $\underline{P}_4O_{10} > \underline{S}O_3 > H_2\underline{O}$

(3) $\underline{N}_2O_5 > \underline{Al}_2O_3 > H_2\underline{S}$

(4) $\underline{Pb}O_2 > \underline{N}_2O_3 > \underline{S}O_3$

80. Consider the reversible processes for 1.0 mol of an ideal gas as shown in the figure.



w_1, w_2, w_3 and w_4 represent work done (in calories) in the processes 1, 2, 3 and 4, respectively; ΔU_2 and ΔU_4 are changes in the internal energy for the processes 2 and 4, respectively.
[use $R = 2\ cal\ K^{-1}\ mol^{-1}$]
The correct option is

(1) $w_1 + w_2 + w_3 + w_4 = 0$

(2) $w_1 + w_3 = -2T_1 \ln\frac{V_2}{V_1} - 2T_2 \ln\frac{V_4}{V_3}$

(3) $w_2 + w_4 = \Delta U_2 - \Delta U_4$

(4) $w_1 + w_2 = 2T_1 \ln\frac{V_2}{V_1}$

81. **Assertion A :**
For an ideal solution formed by mixing liquids **P** and **Q**, $\Delta_{mix} H = 0$ and $\Delta_{mix} V = 0$
Reason R :
No interactions occur between **P** and **Q**
In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

(1) **A** is not correct but **R** is correct

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

(4) **A** is correct but **R** is not correct

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82. Among the species given below, the spin-only magnetic moment is highest for

(Given : Atomic number of Ti = 22, Mn = 25, Fe = 26 and Co = 27)

(1) $[Ti(H_2O)_6]^{3+}$

(2) $[Mn(CN)_6]^{3-}$

(3) $[Fe(CN)_6]^{3-}$

(4) $[Co(NH_3)_6]^{3+}$

83. A protein undergoes reversible thermal denaturation from its initial state **N** to denatured state **D** according to $\mathbf{N} \rightleftharpoons \mathbf{D}$. At 60 °C, the concentrations of both **N** and **D** are equal at equilibrium, and the standard enthalpy change of denaturation is 666 kJ mol^{-1}. The standard entropy change (ΔS°) in kJ $K^{-1}mol^{-1}$ of the protein upon denaturation at 60 °C closest to

(1) 11.1

(2) 2.0

(3) 2000.0

(4) 333.0

84. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Generally, 3*d* transition metals have high melting points.

Reason R : Involvement of 3*d*-electrons in addition to 4*s*-electrons in the interatomic metallic bonding.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **A** is not correct but **R** is correct.

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(3) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

(4) **A** is correct but **R** is not correct.

85. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The first ionization enthalpy of O is lower than that of N and F.

Reason R : The loss of an electron from O leads to stable half-filled *p* orbital.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

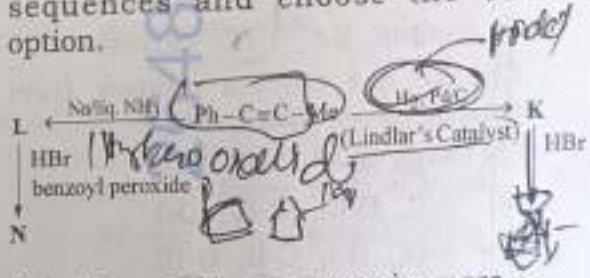
(1) **A** is not correct but **R** is correct.

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(3) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

(4) **A** is correct but **R** is not correct.

86. Consider the following reaction sequences and choose the correct option.



(1) **M** and **N** are stereoisomers

(2) **K** and **L** are geometrical isomers

(3) **K** and **L** are enantiomers

(4) **M** and **N** are geometrical isomers

87. The highest occupied molecular orbital for Ne_2 is

(1) σ^*_{2p}

(2) π_{2p}

(3) σ_{2p}

(4) π^*_{2p}

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88. Match the species in **List I** with their geometry in **List II**

	List I		List II
A.	PCl_5	I.	Tetrahedral
B.	BrF_5	II.	Square Planar
C.	BF_4^-	III.	Trigonal bipyramidal
D.	$[Ni(CN)_4]^{2-}$	IV.	Square pyramidal

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-I, C-II, D-IV

89. Match the vitamins in **List I** with their sources in **List II**

	List I		List II
A.	vitamin A	I.	meat
B.	vitamin B_{12}	II.	sunflower oil
C.	vitamin E	III.	green leafy vegetables
D.	vitamin K	IV.	carrots

Choose the correct answer from the options given below.

(1) A-III, B-I, C-IV, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-I, C-II, D-III

(4) A-IV, B-II, C-I, D-III

90. For the following reaction sequence, choose the correct option

$$\xrightarrow[\text{ii. NaOCl}]{\text{i. } CH_3COCl,\ AlCl_3} P + Q$$

(1) Both **P** and **Q** are carbonyl compounds.

(2) If **P** is the sodium salt of a carboxylic acid, **Q** is a primary alcohol.

(3) **P** and **Q** are aromatic compounds.

(4) If **P** gives a carboxylic acid on acidification, **Q** gives a poisonous gas on exposure to air and light.

91. Given below are two statements :

Statement I : The class name Reptilia refers to creeping or crawling mode of locomotion.

Statement II : All organisms belonging to Reptilia have three chambered heart.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are correct

(3) Both **Statement I** and **Statement II** are incorrect

(4) **Statement I** is correct but **Statement II** is incorrect

92. How many turns of Calvin cycle are required for the formation of three molecules of glucose ?

(1) 18 (2) 6

(3) 3 (4) 1

93. Photorespiration reaction catalyzed by RuBisCo is shown below :

$RuBP + O_2 \rightarrow$ 3-Phosphoglycerate + X

Identify "X" from the given options :

(1) Malate

(2) Phosphoenolpyruvate

(3) 2-Phosphoglycolate

(4) Oxaloacetate

94. Given below are two statements :

Statement I : In gymnosperms, the male and female gametophytes remain within the sporangia.

Statement II : In gymnosperms, seeds are not covered.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are correct

(3) Both **Statement I** and **Statement II** are incorrect

(4) **Statement I** is correct but **Statement II** is incorrect

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95. How many molecules of pyruvic acid are produced at the end of glycolysis from 206 molecules of glucose ?
(1) 412 (2) 206
(3) 309 (4) 103

96. Match **List-I** with **List-II.**

List-I	List-II
A. Fusion of protoplasms between gametes	I. Meiosis
B. Fusion of two nuclei	II. Plasmogamy
C. Generation of haploid spores	III. Karyogamy

Choose the **correct** answer from the options given below :
(1) A-I, B-III, C-II
(2) A-II, B-III, C-I
(3) A-II, B-I, C-III
(4) A-III, B-II, C-I

97. Mitochondrial inner membrane encloses _____.
(1) aqueous humor
(2) matrix
(3) cytosol
(4) mucus

98. Phyllotaxy is the pattern of arrangement of _____.
(1) sepals (2) leaves
(3) flowers (4) fruits

99. Mad cow disease is caused by _____.
(1) *Mycoplasma sp.*
(2) prions
(3) viroids
(4) *Aspergillus sp.*

100. Cell theory was formulated by _____.
(1) Antonie Von Leeuwenhoek
(2) Schleiden and Schwann
(3) Robert Brown
(4) Singer and Nicolson

101. Which of the following plant growth regulators promotes internode elongation prior to flowering in cabbage ?
(1) Ethephon
(2) Abscisic acid
(3) Gibberellin
(4) Indole butyric acid

102. Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700) ?
(1) Carotenoids
(2) Chlorophyll b
(3) Chlorophyll a
(4) Xanthophylls

103. Sphenopsida class belongs to _____.
(1) pteridophytes
(2) bryophytes
(3) angiosperms
(4) gymnosperms

104. Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans ?
(1) Femur-tarsal-patella-tibia
(2) Femur-tibia-patella-tarsal
(3) Patella-femur-tibia-tarsal
(4) Femur-patella-tibia-tarsal

105. Which of the following plant growth regulators is used as herbicide ?
(1) Gibberellin
(2) 2,4-D
(3) Kinetin
(4) Abscisic acid

106. Genus represents _____.
(1) a group of closely related families
(2) an individual plant or animal
(3) a population of plants and animals
(4) a group of closely related species

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107. The plastid that stores xanthophyll is known as _____ .

(1) amyloplast
(2) chloroplast
(3) chromoplast
(4) aleuroplast

108. In water, frogs respire using _____.

(1) trachea
(2) skin
(3) buccal cavity
(4) lungs

109. Which of the following is **not** a characteristic of chordates ?

(1) Presence of post anal part (tail)
(2) Presence of notochord
(3) Central nervous system is dorsal
(4) Absence of gills

110. Smooth endoplasmic reticulum _____.

(1) is a site for the synthesis of carbohydrates
(2) has ribosomes attached to its surface
(3) is the major site for the synthesis of lipids
(4) is actively involved in protein synthesis

111. Which of the following are characteristics of prokaryotic cells ?

(a) Ribosomes are made of 50S and 30S subunits
(b) They can have plasmids
(c) They contain mesosome
(d) They have peroxisomes

Choose the **correct** answer from the options given below :

(1) (a), (b) and (c) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (c) and (d) only

112. Match **List-I** with **List-II**.

List-I	List-II
A. Cristae	I. Flat membrane sacs in stroma of chloroplast
B. Cisternae	II. Infoldings in mitochondria
C. Thylakoids	III. Cell membrane
D. Phospholipid	IV. Disc shaped sacs in the Golgi apparatus

Choose the **correct** answer from the options given below :

(1) A-IV, B-III, C-I, D-II
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-II, B-IV, C-III, D-I

113. Which of the following statements related to pituitary gland are **correct** ?

(a) It is divided anatomically into adenohypophysis and neurohypophysis
(b) It secretes follicle stimulating hormone
(c) It secretes melanocyte stimulating hormone
(d) It does not secrete prolactin

Choose the **correct** answer from the options given below :

(1) (b) and (c) only
(2) (a) and (b) only
(3) (a), (b) and (c) only
(4) (c) and (d) only

114. Which of the following statements regarding photorespiration are correct ?

(a) Do not occur in C3 plants
(b) CO_2 is consumed and O_2 is generated
(c) Phosphoglycolate is formed
(d) No synthesis of ATP and NADPH

Choose the **correct** answer from the options given below :

(1) (a) and (b) only
(2) (a) and (d) only
(3) (c) and (d) only
(4) (b) and (d) only

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115. Which of the following statements is **incorrect** ?
(1) Fibrinogen is produced from fibrin
(2) Blood coagulates in response to an injury
(3) Blood clot consists of fibrins
(4) Fibrin is produced from fibrinogen

116. Arrange the following taxonomic categories in ascending order.
(a) Genus (b) Class
(c) Order (d) Phylum
(e) Family (f) Kingdom
(g) Species
Choose the **correct** answer from the options given below :
(1) (f), (c), (b), (g), (d), (e), (a)
(2) (g), (a), (e), (c), (b), (d), (f)
(3) (a), (c), (d), (g), (f), (b), (e)
(4) (g), (c), (d), (b), (e), (a), (f)

117. Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.
(1) Production of glucose → role of air → release of oxygen → absorption spectra of chlorophyll a and b
(2) Absorption spectra of chlorophyll a and b → production of glucose → release of oxygen → role of air
(3) Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b
(4) Release of oxygen → production of glucose → absorption spectra of chlorophyll a and b → role of air

118. Match **List-I** with **List-II**.

List-I	**List-II**
A. Starch	I. Fights infection
B. Antibody	II. Energy storage
C. Concanavalin A	III. Glucose transport
D. Glut-4	IV. Lectin

Choose the **correct** answer from the options given below :
(1) A-I, B-II, C-III, D-IV
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-II, B-I, C-III, D-IV

119. The number of vertebrae in a human is _____ .
(1) 206 (2) 7
(3) 12 (4) 26

120. Endomembrane system includes _____ .
(1) Golgi complex, chloroplast, peroxisomes and vacuole
(2) endoplasmic reticulum, Golgi complex, lysosomes and vacuole
(3) endoplasmic reticulum, chloroplast, peroxisomes and vacuole
(4) mitochondria, chloroplast, peroxisomes and vacuole

121. Length of the stem at time 0 is 20 cm. The arithmetic growth rate is 30 cm per day. What is the length of the stem at the end of the 7th day ?
(1) 460 cm (2) 50 cm
(3) 170 cm (4) 230 cm

122. Match **List-I** with **List-II**.

List-I	**List-II**
A. Spherical	I. Vibrio
B. Rod	II. Cocci
C. Comma	III. Spirilla
D. Spirillum	IV. Bacilli

Choose the **correct** answer from the options given below :
(1) A-II, B-IV, C-I, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-II, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

123. The number of action potentials generated by sino-arterial node (SAN) in a healthy human is _____ per minute.
(1) 120 - 140 (2) 28 - 30
(3) 70 - 75 (4) 100 - 110

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124. Match **List-I** with **List-II.**

	List-I	List-II
A.	Family	I. Sapindales
B.	Genus	II. Dicotyledonae
C.	Class	III. Anacardiaceae
D.	Phylum	IV. Angiospermae
E.	Order	V. *Mangifera*

Choose the **correct** answer from the options given below :

(1) A-III, B-V, C-II, D-IV, E-I

(2) A-I, B-V, C-II, D-IV, E-III

(3) A-II, B-I, C-III, D-IV, E-V

(4) A-II, B-III, C-V, D-I, E-IV

125. Which of the following is **not** a part of human central neural system ?

(1) Pericardium

(2) Arachnoid

(3) Dura mater

(4) Pia mater

126. Given below are two statements :

Statement I : Chromosomes are fully condensed at the end of prophase I.

Statement II : Meiosis I resembles mitosis.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect, but **Statement II** is true

(2) Both **Statement I** and **Statement II** are true

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is correct, but **Statement II** is false

127. Match **List-I** with **List-II.**

List-I	List-II
A. Marginal placentation	I. *Argemone*
B. Axile placentation	II. Tomato
C. Parietal placentation	III. *Primrose*
D. Free central placentation	IV. Pea

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-I, D-III

(2) A-II, B-IV, C-I, D-III

(3) A-IV, B-II, C-III, D-I

(4) A-IV, B-III, C-I, D-II

128. Symbiotic association between fungi and algae are called _____ .

(1) chrysophytes (2) lichens

(3) sponges (4) mycorrhiza

129. Which of the following is **not** a prokaryote ?

(1) Fungi

(2) Bacteria

(3) Blue green algae

(4) Mycoplasma

130. Arrange the following elements in descending order of their contribution to percentage weight of the human body.

(a) Oxygen (b) Carbon

(c) Hydrogen (d) Nitrogen

Choose the **correct** answer from the options given below :

(1) (b), (a), (c), (d)

(2) (a), (b), (c), (d)

(3) (c), (a), (b), (d)

(4) (b), (c), (d), (a)

131. Which one of the following statements is **incorrect** ?

(1) β-cells of pancreas secrete insulin

(2) α-cells of pancreas secrete glucagon

(3) α-cells of pancreas secrete insulin

(4) Glucagon stimulates glycogenolysis

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132. Which of the following are characteristic features of Solanaceae family ?

(a) Flowers are bisexual and actinomorphic.

(b) Calyx have five sepals and are united

(c) Androecium have five stamens and are epipetalous

(d) Ovary is inferior

Choose the **correct** answer from the options given below :

(1) (b), (c) and (d) only

(2) (a), (b) and (c) only

(3) (d) only

(4) (a) and (b) only

133. Given below are two statements :

Statement I : When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.

Statement II : In phylum Echinodermata, both adults and larvae are radially symmetrical.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are correct

(3) Both **Statement I** and **Statement II** are incorrect

(4) **Statement I** is correct but **Statement II** is incorrect

134. The correct sequence of adult cell cycle phases is _____.

(1) S-M-G2-G1 (2) G1-G2-S-M

(3) G1-M-G2-S (4) G1-S-G2-M

135. In frogs, the number of pairs of cranial nerves arising from the brain are _____.

(1) 12

(3) 9 (2) 6

(4) 10

136. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.

Reason R : Isolation of genetic material needs disruption of cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is not correct but **R** is correct

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(4) **A** is correct but **R** is not correct

137. The method of directly of injecting a sperm into ovum in assisted reproductive technology is called :

(1) Embryo transfer (ET)

(2) Gamete intra fallopian transfer (GIFT)

(3) Zygote intra fallopian transfer (ZIFT)

(4) Intra cytoplasmic sperm injection (ICSI)

138. Adaptive radiation in placental mammals and Australian Marsupials leading to similarity between distant species is an example of _____.

(1) genetic drift

(2) divergent evolution

(3) convergent evolution

(4) founder effect

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139. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarf.

Reason R : Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is not correct but **R** is correct

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(4) **A** is correct but **R** is not correct

140. Arrange the following in descending order of number of species in the Amazonian rain forest.

(a) Plants
(b) Birds
(c) Fishes
(d) Invertebrates
(e) Mammals

Choose the **correct** answer from the options given below :

(1) (b) > (a) > (d) > (c) > (e)
(2) (c) > (b) > (d) > (e) > (a)
(3) (d) > (a) > (c) > (b) > (e)
(4) (e) > (b) > (a) > (c) > (d)

141. Sponges exchange O_2 with CO_2 by _____.

(1) gills
(2) simple diffusion over their entire body surfaces
(3) moist cuticle
(4) tracheal tubes

142. For a person with blood group 'O', which of the following is **not** a possible combination of parents' blood group genotypes ?

(1) Father : I^AI^B and Mother : I^Ai
(2) Father : I^Ai and Mother : I^Bi
(3) Father : I^Ai and Mother : I^Ai
(4) Father : I^Bi and Mother : I^Bi

143. Given below are two statements :

Statement I : Modern *Homo sapiens* arose in Australia and moved across continents.

Statement II : *Homo sapiens* arose around 75000 to 10000 years ago.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are correct

(3) Both **Statement I** and **Statement II** are incorrect

(4) **Statement I** is correct but **Statement II** is incorrect

144. Which of the following is used as an effective sedative and painkiller for treating post-surgery patients ?

(1) Anti-retroviral drugs
(2) Interferon
(3) Antibiotics
(4) Morphine

145. Which of the following plant produces non-albuminous seeds ?

(1) Pea (2) Wheat
(3) Maize (4) Barley

146. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.

Reason R : Goats were more efficient at browsing than Abingdon tortoise.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is not correct but **R** is correct

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(4) **A** is correct but **R** is not correct

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147. The covering of ovum at ovulation is _____.
(1) chorion
(2) endometrium
(3) zona radiata
(4) zona pellucida

148. Which of the following is used as a clot buster ?
(1) Statins
(2) Streptokinase
(3) Penicillin
(4) Cyclosporin A

149. Which of the following structure is **not** a part of the male reproductive system ?
(1) Infundibulum
(2) Rete testis
(3) Epididymis
(4) Vasa efferentia

150. Given below are two statements :
Statement I : Ovulation is caused by LH surge leading to rupture of Graafian follicles.
Statement II : Graafian follicle remaining after ovulation transform into corpus luteum and secretes large amount of estrogen.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) **Statement I** is incorrect but **Statement II** is correct
(2) Both **Statement I** and **Statement II** are correct
(3) Both **Statement I** and **Statement II** are incorrect
(4) **Statement I** is correct but **Statement II** is incorrect

151. The opening between the right atrium and the right ventricle is guarded by _____
(1) sino-atrial node
(2) bicuspid valve
(3) tricuspid valve
(4) semilunar valve

152. Which of the following is **not** evidence for evolution ?
(1) Divergent evolution of anatomical structures such as forelimbs
(2) Convergent evolution of traits like wings of birds and butterflies
(3) Paleontological evidence from fossil records
(4) Embryological support for evolution as proposed by Ernst Heckel

153. The inactive form of Bt toxin is converted to the active form in the insect gut _____.
(1) by nucleases
(2) due to alkaline pH
(3) due to acidic pH
(4) by proteases

154. Colostrum, secreted by mother during initial days of lactation, is abundant in _____.
(1) IgD
(2) IgG
(3) IgM
(4) IgA

155. Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the eggs ?
(1) Polar nucleus
(2) Antipodals
(3) Synergids
(4) Central cells

156. Which of the following disease is **not** sexually transmitted ?
(1) Genital warts
(2) Syphilis
(3) Tuberculosis
(4) Gonorrhoea

157. Which of the following statements about lac-operon is **correct** ?
(1) Galactose can act as an inducer of lac operon
(2) Gene *i* is constitutively expressed
(3) Lactose activates repressor to bind to the operator
(4) Genes *i*, *z*, *y* and *a* share single common promoter

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158. Match **List-I** with **List-II.**

List-I	List-II
A. Transformation	I. Restriction enzyme
B. Cloning site	II. Transfer DNA to host bacteria
C. Selection	III. Replication
D. Ori	IV. Antibiotic

Choose the **correct** answer from the options given below :

(1) A-IV, B-I, C-III, D-II

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-II, C-IV, D-III

(4) A-III, B-IV, C-II, D-I

159. A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1, the frequency of AA is _____.

(1) 0.99 (2) 0.01

(3) 0.02 (4) 0.10

160. Sperm motility is due to _____.

(1) muscular movement

(2) flagellar movement

(3) ciliary movement

(4) amoeboid movement

161. Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the per-capita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is _____.

(1) 100 million

(2) 1 million

(3) 5 million

(4) 10 million

162. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Forelimbs of human and bats are homologous.

Reason R : Forelimbs of humans and bats have similar anatomical structure.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is false but **R** is true

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) Both **A** and **R** are true, but **R** is not the correct explanation of **A**

(4) **A** is true but **R** is false

163. Muscle contraction is initiated by a signal sent by the central nervous system by the release of _____.

(1) cyclic adenine monophosphate

(2) acetyl choline

(3) acetyl coenzyme A

(4) cyclic guanine monophosphate

164. Which of the following hormone is **not** secreted by human placenta ?

(1) LH

(2) hCG

(3) Estrogen

(4) Progesterone

165. Which of the following statements is **correct** about *Plasmodium* ?

(1) Fertilization takes place in mosquito gut

(2) Reproduces sexually in liver cells

(3) Reproduces asexually in RBCs

(4) Gametocytes develop in mosquito gut

166. Which of the following are primary consumers in a food chain ?

(1) Carnivores

(2) Parasites

(3) Predators

(4) Herbivores

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167. Which of the following statements about the reabsorption process in Henle's loop are **correct** ?

(a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.

(b) Urine gets concentrated in Henle's loop.

(c) Reabsorption of Na^+ and water takes place in Henle's loop.

(d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop.

Choose the **correct** answer from the options given below :

(1) (a), (b) and (d) only

(2) (a) and (b) only

(3) (b), (c) and (d) only

(4) (a), (b) and (c) only

168. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The logistic growth model of populations is considered more realistic than the exponential growth model.

Reason R : Resources are finite.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is not correct but **R** is correct

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(4) **A** is correct but **R** is not correct

169. Which of the following is the **correct** order of arrangement of vertebrate column from the head to toe ?

(1) Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

(2) Cervical vertebra, thoracic vertebra, sacrum, lumbar vertebra

(3) Sacrum, lumbar vertebra, thoracic vertebra, cervical vertebra

(4) Cervical vertebra, lumbar vertebra, thoracic vertebra, sacrum

170. Match **List-I** with **List-II**.

List-I	**List-II**
A. Both species are harmed	I. Predation
B. One species is harmed and the other is benefited	II. Mutualism
C. Both species are benefited	III. Competition
D. One is benefited while the other has no effect	IV. Commensalism

Choose the **correct** answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-I, C-IV, D-III

171. If the diploid chromosome number of typical angiosperm is 36, what would be the chromosome number in its endosperm ?

(1) 72 (2) 18

(3) 36 (4) 54

172. Which of the following enzymes synthesizes precursor mRNA ?

(1) DNA polymerase

(2) RNA polymerase I

(3) RNA polymerase II

(4) RNA polymerase III

173. Given below are two statements :

Statement I :

Plasmids are autonomously replicating DNA.

Statement II :

Plasmids are extrachromosomal DNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are correct

(3) Both **Statement I** and **Statement II** are incorrect

(4) **Statement I** is correct but **Statement II** is incorrect

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174. How many theca are present in each lobe of a typical bilobed angiosperm anther ?
(1) 12 (2) 2
(3) 6 (4) 8

175. Natural selection can lead to _____.
(a) stabilisation
(b) genetic drift
(c) directional change
(d) disruption
Choose the **correct** answer from the options given below :
(1) (a) and (c) only
(2) (a) only
(3) (a), (c) and (d) only
(4) (a), (b), (c) and (d)

176. Which of the following statements are **correct** ?
(a) Energy flow from producers to consumers is unidirectional
(b) Energy pyramid can never be inverted
(c) Transfer of energy follows the 1% law
Choose the **correct** answer from the options given below :
(1) (b) and (c) only
(2) (a), (b) and (c)
(3) (a) and (b) only
(4) (a) and (c) only

177. Match **List-I** with **List-II**.

List-I	List-II
A. Excess growth hormone	I. Reabsorption of water and electrolytes in kidney
B. Luteinizing hormone	II. Contraction of uterus during child birth
C. Vasopressin	III. Acromegaly
D. Oxytocin	IV. Ovulation

Choose the **correct** answer from the options given below :
(1) A-IV, B-III, C-I, D-II
(2) A-III, B-IV, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-IV, C-I, D-III

178. Which of the following are secondary lymphoid organs ?
(a) Bone marrow
(b) Tonsils
(c) Spleen
(d) Thymus
Choose the **correct** answer from the options given below :
(1) (a) and (d) only
(2) (a) and (b) only
(3) (b) and (c) only
(4) (b) and (d) only

179. During PCR, primers bind to the DNA strands in the _____ step.
(1) ligation
(2) denaturation
(3) extension
(4) annealing

180. Given below are two statements :
Statement I : Down's syndrome is caused by the absence of one of the X-chromosomes.
Statement II : Turner's syndrome is caused by the presence of an additional copy of the chromosomes.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) **Statement I** is incorrect but **Statement II** is correct
(2) Both **Statement I** and **Statement II** are correct
(3) Both **Statement I** and **Statement II** are incorrect
(4) **Statement I** is correct but **Statement II** is incorrect

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Test Booklet Code

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SUSHRUT

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4. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
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Name of the Candidate (in Capitals) : SWARAKSHI MISHRA

Roll Number : in figures 2302113266

: in words 2 million thirty crore twenty one lakh thirteen thousand two hundred sixty six

Centre of Examination (in Capitals) : GIRLS GOVT GIRLS SR SEC SCHOOL BADSHAHPUR

Candidate's Signature : Swarakshi **Invigilator's Signature :** Babita

Facsimile signature stamp of Centre Superintendent : Prerna Devi Principal GGSSS Badshahpur

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Name of the Candidate (in Capitals) : [redacted]

Roll Number : in figures [redacted]

: in words [redacted]

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Candidate's Signature : [redacted] **Invigilator's Signature :** [redacted]

Facsimile signature stamp of Centre Superintendent : Prerna Devi Principal GGSSS Badshahpur

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1. An ac voltage $V=220\sin(2\times10^3t)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :
(Given : L = 10 mH, C = 25μF, R = 100 Ω)
(1) 11.0 A (2) 22.0 A
(3) 2.2 A (4) 5.5 A

2. The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, then the correct option is :
(1) $n_A=\frac{1}{4}n_B$ (2) $n_A=\frac{1}{2}n_B$
(3) $n_A=n_B$ (4) $n_A=2n_B$

3. A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period 2T. The value of ρ_2/ρ_1 is :
(1) 1/2 (2) 1/4
(3) 4 (4) 2

4. Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is k Nm^{-1}. At a given instant, the extension of the spring is x meter and the speed of the particle is v ms^{-1}. On the x - v plane, if the graph of v as a function of x is a circle, then the correct option is :
(1) $k=m^2$ (2) $k=\sqrt{m}$
(3) $k=\frac{1}{m}$ (4) $k=m$

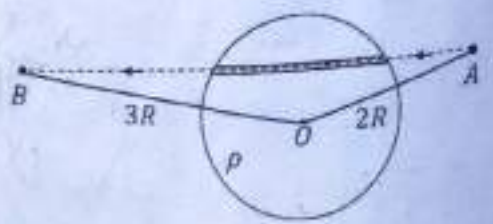
5. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma=5/3$) decreases from 60K to 50K. The work done by the gas in the process is :
(Take the universal gas constant as $R=8.3$ J mol^{-1} K^{-1})
(1) 124.5 J (2) 166 J
(3) 41.5 J (4) 83 J

6. The following table presents the part of the electromagnetic spectrum and the corresponding major applications.

Part of the electromagnetic spectrum		Applications	
P	Microwave	I	For purifying the water
Q	UV rays	II	For warming the food
R	Gamma rays	III	For AM and FM communication systems
S	Radio wave	IV	For treating the Cancer cells

The **correct** option is :
(1) P-II, Q-I, R-IV, S-III
(2) P-II, Q-IV, R-III, S-I
(3) P-I, Q-II, R-III, S-IV
(4) P-I, Q-IV, R-II, S-III

7. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is :
(ϵ_0 is the permittivity of vacuum)



(1) 9 (2) 18
(3) 2 (4) 6

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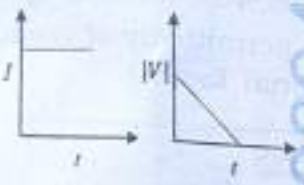
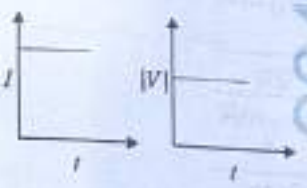
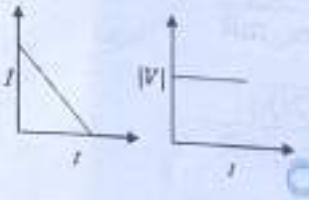
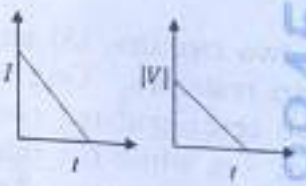
8. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by :

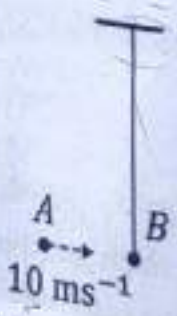
(1) 

(2) 

(3) 

(4) 

9. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed 10 ms^{-1} hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ ms^{-2}$ and neglecting the size of the bob, the value of h is :



(1) 5 (2) 2.5

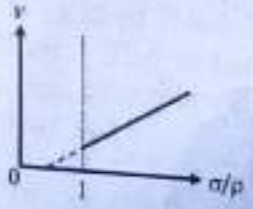
(3) 8 (4) 7

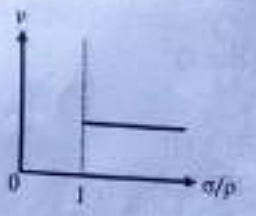
10. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_p/C_v of the gas is 8/7, then the value of f is :

(1) 2 (2) 1

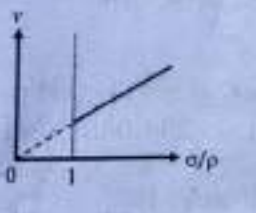
(3) 4 (4) 3

11. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by :

(1) 

(2) 

(3)

(4) 

12. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to :

(1) R^2 (2) R^3

(3) $R^{1/2}$ (4) $R^{3/2}$

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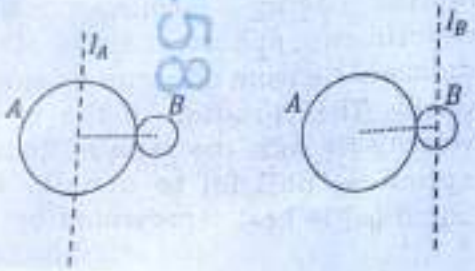
13. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius r < R and mass m < M. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is :



(1) $(m-M)(R-r)^2$

(2) 0

(3) $(M-m)(R+r)^2$

(4) $(m-M)(R+r)^2$

14. Consider the following nuclear reaction :

$$^{238}U \rightarrow {}^{234}Th + {}^{4}He$$

Take masses of ^{238}U, ^{234}Th and ^{4}He as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is :

[Given : 1 u = 931.5 MeV c^{-2}]

(1) 3736 (2) 3740

(3) 3726 (4) 3730

15. Consider a fixed uniformly charg insulating sphere with radius R a total charge +Q. A point charge (q<<Q) with mass m is released fr rest at a distance of 3R from the cen of the charged sphere. When the po charge reaches the surface of t sphere, its speed is :

(ϵ_0 is the permittivity of vacuum, neg gravitational forces).

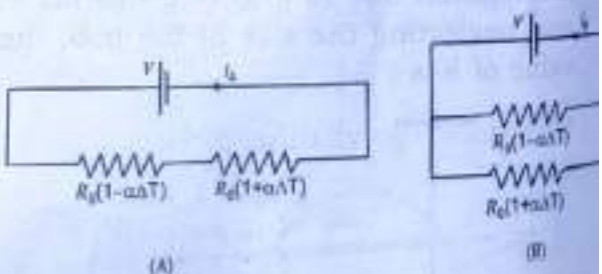
(1) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$

(2) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$

(3) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$

(4) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$

16. Consider two circuits, (A) and (B), ea having two resistors. One of them h a positive temperature coefficient resistance, $+\alpha$, while the other one h a negative temperature of coefficie $-\alpha$, as shown in the figure. The curr through these circuits are denoted I_A and I_B. At initial temperature, t resistance of the two resistors is As the temperature is increased, t correct option that describes t variation of current in these circu is :



(1) I_A increases while I_B decreases

(2) both I_A and I_B remain constant

(3) I_A remains constant while increases

(4) I_A decreases while I_B increases

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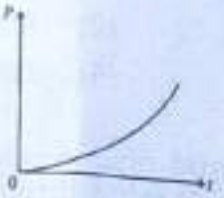
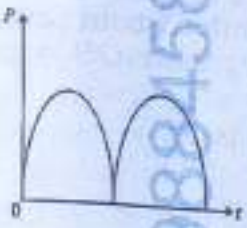
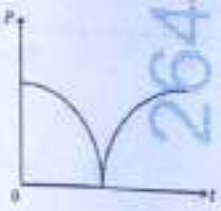
17. A conducting loop of finite resistance lies on the x - y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :

(1)

(2) 

(3) 

(4) 

18. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is :

(Take speed of light $= 3 \times 10^8$ ms^{-1}, charge of electron $= -1.6 \times 10^{-19}$ C and mass of electron $= 9 \times 10^{-31}$ kg)

(1) 225 (2) 275

(3) $\frac{2}{450}$ (4) $\frac{1}{250}$

19. Consider that σ_s, k_B & b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is :

(1) $[L^{-1}T^{-1}K^{-3}]$

(2) $[L^{-1}T^{-1}K^{-4}]$

(3) $[L^{-1}T^{-1}K^{-2}]$

(4) $[L^{-1}K^{-2}]$

20. Two infinitely long parallel conducting wires A and B carry currents I and 2I, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is :

[g is the acceleration due to gravity and μ_0 is the permeability of free space.]

(1) $\frac{2\mu_0 I^2}{\pi\lambda g}$ (2) $\frac{4\mu_0 I^2}{\pi\lambda g}$

(3) $\frac{\mu_0 I^2}{2\pi\lambda g}$ (4) $\frac{\mu_0 I^2}{\pi\lambda g}$

21. Which of the following measurements require 'index correction' ?

(1) Measurement of focal length of lenses using optical bench

(2) Measurement of speed of sound using resonance tube

(3) Measurement of resistance of a wire using meter bridge

(4) Measurement of gravitational acceleration using simple pendulum

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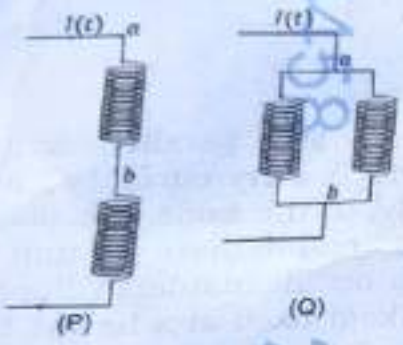
22. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is :

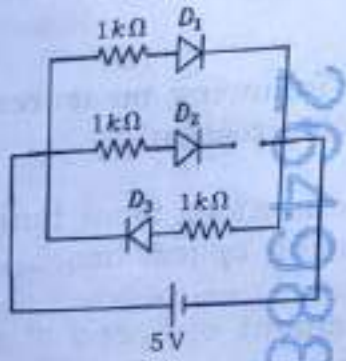
[Neglect the effect of mutual inductance.]



(1) 1 (2) 2

(3) 1/4 (4) 1/2

23. Three identical p-n junction diodes D_1, D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is :



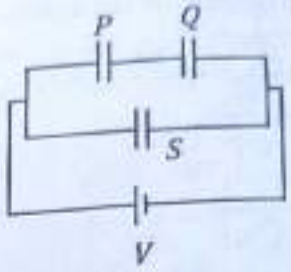
(1) $W_3 > W_2 > W_1$

(2) $W_2 > W_1 = W_3$

(3) $W_1 > W_2 > W_3$

(4) $W_3 = W_1 > W_2$

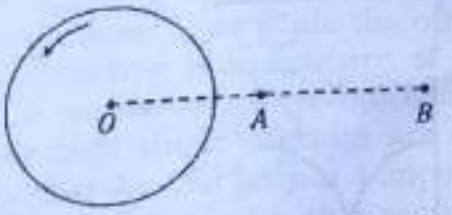
24. Three identical capacitors P, Q and S each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is



(1) 1/2 (2) 1/6

(3) 2/3 (4) 1/3

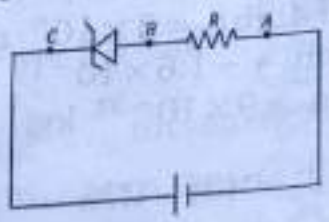
25. A thin horizontal disc is rotating abo a vertical axis passing through its fi centre O. Its angular momentum is and L_B computed about points A and respectively, with $OB = 2 \times OA$. T value of $\frac{L_A}{L_B}$ is :



(1) 1 (2) 2

(3) $\frac{1}{4}$ (4) $\frac{1}{2}$

26. An ideal Zener diode with breakd voltage of -3 V is reverse biased wi negative input voltage $V_i = -5$ V. magnitude of voltage difference betw points B and A is :



(1) 1 V (2) 0 V

(3) 3 V (4) 2 V

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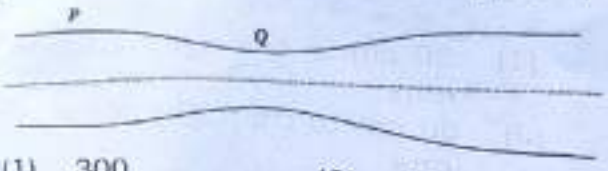
27. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is 15 Nm^{-2}. The area of cross-section at P and Q are 40 cm^2 and 20 cm^2, respectively. The rate of flow of water through the pipe, in cm^3s^{-1}, is :

[Take density of water = 1000 $kg\ m^{-3}$]



(1) 300 (2) 400

(3) 100 (4) 200

28. A particle of mass M moves along a horizontal x axis from $x=0$ to $x=L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x)=\mu_0-\alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L)=0$. The total work done by the frictional force during the motion is $n\mu_0MgL$, where g is the acceleration due to gravity. The value of n is :

(1) $\frac{1}{3}$ (2) $\frac{1}{2}$

(3) 3 (4) 1

29. A ray of light with wavelength λ is incident on three different photo-electric cells namely 1, 2 and 3. The threshold wavelength of these photo-electric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ and $\lambda_3 >> \lambda$. The correct option is :

(1) $V_1 > V_2, V_3 = 0$

(2) $V_1 < V_2, V_3 = 0$

(3) $V_1 = 0, V_2 < V_3$

(4) $V_1 = 0, V_2 > V_3$

30. Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6} \times 10^5\ ms^{-1}$. The radius of the orbit is $x \times 10^{-9}$ m. The value of x is :

[Take the mass of electron to be 9×10^{-31} kg, charge of electron $= -1.6 \times 10^{-19}$ C and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ N $m^2\ C^{-2}$]

(1) 2 (2) 1

(3) 4 (4) 3

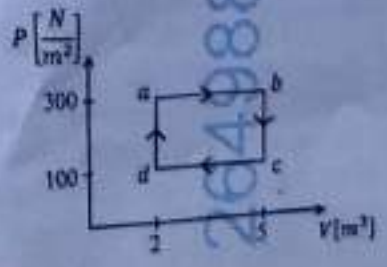
Speed = 10 m/s. $g = 10\ m/s^2$

31. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed 10 ms^{-1}, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be 10 ms^{-2}, the value of θ is :

(1) $\tan^{-1}(\sqrt{3}/2)$ (2) $\tan^{-1}(2\sqrt{3})$

(3) $\tan^{-1}\left(\frac{1}{5}\right)$ (4) $\tan^{-1}\left(\frac{2}{5}\right)$

32. One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is :



(1) 600 J (2) 800 J

(3) 400 J (4) 500 J

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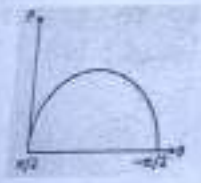
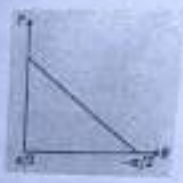
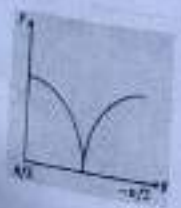
33. A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point $A\left(\theta=\frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point $B\left(\theta=-\frac{\pi}{2}\right)$. During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ ?

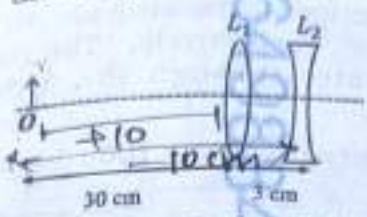
(1) 

(2) 

(3)

(4) 

34. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths $+10$ cm and -10 cm, respectively. The position of the image formed is :



(1) 30 cm to the right of the concave lens

(2) 60 cm to the right of the concave lens

(3) 20 cm to the left of the concave lens

(4) 60 cm to the left of the concave lens

35. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2=\frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is :

(1) $\sqrt{2}$ (2) 2

(3) $\frac{1}{\sqrt{2}}$ (4) 1

36. An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\epsilon_r=9$, has the electric field, $E_x=E_0 \sin(kz-2\pi\times10^6 t)$ Vm^{-1} where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is :

(1) The magnetic field is given by the relation $B_y=\frac{B_0}{v}\sin(kz-2\pi\times10^6 t)$ where v is the speed of the electromagnetic wave inside the medium

(2) The direction of propagation of the electromagnetic wave is along $+z$

(3) The speed of the electromagnetic wave inside the medium is 10^8 ms^{-1}

(4) The wavelength of the electromagnetic wave inside the medium is 300 m

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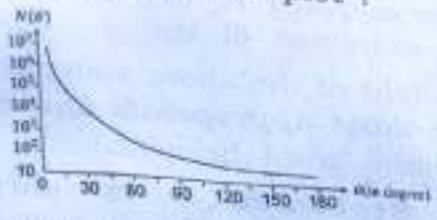
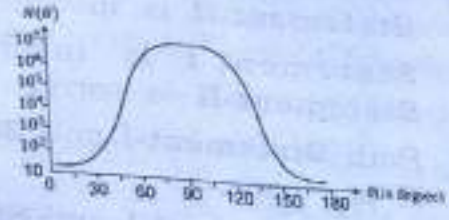
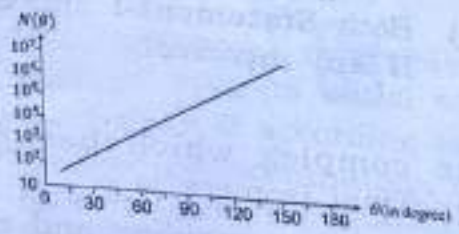
37. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1\ ms^{-2}$, $\beta = 6\ ms^{-1}$ and $\gamma = 5$ m. The average speed of the particle, in ms^{-1}, from $t = 0$ to $t = 6$ s is :

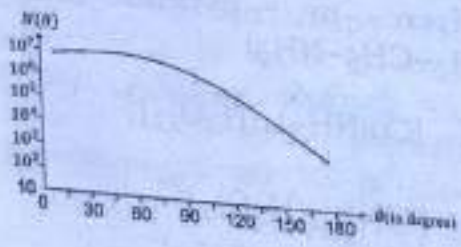
(1) 3 (2) 0
(3) 12 (4) 6

38. In Geiger-Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the correct plot ?

(1) 

(2) 

(3) 

(4) 

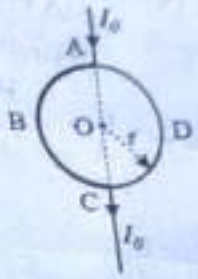
39. For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is :

(1) 1 (2) $\frac{3}{5}$
(3) $\frac{5}{9}$ (4) $\frac{9}{5}$

40. Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

(1) $(\mu_0/2\pi)n^2r^2l$ (2) $2\mu_0\pi n^2r^2l$
(3) $\mu_0\pi n^2r^2l$ (4) $\mu_0 n^2r^2l$

41. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the center O of the loop is :



(1) $\frac{\mu_0 I_0}{2r}$ (2) $\frac{\mu_0 I_0}{2\pi r}$
(3) $\frac{\mu_0 I_0}{12r}$ (4) $\frac{\mu_0 I_0}{4r}$

42. The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is :

(1) $4\pi R^3\alpha\Delta T$ (2) $6\pi R^3\alpha\Delta T$
(3) $2\pi R^3\alpha\Delta T$ (4) $3\pi R^3\alpha\Delta T$

43. One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of a wire to be 1 cm, the actual length of the wire is :

(1) 1.00 cm (2) 1.04 cm
(3) 0.60 cm (4) 0.96 cm

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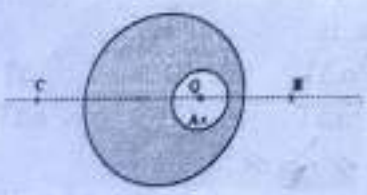
44. Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5, respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm, is :

(1) 2 (2) 3
(3) 0 (4) 1

45. A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B and E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is :



(1) $E_A = 0,\ E_B > E_C$
(2) $E_A \neq 0,\ E_B < E_C$
(3) $E_A = 0,\ E_B = E_C$
(4) $E_A \neq 0,\ E_B = E_C$

46. The lanthanide ion having four unpaired electrons is

(Given : Atomic numbers of Ce = 58, Nd = 60, Tb = 65 and Ho = 67)

(1) Tb^{3+} (2) Ho^{3+}
(3) Nd^{3+} (4) Ce^{3+}

47. The standard electrode potential for the half-cell reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$ at 298 K is

(Given: $E^\circ(Fe^{3+}/Fe) = -0.04$ V
$E^\circ(Fe^{2+}/Fe) = -0.44$ V at 298 K)

(1) −0.48 V (2) +0.92 V
(3) +0.40 V (4) +0.76 V

48. Given below are two statements:

Statement-I : Heating NaCl with concentrated H_2SO_4 and MnO_2 results in oxidation of Mn.

Statement-II : Heating NaI with concentrated H_2SO_4 and MnO_2 results in reduction of Mn.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **Statement-I** is correct but **Statement-II** is incorrect.
(2) **Statement-I** is incorrect but **Statement-II** is correct.
(3) Both **Statement-I** and **Statement-II** are correct.
(4) Both **Statement-I** and **Statement-II** are incorrect.

49. The complex which has *facial* and *meridional* isomers is

(Given : py = pyridine and en = CH_2–CH_2–NH_2)

(1) $[Co(NH_3)_4(H_2O)_2]^{3+}$
(2) $[Ni(en)_2(H_2O)_2]^{2+}$
(3) $[Cr(py)_3(Cl)_3]$
(4) $[Cr(H_2O)_6]^{3+}$

50. Among the following, the compound having conjugated double bonds is

(1) hepta-1,5-diene
(2) hepta-1,6-diene
(3) hepta-1,3-diene
(4) hepta-1,4-diene

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51. Match the species in **List I** with their geometry in **List II**

	List I	List II
A.	PCl_5	I. Tetrahedral
B.	BrF_5	II. Square Planar
C.	BF_4^-	III. Trigonal bipyramidal
D.	$[Ni(CN)_4]^{2-}$	IV. Square pyramidal

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-IV, C-I, D-II

52. The amino acid that gives a red-blood colour on treating its sodium fusion extract with sodium nitroprusside is

(1) methionine (2) serine

(3) leucine (4) threonine

53. A protein undergoes reversible thermal denaturation from its initial state **N** to denatured state **D** according to $\mathbf{N} \rightleftharpoons \mathbf{D}$. At 60 °C, the concentrations of both **N** and **D** are equal at equilibrium, and the standard enthalpy change of denaturation is 666 kJ mol^{-1}. The standard entropy change (ΔS° in kJ $K^{-1}mol^{-1}$) of the protein upon denaturation at 60 °C is closest to

(1) 333.0 (2) 11.1

(3) 2.0 (4) 2000.0

54. $2A \xrightarrow{k} B$ is a zero-order reaction, where $k = 1.0$ mol L^{-1} min^{-1}. If the initial concentration of A is 2 M, then the time taken to complete 75% of the reaction will be

(1) 1.0 min (2) 2.0 min

(3) 1.5 min (4) 0.75 min

55. Given below are two statements:

Statement-I : $[Fe(ox)_3]^{3-}$ is chiral.

Statement-II : *trans*-$[Cr(H_2O)_2(ox)_2]^{0}$ is chiral.

(Given : oxH_2 = HOOC – COOH)

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **Statement-I** is correct but **Statement-II** is incorrect.

(2) **Statement-I** is incorrect but **Statement-II** is correct.

(3) Both **Statement-I** and **Statement-II** are correct.

(4) Both **Statement-I** and **Statement-II** are incorrect.

56. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Generally, 3*d* transition metals have high melting points.

Reason R : Involvement of 3*d*-electrons in addition to 4*s*-electrons in the interatomic metallic bonding.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **A** is correct but **R** is not correct.

(2) **A** is not correct but **R** is correct.

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

57. Among the species given below, the spin-only magnetic moment is highest for

(Given : Atomic number of Ti = 22, Mn = 25, Fe = 26 and Co = 27)

(1) $[Co(NH_3)_6]^{3+}$ (2) $[Ti(H_2O)_6]^{3+}$

(3) $[Mn(CN)_6]^{3-}$ (4) $[Fe(CN)_6]^{3-}$

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58. According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is

(1) $Cl^- > H_2O > NH_3 > CO$
(2) $Cl^- > NH_3 > H_2O > CO$
(3) $CO > NH_3 > H_2O > Cl^-$
(4) $CO > H_2O > NH_3 > Cl^-$

59. In potash alum, the ratio of K^+ and SO_4^{2-} ions is

(1) 2:3 (2) 3:2
(3) 1:2 (4) 2:1

60. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The first ionization enthalpy of O is lower than that of N and F.

Reason R : The loss of an electron from O leads to stable half-filled *p* orbital.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) **A** is correct but **R** is not correct.
(2) **A** is not correct but **R** is correct.
(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

61. Given below are two statements:

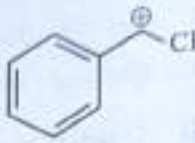
Statement-I : Oxidation of *p*-nitrotoluene with acidic $KMnO_4$ gives an acid that is stronger than benzoic acid.

Statement-II : Reduction of *p*-nitrotoluene with Sn/HCl followed by neutralization gives an amine that is more basic than aniline.

In light of the above statements, choose the ***most appropriate*** answer from the options given below.

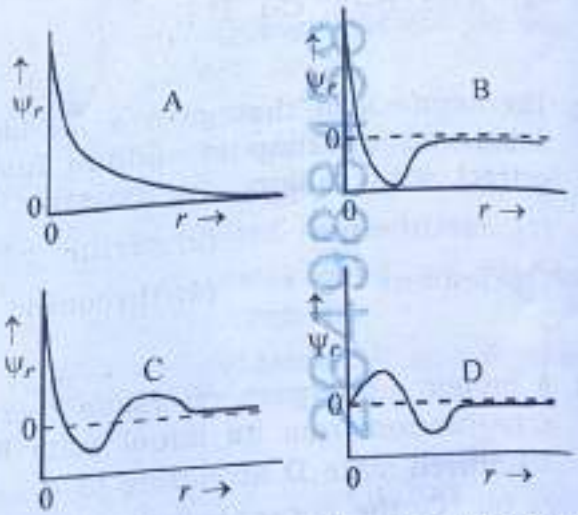
(1) **Statement-I** is correct but **Statement-II** is incorrect.
(2) **Statement-I** is incorrect but **Statement-II** is correct.
(3) Both **Statement-I** and **Statement-II** are correct.
(4) Both **Statement-I** and **Statement-II** are incorrect.

62. The following carbocation is stabilized by the interaction of the empty *p* orbital with



(1) empty σ^* and filled π orbitals
(2) empty σ^* and empty π^* orbitals
(3) filled σ and filled π orbitals
(4) empty σ and empty π^* orbitals

63. Consider the following schematic plots of orbital wavefunction (ψ_r) against distance (r) from the nucleus.



The figure representing two radial nod in the orbital is

(1) C (2) D
(3) A (4) B

64. The correct order of solubility of t given salts in water at 298 K is

Salt	K_{sp} at 298 K
AgBr	5.0×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

(1) $Hg_2Cl_2 > AgBr > Zn(OH)_2$
(2) $Zn(OH)_2 > AgBr > Hg_2Cl_2$
(3) $Hg_2Cl_2 > Zn(OH)_2 > AgBr$
(4) $AgBr > Zn(OH)_2 > Hg_2Cl_2$

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65. A 1:3 electrolyte in an aqueous solution is

(1) $[Co(NH_3)_6]Cl_3$

(2) $[Co(NH_3)_3(NO_2)_3]$

(3) $[CoCl_2(NH_3)_4]Cl$

(4) $[CoCl(NH_3)_5]Cl_2$

66. The correct statement about peptides and proteins is

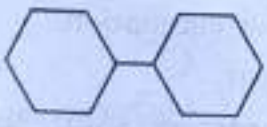
(1) In β-pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

(2) In α-helices, the polypeptide chain is twisted into a left-handed screw (helix) through intramolecular hydrogen bonds.

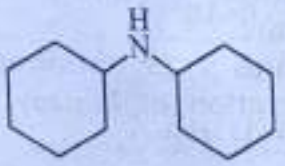
(3) Tertiary structure of proteins has two or more polypeptide subunits.

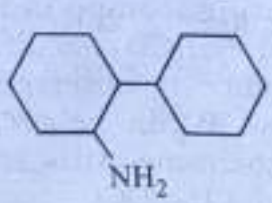
(4) Only the proteins having a quaternary structure are biologically active.

67. One of the products formed in the following reaction is

(1) 

(2)

(3) 

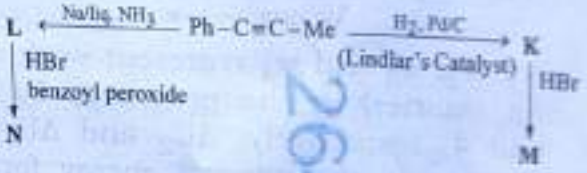
(4) 

68. In an acidic medium, 10 mL of 0.25 M oxalic acid is titrated with $KMnO_4$ solution. If the volume of $KMnO_4$ solution required to reach end point is 10 mL, the strength of the $KMnO_4$ solution is

(1) 0.25 M (2) 0.15 M

(3) 0.10 M (4) 0.20 M

69. Consider the following reaction sequences and choose the correct option.



(1) **M** and **N** are geometrical isomers

(2) **M** and **N** are stereoisomers

(3) **K** and **L** are geometrical isomers

(4) **K** and **L** are enantiomers

70. Arrange the following compounds in the increasing order of polarity

A. $CH_3CH_2OCH_2CH_3$

B. CH_3CH_2OH

C. CH_3COCH_3

D. CH_3COOH

Choose the correct answer from the options given below.

(1) C < A < B < D

(2) A < C < B < D

(3) A < B < C < D

(4) C < A < D < B

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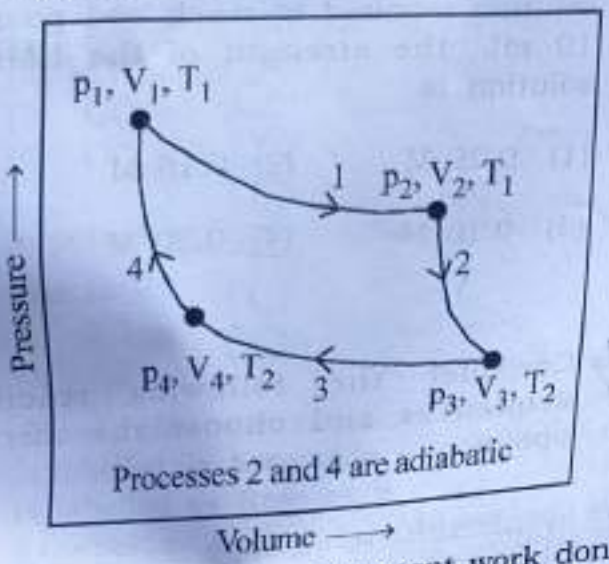
71. Consider the reversible processes for 1.0 mol of an ideal gas as shown in the figure.



w_1, w_2, w_3 and w_4 represent work done (in calories) in the processes 1, 2, 3 and 4, respectively; ΔU_2 and ΔU_4 are changes in the internal energy for the processes 2 and 4, respectively.

[use $R = 2$ cal K^{-1} mol^{-1}]

The correct option is

(1) $w_1 + w_2 = 2T_1 \ln \frac{V_2}{V_1}$

(2) $w_1 + w_2 + w_3 + w_4 = 0$

(3) $w_1 + w_3 = -2T_1 \ln \frac{V_2}{V_1} - 2T_2 \ln \frac{V_4}{V_3}$

(4) $w_2 + w_4 = \Delta U_2 - \Delta U_4$

72. The green paramagnetic species formed by heating $KMnO_4$ at 513 K is

(1) MnO (2) KO_2

(3) K_2MnO_4 (4) Mn_3O_4

73. The numbers 17.0145 and 21.0235 were rounded to three figures after the decimal point. The resulting numbers, respectively, are

(1) 17.014 and 21.024

(2) 17.015 and 21.024

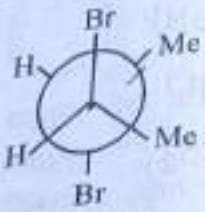
(3) 17.014 and 21.023

(4) 17.015 and 21.023

74. Given below are two statements:

Statement I:

trans-But-2-ene upon treatment with Br_2 in CCl_4 gives the following product



Statement II:

cis-But-2-ene upon treatment with alkaline $KMnO_4$ gives the following product

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

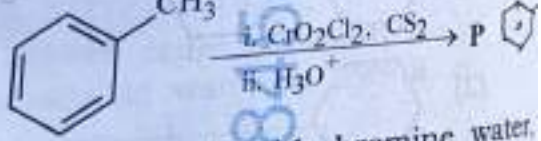
(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

75. Consider the following reaction, and choose the correct option.



(1) On treatment with bromine water, compound **P** gives a white precipitate.

(2) Compound **P** is obtained by the hydrogenation of benzoyl chloride with Pd on $BaSO_4$.

(3) On treating compound **P** with saturated $NaHCO_3$ solution, brisk effervescence is observed.

(4) Compound **P** can be prepared by treating benzene with anhydrous $AlCl_3$ and CH_3COCl.

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76. Match the vitamins in **List I** with their sources in **List II**

List I	List II
A. vitamin A	I. meat
B. vitamin B_{12}	II. sunflower oil
C. vitamin E	III. green leafy vegetables
D. vitamin K	IV. carrots

Choose the correct answer from the options given below.

(1) A-IV, B-II, C-I, D-III

(2) A-III, B-I, C-IV, D-II

(3) A-II, B-III, C-IV, D-I

(4) A-IV, B-I, C-II, D-III

77. The amount of carbon dioxide evolved upon complete combustion of 116 g of *n*-butane is

(Given: atomic mass in amu H = 1, C = 12 and O = 16)

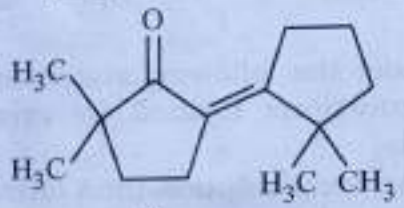
(1) 176 g

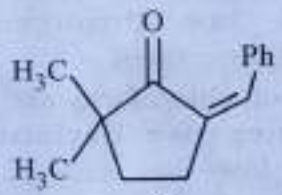
(2) 362 g

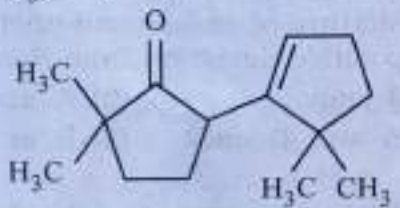
(3) 352 g

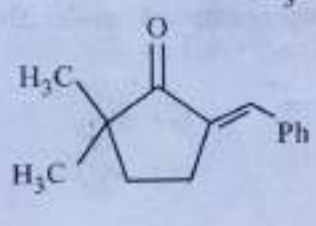
(4) 322 g

78. The compound that **CANNOT** be obtained from the aldol condensation reaction shown below, is

(1)


(2)


(3)


(4)


79. The correct statement is

(1) Magnesium has a maximum covalency of four.

(2) Aluminium has five valence orbitals.

(3) Boron has a maximum covalency of four.

(4) Beryllium has three valence orbitals.

80. The formula of tetraammineaquachloridocobalt(III) chloride is

(1) $[Co(NH_3)_4(H_2O)Cl]Cl$

(2) $[Co(NH_3)_4(H_2O)Cl]Cl_2$

(3) $[Co(NH_3)_4Cl_2] \times H_2O$

(4) $[Co(NH_3)_4]Cl_3 \times H_2O$

81. Assertion A :

For an ideal solution formed by mixing liquids **P** and **Q**, Δ_{mix} H = 0 and Δ_{mix} V = 0

Reason R :

No interactions occur between **P** and **Q**

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

(1) **A** is correct but **R** is not correct

(2) **A** is not correct but **R** is correct

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

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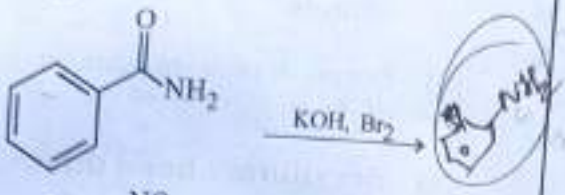
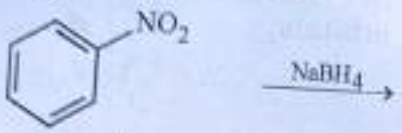
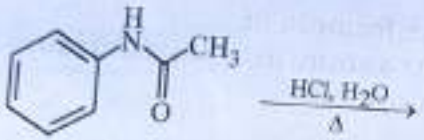
82. Identify the reactions which give aniline as the major product.

A.

B. 

C. 

D. 

Choose the correct answer from the options given below.

(1) A and C only (2) C and D only
(3) A and B only (4) B and D only

83. The correct decreasing order of oxidation state of the underlined atom in each molecule is

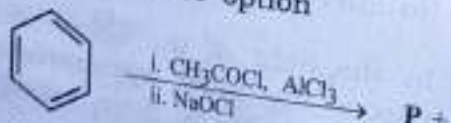
(1) $\underline{PbO_2} > \underline{N_2O_3} > \underline{SO_3}$

(2) $\underline{P_4O_6} > \underline{Cl_2O_7} > \underline{AlH_3}$

(3) $\underline{P_4O_{10}} > \underline{SO_3} > \underline{H_2O}$

(4) $\underline{N_2O_5} > \underline{Al_2O_3} > \underline{H_2S}$

84. For the following reaction sequence, choose the correct option

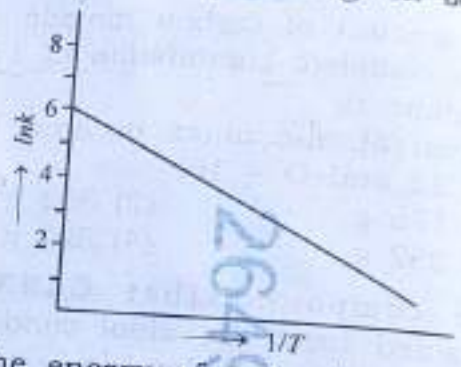


(1) If **P** gives a carboxylic acid on acidification, **Q** gives a poisonous gas on exposure to air and light.
(2) Both **P** and **Q** are carbonyl compounds.
(3) If **P** is the sodium salt of a carboxylic acid, **Q** is a primary alcohol.
(4) **P** and **Q** are aromatic compounds.

85. Two moles of an ideal gas undergo isothermal expansion from 10 L to 100 L at 300 K. The values of ΔS_{system} and $\Delta S_{surroundings}$ are
(R is universal gas constant)

(1) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 4.606\ R$
(2) $\Delta S_{system} = 4.606\ R$; $\Delta S_{surroundings} = -4.606\ R$
(3) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 0$
(4) $\Delta S_{system} = 4.606\ R$; $\Delta S_{surroundings} = 4.606\ R$

86. For an elementary chemical reaction the Arrhenius plot is given below.



If the energy of activation is 6.64 kJ mol^{-1} and R = 8.3 J K^{-1} mol^{-1}, the temperature at which the rate constant becomes e^2 min^{-1}, is

(1) 200 K (2) 250 K
(3) 125 K (4) 150 K

87. Consider the following statements about the solutions formed by mixing two liquids.

A. An ideal solution thus formed obeys Raoult's law throughout the composition range.
B. Mixture of chloroform and acetone shows negative deviation from Raoult's law.
C. Mixture of aniline and phenol shows positive deviation from Raoult's law.

(1) A only (2) A and C only
(3) A and B only (4) B and C only

88. The highest occupied molecular orbital for Ne_2 is

(1) π^*_{2p} (2) σ^*_{2p}
(3) π_{2p} (4) σ_{2p}

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89. For a salt **XY**, which is a strong electrolyte, the plot of Λ_m versus $\sqrt{c}$ has a slope of $-90.0\ S\ cm^2\ mol^{-3/2}\ L^{1/2}$ at 298 K. At 0.01 M concentration of **XY**, the value of Λ_m is $145.0\ S\ cm^2\ mol^{-1}$. The limiting molar conductivity of Y^- ion ($\lambda^0_{Y^-}$, in $S\ cm^2\ mol^{-1}$) at 298 K will be

(Given: $\lambda^0_{X^+} = 74.0\ S\ cm^2\ mol^{-1}$)

(1) 90.0 (2) 76.0
(3) 80.0 (4) 100.0

90. Among the following options, the correct trend in the electron gain enthalpy is,

(1) Cl > F > Br > I
(2) I > Br > Cl > F
(3) F > Cl > Br > I
(4) Br > Cl > F > I

91. Given below are two statements :

Statement I : Chromosomes are fully condensed at the end of prophase I.

Statement II : Meiosis I resembles mitosis.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct, but **Statement II** is false
(2) **Statement I** is incorrect, but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false

92. Which of the following is **not** a characteristic of chordates ?

(1) Absence of gills
(2) Presence of post anal part (tail)
(3) Presence of notochord
(4) Central nervous system is dorsal

93. Phyllotaxy is the pattern of arrangement of _____.

(1) fruits (2) sepals
(3) leaves (4) flowers

94. Which one of the following statements is **incorrect** ?

(1) Glucagon stimulates glycogenolysis
(2) β-cells of pancreas secrete insulin
(3) α-cells of pancreas secrete glucagon
(4) α-cells of pancreas secrete insulin

95. The plastid that stores xanthophyll is known as _____.

(1) aleuroplast (2) amyloplast
(3) chloroplast (4) chromoplast

96. Match **List-I** with **List-II.**

List-I	**List-II**
A. Fusion of protoplasms between gametes	I. Meiosis
B. Fusion of two nuclei	II. Plasmogamy
C. Generation of haploid spores	III. Karyogamy

Choose the **correct** answer from the options given below :

(1) A-III, B-II, C-I (2) A-I, B-III, C-II
(3) A-II, B-III, C-I (4) A-II, B-I, C-III

97. Given below are two statements :

Statement I : In gymnosperms, the male and female gametophytes remain within the sporangia.

Statement II : In gymnosperms, seeds are not covered.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect
(2) **Statement I** is incorrect but **Statement II** is correct
(3) Both **Statement I** and **Statement II** are correct
(4) Both **Statement I** and **Statement II** are incorrect

98. In water, frogs respire using _____.

(1) lungs (2) trachea
(3) skin (4) buccal cavity

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99. Mad cow disease is caused by _____.
(1) *Aspergillus sp.*
(2) *Mycoplasma sp.*
(3) prions
(4) viroids

100. Which of the following statements regarding photorespiration are correct ?
(a) Do not occur in C3 plants
(b) CO_2 is consumed and O_2 is generated
(c) Phosphoglycolate is formed
(d) No synthesis of ATP and NADPH
Choose the **correct** answer from the options given below :
(1) (b) and (d) only
(2) (a) and (b) only
(3) (a) and (d) only
(4) (c) and (d) only

101. Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.
(1) Release of oxygen → production of glucose → absorption spectra of chlorophyll a and b → role of air
(2) Production of glucose → role of air → release of oxygen → absorption spectra of chlorophyll a and b
(3) Absorption spectra of chlorophyll a and b → production of glucose → release of oxygen → role of air
(4) Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b

102. How many turns of Calvin cycle are required for the formation of three molecules of glucose ?
(1) 1 (2) 18
(3) 6 (4) 3

103. Mitochondrial inner membrane encloses_____.
(1) mucus
(2) aqueous humor
(3) matrix
(4) cytosol

104. Which of the following statements is **incorrect** ?
(1) Fibrin is produced from fibrinogen
(2) Fibrinogen is produced from fibrin
(3) Blood coagulates in response to injury
(4) Blood clot consists of fibrins

105. Sphenopsida class belongs _____.
(1) gymnosperms
(2) pteridophytes
(3) bryophytes
(4) angiosperms

106. Match **List-I** with **List-II.**

List-I		**List-II**
A.	Spherical	I. Vibrio
B.	Rod	II. Cocci
C.	Comma	III. Spirilla
D.	Spirillum	IV. Bacilli

Choose the **correct** answer from the options given below :
(1) A-II, B-I, C-IV, D-III
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-II, C-I, D-IV

107. Which of the following are characteristics of prokaryotic cells ?
(a) Ribosomes are made of 50S and 30S subunits
(b) They can have plasmids
(c) They contain mesosome
(d) They have peroxisomes
Choose the **correct** answer from the options given below :
(1) (a), (c) and (d) only
(2) (a), (b) and (c) only
(3) (b) and (c) only
(4) (a) and (c) only

108. Smooth endoplasmic reticulum _____.
(1) is actively involved in protein synthesis
(2) is a site for the synthesis of carbohydrates
(3) has ribosomes attached to its surface
(4) is the major site for the synthesis of lipids

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109. Given below are two statements :

Statement I : The class name Reptilia refers to creeping or crawling mode of locomotion.

Statement II : All organisms belonging to Reptilia have three chambered heart.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

110. In frogs, the number of pairs of cranial nerves arising from the brain are _____.

(1) 10 (2) 12

(3) 6 (4) 9

111. Cell theory was formulated by _____.

(1) Singer and Nicolson

(2) Antonie Von Leeuwenhoek

(3) Schleiden and Schwann

(4) Robert Brown

112. Which of the following statements related to pituitary gland are **correct** ?

(a) It is divided anatomically into adenohypophysis and neurohypophysis

(b) It secretes follicle stimulating hormone

(c) It secretes melanocyte stimulating hormone

(d) It does not secrete prolactin

Choose the **correct** answer from the options given below :

(1) (c) and (d) only

(2) (b) and (c) only

(3) (a) and (b) only

(4) (a), (b) and (c) only

113. Which of the following is **not** a prokaryote ?

(1) Mycoplasma

(2) Fungi

(3) Bacteria

(4) Blue green algae

114. Length of the stem at time 0 is 20 cm. The arithmetic growth rate is 30 cm per day. What is the length of the stem at the end of the 7th day ?

(1) 230 cm (2) 460 cm

(3) 50 cm (4) 170 cm

115. Which of the following is **not** a part of human central neural system ?

(1) Pia mater (2) Pericardium

(3) Arachnoid (4) Dura mater

116. Which of the following plant growth regulators promotes internode elongation prior to flowering in cabbage ?

(1) Indole butyric acid

(2) Ethephon

(3) Abscisic acid

(4) Gibberellin

117. Match **List-I** with **List-II.**

List-I	**List-II**
A. Marginal placentation	I. *Argemone*
B. Axile placentation	II. Tomato
C. Parietal placentation	III. *Primrose*
D. Free central placentation	IV. Pea

Choose the **correct** answer from the options given below :

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-II, C-I, D-III

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-II, C-III, D-I

118. Which of the following plant growth regulators is used as herbicide ?

(1) Abscisic acid (2) Gibberellin

(3) 2,4-D (4) Kinetin

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119. Given below are two statements :

Statement I : When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.

Statement II : In phylum Echinodermata, both adults and larvae are radially symmetrical.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

120. Match **List-I** with **List-II.**

List-I	**List-II**
A. Starch	I. Fights infection
B. Antibody	II. Energy storage
C. Concanavalin A	III. Glucose transport
D. Glut-4	IV. Lectin

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-I, B-II, C-III, D-IV

(3) A-I, B-II, C-IV, D-III

(4) A-II, B-I, C-IV, D-III

121. The number of vertebrae in a human is _____ .

(1) 26 (2) 206

(3) 7 (4) 12

122. Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700) ?

(1) Xanthophylls

(2) Carotenoids

(3) Chlorophyll b

(4) Chlorophyll a

123. Arrange the following taxonomic categories in ascending order.

(a) Genus (b) Class

(c) Order (d) Phylum

(e) Family (f) Kingdom

(g) Species

Choose the **correct** answer from the options given below :

(1) (g), (c), (d), (b), (e), (a), (f)

(2) (f), (c), (b), (g), (d), (e), (a)

(3) (g), (a), (e), (c), (b), (d), (f)

(4) (a), (c), (d), (g), (f), (b), (c)

124. Match **List-I** with **List-II.**

List-I	**List-II**
A. Family	I. Sapindales
B. Genus	II. Dicotyledonae
C. Class	III. Anacardiaceae
D. Phylum	IV. Angiospermae
E. Order	V. *Mangifera*

Choose the **correct** answer from the options given below :

(1) A-II, B-III, C-V, D-I, E-IV

(2) A-III, B-V, C-II, D-IV, E-I

(3) A-I, B-V, C-II, D-IV, E-III

(4) A-II, B-I, C-III, D-IV, E-V

125. Arrange the following elements in descending order of their contribution to percentage weight of the human body.

(a) Oxygen (b) Carbon

(c) Hydrogen (d) Nitrogen

Choose the **correct** answer from the options given below :

(1) (b), (c), (d), (a)

(2) (b), (a), (c), (d)

(3) (a), (b), (c), (d)

(4) (c), (a), (b), (d)

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126. Match **List-I** with **List-II.**

List-I		List-II
A. Cristae	I.	Flat membrane sacs in stroma of chloroplast
B. Cisternae	II.	Infoldings in mitochondria
C. Thylakoids	III.	Cell membrane
D. Phospholipid	IV.	Disc shaped sacs in the Golgi apparatus

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-III, D-I

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-IV, C-I, D-III

127. The number of action potentials generated by sino-arterial node (SAN) in a healthy human is _____ per minute.

(1) 100 - 110 (2) 120 - 140

(3) 28 - 30 (4) 70 - 75

128. Symbiotic association between fungi and algae are called _____.

(1) mycorrhiza (2) chrysophytes

(3) lichens (4) sponges

129. How many molecules of pyruvic acid are produced at the end of glycolysis from 206 molecules of glucose ?

(1) 103 (2) 412

(3) 206 (4) 309

130. Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans ?

(1) Femur-patella-tibia-tarsal

(2) Femur-tarsal-patella-tibia

(3) Femur-tibia-patella-tarsal

(4) Patella-femur-tibia-tarsal

131. Genus represents _____.

(1) a group of closely related species

(2) a group of closely related families

(3) an individual plant or animal

(4) a population of plants and animals

132. The correct sequence of adult cell cycle phases is _____.

(1) G1-S-G2-M (2) S-M-G2-G1

(3) G1-G2-S-M (4) G1-M-G2-S

133. Endomembrane system includes _____.

(1) mitochondria, chloroplast, peroxisomes and vacuole

(2) Golgi complex, chloroplast, peroxisomes and vacuole

(3) endoplasmic reticulum, Golgi complex, lysosomes and vacuole

(4) endoplasmic reticulum, chloroplast, peroxisomes and vacuole

134. Photorespiration reaction catalyzed by RuBisCo is shown below :

$RuBP + O_2 \rightarrow$ 3-Phosphoglycerate + X

Identify "X" from the given options :

(1) Oxaloacetate

(2) Malate

(3) Phosphoenolpyruvate

(4) 2-Phosphoglycolate

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135. Which of the following are characteristic features of Solanaceae family ?

(a) Flowers are bisexual and actinomorphic

(b) Calyx have five sepals and are united

(c) Androecium have five stamens and are epipetalous

(d) Ovary is inferior

Choose the **correct** answer from the options given below :

(1) (a) and (b) only

(2) (b), (c) and (d) only

(3) (a), (b) and (c) only

(4) (d) only

136. Sperm motility is due to _____.

(1) amoeboid movement

(2) muscular movement

(3) flagellar movement

(4) ciliary movement

137. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.

Reason R : Goats were more efficient at browsing than Abingdon tortoise.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is correct but **R** is not correct

(2) **A** is not correct but **R** is correct

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

138. Match **List-I** with **List-II**.

List-I	List-II
A. Both species are harmed	I. Predation
B. One species is harmed and the other is benefited	II. Mutualism
C. Both species are benefited	III. Competition
D. One is benefited while the other has no effect	IV. Commensalism

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-IV, C-II, D-I

(4) A-I, B-II, C-III, D-IV

139. Which of the following structure is **not** a part of the male reproductive system ?

(1) Vasa efferentia

(2) Infundibulum

(3) Rete testis

(4) Epididymis

140. Which of the following plant produces non-albuminous seeds ?

(1) Barley (2) Pea

(3) Wheat (4) Maize

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141. Which of the following statements about lac-operon is **correct** ?

(1) Genes *i*, *z*, *y* and *a* share single common promoter

(2) Galactose can act as an inducer of lac operon

(3) Gene *i* is constitutively expressed

(4) Lactose activates repressor to bind to the operator

142. Which of the following is used as an effective sedative and painkiller for treating post-surgery patients ?

(1) Morphine

(2) Anti-retroviral drugs

(3) Interferon

(4) Antibiotics

143. Which of the following is the **correct** order of arrangement of vertebrate column from the head to toe ?

(1) Cervical vertebra, lumbar vertebra, thoracic vertebra, sacrum

(2) Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

(3) Cervical vertebra, thoracic vertebra, sacrum, lumbar vertebra

(4) Sacrum, lumbar vertebra, thoracic vertebra, cervical vertebra

144. Muscle contraction is initiated by a signal sent by the central nervous system by the release of _____.

(1) cyclic guanine monophosphate

(2) cyclic adenine monophosphate

(3) acetyl choline

(4) acetyl coenzyme A

145. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Forelimbs of human and bats are homologous.

Reason R : Forelimbs of humans and bats have similar anatomical structure.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is true but **R** is false

(2) **A** is false but **R** is true

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) Both **A** and **R** are true, but **R** is not the correct explanation of **A**

pair of X chromosome

146. Given below are two statements :

Statement I : Down's syndrome is caused by the absence of one of the X-chromosomes.

Statement II : Turner's syndrome is caused by the presence of an additional copy of the chromosomes.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

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147. Which of the following statements about the reabsorption process in Henle's loop are **correct** ?

(a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.

(b) Urine gets concentrated in Henle's loop.

(c) Reabsorption of Na^+ and water takes place in Henle's loop.

(d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop.

Choose the **correct** answer from the options given below :

(1) (a), (b) and (c) only

(2) (a), (b) and (d) only

(3) (a) and (b) only

(4) (b), (c) and (d) only

148. If the diploid chromosome number of typical angiosperm is 36, what would be the chromosome number in its endosperm ?

(1) 54 (2) 72

(3) 18 (4) 36

149. Arrange the following in descending order of number of species in the Amazonian rain forest.

(a) Plants

(b) Birds

(c) Fishes

(d) Invertebrates

(e) Mammals

Choose the **correct** answer from the options given below :

(1) (e) > (b) > (a) > (c) > (d)

(2) (b) > (a) > (d) > (c) > (e)

(3) (c) > (b) > (d) > (e) > (a)

(4) (d) > (a) > (c) > (b) > (e)

150. A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1, the frequency of AA is _____.

(1) 0.10 (2) 0.99

(3) 0.01 (4) 0.02

151. The opening between the right atrium and the right ventricle is guarded by _____.

(1) semilunar valve

(2) sino-atrial node

(3) bicuspid valve

(4) tricuspid valve

152. Which of the following enzymes synthesizes precursor mRNA ?

(1) RNA polymerase III

(2) DNA polymerase

(3) RNA polymerase I

(4) RNA polymerase II

153. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.

Reason R : Isolation of genetic material needs disruption of cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is correct but **R** is not correct

(2) **A** is not correct but **R** is correct

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

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154. Which of the following hormone is **not** secreted by human placenta ?

(1) Progesterone
(2) LH
(3) hCG
(4) Estrogen

155. Colostrum, secreted by mother during initial days of lactation, is abundant in _____.

(1) IgA (2) IgD
(3) IgG (4) IgM

156. Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the per-capita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is _____.

(1) 10 million (2) 100 million
(3) 1 million (4) 5 million

157. Given below are two statements :

Statement I :

Plasmids are autonomously replicating DNA.

Statement II :

Plasmids are extrachromosomal DNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect
(2) **Statement I** is incorrect but **Statement II** is correct
(3) Both **Statement I** and **Statement II** are correct
(4) Both **Statement I** and **Statement II** are incorrect

158. During PCR, primers bind to the DNA strands in the _____ step.

(1) annealing
(2) ligation
(3) denaturation
(4) extension

159. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Transformation	I.	Restriction enzyme
B.	Cloning site	II.	Transfer DNA to host bacteria
C.	Selection	III.	Replication
D.	Ori	IV.	Antibiotic

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-I, B-II, C-IV, D-III

160. Adaptive radiation in placental mammals and Australian Marsupials leading to similarity between distant species is an example of _____.

(1) founder effect
(2) genetic drift
(3) divergent evolution
(4) convergent evolution

161. The covering of ovum at ovulation is _____.

(1) zona pellucida
(2) chorion
(3) endometrium
(4) zona radiata

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162. Which of the following statements are **correct** ?

(a) Energy flow from producers to consumers is unidirectional

(b) Energy pyramid can never be inverted

(c) Transfer of energy follows the 1% law

Choose the **correct** answer from the options given below :

(1) (a) and (c) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (b) only

163. How many theca are present in each lobe of a typical bilobed angiosperm anther ?

(1) 8 (2) 12

(3) 2 (4) 6

164. Which of the following statements is **correct** about *Plasmodium* ?

(1) Gametocytes develop in mosquito gut

(2) Fertilization takes place in mosquito gut

(3) Reproduces sexually in liver cells

(4) Reproduces sexually in RBCs

165. For a person with blood group 'O', which of the following is **not** a possible combination of parents' blood group genotypes ?

(1) Father : I^Bi and Mother : I^Bi

(2) Father : I^AI^B and Mother : I^Ai

(3) Father : I^Ai and Mother : I^Bi

(4) Father : I^Ai and Mother : I^Ai

166. Which of the following is used as a clot buster ?

(1) Cyclosporin A

(2) Statins

(3) Streptokinase

(4) Penicillin

167. Which of the following disease is **not** sexually transmitted ?

(1) Gonorrhoea

(2) Genital warts

(3) Syphilis

(4) Tuberculosis

168. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarf.

Reason R : Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is correct but **R** is not correct

(2) **A** is not correct but **R** is correct

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

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169. The method of directly of injecting a sperm into ovum in assisted reproductive technology is called :

(1) Intra cytoplasmic sperm injection (ICSI)

(2) Embryo transfer (ET)

(3) Gamete intra fallopian transfer (GIFT)

(4) Zygote intra fallopian transfer (ZIFT)

170. The inactive form of Bt toxin is converted to the active form in the insect gut _____.

(1) by proteases

(2) by nucleases

(3) due to alkaline pH

(4) due to acidic pH

171. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The logistic growth model of populations is considered more realistic than the exponential growth model.

Reason R : Resources are finite.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **A** is correct but **R** is not correct

(2) **A** is not correct but **R** is correct

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

172. Which of the following are secondary lymphoid organs ?

(a) Bone marrow

(b) Tonsils

(c) Spleen

(d) Thymus

Choose the **correct** answer from the options given below :

(1) (b) and (d) only

(2) (a) and (d) only

(3) (a) and (b) only

(4) (b) and (c) only

173. Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the eggs ?

(1) Central cells

(2) Polar nucleus

(3) Antipodals

(4) Synergids

174. Given below are two statements :

Statement I : Ovulation is caused by LH surge leading to rupture of Graafian follicles.

Statement II : Graafian follicle remaining after ovulation transform into corpus luteum and secretes large amount of estrogen.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

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175. Given below are two statements :

Statement I : Modern *Homo sapiens* arose in Australia and moved across continents.

Statement II : *Homo sapiens* arose around 75000 to 10000 years ago.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

176. Which of the following are primary consumers in a food chain ?

(1) Herbivores

(2) Carnivores

(3) Parasites

(4) Predators

177. Which of the following is **not** evidence for evolution ?

(1) Embryological support for evolution as proposed by Ernst Heckel

(2) Divergent evolution of anatomical structures such as forelimbs

(3) Convergent evolution of traits like wings of birds and butterflies

(4) Paleontological evidence from fossil records

178. Sponges exchange O_2 with CO_2 by _____.

(1) tracheal tubes

(2) gills

(3) simple diffusion over their entire body surfaces

(4) moist cuticle

179. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Excess growth hormone	I.	Reabsorption of water and electrolytes in kidney
B.	Luteinizing hormone	II.	Contraction of uterus during child birth
C.	Vasopressin	III.	Acromegaly
D.	Oxytocin	IV.	Ovulation

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-IV, C-I, D-II

180. Natural selection can lead to _____.

(a) stabilisation

(b) genetic drift

(c) directional change

(d) disruption

Choose the **correct** answer from the options given below :

(1) (a), (b), (c) and (d)

(2) (a) and (c) only

(3) (a) only

(4) (a), (c) and (d) only

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परीक्षा पुस्तिका संकेत
Test Booklet Code

SUSHRUT

No.: 274744749
सं. :

Hindi+English

50

इस परीक्षा पुस्तिका को तब तक न खोलें जब तक कहा न जाए।
Do not open this Test Booklet until you are asked to do so.

इस पुस्तिका में 64 पृष्ठ हैं।
This Booklet contains 64 pages.

इस परीक्षा पुस्तिका के पिछले आवरण पर दिए निर्देशों को ध्यान से पढ़ें।
Read carefully the Instructions on the Back Cover of this Test Booklet.

महत्वपूर्ण निर्देश :

1. उज़र पत्र इस परीक्षा पुस्तिका के अन्दर रखा है। जब आपको परीक्षा पुस्तिका खोलने को कहा जाए, तो उज़र पत्र निकाल कर ध्यानपूर्वक मूल प्रतिलिपि पर केवल **नीले/काले** बॉल पॉइंट पेन से विवरण भरें।
2. परीक्षा की अवधि 3 **घंटा 15 मिनट** है एवं परीक्षा पुस्तिका में **भौतिकी, रसायनशास्त्र एवं जीवविज्ञान (वनस्पतिविज्ञान एवं प्राणिविज्ञान)** विषयों से 180 बहुविकल्पीय प्रश्न हैं (4 विकल्पों में से एक सही उज़र है)।
3. जहाँ कहीं भी प्रतिकों/स्थिरांकों का उल्लेख नहीं किया गया हो, उन्हें उनके मानक अर्थ/मूल्य के अनुसार माना जाए।
4. प्रत्येक प्रश्न 4 अंक का है। प्रत्येक सही उज़र के लिए परीक्षार्थी को 4 अंक दिए जाएंगे। प्रत्येक गलत उज़र के लिए कुल योग में से **एक** अंक घटाया जाएगा। **अधिकतम अंक 720 हैं।**
5. इस पृष्ठ पर विवरण अंकित करने एवं उज़र पत्र पर निशान लगाने के लिए **केवल नीले/काले बॉल पॉइंट पेन** का प्रयोग करें।
6. रफ कार्य इस परीक्षा पुस्तिका में निर्धारित स्थान पर ही करें।
7. परीक्षा सज़्पन्न होने पर, परीक्षार्थी कक्ष/हॉल छोड़ने से पूर्व **उज़र पत्र (मूल प्रतिलिपि एवं कार्यालय प्रतिलिपि) कक्ष निरीक्षक को अवश्य सौंप दें।** परीक्षार्थी अपने साथ प्रश्न पुस्तिका ले जा सकते हैं।
8. **इस पुस्तिका का संकेत 50 है। OMR उज़रपत्र में इस पुस्तिका के संकेत को दर्ज करना आवश्यक है।**

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**.
3. Wherever the symbols/constants are not mentioned, they are to be considered as per their standard meaning/value.
4. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate **must handover the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
8. **The CODE for this Booklet is 50. Make sure to enter this code in the OMR answer sheet.**

प्रश्नों के अनुवाद में किसी अस्पष्टता की स्थिति में, अंग्रेजी संस्करण को ही अंतिम माना जायेगा।
In case of any ambiguity in translation of any question, English version shall be treated as final.

परीक्षार्थी का नाम (बड़े अक्षरों में) :
Name of the Candidate (in Capitals) : SITA RAM MEGHAWAL

अनुक्रमांक : अंकों में
Roll Number : in figures 390210 1445
: शब्दों में
: in words

परीक्षा केन्द्र (बड़े अक्षरों में) :
Centre of Examination (in Capitals) : SMPS FORT School, Bikaner.

परीक्षार्थी के हस्ताक्षर :
Candidate's Signature : Sitaram

निरीक्षक के हस्ताक्षर :
Invigilator's Signature :

केन्द्र अधीक्षक की प्रतिकृति हस्ताक्षर मोहर :
Facsimile signature stamp of
Centre Superintendent :

1. द्रव्यमान M का एक कण एक क्षैतिज x अक्ष पर $x=0$ से $x=L$ तक गति करता है। गतिज घर्षण गुणांक x के फलन के रूप में $\mu_k(x)=\mu_0-\alpha x$ के अनुसार बदलता है, जहाँ μ_0, α उचित विमाओं वाले नियतांक हैं, ताकि $\mu_k(L)=0$ हो। गति के दौरान घर्षण बल द्वारा किया गया कुल कार्य $n\mu_0 MgL$ है, जहाँ g गुरुत्वीय त्वरण है। n का मान है :

(1) 3 (2) 1

(3) $\frac{1}{3}$ (4) $\frac{1}{2}$

1. A particle of mass M moves along a horizontal x axis from $x=0$ to $x=L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x)=\mu_0-\alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L)=0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is :

(1) 3 (2) 1

(3) $\frac{1}{3}$ (4) $\frac{1}{2}$

2. एक आदर्श गैस A में अणुओं का माध्य मुक्त पथ, एक अन्य आदर्श गैस B के माध्य मुक्त पथ का आधा है। गैस A के गोलाकार अणुओं का व्यास, गैस B के अणुओं के व्यास का दोगुना है। यदि गैसों A और B के संख्या घनत्व क्रमशः n_A और n_B हैं, तो सही विकल्प है :

(1) $n_A=n_B$ (2) $n_A=2n_B$

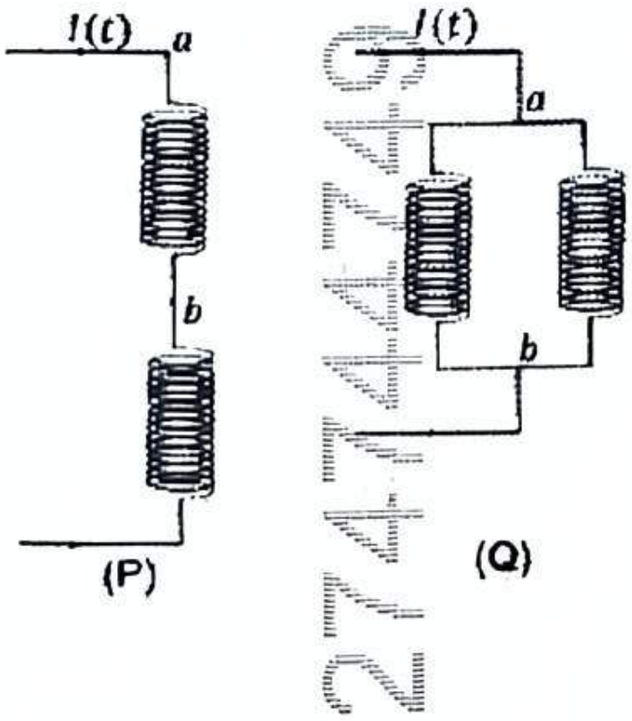
(3) $n_A=\frac{1}{4}n_B$ (4) $n_A=\frac{1}{2}n_B$

2. The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, then the correct option is :

(1) $n_A=n_B$ (2) $n_A=2n_B$

(3) $n_A=\frac{1}{4}n_B$ (4) $n_A=\frac{1}{2}n_B$

3. दो समान प्रेरकों को दो अलग-अलग विन्यासों P और Q में जोड़ा गया है, जहाँ एक समय-परिवर्ती धारा $I(t)$ प्रवाहित हो रही है, जैसा कि चित्र में दिखाया गया है। विन्यास P के लिए बिंदुओं a और b के बीच प्रेरित विद्युत वाहक बल E_P है और विन्यास Q के लिए यह E_Q है। अनुपात E_P/E_Q है : [पारस्परिक प्रेरकत्व के प्रभाव की उपेक्षा करें।]

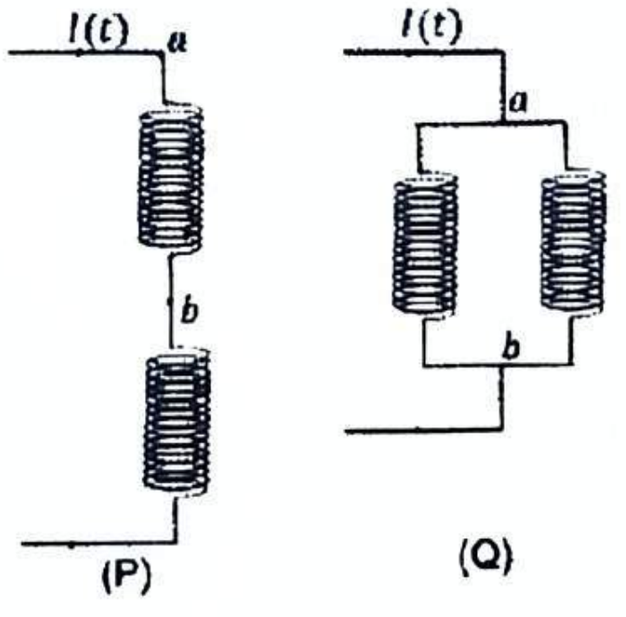


(1) 1/4 (2) 1/2

(3) 1 (4) 2

3. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is :

[Neglect the effect of mutual inductance.]



(1) 1/4 (2) 1/2

(3) 1 (4) 2

4. ध्वनि तरंगों के लिए, यदि एक खुले सिरे वाली नली के 5^{th} हार्मोनिक के लिए नोड्स (nodes) की संख्या n है और उसी नली के एक सिरा बंद होने पर उसके 9^{th} हार्मोनिक के लिए नोड्स की संख्या m है, तो अनुपात $\frac{n}{m}$ है :

(1) $\frac{5}{9}$ (2) $\frac{9}{5}$

(3) 1 (4) $\frac{3}{5}$

4. For sound waves, if the number of nodes for the 5^{th} harmonic of an open-ended pipe is n and that for the 9^{th} harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is :

(1) $\frac{5}{9}$ (2) $\frac{9}{5}$

(3) 1 (4) $\frac{3}{5}$

5. लंबाई l और त्रिज्या r के एक लंबे परिनालिका (solenoid) पर विचार कीजिए। यदि n प्रति इकाई लंबाई में फेरों की संख्या है और μ_0 निर्वात की चुम्बकीय पारगम्यता है, तो परिनालिका का प्रेरकत्व (inductance) है :

(1) $\mu_0\pi n^2r^2l$

(2) $\mu_0 n^2r^2l$

(3) $(\mu_0/2\pi)n^2r^2l$

(4) $2\mu_0\pi n^2r^2l$

5. Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

(1) $\mu_0\pi n^2r^2l$

(2) $\mu_0 n^2r^2l$

(3) $(\mu_0/2\pi)n^2r^2l$

(4) $2\mu_0\pi n^2r^2l$

6. एक सीधी रेखा में गतिमान एक कण पर विचार कीजिए, जिसकी समय के फलन के रूप में स्थिति $s(t)=\alpha t^2-\beta t+\gamma$ द्वारा दी गई है, जहाँ $\alpha=1\ ms^{-2}$, $\beta=6\ ms^{-1}$ और $\gamma=5$ m है। t = 0 s से t = 6 s तक कण की औसत चाल, ms^{-1} में है :

(1) 12

(2) 6

(3) 3

(4) 0

6. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t)=\alpha t^2-\beta t+\gamma$, where $\alpha=1\ ms^{-2}$, $\beta=6\ ms^{-1}$ and $\gamma=5$ m. The average speed of the particle, in ms^{-1}, from t = 0 to t = 6 s is :

(1) 12

(2) 6

(3) 3

(4) 0

7. निम्नलिखित नाभिकीय अभिक्रिया पर विचार कीजिए :

$$^{238}U \rightarrow {}^{234}Th + {}^{4}He$$

^{238}U, ^{234}Th और ^{4}He के द्रव्यमान क्रमशः 238.050 u, 234.043 u और 4.003 u लें। अभिक्रिया के लिए Q मान, keV में, है :

[दिया है : 1 u = 931.5 MeV c^{-2}]

(1) 3726 (2) 3730

(3) 3736 (4) 3740

7. Consider the following nuclear reaction :

$$^{238}U \rightarrow {}^{234}Th + {}^{4}He$$

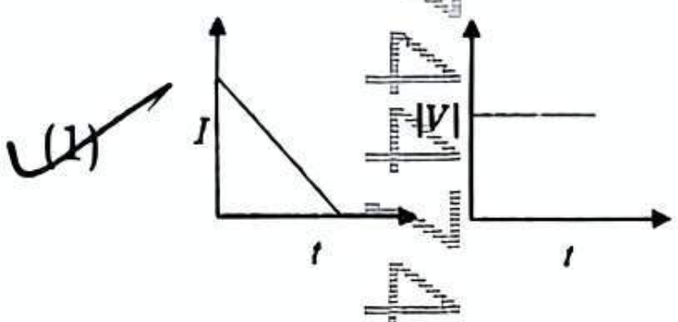
Take masses of ^{238}U, ^{234}Th and ^{4}He as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is :

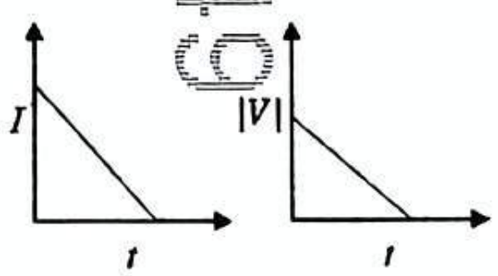
[Given : 1 u = 931.5 MeV c^{-2}]

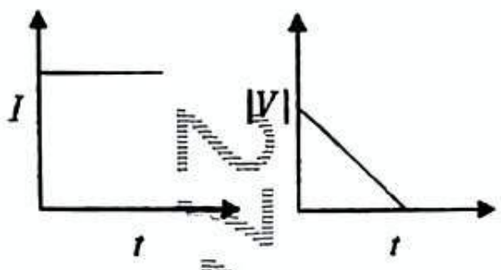
(1) 3726 (2) 3730

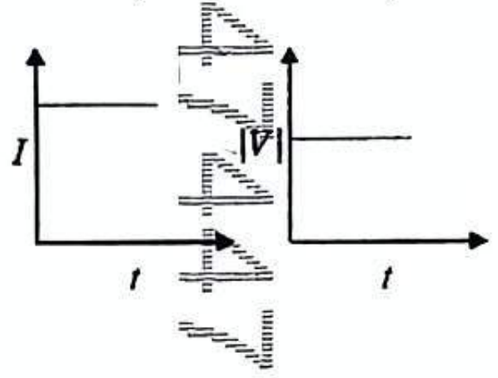
(3) 3736 (4) 3740

8. प्रकाश की एक किरण एक धातु की सतह पर इस प्रकार गिरती है कि प्रकाश-इलेक्ट्रॉन उत्पन्न होते हैं। यदि प्रकाश स्रोत की शक्ति समय t के साथ रैखिक रूप से घटने लगती है, तो प्रकाश-धारा I और निरोधी विभव के परिमाण $|V|$ का समय के साथ परिवर्तन सबसे अच्छी तरह से किसके द्वारा निरूपित किया गया है :

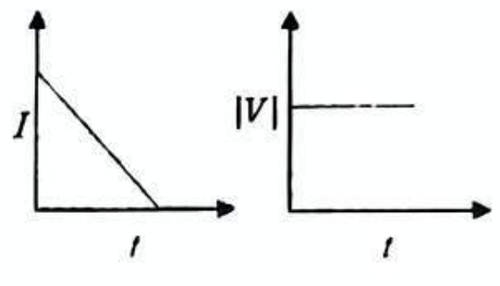
(1) 

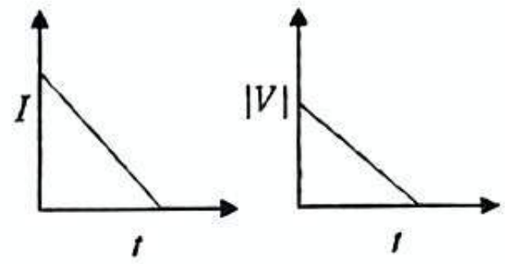
(2) 

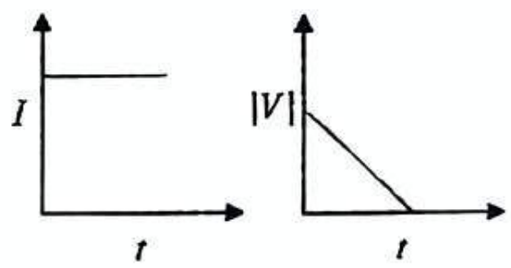
(3) 

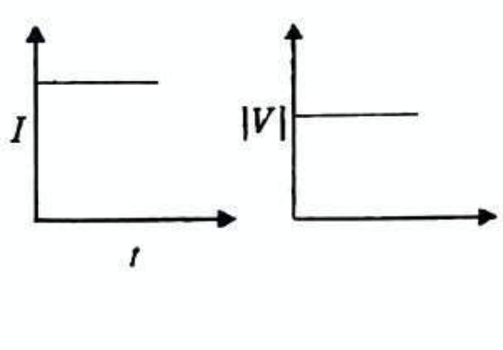
(4) 

8. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by :

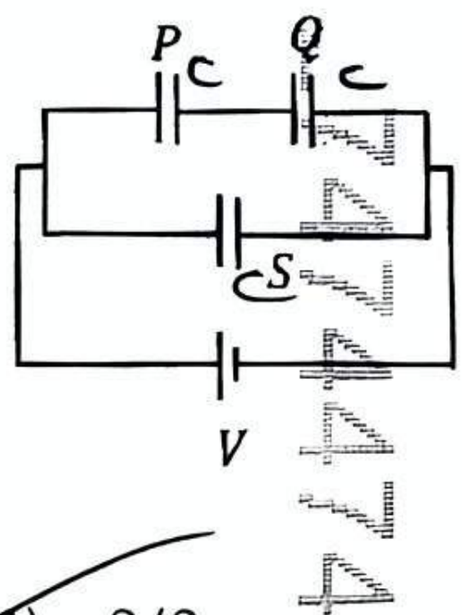
(1) 

(2) 

(3) 

(4) 

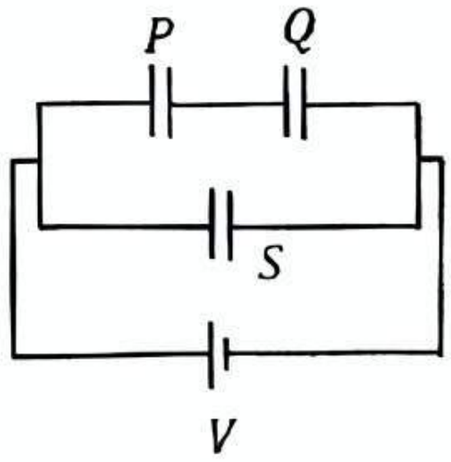
9. तीन समान संधारित्र P, Q और S, जिनमें से प्रत्येक की धारिता C है, उन्हें V वोल्टेज की एक बैटरी से जोड़ा गया है, जैसा कि चित्र में दिखाया गया है। यदि संधारित्र P में संचित ऊर्जा और निकाय में संचित कुल ऊर्जा क्रमश: U_P और U_T हैं, तो अनुपात $\frac{U_P}{U_T}$ है :



(1) 2/3 (2) 1/3

(3) 1/2 (4) 1/6

9. Three identical capacitors P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is :

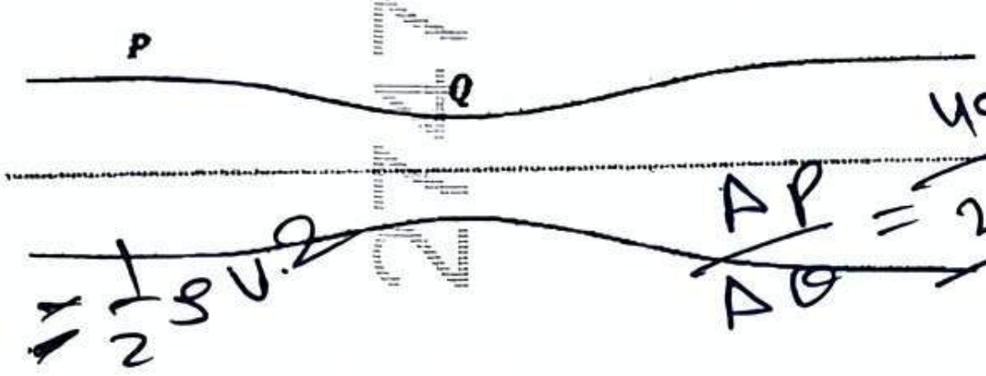


(1) 2/3 (2) 1/3

(3) 1/2 (4) 1/6

10. जल (प्रवाह), दिखाए गये चित्र के अनुसार वृत्ताकार अनुप्रस्थ-काट वाली एक क्षैतिज नली के माध्यम से धारा-रेखीय गति में प्रवाहित होता है। P और Q के बीच जल का दाब अंतर $15\ Nm^{-2}$ है। P और Q पर अनुप्रस्थ-काट का क्षेत्रफल क्रमशः $40\ cm^2$ और $20\ cm^2$ है। नली के माध्यम से जल के प्रवाह की दर, cm^3s^{-1} में, है :

[जल का घनत्व = $1000\ kg\ m^{-3}$ लें]

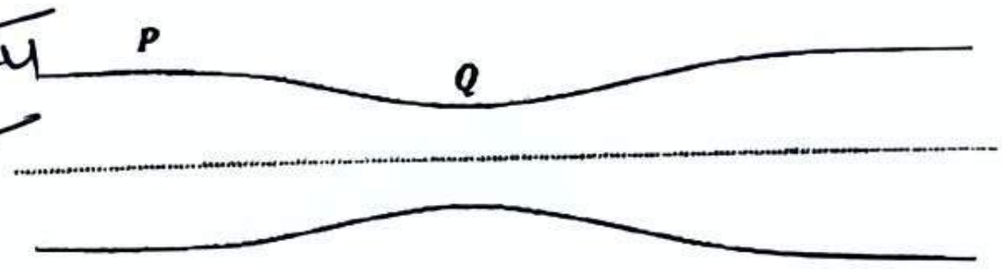


(1) 100 (2) 200

(3) 300 (4) 400

10. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is $15\ Nm^{-2}$. The area of cross-section at P and Q are $40\ cm^2$ and $20\ cm^2$, respectively. The rate of flow of water through the pipe, in cm^3s^{-1}, is :

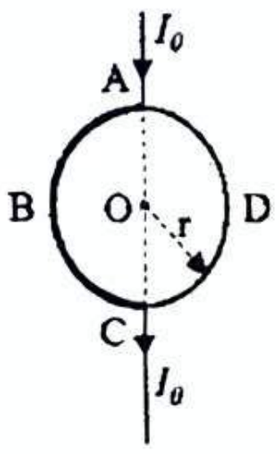
[Take density of water = $1000\ kg\ m^{-3}$]



(1) 100 (2) 200

(3) 300 (4) 400

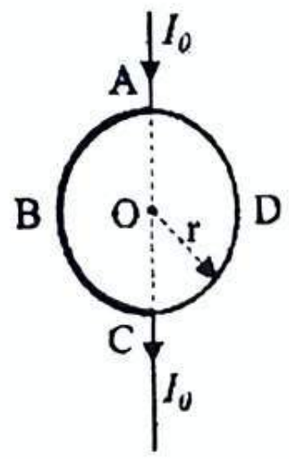
11. जैसा कि चित्र में दिखाया गया है, r त्रिज्या के एक धात्विक वृत्ताकार छल्ला से एक धारा I_0 प्रवाहित हो रही है। खंड ABC का प्रतिरोध, ADC के प्रतिरोध का आधा है। लूप के केन्द्र O पर चुम्बकीय क्षेत्र का परिमाण है :



(1) $\frac{\mu_0 I_0}{12r}$ (2) $\frac{\mu_0 I_0}{4r}$

(3) $\frac{\mu_0 I_0}{2r}$ (4) $\frac{\mu_0 I_0}{2\pi r}$

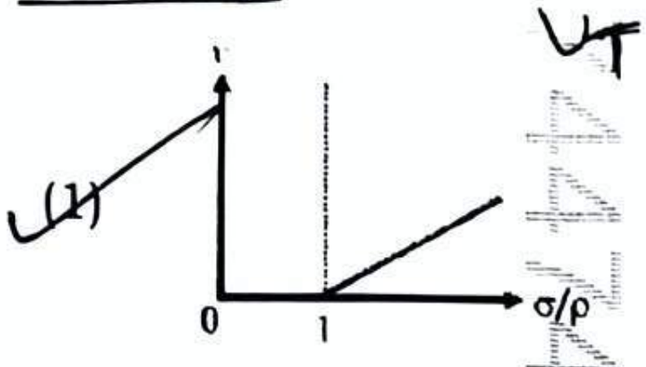
11. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the center O of the loop is :

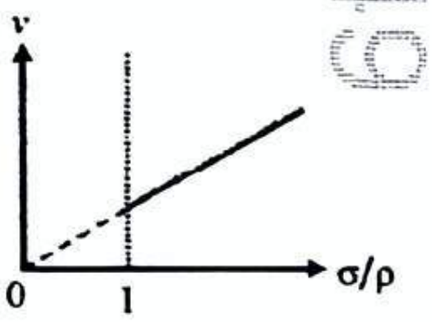


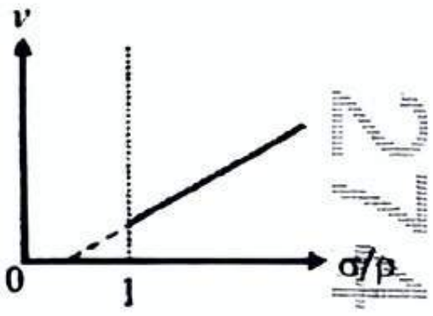
(1) $\frac{\mu_0 I_0}{12r}$ (2) $\frac{\mu_0 I_0}{4r}$

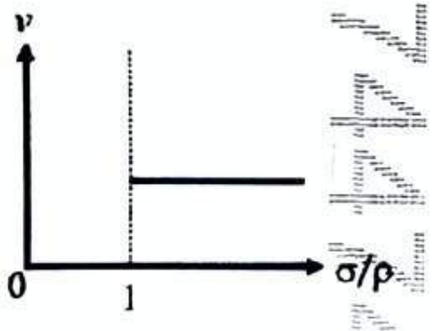
(3) $\frac{\mu_0 I_0}{2r}$ (4) $\frac{\mu_0 I_0}{2\pi r}$

12. सीमांत वेग प्रयोग का उपयोग करके द्रवों की श्यानता के मापन में, समान त्रिज्या वाली परंतु भिन्न घनत्व वाली गोलाकार गेंदों का उपयोग किया जाता है। गोलाकार गेंद के घनत्व (σ) और द्रव के घनत्व (ρ) के अनुपात के साथ सीमांत वेग (v) के परिवर्तन को सबसे अच्छी तरह से किसके द्वारा निरूपित किया गया है :

(1) 

(2) 

(3) 

(4) 

13. समान द्रव्यमान वाले दो ग्रहों P_1 और P_2 की त्रिज्याएँ क्रमशः R_1 और R_2 हैं, जहाँ $R_2=\frac{R_1}{2}$ है। P_1 और P_2 के पलायन वेग क्रमशः v_1 और v_2 हैं। तब $\frac{v_2}{v_1}$ है :

(1) $\frac{1}{\sqrt{2}}$ (2) 1

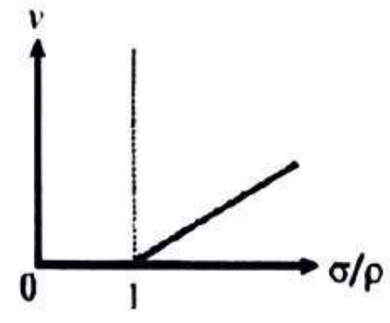
(3) $\sqrt{2}$ (4) 2

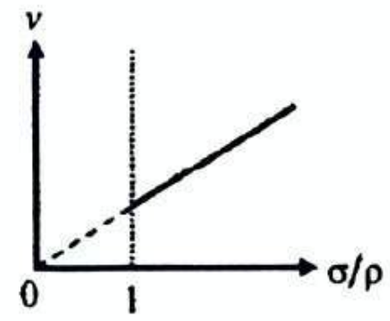
14. एक सौर मंडल में, त्रिज्या R की वृत्ताकार कक्षा का अनुसरण करने वाले एक ग्रह के परिक्रमण का आवर्तकाल किसके समानुपाती होता है :

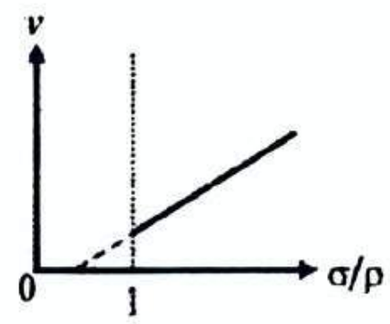
(1) $R^{1/2}$ (2) $R^{3/2}$

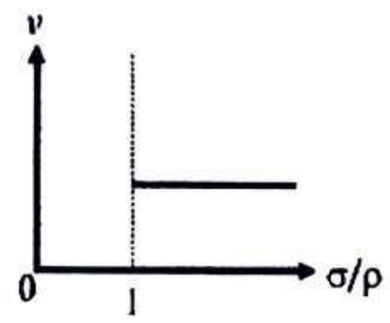
(3) R^2 (4) R^3

12. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by :

(1) 

(2) 

(3) 

(4) 

13. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2=\frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is :

(1) $\frac{1}{\sqrt{2}}$ (2) 1

(3) $\sqrt{2}$ (4) 2

14. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to :

(1) $R^{1/2}$ (2) $R^{3/2}$

(3) R^2 (4) R^3

15. दो अनंत लंबाई के समान्तर चालक तार A और B, क्रमशः I और 2I धारा एक ही दिशा में ले जा रहे हैं। तार A का प्रति इकाई लंबाई का द्रव्यमान λ एकसमान है और यह एक विद्युत-रोधी/रोधित फर्श पर स्थित है। तार B को फर्श से h ऊँचाई पर स्थिर रखा गया है। h का न्यूनतम परिमाण ताकि तार A फर्श से ऊपर न उठे, है : [g गुरुत्वीय त्वरण है और μ_0 निर्वात की चुम्बकीय पारगम्यता है।]

(1) $\frac{\mu_0 I^2}{2\pi\lambda g}$ (2) $\frac{\mu_0 I^2}{\pi\lambda g}$

(3) $\frac{2\mu_0 I^2}{\pi\lambda g}$ (4) $\frac{4\mu_0 I^2}{\pi\lambda g}$

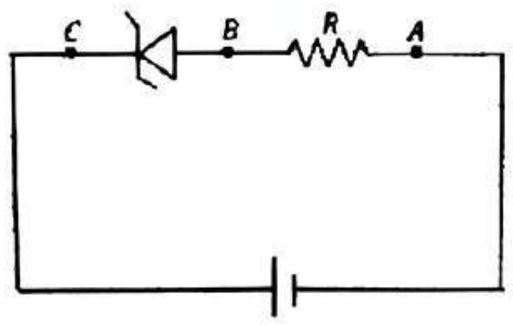
15. Two infinitely long parallel conducting wires A and B carry currents I and 2I, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is :

[g is the acceleration due to gravity and μ_0 is the permeability of free space.]

(1) $\frac{\mu_0 I^2}{2\pi\lambda g}$ (2) $\frac{\mu_0 I^2}{\pi\lambda g}$

(3) $\frac{2\mu_0 I^2}{\pi\lambda g}$ (4) $\frac{4\mu_0 I^2}{\pi\lambda g}$

16. -3 V के ब्रेकडाउन वोल्टेज वाले एक आदर्श जेनर डायोड को -5 V के ऋणात्मक निवेश वोल्टेज V_i के साथ पश्चदिशिक बायसित (reverse biased) किया गया है। बिंदुओं B और A के बीच वोल्टेज अंतर का परिमाण है :



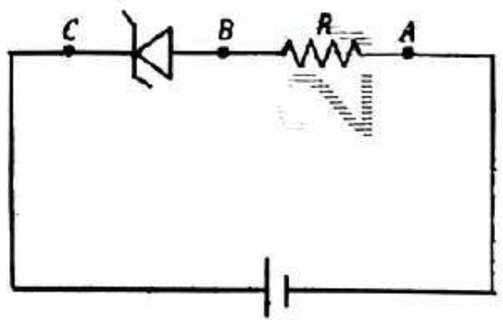
(1) 3 V

(2) 2 V

(3) 1 V

(4) 0 V

16. An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between points B and A is :



(1) 3 V

(2) 2 V

(3) 1 V

(4) 0 V

17. एक रुद्धोष्म प्रसार में, एक मोल एकपरमाणुक आदर्श गैस ($\gamma = 5/3$) का तापमान 60K से घटकर 50K हो जाता है। इस प्रक्रम में गैस द्वारा किया गया कार्य है : (सार्वत्रिक गैस नियतांक $R = 8.3$ J mol^{-1} K^{-1} लें)

(1) 41.5 J (2) 83 J

(3) 124.5 J (4) 166 J

17. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60K to 50K. The work done by the gas in the process is :

(Take the universal gas constant as $R = 8.3$ J mol^{-1} K^{-1})

(1) 41.5 J (2) 83 J

(3) 124.5 J (4) 166 J

18. तरंगदैर्ध्य λ की प्रकाश की एक किरण तीन अलग-अलग प्रकाश-वैद्युत सेल, क्रमशः 1, 2 और 3 पर आपतित होती है। इन प्रकाश-वैद्युत सेलों की देहली तरंगदैर्ध्य क्रमशः λ_1, λ_2, और λ_3, हैं और इन सेलों के निरोधात्मक विभवों का परिमाण क्रमशः V_1, V_2 और V_3, है। λ और देहली तरंगदैर्ध्यों के बीच संबंध $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ और $\lambda_3 >> \lambda$ हैं। सही विकल्प है :

(1) $V_1 = 0, V_2 < V_3$

(2) $V_1 = 0, V_2 > V_3$

(3) $V_1 > V_2, V_3 = 0$

(4) $V_1 < V_2, V_3 = 0$

18. A ray of light with wavelength λ is incident on three different photo-electric cells namely 1, 2 and 3. The threshold wavelength of these photo-electric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ and $\lambda_3 >> \lambda$. The correct option is :

(1) $V_1 = 0, V_2 < V_3$

(2) $V_1 = 0, V_2 > V_3$

(3) $V_1 > V_2, V_3 = 0$

(4) $V_1 < V_2, V_3 = 0$

19. एक फोटॉन और एक इलेक्ट्रॉन, जिनमें से प्रत्येक की ऊर्जा 20 eV है, मुक्त आकाश में गति करते हैं। इलेक्ट्रॉन के रेखीय संवेग p_e तथा फोटॉन के रेखीय संवेग p_{Ph} का अनुपात, $\frac{p_e}{p_{Ph}}$ है :

(प्रकाश की चाल $= 3 \times 10^8\ ms^{-1}$, इलेक्ट्रॉन का आवेश $= -1.6 \times 10^{-19}$ C और इलेक्ट्रॉन का द्रव्यमान $= 9 \times 10^{-31}$ kg लें)

(1) $\frac{2}{450}$ (2) $\frac{1}{250}$

(3) 225 (4) 275

19. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is :

(Take speed of light $= 3 \times 10^8\ ms^{-1}$, charge of electron $= -1.6 \times 10^{-19}$ C and mass of electron $= 9 \times 10^{-31}$ kg)

(1) $\frac{2}{450}$ (2) $\frac{1}{250}$

(3) 225 (4) 275

20. निम्नलिखित में से किस माप के लिए 'सूचकांक संशोधन' (index correction) की आवश्यकता होती है?

(1) मीटर सेतु का उपयोग करके एक तार के प्रतिरोध का मापन

(2) सरल लोलक का उपयोग करके गुरुत्वीय त्वरण का मापन

(3) ऑप्टिकल बेंच का उपयोग करके लेन्सों की फोकस दूरी का मापन

(4) अनुनाद नली का उपयोग करके ध्वनि की चाल का मापन

20. Which of the following measurements require 'index correction' ?

(1) Measurement of resistance of a wire using meter bridge

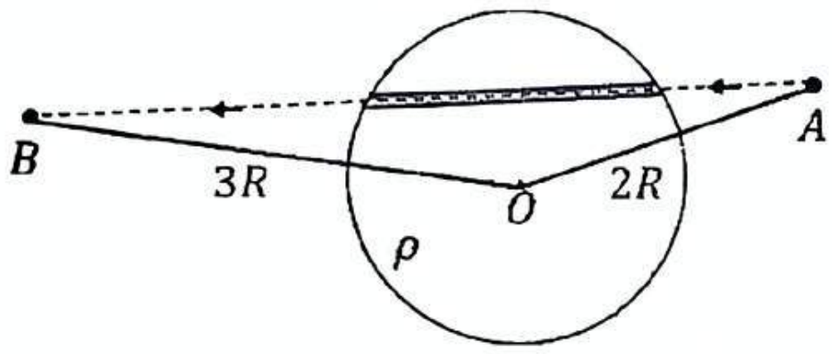
(2) Measurement of gravitational acceleration using simple pendulum

(3) Measurement of focal length of lenses using optical bench

(4) Measurement of speed of sound using resonance tube

21. एक इकाई धनात्मक बिंदु आवेश को धीरे-धीरे एक अत्यंत पतली नली के माध्यम से ले जाया जाता है जो त्रिज्या R के एक आवेशित परावैद्युत गोले के अंदर है, जिसका एकसमान धनात्मक आवेश घनत्व ρ है, जैसा कि चित्र में दिखाया गया है। आवेश की प्रारंभिक और अंतिम स्थितियाँ गोले के केंद्र से क्रमशः $2R$ और $3R$ की दूरी पर A और B द्वारा चिह्नित हैं। इस प्रक्रिया में, बिंदु आवेश पर किए गए कुल कार्य का परिमाण $\frac{\rho R^2}{n\epsilon_0}$ है। n का मान है :

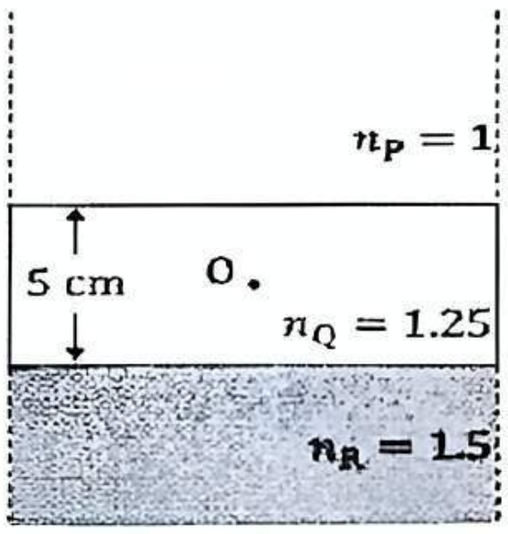
(ϵ_0 निर्वात की विद्युतशीलता है)



(1) 2 (2) 6

(3) 9 (4) 18

22. क्रमशः अपवर्तनांक 1, 1.25, और 1.5 वाले तीन माध्यमों P, Q और R पर विचार कीजिए। 5 cm मोटाई वाला माध्यम Q, विस्तृत माध्यमों P और R के बीच इस प्रकार रखा गया है जैसा कि चित्र में दिखाया गया है। एक वस्तु O को माध्यम Q के केंद्र पर रखा गया है। यदि माध्यम P से अभिलम्ब दिशा के निकट देखा जाए, तो O की आभासी गहराई h_1 है। माध्यम R से इसी प्रकार के अवलोकन के लिए, आभासी गहराई h_2 है। $|h_1 - h_2|$ का मान, cm में, है :

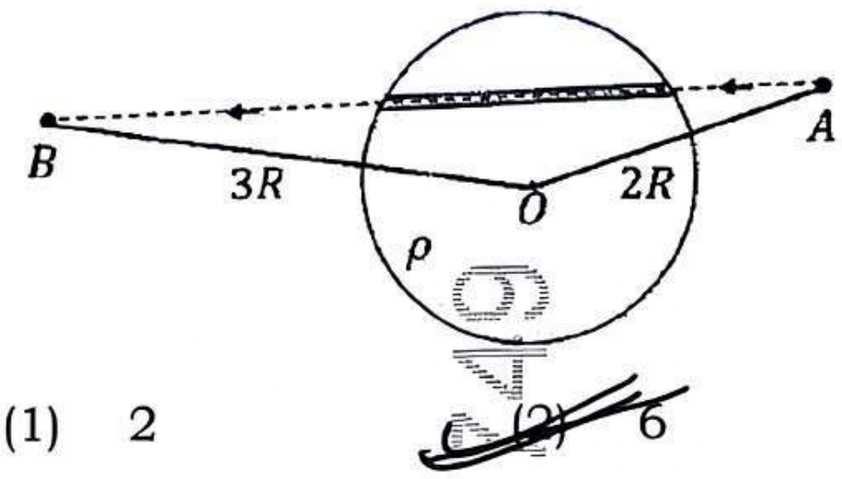


(1) 0 (2) 1

(3) 2 (4) 3

21. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is :

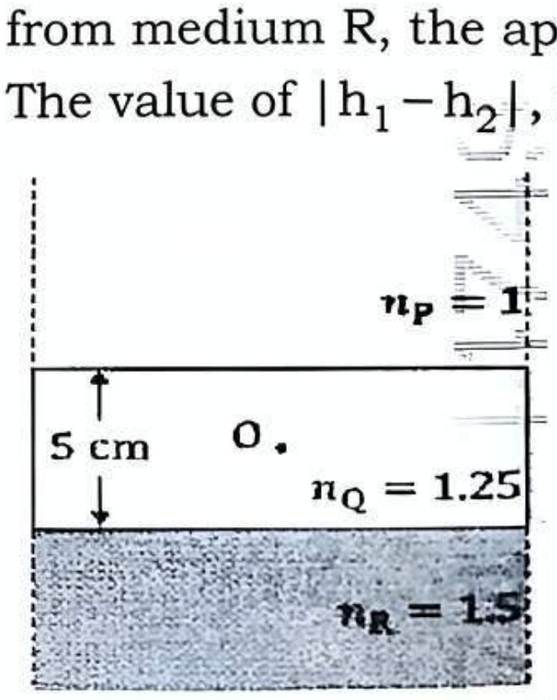
(ϵ_0 is the permittivity of vacuum)



(1) 2 (2) 6

(3) 9 (4) 18

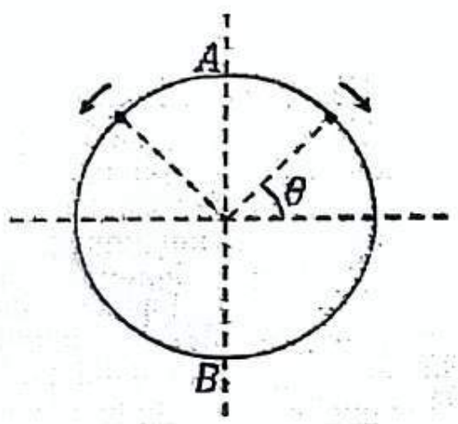
22. Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5, respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm, is :

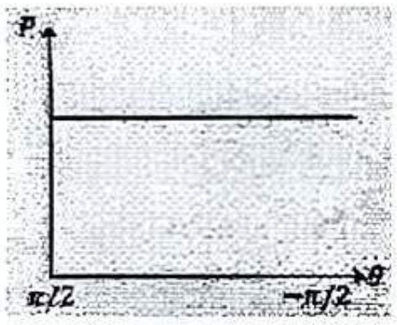


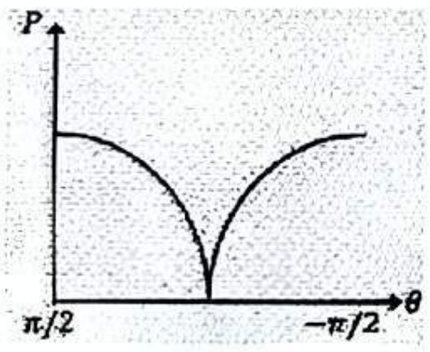
(1) 0 (2) 1

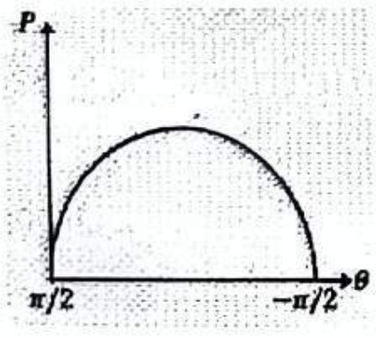
(3) 2 (4) 3

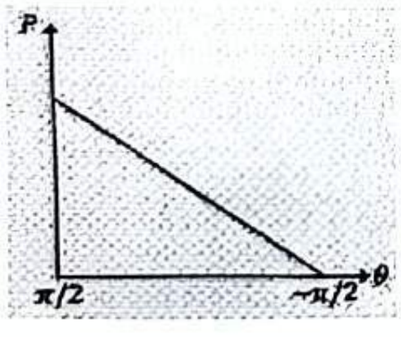
23. इकाई त्रिज्या का एक घर्षणरहित वृत्ताकार तार क्षैतिज तल पर स्थिर है। इकाई द्रव्यमान के दो बिंदु कण बिंदु $A\left(\theta=\frac{\pi}{2}\right)$ से एक साथ समान एकसमान कोणीय चालों के साथ विपरीत दिशाओं में गति करना प्रारंभ करते हैं, और बिंदु $B\left(\theta=-\frac{\pi}{2}\right)$ पर पुनः मिलते हैं। इस समय के दौरान, निम्नलिखित में से कौन सा चित्र θ के फलन के रूप में निकाय के कुल रैखिक संवेग P के परिमाण को योजनाबद्ध रूप से निरूपित करता है?



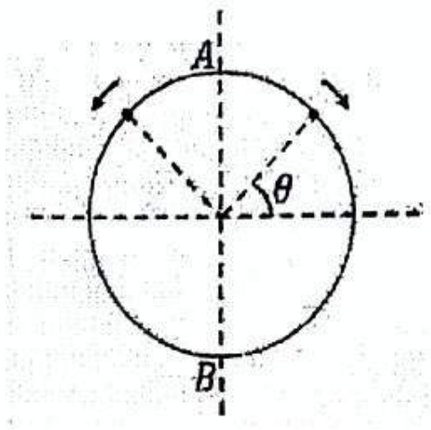
(1)


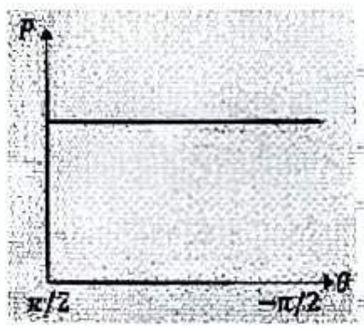
(2)


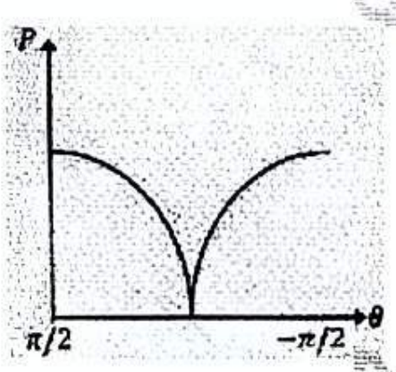
(3)


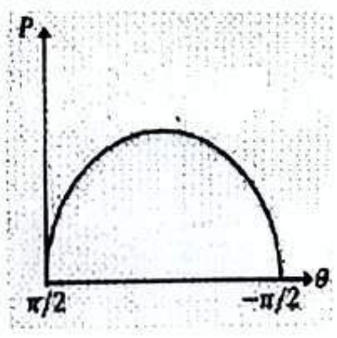
(4)


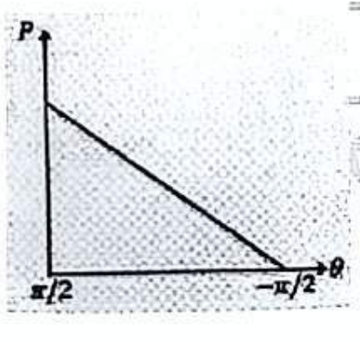
23. A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point $A\left(\theta=\frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point $B\left(\theta=-\frac{\pi}{2}\right)$. During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ ?



(1)


(2)


(3)


(4)


24. त्रिज्या R के एक धात्विक गोले के तापमान में ΔT की अल्प वृद्धि की जाती है। यदि धातु का रैखिक तापीय प्रसार गुणांक α है, तो गोले के आयतन में अनुमानित वृद्धि है :

(1) $2\pi R^3\alpha\Delta T$ (2) $3\pi R^3\alpha\Delta T$

(3) $4\pi R^3\alpha\Delta T$ (4) $6\pi R^3\alpha\Delta T$

24. The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is :

(1) $2\pi R^3\alpha\Delta T$ (2) $3\pi R^3\alpha\Delta T$

(3) $4\pi R^3\alpha\Delta T$ (4) $6\pi R^3\alpha\Delta T$

25. एकसमान घनत्व का एक बेलनाकार कॉर्क ρ_1 घनत्व के एक द्रव में तैरता है। यदि कॉर्क को थोड़ा दबाकर छोड़ दिया जाए, तो यह आवर्तकाल T के साथ सरल आवर्त गति करता है। यदि वही कॉर्क ρ_2 घनत्व के किसी अन्य द्रव में तैरता है, तो इसी प्रकार के दोलन का आवर्तकाल 2T होता है। ρ_2/ρ_1 का मान है :

(1) 4 (2) 2

(3) 1/2 (4) 1/4

25. A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period 2T. The value of ρ_2/ρ_1 is :

(1) 4 (2) 2

(3) 1/2 (4) 1/4

26. एक वर्नियर कैलीपर के मुख्य पैमाने का एक भाग 1 mm के बराबर है और वर्नियर पैमाने पर भागों की संख्या 10 है। जब दोनों जबड़े एक-दूसरे को स्पर्श करते हैं, तो वर्नियर पैमाना मुख्य पैमाने के शून्य के बाईं ओर इस प्रकार खिसक जाता है कि वर्नियर पैमाने का 4^{th} भाग मुख्य पैमाने के एक भाग के साथ संपाती हो जाता है। यदि यह वर्नियर कैलीपर एक तार की लंबाई 1 cm मापता है, तो तार की वास्तविक लंबाई है :

(1) 0.60 cm

(2) 0.96 cm

(3) 1.00 cm

(4) 1.04 cm

26. One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4^{th} Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of a wire to be 1 cm, the actual length of the wire is :

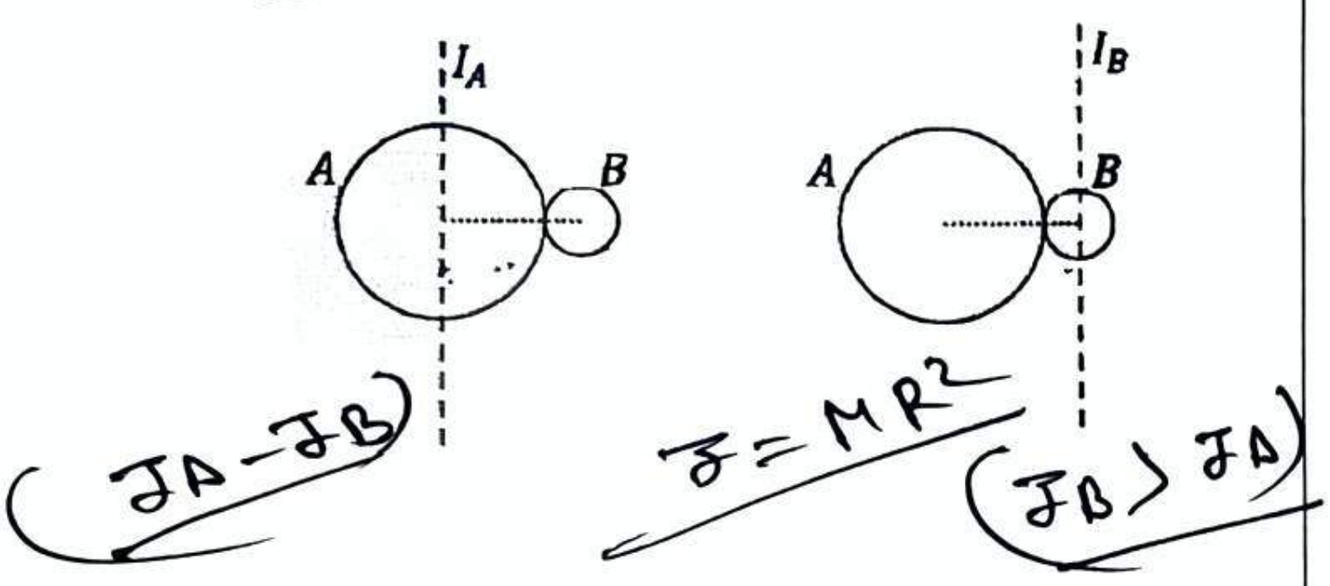
(1) 0.60 cm

(2) 0.96 cm

(3) 1.00 cm

(4) 1.04 cm

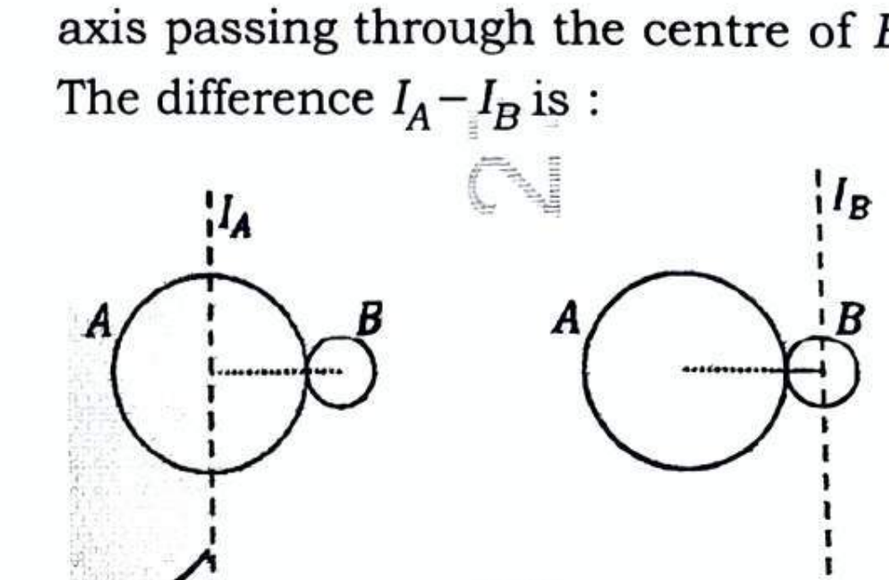
27. त्रिज्या R और द्रव्यमान M के एक ठोस गोले A को एक बिंदु पर त्रिज्या $r < R$ और द्रव्यमान $m < M$ के एक छोटे ठोस गोले B से जोड़ा गया है। मान लें कि उनके केन्द्रों को जोड़ने वाली रेखा क्षैतिज है। A के केन्द्र से गुजरने वाले एक ऊर्ध्वाधर अक्ष के परितः निकाय का जड़त्व आघूर्ण I_A है और B के केन्द्र से गुजरने वाले एक ऊर्ध्वाधर अक्ष के परितः गणना किया गया जड़त्व आघूर्ण I_B है। अंतर $I_A - I_B$ है :



(1) $(M-m)(R+r)^2$ (2) $(m-M)(R+r)^2$

(3) $(m-M)(R-r)^2$ (4) 0

27. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is :



(1) $(M-m)(R+r)^2$ (2) $(m-M)(R+r)^2$

(3) $(m-M)(R-r)^2$ (4) 0

28. एक विमीय सरल आवर्त कंपन करने वाले एक स्प्रिंग-द्रव्यमान दोलक पर विचार कीजिए। कण का द्रव्यमान m kg है और स्प्रिंग नियतांक k Nm^{-1} है। किसी दिए गए क्षण पर, स्प्रिंग का विस्तार x मीटर है और कण की चाल v ms^{-1} है। x - v तल पर, यदि x के फलन के रूप में v का आलेख एक वृत्त है, तो सही विकल्प है :

(1) $k = \frac{1}{m}$ (2) $k = m$

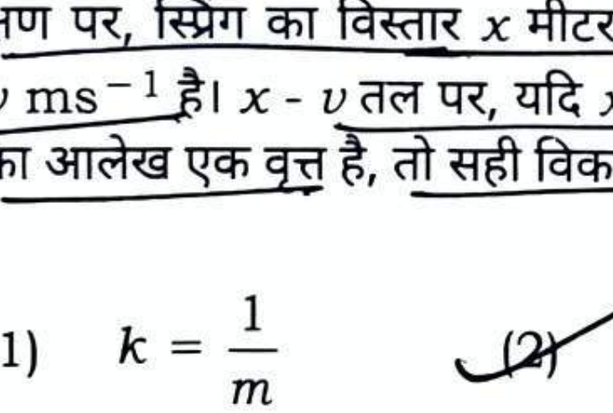
(3) $k = m^2$ (4) $k = \sqrt{m}$

28. Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is k Nm^{-1}. At a given instant, the extension of the spring is x meter and the speed of the particle is v ms^{-1}. On the x - v plane, if the graph of v as a function of x is a circle, then the correct option is :

(1) $k = \frac{1}{m}$ (2) $k = m$

(3) $k = m^2$ (4) $k = \sqrt{m}$

29. चित्र में दिखाए गए लेन्स संयोजन में, दो लेन्स L_1 और L_2 हैं, जिनकी फोकस दूरियाँ क्रमशः +10 cm और −10 cm हैं। बनने वाले प्रतिबिम्ब की स्थिति है :



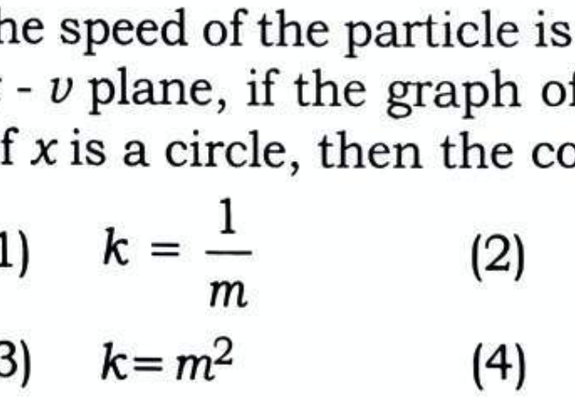
(1) अवतल लेन्स के बाईं ओर 20 cm

(2) अवतल लेन्स के बाईं ओर 60 cm

(3) अवतल लेन्स के दाईं ओर 30 cm

(4) अवतल लेन्स के दाईं ओर 60 cm

29. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and −10 cm, respectively. The position of the image formed is :



(1) 20 cm to the left of the concave lens

(2) 60 cm to the left of the concave lens

(3) 30 cm to the right of the concave lens

(4) 60 cm to the right of the concave lens

30. एक श्रेणी-क्रम LCR परिपथ में एक ac वोल्टेज $V = 220 \sin(2 \times 10^3 t)$ Volt अनुप्रयुक्त किया जाता है। तब इस परिपथ में धारा का आयाम है : (दिया है: L = 10 mH, C = 25μF, R = 100 Ω)

(1) 2.2 A (2) 5.5 A

(3) 11.0 A (4) 22.0 A

30. An ac voltage $V = 220 \sin(2 \times 10^3 t)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :

(Given : L = 10 mH, C = 25μF, R = 100 Ω)

(1) 2.2 A (2) 5.5 A

(3) 11.0 A (4) 22.0 A

31. एक पतली क्षैतिज डिस्क अपने स्थिर केन्द्र *O* से गुजरने वाले एक ऊर्ध्वाधर अक्ष के परित: घूर्णन कर रही है। क्रमश: बिंदुओं *A* और *B* के परित: गणना किए गए इसके कोणीय संवेग L_A और L_B हैं,

जहाँ $OB = 2 \times OA$ है। $\frac{L_A}{L_B}$ का मान है :

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$

(3) 1 (4) 2

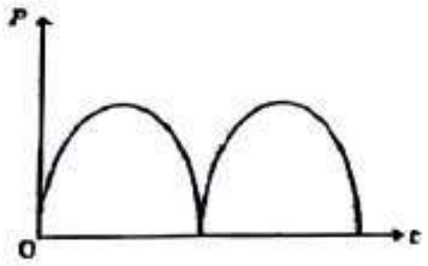
31. A thin horizontal disc is rotating about a vertical axis passing through its fixed centre *O*. Its angular momentum is L_A and L_B computed about points *A* and *B*, respectively, with $OB = 2 \times OA$. The value of $\frac{L_A}{L_B}$ is :

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$

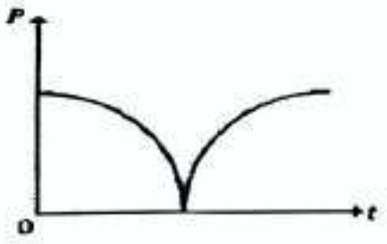
(3) 1 (4) 2

32. परिमित प्रतिरोध का एक चालक लूप x - y तल पर स्थित है। z दिशा में एक नियत चुम्बकीय क्षेत्र है। लूप का क्षेत्रफल समय t के साथ, उचित इकाइयों में $A = A_0(1 + \sin t)$ के रूप में परिवर्तित होता है। वह चित्र जो समय के फलन के रूप में लूप में क्षय होने वाली शक्ति P के गुणात्मक व्यवहार को सही ढंग से दर्शाता है, वह है :

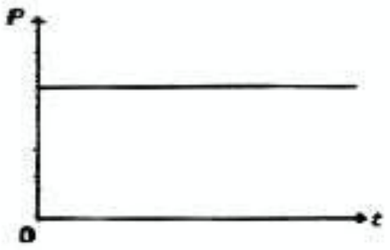
(1)



(2)



(3)

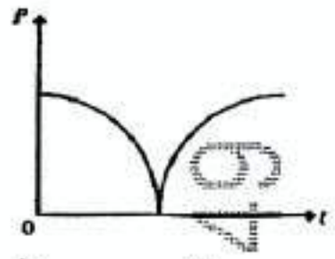


(4)

32. A conducting loop of finite resistance lies on the x - y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :

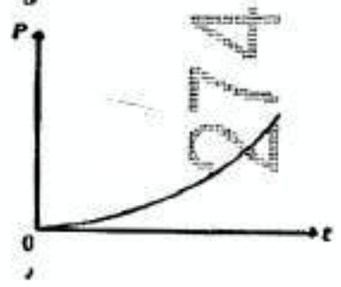
(1)

(2)

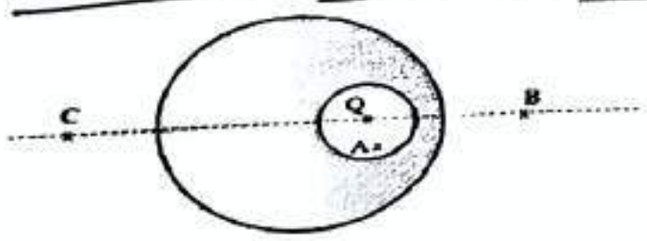


(3)

(4)

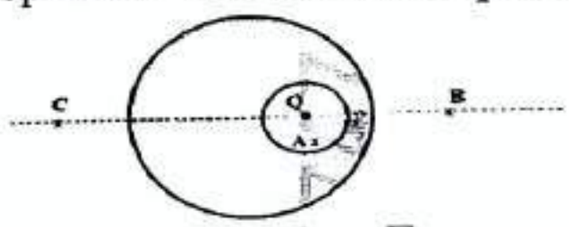


33. एक बिंदु आवेश Q को एक ठोस विलगित चालक गोले के भीतर एक कोटर (cavity) में रखा गया है। चित्र में दर्शाए गए बिंदुओं A, B और C पर विचार करें, जहाँ वैद्युत क्षेत्रों के परिमाण क्रमशः E_A, E_B और E_C हैं। बिंदु B और C ठोस गोले के केंद्र से समान दूरी पर हैं। सही विकल्प है :



(1) $E_A = 0$, $E_B = E_C$
(2) $E_A \neq 0$, $E_B = E_C$
(3) $E_A = 0$, $E_B > E_C$
(4) $E_A \neq 0$, $E_B < E_C$

33. A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B and E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is :



(1) $E_A = 0$, $E_B = E_C$
(2) $E_A \neq 0$, $E_B = E_C$
(3) $E_A = 0$, $E_B > E_C$
(4) $E_A \neq 0$, $E_B < E_C$

34. त्रिज्या R और कुल आवेश +Q वाले एक स्थिर एकसमान आवेशित कुचालक गोले पर विचार कीजिए। द्रव्यमान m वाले एक बिंदु आवेश −q (q<<Q) को आवेशित गोले के केन्द्र से 3R की दूरी पर विराम अवस्था से मुक्त किया जाता है। जब बिंदु आवेश गोले की सतह पर पहुँचता है, तो इसकी चाल होती है : (ϵ_0 निर्वात की विद्युतशीलता है, गुरुत्वाकर्षण बलों की उपेक्षा करें)।

(1) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$ (2) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$

(3) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$ (4) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$

34. Consider a fixed uniformly charged insulating sphere with radius R and total charge +Q. A point charge −q (q<<Q) with mass m is released from rest at a distance of 3R from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is :

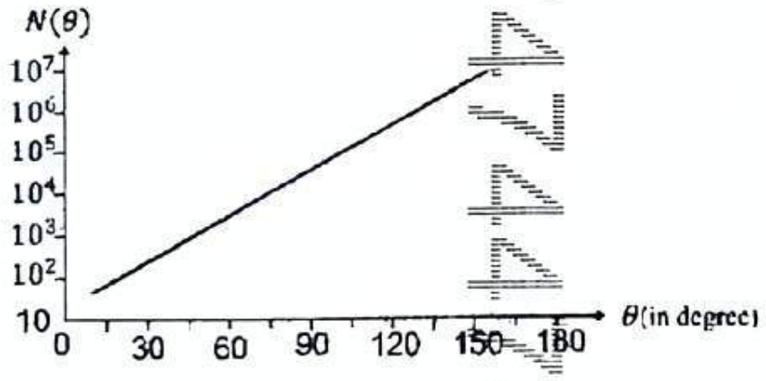
(ϵ_0 is the permittivity of vacuum, neglect gravitational forces).

(1) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$ (2) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$

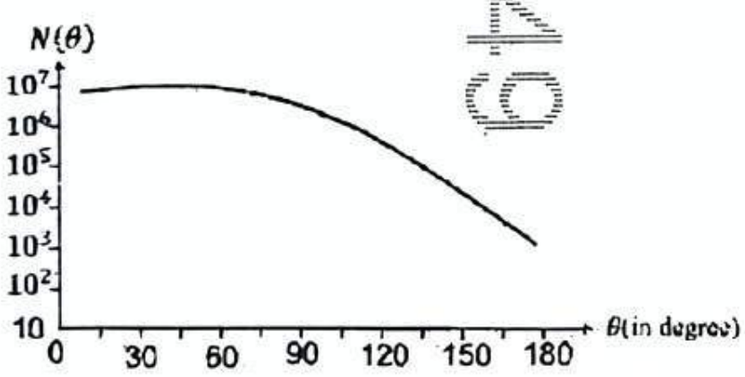
(3) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$ (4) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$

35. गीगर-मार्सडेन प्रयोग में, प्रकीर्णित α-कणों की संख्या $N(\theta)$ को प्रकीर्णन कोण θ के फलन के रूप में अंकित किया गया है। निम्नलिखित में से कौन सा विकल्प सही आलेख को निरूपित करता है ?

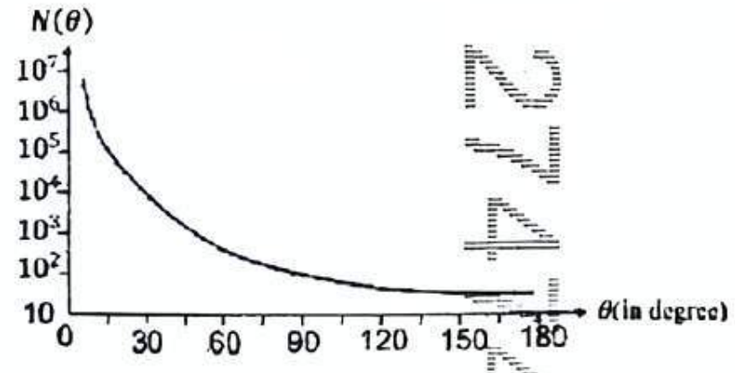
(1)



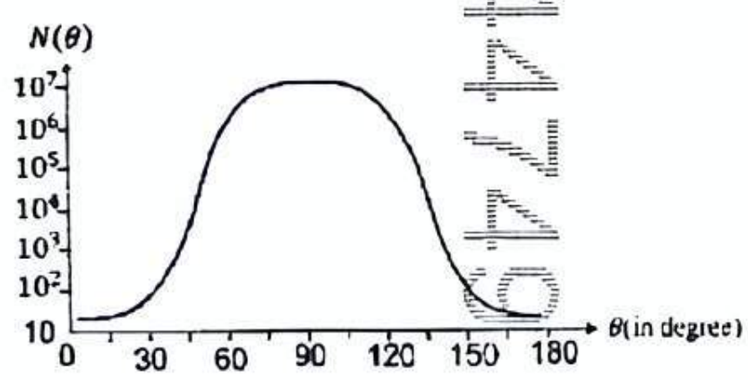
(2)



(3)

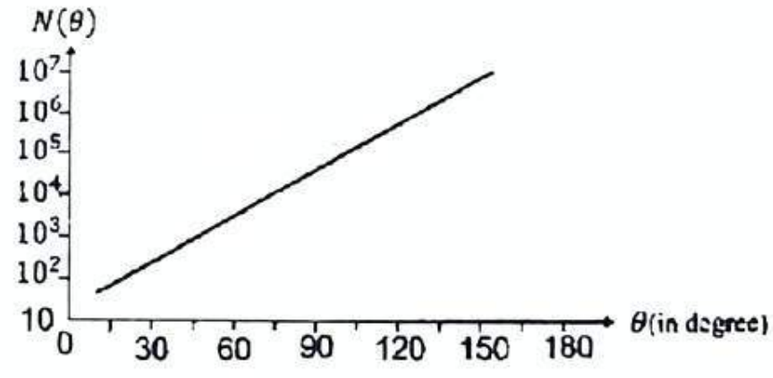


(4)

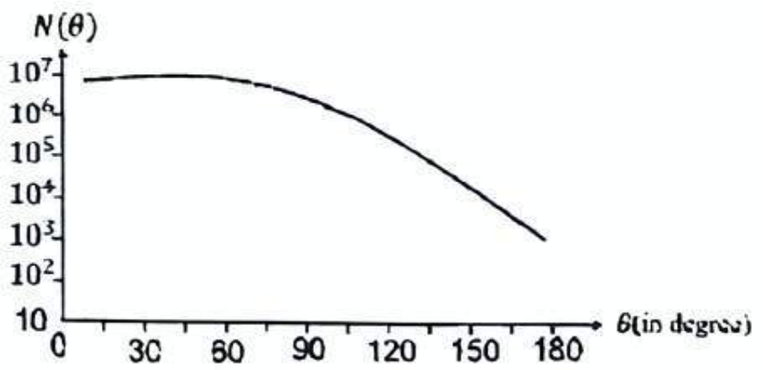


35. In Geiger-Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the correct plot ?

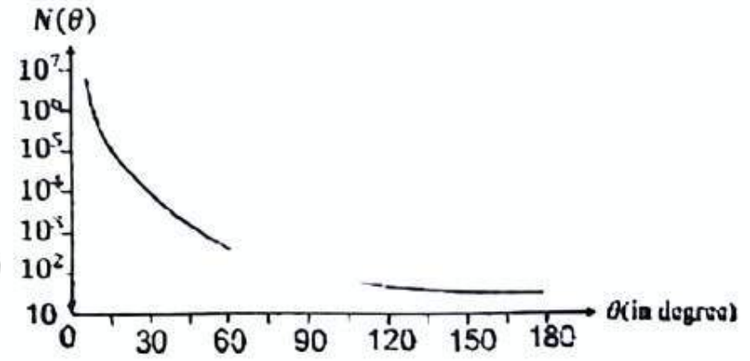
(1)



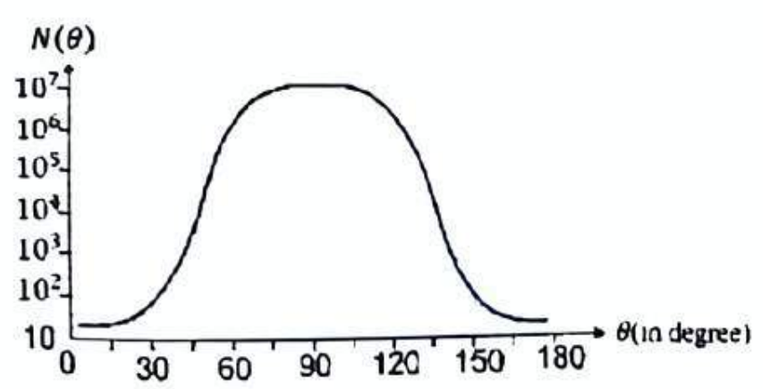
(2)



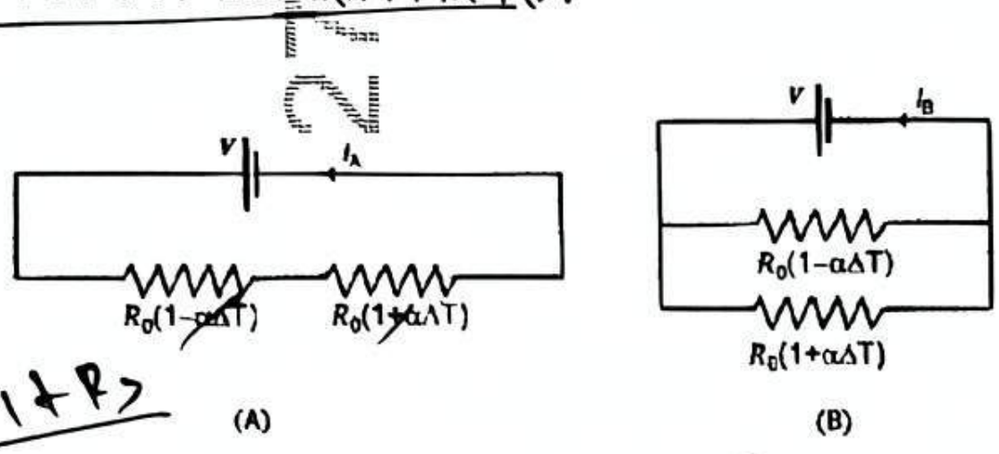
(3)



(4)



36. दो परिपथों, (A) और (B) पर विचार करें, जिनमें से प्रत्येक में दो प्रतिरोधक हैं। जैसा कि चित्र में दिखाया गया है, उनमें से एक का प्रतिरोध ताप गुणांक धनात्मक, $+\alpha$ है, जबकि दूसरे का प्रतिरोध ताप गुणांक ऋणात्मक, $-\alpha$ है। इन परिपथों में प्रवाहित धारा को I_A और I_B द्वारा दर्शाया गया है। प्रारंभिक तापमान पर, दोनों प्रतिरोधकों का प्रतिरोध R_0 है। जैसे-जैसे तापमान बढ़ाया जाता है, इन परिपथों में धारा के परिवर्तन का वर्णन करने वाला सही विकल्प है :



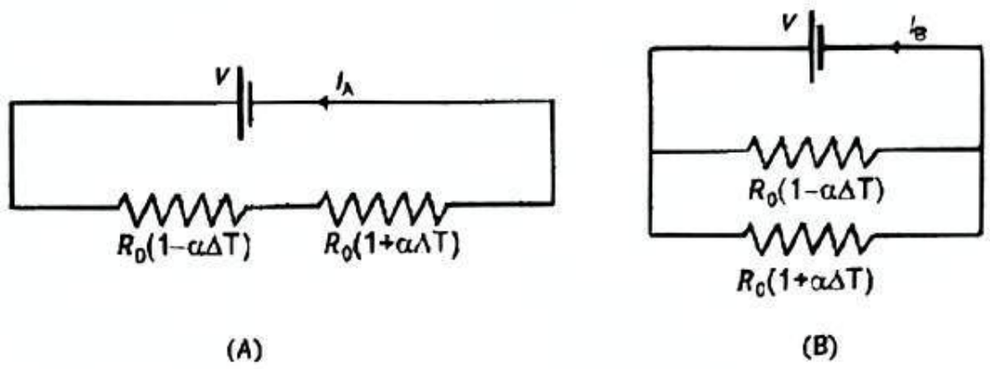
(1) I_A स्थिर रहता है जबकि I_B बढ़ता है

(2) I_A घटता है जबकि I_B बढ़ता है

(3) I_A बढ़ता है जबकि I_B घटता है

(4) I_A और I_B दोनों स्थिर रहते हैं

36. Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature of coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is :



(1) I_A remains constant while I_B increases

(2) I_A decreases while I_B increases

(3) I_A increases while I_B decreases

(4) both I_A and I_B remain constant

37. मान लीजिए कि σ_s, k_B, b क्रमशः स्टीफन-बोल्ट्ज़मैन नियतांक, बोल्ट्ज़मैन नियतांक और वीन के विस्थापन नियम के नियतांक को निरूपित करते हैं। $\sigma_s k_B^{-1} b$ की विमा है :

(1) $[L^{-1}T^{-1}K^{-2}]$

(2) $[L^{-1}K^{-2}]$

(3) $[L^{-1}T^{-1}K^{-3}]$

(4) $[L^{-1}T^{-1}K^{-4}]$

37. Consider that σ_s, k_B, b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is :

(1) $[L^{-1}T^{-1}K^{-2}]$

(2) $[L^{-1}K^{-2}]$

(3) $[L^{-1}T^{-1}K^{-3}]$

(4) $[L^{-1}T^{-1}K^{-4}]$

38. एक हानिरहित परावैद्युत माध्यम में गमन करने वाली एक वैद्युतचुम्बकीय तरंग, जिसका परावैद्युत नियतांक $\epsilon_r = 9$ है, का वैद्युत क्षेत्र $E_x = E_0 \sin (kz - 2\pi \times 10^6 t)\ Vm^{-1}$ है, जहाँ E_0 आयाम है और k तरंग सदिश है। निम्नलिखित विकल्पों में से, **गलत** विकल्प है :

(1) माध्यम के भीतर वैद्युतचुम्बकीय तरंग की चाल $10^8\ ms^{-1}$ है

(2) माध्यम के भीतर वैद्युतचुम्बकीय तरंग की तरंगदैर्ध्य 300 m है

(3) चुम्बकीय क्षेत्र संबंध $B_y = \frac{B_0}{\nu} \sin (kz - 2\pi \times 10^6\ t)$ द्वारा दिया गया है जहाँ ν माध्यम के भीतर वैद्युतचुम्बकीय तरंग की चाल है

(4) वैद्युतचुम्बकीय तरंग के संचरण की दिशा $+z$ के अनुदिश है

38. An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\epsilon_r = 9$, has the electric field, $E_x = E_0 \sin (kz - 2\pi \times 10^6 t)\ Vm^{-1}$ where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is :

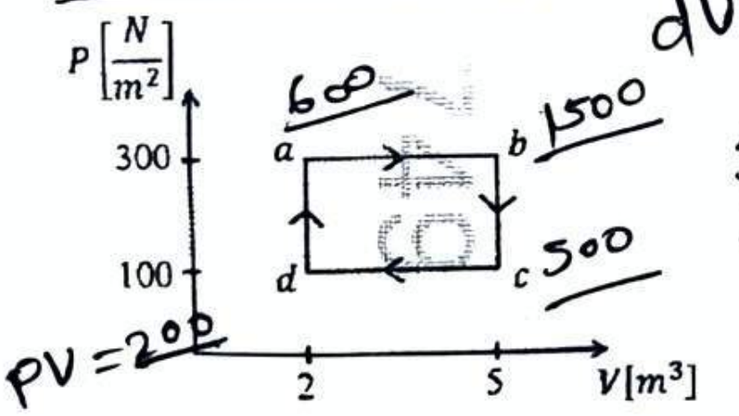
(1) The speed of the electromagnetic wave inside the medium is $10^8\ ms^{-1}$

(2) The wavelength of the electromagnetic wave inside the medium is 300 m

(3) The magnetic field is given by the relation $B_y = \frac{B_0}{\nu} \sin (kz - 2\pi \times 10^6\ t)$ where ν is the speed of the electromagnetic wave inside the medium

(4) The direction of propagation of the electromagnetic wave is along $+z$

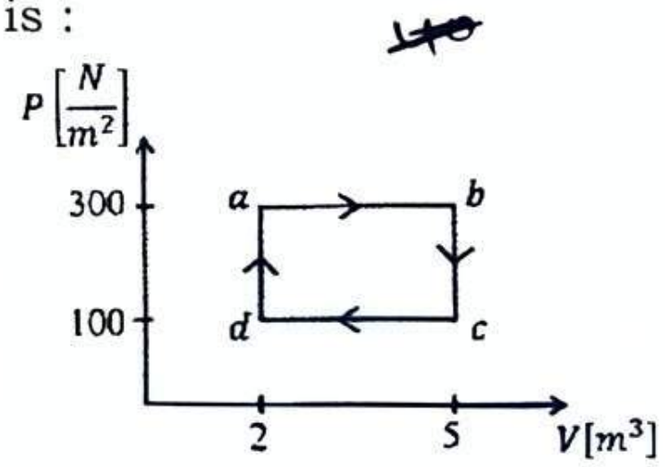
39. एक मोल एकपरमाणुक आदर्श गैस एक चक्रीय प्रक्रम से गुजरती है जैसा कि चित्र में दिखाया गया है। गैस को दी गयी (supplied) कुल ऊष्मा है :



(1) 400 J (2) 500 J

(3) 600 J (4) 800 J

39. One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is :



(1) 400 J (2) 500 J

(3) 600 J (4) 800 J

40. मान लीजिए कि एक इलेक्ट्रॉन हाइड्रोजन परमाणु की एक उत्तेजित अवस्था में $\sqrt{25.6} \times 10^5\ ms^{-1}$ के वेग से परिक्रमा कर रहा है। कक्षा की त्रिज्या $x \times 10^{-9}$ m है। x का मान है : [इलेक्ट्रॉन का द्रव्यमान 9×10^{-31} kg, इलेक्ट्रॉन का आवेश $= -1.6 \times 10^{-19}$ C और $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ N\ m^2\ C^{-2}$ लें]

(1) 4 (2) 3

(3) 2 (4) 1

40. Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6} \times 10^5\ ms^{-1}$. The radius of the orbit is $x \times 10^{-9}$ m. The value of x is : [Take the mass of electron to be 9×10^{-31} kg, charge of electron $= -1.6 \times 10^{-19}$ C and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ N\ m^2\ C^{-2}$]

(1) 4 (2) 3

(3) 2 (4) 1

41. एक कार 50 m त्रिज्या के एक वृत्ताकार दौड़-पथ पर चलती है, जो θ कोण पर बैंकड (banked) है। यदि कार $10\ ms^{-1}$ की चाल से चलती है, तो उसके टायरों का घिसाव न्यूनतम होता है। गुरुत्वीय त्वरण को $10\ ms^{-2}$ मानते हुए, θ का मान है :

(1) $\tan^{-1}\left(\frac{1}{5}\right)$ (2) $\tan^{-1}\left(\frac{2}{5}\right)$

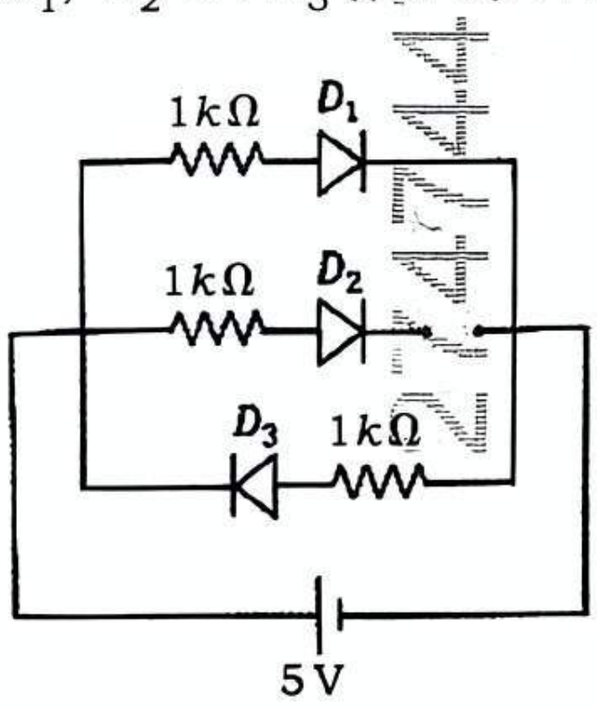
(3) $\tan^{-1}\left(\sqrt{3}/2\right)$ (4) $\tan^{-1}\left(2\sqrt{3}\right)$

41. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed $10\ ms^{-1}$, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be $10\ ms^{-2}$, the value of θ is :

(1) $\tan^{-1}\left(\frac{1}{5}\right)$ (2) $\tan^{-1}\left(\frac{2}{5}\right)$

(3) $\tan^{-1}\left(\sqrt{3}/2\right)$ (4) $\tan^{-1}\left(2\sqrt{3}\right)$

42. तीन समान p-n संधि डायोड D_1, D_2 और D_3 को एक बैटरी के आर-पार चित्र में दिखाए अनुसार जोड़ा गया है। यदि D_1, D_2 और D_3 के क्षय क्षेत्रों की चौड़ाई क्रमशः W_1, W_2 और W_3 है, तो सही विकल्प है :



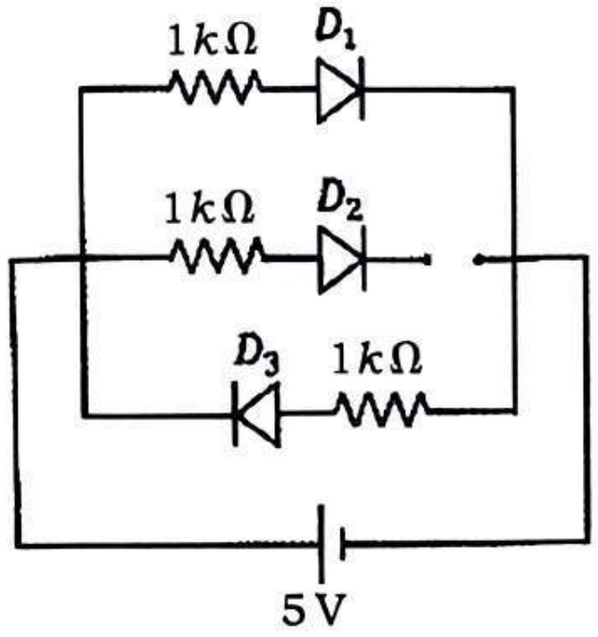
(1) $W_1 > W_2 > W_3$

(2) $W_3 = W_1 > W_2$

(3) $W_3 > W_2 > W_1$

(4) $W_2 > W_1 = W_3$

42. Three identical p-n junction diodes D_1, D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is :



(1) $W_1 > W_2 > W_3$

(2) $W_3 = W_1 > W_2$

(3) $W_3 > W_2 > W_1$

(4) $W_2 > W_1 = W_3$

43. निम्नलिखित तालिका विद्युत-चुम्बकीय स्पेक्ट्रम के भागों और उनके संगत प्रमुख अनुप्रयोगों को प्रस्तुत करती है।

विद्युत-चुम्बकीय स्पेक्ट्रम का भाग		अनुप्रयोग	
P	माइक्रोवेव	I	जल को शुद्ध करने के लिए
Q	UV किरणें	II	भोजन को गर्म करने के लिए
R	गामा किरणें	III	AM और FM संचार प्रणालियों के लिए
S	रेडियो तरंग	IV	कैंसर कोशिकाओं के उपचार के लिए

सही विकल्प है :

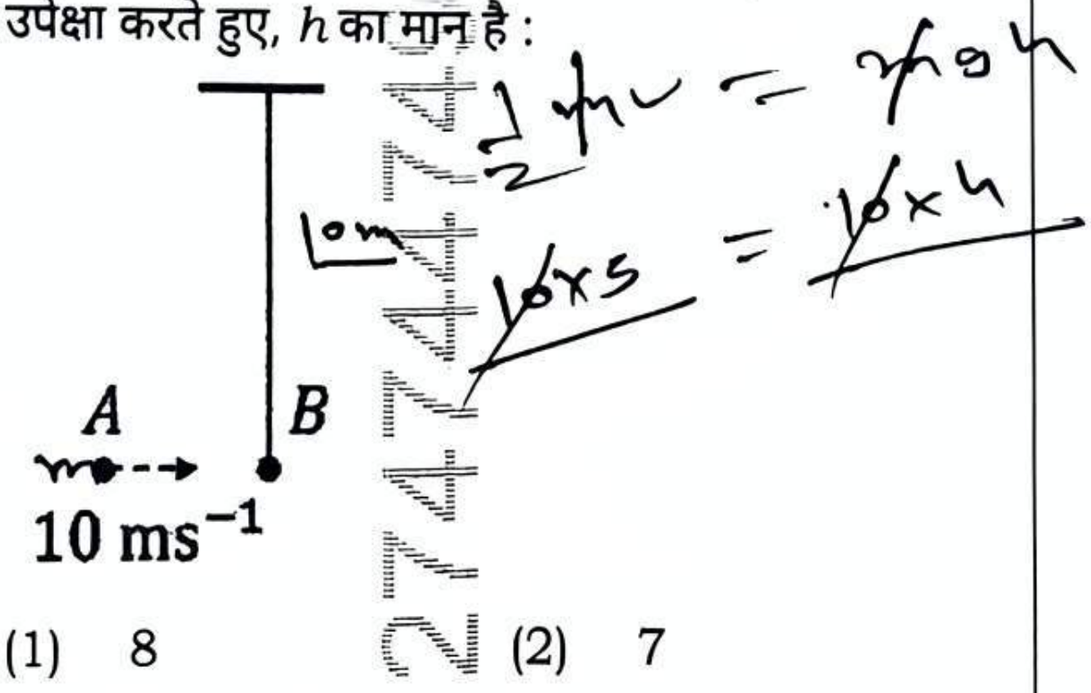
(1) P-I, Q-II, R-III, S-IV

(2) P-I, Q-IV, R-II, S-III

(3) P-II, Q-I, R-IV, S-III

(4) P-II, Q-IV, R-III, S-I

44. द्रव्यमान m का एक बॉब (गोलक) B विराम अवस्था में है और जैसा कि चित्र में दिखाया गया है, 10 m लंबाई की एक द्रव्यमानरहित डोरी के माध्यम से छत से लम्बवत् लटका हुआ है। द्रव्यमान m का एक बिंदु द्रव्यमान A, $10\ ms^{-1}$ की चाल से क्षैतिज रूप से गति करते हुए बॉब (गोलक) B से प्रत्यास्थ रूप से टकराता है। टक्कर के बाद बॉब (गोलक) B, h मीटर की ऊँचाई तक उठता है। गुरुत्वीय त्वरण $g = 10\ ms^{-2}$ मानकर और बॉब (गोलक) के आकार की उपेक्षा करते हुए, h का मान है :



(1) 8

(2) 7

(3) 5

(4) 2.5

43. The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the electromagnetic spectrum		Applications	
P	Microwave	I	For purifying the water
Q	UV rays	II	For warming the food
R	Gamma rays	III	For AM and FM communication systems
S	Radio wave	IV	For treating the Cancer cells

The **correct** option is :

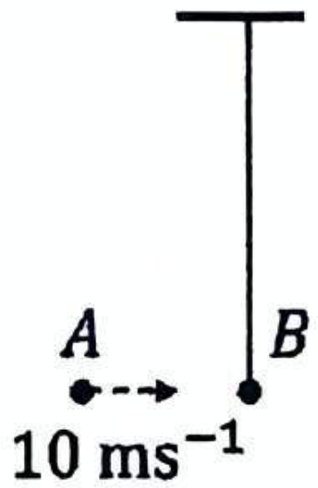
(1) P-I, Q-II, R-III, S-IV

(2) P-I, Q-IV, R-II, S-III

(3) P-II, Q-I, R-IV, S-III

(4) P-II, Q-IV, R-III, S-I

44. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed $10\ ms^{-1}$ hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ ms^{-2}$ and neglecting the size of the bob, the value of h is :



(1) 8

(2) 7

(3) 5

(4) 2.5

45. एक आदर्श गैस बहुपरमाणुक अणुओं से बनी है। प्रत्येक अणु के पास तीन स्थानांतरण, तीन घूर्णन और f संख्या में कंपन मोड हैं। यदि गैस की ऊष्मा धारिताओं का अनुपात C_P/C_V, 8/7 है, तो f का मान है :

(1) 4 (2) 3

(3) 2 (4) 1

45. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_P/C_V of the gas is 8/7, then the value of f is :

(1) 4 (2) 3

(3) 2 (4) 1

46. निम्नलिखित अभिक्रिया अनुक्रम के लिए, सही विकल्प चुनिए-

$\xrightarrow[\text{ii. NaOCl}]{\text{i. } CH_3COCl,\ AlCl_3}$ **P + Q**

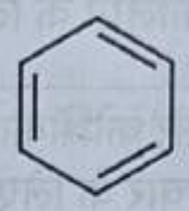
(1) यदि **P** एक कार्बोक्सिलिक अम्ल का सोडियम लवण है, तो **Q** एक प्राथमिक ऐल्कोहॉल है।

(2) **P** और **Q** एरोमैटिक यौगिक हैं।

(3) यदि **P** अम्लीकरण पर एक कार्बोक्सिलिक अम्ल देता है, तो **Q** हवा और प्रकाश के संपर्क में आने पर एक विषैली गैस देता है।

(4) **P** और **Q** दोनों कार्बोनिल यौगिक हैं।

46. For the following reaction sequence, choose the correct option



$\xrightarrow[\text{ii. NaOCl}]{\text{i. } CH_3COCl,\ AlCl_3}$ **P + Q**

(1) If **P** is the sodium salt of a carboxylic acid, **Q** is a primary alcohol.

(2) **P** and **Q** are aromatic compounds.

(3) If **P** gives a carboxylic acid on acidification, **Q** gives a poisonous gas on exposure to air and light.

(4) Both **P** and **Q** are carbonyl compounds.

47. नीचे दो कथन दिए गए हैं :

कथन-I : $[Fe(ox)_3]^{3-}$ कायरल (chiral) है।

कथन-II : विपक्ष (ट्रान्स) $-[Cr(H_2O)_2(ox)_2]^-$ कायरल (chiral) है।

(दिया है : oxH_2 = HOOC – COOH)

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए :

(1) **कथन-I** और **कथन-II**, दोनों सही हैं।

(2) **कथन-I** और **कथन-II**, दोनों गलत हैं।

(3) **कथन-I** सही है, परन्तु **कथन-II** गलत है।

(4) **कथन-I** गलत है, परन्तु **कथन-II** सही है।

47. Given below are two statements:

Statement-I : $[Fe(ox)_3]^{3-}$ is chiral.

Statement-II : *trans* $-[Cr(H_2O)_2(ox)_2]^-$ is chiral.

(Given : oxH_2 = HOOC – COOH)

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

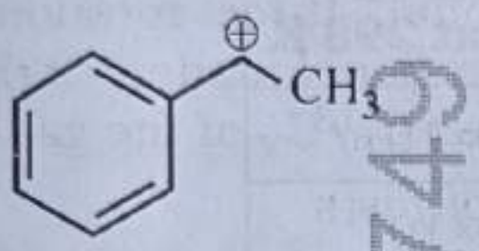
(1) Both **Statement-I** and **Statement-II** are correct.

(2) Both **Statement-I** and **Statement-II** are incorrect.

(3) **Statement-I** is correct but **Statement-II** is incorrect.

(4) **Statement-I** is incorrect but **Statement-II** is correct.

48. निम्नलिखित कार्बधनायन (carbocation) खाली p कक्षक की किसके साथ अन्योन्यक्रिया द्वारा स्थायी होता है



(1) भरे हुए σ और भरे हुए π कक्षकों

(2) खाली σ और खाली π* कक्षकों

(3) खाली σ* और भरे हुए π कक्षकों

(4) खाली σ* और खाली π* कक्षकों

49. फिटकरी (potash alum) में, K^+ और SO_4^{2-} आयनों का अनुपात है

(1) 1:2 (2) 2:1

(3) 2:3 (4) 3:2

50. पेप्टाइड्स और प्रोटीनों के बारे में सही कथन है

(1) प्रोटीनों की तृतीयक संरचना में दो या दो से अधिक पॉलीपेप्टाइड उप-इकाइयाँ होती हैं।

(2) केवल वही प्रोटीन जैविक रूप से सक्रिय होते हैं जिनमें चतुर्थक संरचना होती है।

(3) β-प्लीटेड शीट संरचनाओं में, पेप्टाइड श्रृंखलाएँ अंतर-आणविक हाइड्रोजन बंधों द्वारा एक साथ जुड़ी रहती हैं।

(4) α-हेलिसेस में, पॉलीपेप्टाइड श्रृंखला अंतः-आणविक हाइड्रोजन बंधों के माध्यम से एक वाम-हस्त स्क्रू (हेलिक्स) के रूप में मुड़ी होती है।

51. संख्याओं 17.0145 और 21.0235 को दशमलव बिंदु के बाद तीन अंकों तक निकटित किया गया। परिणामी संख्याएँ, क्रमशः हैं :

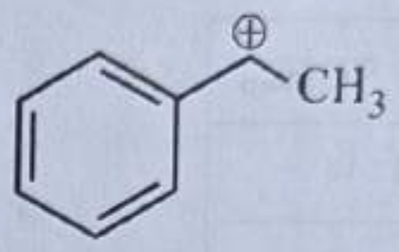
(1) 17.014 और 21.023

(2) 17.015 और 21.023

(3) 17.014 और 21.024

(4) 17.015 और 21.024

48. The following carbocation is stabilized by the interaction of the empty p orbital with



(1) filled σ and filled π orbitals

(2) empty σ and empty π* orbitals

(3) empty σ* and filled π orbitals

(4) empty σ* and empty π* orbitals

49. In potash alum, the ratio of K^+ and SO_4^{2-} ions is

(1) 1:2 (2) 2:1

(3) 2:3 (4) 3:2

50. The correct statement about peptides and proteins is

(1) Tertiary structure of proteins has two or more polypeptide subunits.

(2) Only the proteins having a quaternary structure are biologically active.

(3) In β-pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

(4) In α-helices, the polypeptide chain is twisted into a left-handed screw (helix) through intramolecular hydrogen bonds.

51. The numbers 17.0145 and 21.0235 were rounded to three figures after the decimal point. The resulting numbers, respectively, are

(1) 17.014 and 21.023

(2) 17.015 and 21.023

(3) 17.014 and 21.024

(4) 17.015 and 21.024

52. 298 K पर जल में दिए गए लवणों की विलेयता का सही क्रम है

लवण	298 K पर K_{sp}
AgBr	5.0×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

(1) $Hg_2Cl_2 > Zn(OH)_2 > AgBr$

(2) $AgBr > Zn(OH)_2 > Hg_2Cl_2$

(3) $Hg_2Cl_2 > AgBr > Zn(OH)_2$

(4) $Zn(OH)_2 > AgBr > Hg_2Cl_2$

52. The correct order of solubility of the given salts in water at 298 K is

Salt	K_{sp} at 298 K
AgBr	5.0×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

(1) $Hg_2Cl_2 > Zn(OH)_2 > AgBr$

(2) $AgBr > Zn(OH)_2 > Hg_2Cl_2$

(3) $Hg_2Cl_2 > AgBr > Zn(OH)_2$

(4) $Zn(OH)_2 > AgBr > Hg_2Cl_2$

53. निम्नलिखित विकल्पों में से, इलेक्ट्रॉन लब्धि एन्थैल्पी का सही क्रम है

(1) F > Cl > Br > I

(2) Br > Cl > F > I

(3) Cl > F > Br > I

(4) I > Br > Cl > F

53. Among the following options, the correct trend in the electron gain enthalpy is

(1) F > Cl > Br > I

(2) Br > Cl > F > I

(3) Cl > F > Br > I

(4) I > Br > Cl > F

54. अभिकथन A :

तरल **P** और **Q** के मिश्रण से बने एक आदर्श विलयन के लिए, $\Delta_{mix} H = 0$ और $\Delta_{mix} V = 0$ होता है।

कारण R :

P और **Q** के बीच कोई अन्योन्यक्रिया नहीं होती है। उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए।

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है।

(2) **A** और **R** दोनों सही हैं परंतु **R**, **A** की सही व्याख्या **नहीं** है।

(3) **A** सही है, परंतु **R** सही नहीं है।

(4) **A** सही नहीं है, परंतु **R** सही है।

54. Assertion A:

For an ideal solution formed by mixing liquids **P** and **Q**, $\Delta_{mix} H = 0$ and $\Delta_{mix} V = 0$

Reason R:

No interactions occur between **P** and **Q**

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(2) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

(3) **A** is correct but **R** is not correct

(4) **A** is not correct but **R** is correct

55. वह अमीनो अम्ल जिसका सोडियम फ्यूजन निष्कर्ष (sodium fusion extract), सोडियम नाइट्रोप्रुसाइड के साथ उपचार करने पर रक्त-लाल रंग देता है, वह है

(1) ल्यूसीन

(2) थ्रिऑनीन

(3) मेथियोनाइन

(4) सेरीन

55. The amino acid that gives a red-blood colour on treating its sodium fusion extract with sodium nitroprusside is

(1) leucine

(2) threonine

(3) methionine

(4) serine

56. 298 K पर अर्ध-सेल अभिक्रिया $Fe^{3+} + e^- \rightarrow Fe^{2+}$ के लिए मानक इलेक्ट्रोड विभव (E°) है
(दिया है : 298 K पर $E^\circ(Fe^{3+}/Fe) = -0.04$ V और $E^\circ(Fe^{2+}/Fe) = -0.44$ V)

(1) +0.40 V (2) +0.76 V
(3) −0.48 V (4) +0.92 V

56. The standard electrode potential (E°) for the half-cell reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$ at 298 K is
(Given : $E^\circ(Fe^{3+}/Fe) = -0.04$ V and $E^\circ(Fe^{2+}/Fe) = -0.44$ V at 298 K)

(1) +0.40 V (2) +0.76 V
(3) −0.48 V (4) +0.92 V

57. एक अम्लीय माध्यम में, 0.25 M ऑक्सैलिक अम्ल के 10 mL का $KMnO_4$ विलयन के साथ अनुमापन किया जाता है। यदि अंत बिंदु तक पहुँचने के लिए आवश्यक $KMnO_4$ विलयन का आयतन 10 mL है, तो $KMnO_4$ विलयन की सांद्रता है

(1) 0.10 M (2) 0.20 M
(3) 0.25 M (4) 0.15 M

57. In an acidic medium, 10 mL of 0.25 M oxalic acid is titrated with $KMnO_4$ solution. If the volume of $KMnO_4$ solution required to reach end point is 10 mL, the strength of the $KMnO_4$ solution is

(1) 0.10 M (2) 0.20 M
(3) 0.25 M (4) 0.15 M

58. क्रिस्टल क्षेत्र सिद्धांत के अनुसार, लिगेंड्स की उनकी क्षेत्र शक्ति के घटते क्रम के संबंध में सही क्रम है

(1) $CO > NH_3 > H_2O > Cl^-$
(2) $CO > H_2O > NH_3 > Cl^-$
(3) $Cl^- > H_2O > NH_3 > CO$
(4) $Cl^- > NH_3 > H_2O > CO$

58. According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is

(1) $CO > NH_3 > H_2O > Cl^-$
(2) $CO > H_2O > NH_3 > Cl^-$
(3) $Cl^- > H_2O > NH_3 > CO$
(4) $Cl^- > NH_3 > H_2O > CO$

59. एक आदर्श गैस के दो मोल 300 K पर 10 L से 100 L तक मुक्त प्रसार (free expansion) करते हैं। ΔS_{system} और $\Delta S_{surroundings}$ के मान हैं
(R सार्वत्रिक गैस नियतांक है)

(1) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 0$
(2) $\Delta S_{system} = 4.606$ R; $\Delta S_{surroundings} = -4.606$ R
(3) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 4.606$ R
(4) $\Delta S_{system} = 4.606$ R; $\Delta S_{surroundings} = 0$

59. Two moles of an ideal gas undergo free expansion from 10 L to 100 L at 300 K. The values of ΔS_{system} and $\Delta S_{surroundings}$ are
(R is universal gas constant)

(1) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 0$
(2) $\Delta S_{system} = 4.606$ R; $\Delta S_{surroundings} = -4.606$ R
(3) $\Delta S_{system} = 0$; $\Delta S_{surroundings} = 4.606$ R
(4) $\Delta S_{system} = 4.606$ R; $\Delta S_{surroundings} = 0$

60. $2A \xrightarrow{k} B$ एक शून्य-कोटि की अभिक्रिया है, जहाँ $k = 1.0$ mol L^{-1} min^{-1} है। यदि A की प्रारंभिक सांद्रता 2 M है, तो अभिक्रिया के 75% पूरा होने में लगा समय होगा

(1) 1.5 min
(2) 0.75 min
(3) 1.0 min
(4) 2.0 min

60. $2A \xrightarrow{k} B$ is a zero-order reaction, where $k = 1.0$ mol L^{-1} min^{-1}. If the initial concentration of A is 2 M, then the time taken to complete 75% of the reaction will be

(1) 1.5 min
(2) 0.75 min
(3) 1.0 min
(4) 2.0 min

61. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **कारण R** चिह्नित किया गया है।

अभिकथन A : सामान्यतः, $3d$ संक्रमण धातुओं के गलनांक उच्च होते हैं।

कारण R : अंतर-परमाणु धात्विक आबंध में $4s$ -इलेक्ट्रॉनों के अतिरिक्त $3d$-इलेक्ट्रॉनों की भागीदारी।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है।

(2) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या **नहीं** है।

(3) **A** सही है परन्तु **R** सही नहीं है।

(4) **A** सही नहीं है परन्तु **R** सही है।

61. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Generally, $3d$ transition metals have high melting points.

Reason R : Involvement of $3d$-electrons in addition to $4s$-electrons in the interatomic metallic bonding.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

(3) **A** is correct but **R** is not correct.

(4) **A** is not correct but **R** is correct.

62. एक लवण **XY** के लिए, जो एक प्रबल विद्युत अपघट्य है, 298 K पर Λ_m बनाम $\sqrt{c}$ के आलेख का ढाल $-90.0\ S\ cm^2\ mol^{-3/2}\ L^{1/2}$ है। **XY** की 0.01 M सांद्रता पर, Λ_m का मान $145.0\ S\ cm^2\ mol^{-1}$ है। 298 K पर Y^- आयन की सीमांत मोलर चालकता ($\lambda^0_{Y^-}$ $S\ cm^2\ mol^{-1}$ में) होगी

(दिया है : $\lambda^0_{X^+} = 74.0\ S\ cm^2\ mol^{-1}$)

(1) 80.0 (2) 100.0

(3) 90.0 (4) 76.0

62. For a salt **XY**, which is a strong electrolyte, the plot of Λ_m versus $\sqrt{c}$ has a slope of $-90.0\ S\ cm^2\ mol^{-3/2}\ L^{1/2}$ at 298 K. At 0.01 M concentration of **XY**, the value of Λ_m is $145.0\ S\ cm^2\ mol^{-1}$. The limiting molar conductivity of Y^- ion ($\lambda^0_{Y^-}$, in $S\ cm^2\ mol^{-1}$) at 298 K will be

(Given : $\lambda^0_{X^+} = 74.0\ S\ cm^2\ mol^{-1}$)

(1) 80.0 (2) 100.0

(3) 90.0 (4) 76.0

63. 116 g n-ब्यूटेन के पूर्ण दहन पर उत्पन्न कार्बन डाइऑक्साइड की मात्रा है

(दिया है : amu में परमाणु द्रव्यमान H = 1, C = 12 और O = 16)

(1) 352 g (2) 322 g

(3) 176 g (4) 362 g

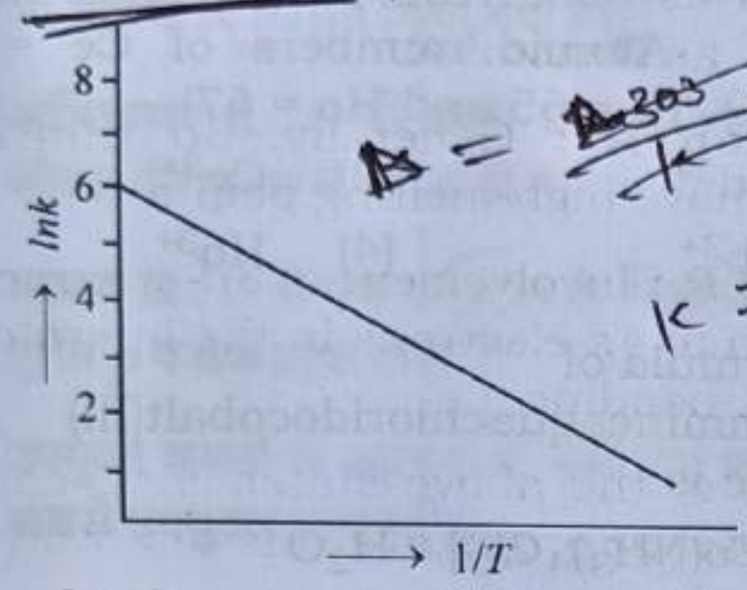
63. The amount of carbon dioxide evolved upon complete combustion of 116 g of n-butane is

(Given: atomic mass in amu H = 1, C = 12 and O = 16)

(1) 352 g (2) 322 g

(3) 176 g (4) 362 g

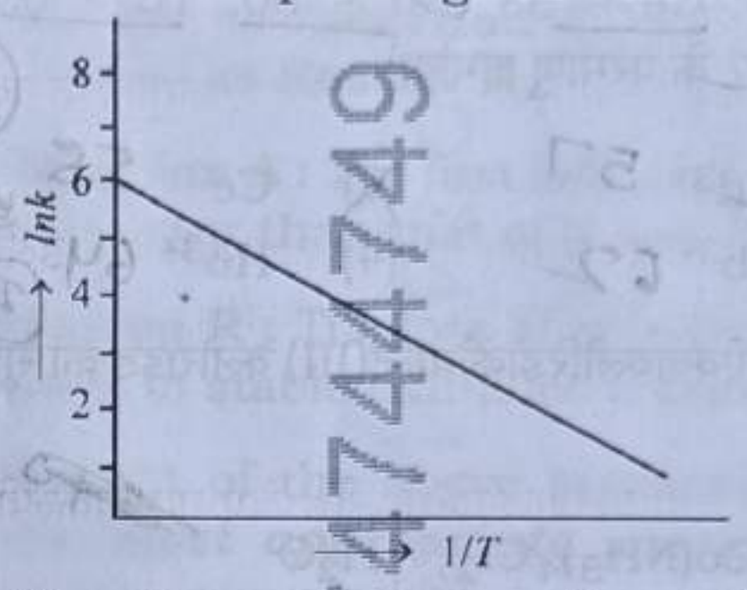
64. एक प्राथमिक रासायनिक अभिक्रिया के लिए, अर्हिनियस प्लॉट नीचे दिया गया है।



यदि सक्रियण ऊर्जा 6.64 kJ mol^{-1} है और R = 8.3 J K^{-1} mol^{-1} है, तो वह तापमान जिस पर वेग स्थिरांक e^2 min^{-1} हो जाता है, वह है

(1) 125 K (2) 150 K
(3) 200 K (4) 250 K

64. For an elementary chemical reaction, the Arrhenius plot is given below.



If the energy of activation is 6.64 kJ mol^{-1} and R = 8.3 J K^{-1} mol^{-1}, the temperature at which the rate constant becomes e^2 min^{-1}, is

(1) 125 K (2) 150 K
(3) 200 K (4) 250 K

65. नीचे दो कथन दिए गए हैं:

कथन-I : NaCl के साथ सांद्र H_2SO_4 और MnO_2 को गर्म करने से Mn का ऑक्सीकरण होता है।

कथन-II : NaI को सांद्र H_2SO_4 और MnO_2 के साथ गर्म करने से Mn का अपचयन होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए:

(1) **कथन-I** और **कथन-II**, दोनों सही हैं।

(2) **कथन-I** और **कथन-II**, दोनों गलत हैं।

(3) **कथन-I** सही है, परन्तु **कथन-II** गलत है।

(4) **कथन-I** गलत है, परन्तु **कथन-II** सही है।

65. Given below are two statements:

Statement-I : Heating NaCl with concentrated H_2SO_4 and MnO_2 results in oxidation of Mn.

Statement-II : Heating NaI with concentrated H_2SO_4 and MnO_2 results in reduction of Mn.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) Both **Statement-I** and **Statement-II** are correct.

(2) Both **Statement-I** and **Statement-II** are incorrect.

(3) **Statement-I** is correct but **Statement-II** is incorrect.

(4) **Statement-I** is incorrect but **Statement-II** is correct.

66. नीचे दी गई प्रजातियों (स्पीशीज) में से, किसका केवल-चक्रण चुंबकीय आघूर्ण (spin-only magnetic moment) उच्चतम है?

(दिया है : Ti की परमाणु संख्या = 22, Mn = 25, Fe = 26 और Co = 27)

(1) $[Mn(CN)_6]^{3-}$ (2) $[Fe(CN)_6]^{3-}$

(3) $[Co(NH_3)_6]^{3+}$ (4) $[Ti(H_2O)_6]^{3+}$

66. Among the species given below, the spin-only magnetic moment is highest for

(Given : Atomic number of Ti = 22, Mn = 25, Fe = 26 and Co = 27)

(1) $[Mn(CN)_6]^{3-}$ (2) $[Fe(CN)_6]^{3-}$

(3) $[Co(NH_3)_6]^{3+}$ (4) $[Ti(H_2O)_6]^{3+}$

67. चार अयुग्मित इलेक्ट्रॉनों वाला लैंथेनाइड आयन है
(दिया है : Ce = 58, Nd = 60, Tb = 65 और Ho = 67 के परमाणु क्रमांक)

(1) Nd^{3+}
(2) Ce^{3+}
(3) Tb^{3+}
(4) Ho^{3+}

67. The lanthanide ion having four unpaired electrons is
(Given : Atomic numbers of Ce = 58, Nd = 60, Tb = 65 and Ho = 67)

(1) Nd^{3+}
(2) Ce^{3+}
(3) Tb^{3+}
(4) Ho^{3+}

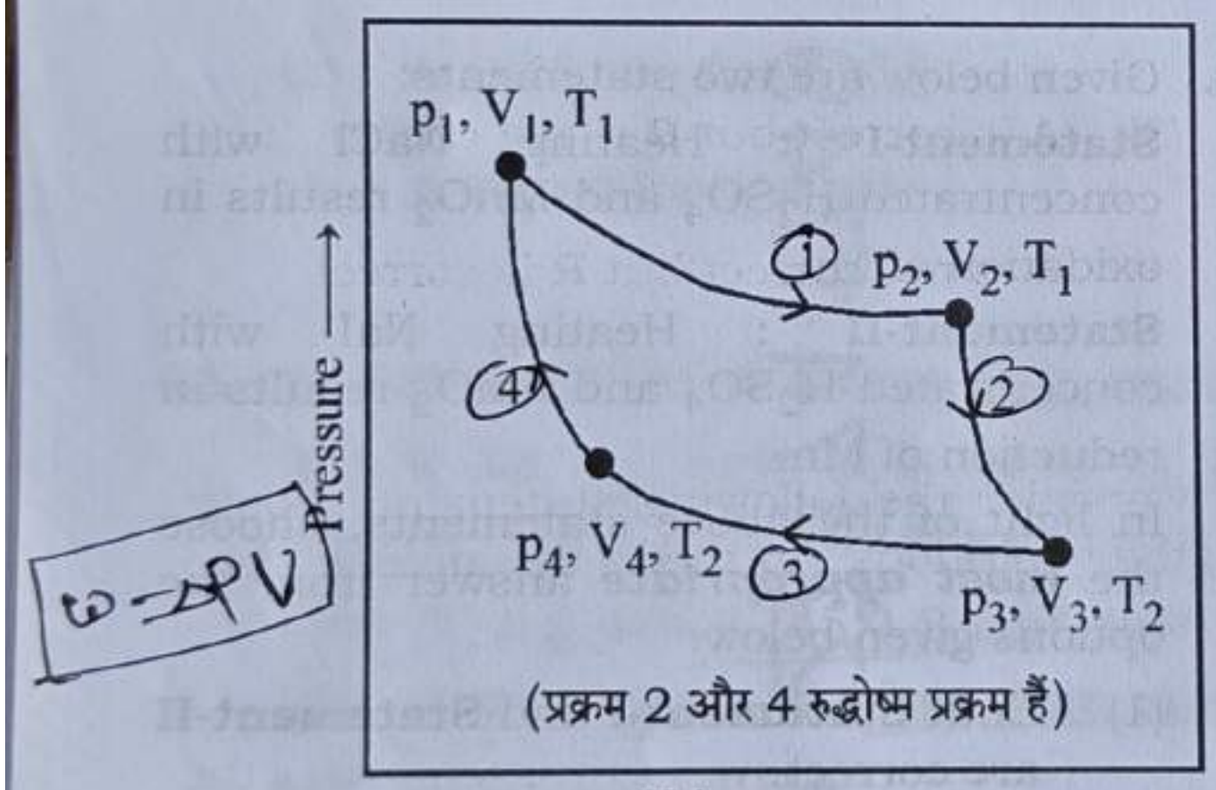
68. टेट्राऐमीनऐक्वाक्लोरिडोकोबाल्ट(III) क्लोराइड का सूत्र है

(1) $[Co(NH_3)_4Cl_2] \times H_2O$
(2) $[Co(NH_3)_4]Cl_3 \times H_2O$
(3) $[Co(NH_3)_4(H_2O)Cl]Cl$
(4) $[Co(NH_3)_4(H_2O)Cl]Cl_2$

68. The formula of tetraammineaquachloridocobalt(III) chloride is

(1) $[Co(NH_3)_4Cl_2] \times H_2O$
(2) $[Co(NH_3)_4]Cl_3 \times H_2O$
(3) $[Co(NH_3)_4(H_2O)Cl]Cl$
(4) $[Co(NH_3)_4(H_2O)Cl]Cl_2$

69. 1.0 mol आदर्श गैस के लिए चित्र में दिखाई गई उत्क्रमणीय प्रक्रियाओं पर विचार कीजिए।



w_1, w_2, w_3 और w_4 क्रमशः प्रक्रिया 1, 2, 3 और 4 में किए गए कार्य (कैलोरी में) को दर्शाते हैं; ΔU_2 और ΔU_4 क्रमशः प्रक्रिया 2 और 4 के लिए आंतरिक ऊर्जा में परिवर्तन हैं।

[$R = 2$ cal K^{-1} mol^{-1} उपयोग करें]

सही विकल्प है

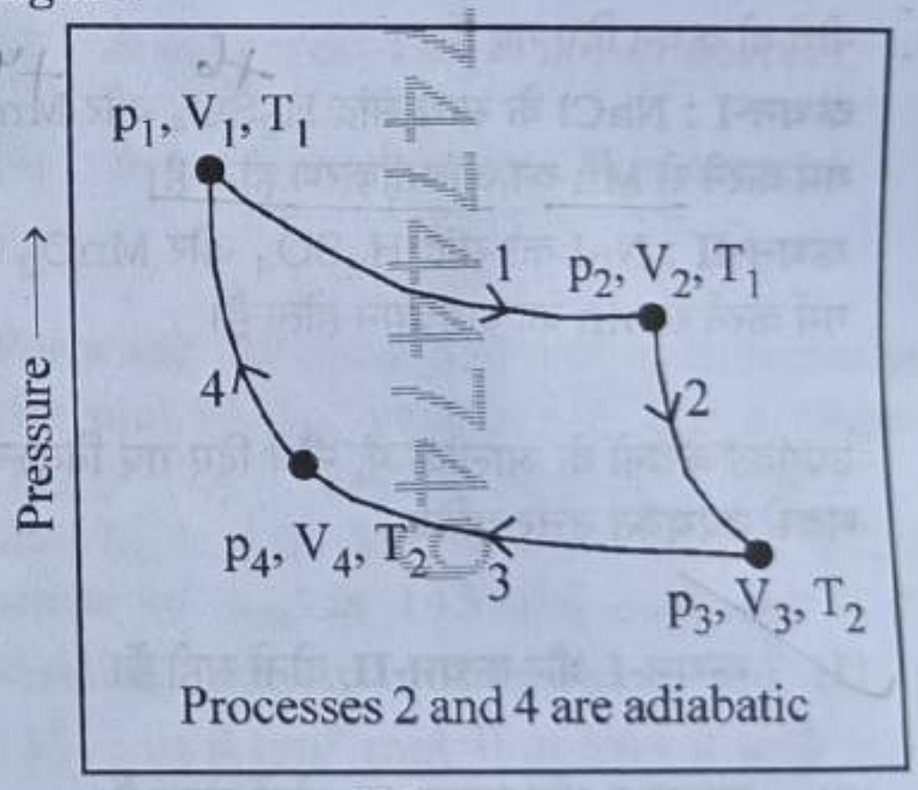
(1) $w_1 + w_3 = -2T_1 \ln\frac{V_2}{V_1} - 2T_2 \ln\frac{V_4}{V_3}$

(2) $w_2 + w_4 = \Delta U_2 - \Delta U_4$

(3) $w_1 + w_2 = 2T_1 \ln\frac{V_2}{V_1}$

(4) $w_1 + w_2 + w_3 + w_4 = 0$

69. Consider the reversible processes for 1.0 mol of an ideal gas as shown in the figure.



w_1, w_2, w_3 and w_4 represent work done (in calories) in the processes 1, 2, 3 and 4, respectively; ΔU_2 and ΔU_4 are changes in the internal energy for the processes 2 and 4, respectively.

[use $R = 2$ cal K^{-1} mol^{-1}]

The correct option is

(1) $w_1 + w_3 = -2T_1 \ln\frac{V_2}{V_1} - 2T_2 \ln\frac{V_4}{V_3}$

(2) $w_2 + w_4 = \Delta U_2 - \Delta U_4$

(3) $w_1 + w_2 = 2T_1 \ln\frac{V_2}{V_1}$

(4) $w_1 + w_2 + w_3 + w_4 = 0$

70. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **कारण R** चिह्नित किया गया है।

अभिकथन A : O की प्रथम आयनन ऊर्जा (एन्थैल्पी), N और F की तुलना में कम होती है।

कारण R : O से एक इलेक्ट्रॉन के निकलने से स्थायी अर्ध-पूरित p कक्षक प्राप्त होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है।

(2) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या **नहीं** है।

(3) **A** सही है परन्तु **R** सही नहीं है।

(4) **A** सही नहीं है परन्तु **R** सही है।

70. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The first ionization enthalpy of O is lower than that of N and F.

Reason R : The loss of an electron from O leads to stable half-filled p orbital.

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

(3) **A** is correct but **R** is not correct.

(4) **A** is not correct but **R** is correct.

71. दो द्रवों के मिश्रण से बने विलयनों के बारे में निम्नलिखित कथनों पर विचार कीजिए।

A. इस प्रकार बना एक आदर्श विलयन संपूर्ण सांद्रता सीमा में राउल्ट के नियम का पालन करता है।

B. क्लोरोफॉर्म और एसीटोन का मिश्रण राउल्ट के नियम से ऋणात्मक विचलन प्रदर्शित करता है।

C. ऐनिलीन और फिनोल का मिश्रण राउल्ट के नियम से धनात्मक विचलन प्रदर्शित करता है।

(1) केवल A और B

(2) केवल B और C

(3) केवल A

(4) केवल A और C

71. Consider the following statements about the solutions formed by mixing two liquids.

A. An ideal solution thus formed obeys Raoult's law throughout the composition range.

B. Mixture of chloroform and acetone shows negative deviation from Raoult's law.

C. Mixture of aniline and phenol shows positive deviation from Raoult's law.

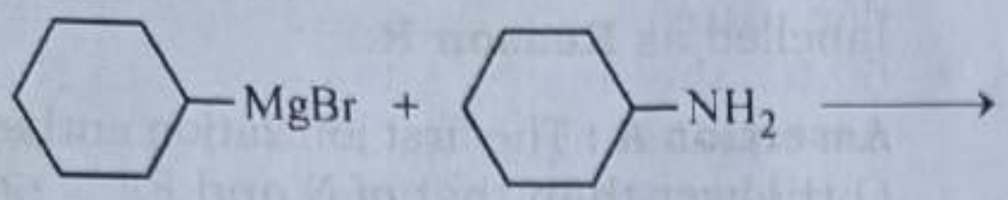
(1) A and B only

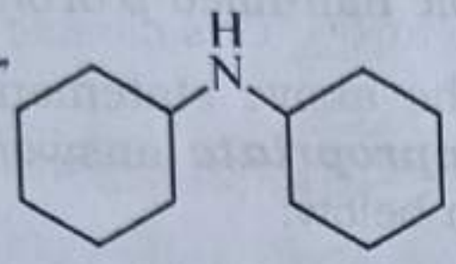
(2) B and C only

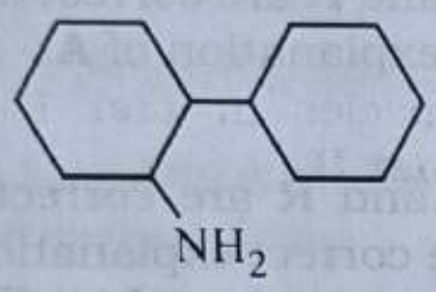
(3) A only

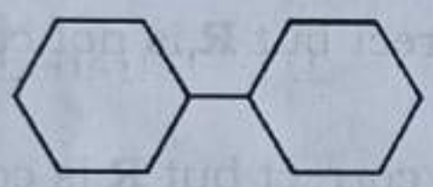
(4) A and C only

72. निम्नलिखित अभिक्रिया में बनने वाले उत्पादों में से एक है



(1) 

(2) 

(3) 

(4)

73. सही कथन है

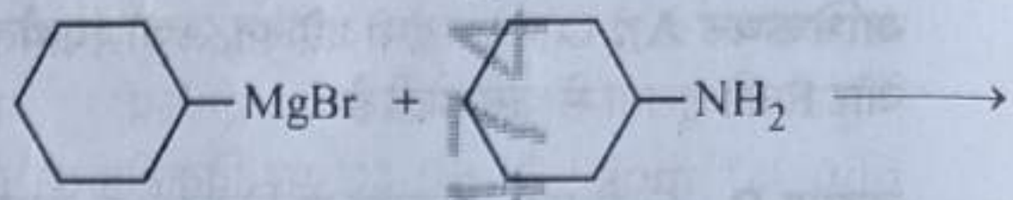
(1) बोरॉन की अधिकतम सहसंयोजकता चार होती है।

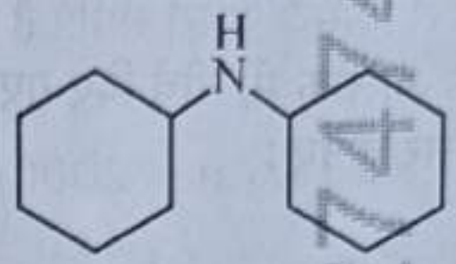
(2) बेरिलियम के तीन संयोजकता कक्षक होते हैं।

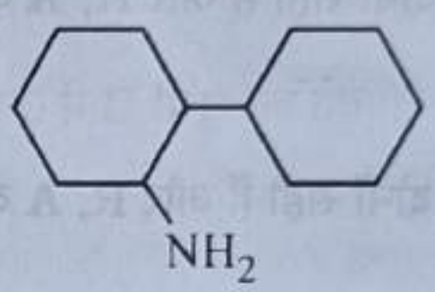
(3) मैग्नीशियम की अधिकतम सहसंयोजकता चार होती है।

(4) एल्युमीनियम के पाँच संयोजकता कक्षक होते हैं।

72. One of the products formed in the following reaction is



(1) 

(2) 

(3)

(4)

73. The correct statement is

(1) Boron has a maximum covalency of four.

(2) Beryllium has three valence orbitals.

(3) Magnesium has a maximum covalency of four.

(4) Aluminium has five valence orbitals.

74. एक प्रोटीन अपनी प्रारंभिक अवस्था **N** से विकृत अवस्था **D** में **N** ⇌ **D** के अनुसार उत्क्रमणीय तापीय विकृतीकरण (thermal denaturation) से गुजरता है। 60 °C पर, साम्यावस्था में **N** और **D** दोनों की सांद्रताएँ समान हैं, और विकृतीकरण का मानक एन्थैल्पी परिवर्तन 666 kJ mol^{-1} है। 60 °C पर विकृतीकरण होने पर प्रोटीन का मानक एन्ट्रॉपी परिवर्तन (ΔS° kJ $K^{-1}mol^{-1}$ में) निम्नलिखित में से किसके सबसे समीप है ?

(1) 2.0 (2) 2000.0

(3) 333.0 (4) 11.1

74. A protein undergoes reversible thermal denaturation from its initial state **N** to denatured state **D** according to **N** ⇌ **D**. At 60 °C, the concentrations of both **N** and **D** are equal at equilibrium, and the standard enthalpy change of denaturation is 666 kJ mol^{-1}. The standard entropy change (ΔS° in kJ $K^{-1}mol^{-1}$) of the protein upon denaturation at 60 °C is closest to

(1) 2.0 (2) 2000.0

(3) 333.0 (4) 11.1

75. **सूची I** में दी गई प्रजातियों का **सूची II** में उनकी ज्यामिति के साथ मिलान कीजिए

सूची I		**सूची II**
A. PCl_5	I.	चतुष्फलकीय (Tetrahedral)
B. BrF_5	II.	वर्ग समतलीय (Square Planar)
C. BF_4^-	III.	त्रिकोणीय द्विपिरामिडीय (Trigonal bipyramidal)
D. $[Ni(CN)_4]^{2-}$	IV.	वर्ग पिरामिडीय (Square pyramidal)

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनिए:

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-II, D-IV

(4) A-III, B-II, C-I, D-IV

75. Match the species in **List I** with their geometry in **List II**

List I		**List II**
A. PCl_5	I.	Tetrahedral
B. BrF_5	II.	Square Planar
C. BF_4^-	III.	Trigonal bipyramidal
D. $[Ni(CN)_4]^{2-}$	IV.	Square pyramidal

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

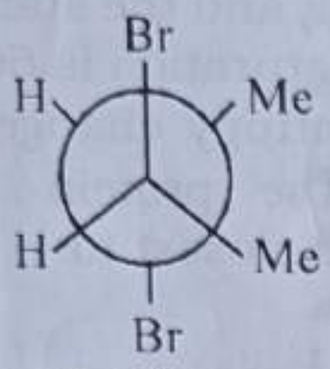
(3) A-III, B-I, C-II, D-IV

(4) A-III, B-II, C-I, D-IV

76. नीचे दो कथन दिए गए हैं:

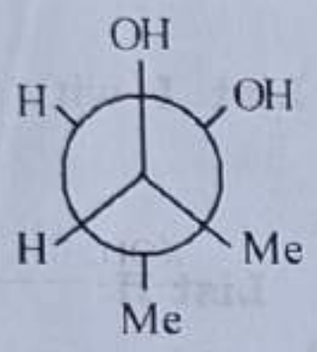
कथन I :

CCl_4 में Br_2 के साथ उपचार करने पर *विपक्ष*-ब्यूट-2-ईन निम्नलिखित उत्पाद देता है



कथन II :

क्षारीय $KMnO_4$ के साथ उपचार करने पर *समपक्ष*-ब्यूट-2-ईन निम्नलिखित उत्पाद देता है



उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए।

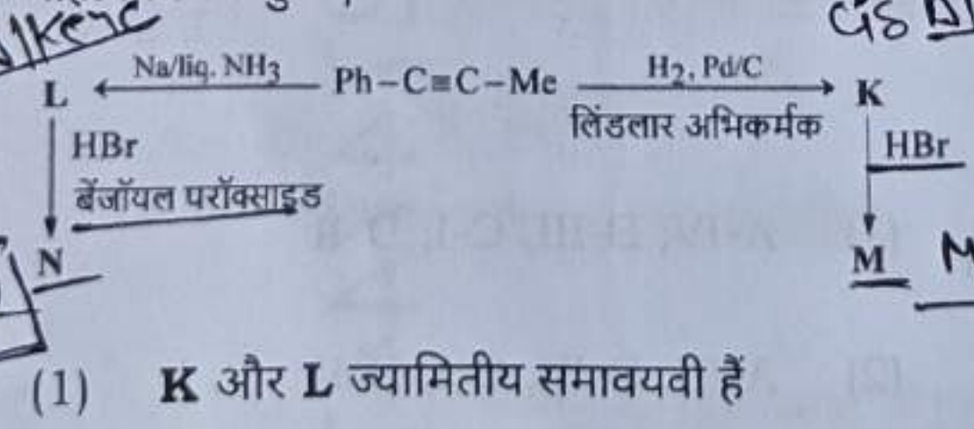
(1) **कथन-I** और **कथन-II**, दोनों सही हैं।

(2) **कथन-I** और **कथन-II**, दोनों गलत हैं।

(3) **कथन-I** सही है, परन्तु **कथन-II** गलत है।

(4) **कथन-I** गलत है, परन्तु **कथन-II** सही है।

77. निम्नलिखित अभिक्रिया अनुक्रमों पर विचार कीजिए और सही विकल्प चुनिए।



(1) **K** और **L** ज्यामितीय समावयवी हैं

(2) **K** और **L** प्रतिबिंब रूप (enantiomers) हैं

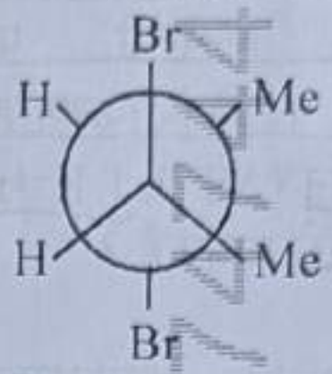
(3) **M** और **N** ज्यामितीय समावयवी हैं

(4) **M** और **N** त्रिविम समावयवी (stereoisomers) हैं

76. Given below are two statements:

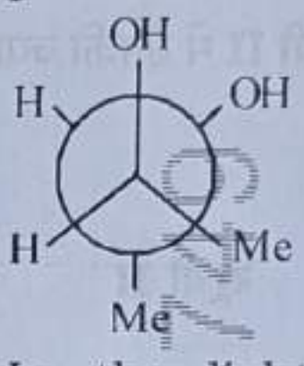
Statement I :

trans-But-2-ene upon treatment with Br_2 in CCl_4 gives the following product



Statement II :

cis-But-2-ene upon treatment with alkaline $KMnO_4$ gives the following product



In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

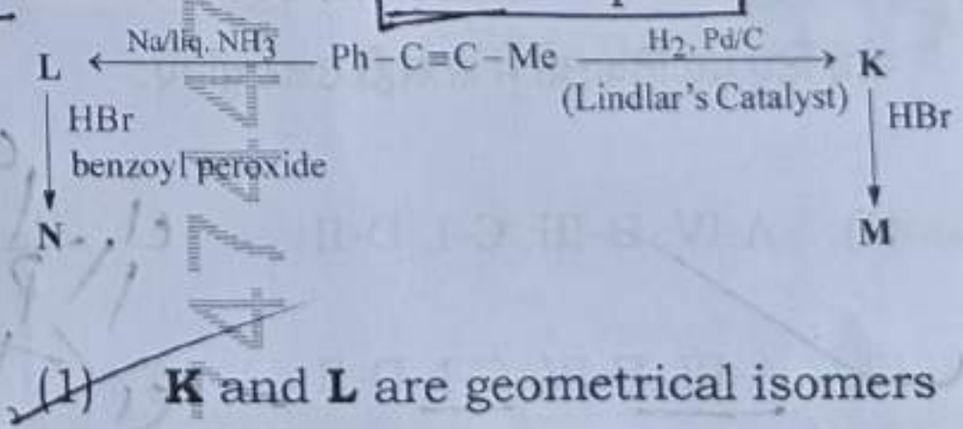
(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

77. Consider the following reaction sequences and choose the correct option.



(1) **K** and **L** are geometrical isomers

(2) **K** and **L** are enantiomers

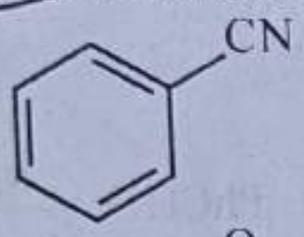
(3) **M** and **N** are geometrical isomers

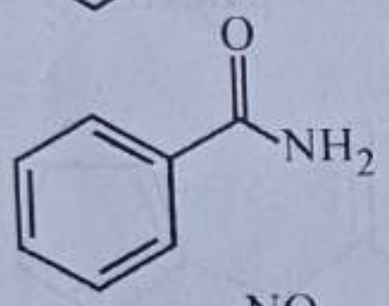
(4) **M** and **N** are stereoisomers

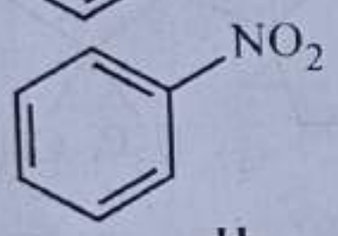
78. वह संकुल जिसमें *facial* (फलकीय) और *meridional* (रेखांशिक) समावयवी होते हैं, है
(दिया है : py = पिरिडीन और en = $H_2N–CH_2–CH_2–NH_2$)
(1) $[Cr(py)_3(Cl)_3]$
(2) $[Cr(H_2O)_6]^{3+}$
(3) $[Co(NH_3)_4(H_2O)_2]^{3+}$
(4) $[Ni(en)_2(H_2O)_2]^{2+}$

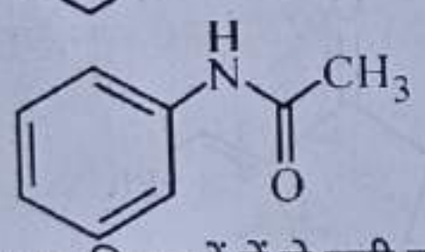
78. The complex which has *facial* and *meridional* isomers is
(Given : py = pyridine and en = $H_2N–CH_2–CH_2–NH_2$)
(1) $[Cr(py)_3(Cl)_3]$
(2) $[Cr(H_2O)_6]^{3+}$
(3) $[Co(NH_3)_4(H_2O)_2]^{3+}$
(4) $[Ni(en)_2(H_2O)_2]^{2+}$

79. उन अभिक्रियाओं की पहचान कीजिए जिनमें मुख्य उत्पाद के रूप में ऐनिलीन प्राप्त होता है।

A.  $\xrightarrow{LiAlH_4}$

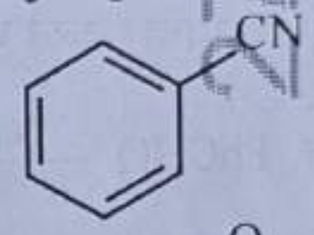
B.  $\xrightarrow{KOH,\ Br_2}$

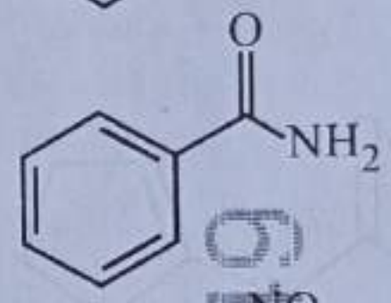
C.  $\xrightarrow{NaBH_4}$

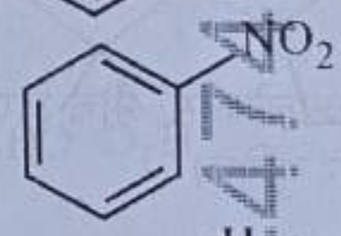
D.  $\xrightarrow[\Delta]{HCl,\ H_2O}$

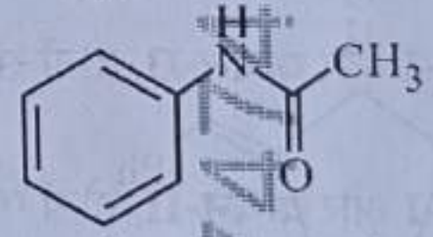
नीचे दिए गए विकल्पों में से सही उत्तर चुनिए।
(1) केवल A और B
(2) केवल B और D
(3) केवल A और C
(4) केवल C और D

79. Identify the reactions which give aniline as the major product.

A.  $\xrightarrow{LiAlH_4}$

B.  $\xrightarrow{KOH,\ Br_2}$

C.  $\xrightarrow{NaBH_4}$

D.  $\xrightarrow[\Delta]{HCl,\ H_2O}$

Choose the correct answer from the options given below.
(1) A and B only
(2) B and D only
(3) A and C only
(4) C and D only

80. **सूची-I** में दिए गए विटामिनों का **सूची-II** में दिए गए उनके स्रोतों के साथ मिलान कीजिए

	सूची-I		**सूची-II**
A.	विटामिन A	I.	मांस
B.	विटामिन B_{12}	II.	सूरजमुखी का तेल
C.	विटामिन E	III.	हरी पत्तेदार सब्जियाँ
D.	विटामिन K	IV.	गाजर

नीचे दिए गए विकल्पों में से सही उत्तर चुनिए।
(1) A-II, B-III, C-IV, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

80. Match the vitamins in **List-I** with their sources in **List-II**

	List-I		**List-II**
A.	vitamin A	I.	meat
B.	vitamin B_{12}	II.	sunflower oil
C.	vitamin E	III.	green leafy vegetables
D.	vitamin K	IV.	carrots

Choose the correct answer from the options given below.
(1) A-II, B-III, C-IV, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

81. प्रत्येक अणु में रेखांकित परमाणु की ऑक्सीकरण अवस्था का सही घटता क्रम है

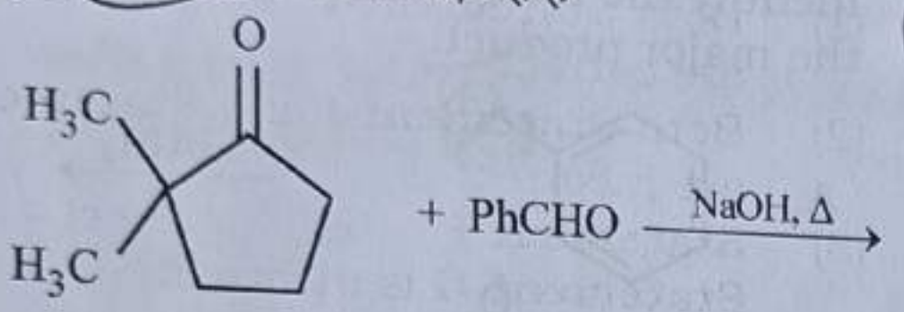
(1) $\underline{P_4O_{10}} > \underline{SO_3} > H_2\underline{O}$

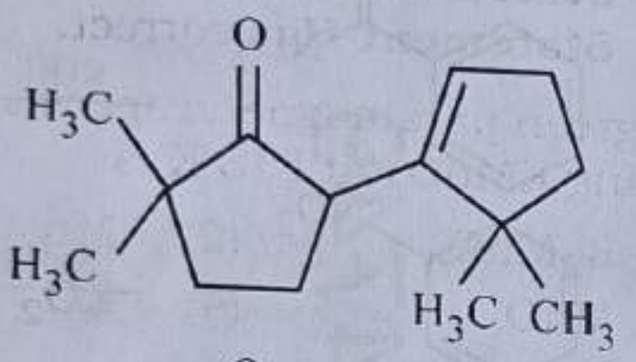
(2) $\underline{N_2O_5} > \underline{Al_2O_3} > H_2\underline{S}$

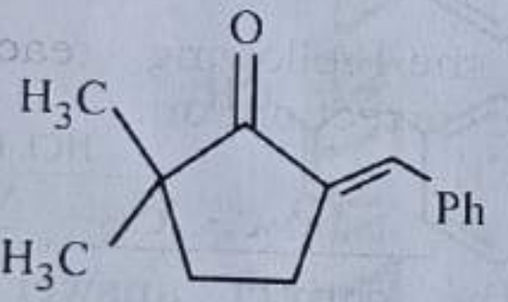
(3) $\underline{PbO_2} > \underline{N_2O_3} > \underline{SO_3}$

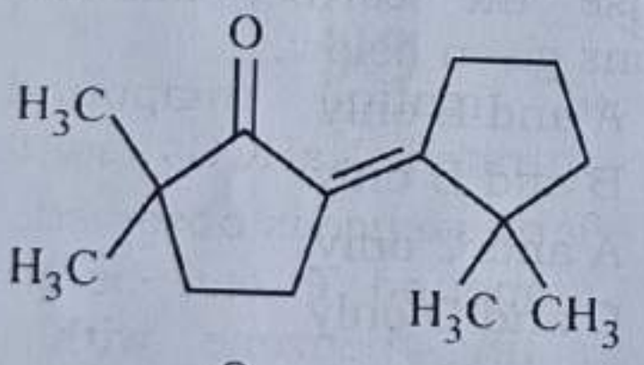
(4) $\underline{P_4O_6} > \underline{Cl_2O_7} > \underline{AlH_3}$

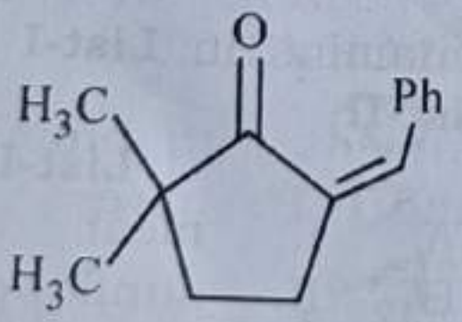
82. वह यौगिक जो नीचे दिखाई गई एल्डोल संघनन अभिक्रिया से प्राप्त नहीं किया जा सकता, वह है



(1) 

(2) 

(3) 

(4) 

83. निम्नलिखित में से, संयुग्मित द्वि-आबंध (conjugated double bonds) वाला यौगिक है

(1) हेप्टा -1,3- डाईईन

(2) हेप्टा -1,4- डाईईन

(3) हेप्टा -1,5- डाईईन

(4) हेप्टा -1,6- डाईईन

81. The correct decreasing order of oxidation state of the underlined atom in each molecule is

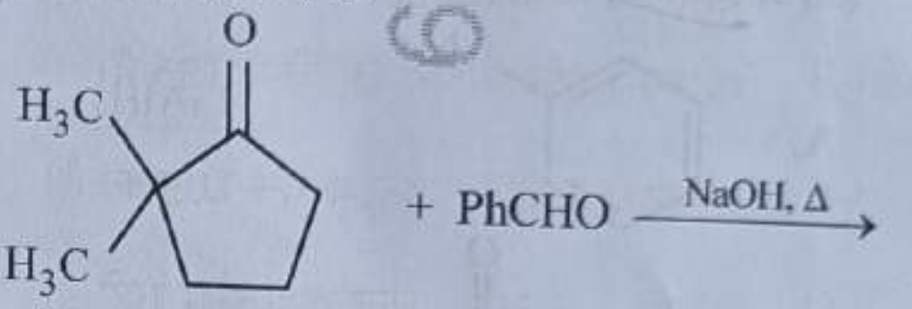
(1) $\underline{P_4O_{10}} > \underline{SO_3} > H_2\underline{O}$

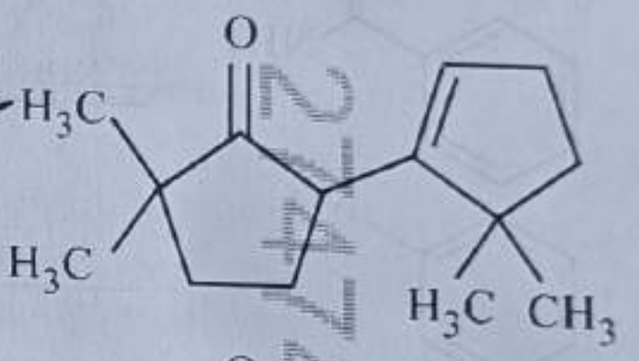
(2) $\underline{N_2O_5} > \underline{Al_2O_3} > H_2\underline{S}$

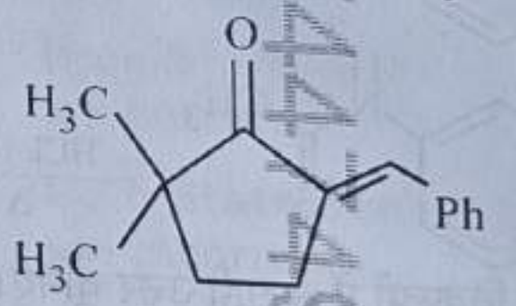
(3) $\underline{PbO_2} > \underline{N_2O_3} > \underline{SO_3}$

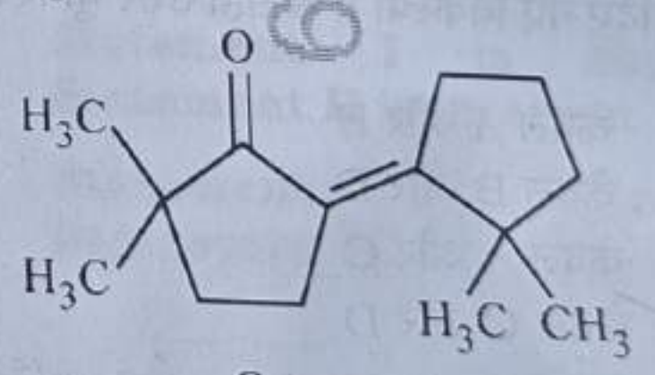
(4) $\underline{P_4O_6} > \underline{Cl_2O_7} > \underline{AlH_3}$

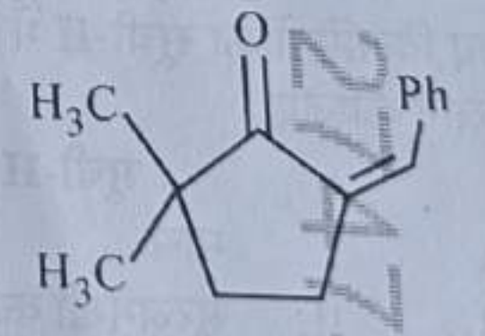
82. The compound that **CANNOT** be obtained from the aldol condensation reaction shown below, is



(1) 

(2) 

(3) 

(4) 

83. Among the following, the compound having conjugated double bonds is

(1) hepta-1,3-diene

(2) hepta-1,4-diene

(3) hepta-1,5-diene

(4) hepta-1,6-diene

84. नीचे दो कथन दिए गए हैं:

कथन-I : अम्लीय $KMnO_4$ के साथ *p*-नाइट्रोटोलुईन का ऑक्सीकरण एक ऐसा अम्ल देता है जो बेन्जोइक अम्ल से अधिक प्रबल होता है।

कथन-II : Sn/HCl के साथ *p*-नाइट्रोटोलुईन का अपचयन और उसके बाद उदासीनीकरण एक ऐसा ऐमीन देता है जो ऐनिलीन से अधिक क्षारीय होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनिए।

(1) **कथन-I** और **कथन-II**, दोनों सही हैं।

(2) **कथन-I** और **कथन-II**, दोनों गलत हैं।

(3) **कथन-I** सही है, परन्तु **कथन-II** गलत है।

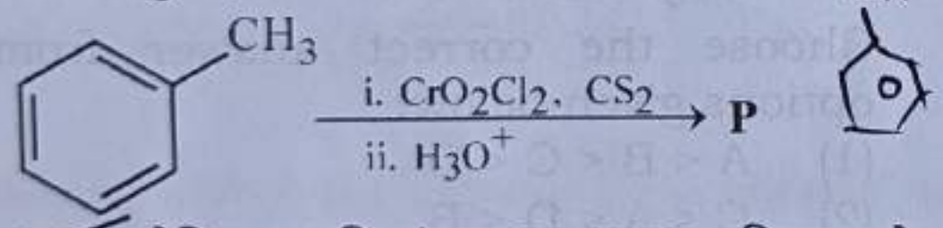
(4) **कथन-I** गलत है, परन्तु **कथन-II** सही है।

85. 513 K पर $KMnO_4$ को गर्म करने पर बनने वाली हरी अनुचुंबकीय प्रजाति (स्पीशीज) है

(1) K_2MnO_4 (2) Mn_3O_4

(3) MnO (4) KO_2

86. निम्नलिखित अभिक्रिया पर विचार कीजिए, और सही विकल्प चुनिए।



(1) यौगिक **P** की संतृप्त $NaHCO_3$ विलयन के साथ अभिक्रिया कराने पर, तीव्र बुदबुदाहट देखी जाती है।

(2) यौगिक **P** को बेन्जीन की निर्जल $AlCl_3$ और CH_3COCl के साथ अभिक्रिया द्वारा बनाया जा सकता है।

(3) ब्रोमीन जल के साथ उपचार करने पर, यौगिक **P** एक सफेद अवक्षेप देता है।

(4) यौगिक **P** को $BaSO_4$ पर Pd की उपस्थिति में बेन्ज़ॉयल क्लोराइड के हाइड्रोजनीकरण द्वारा प्राप्त किया जाता है।

87. जलीय विलयन में 1:3 विद्युत अपघट्य (electrolyte) है

(1) $[CoCl_2(NH_3)_4]Cl$

(2) $[CoCl(NH_3)_5]Cl_2$

(3) $[Co(NH_3)_6]Cl_3$

(4) $[Co(NH_3)_3(NO_2)_3]$

84. Given below are two statements:

Statement-I : Oxidation of *p*-nitrotoluene with acidic $KMnO_4$ gives an acid that is stronger than benzoic acid.

Statement-II : Reduction of *p*-nitrotoluene with Sn/HCl followed by neutralization gives an amine that is more basic than aniline.

In light of the above statements, choose the ***most appropriate*** answer from the options given below.

(1) Both **Statement-I** and **Statement-II** are correct.

(2) Both **Statement-I** and **Statement-II** are incorrect.

(3) **Statement-I** is correct but **Statement-II** is incorrect.

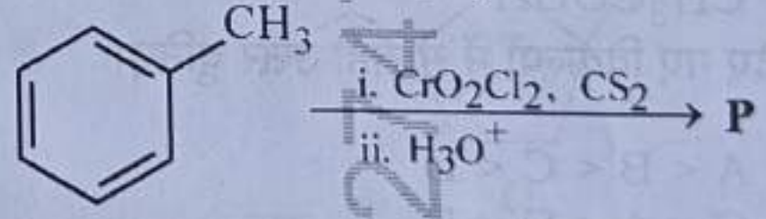
(4) **Statement-I** is incorrect but **Statement-II** is correct.

85. The green paramagnetic species formed by heating $KMnO_4$ at 513 K is

(1) K_2MnO_4 (2) Mn_3O_4

(3) MnO (4) KO_2

86. Consider the following reaction, and choose the correct option.



(1) On treating compound **P** with saturated $NaHCO_3$ solution, brisk effervescence is observed.

(2) Compound **P** can be prepared by treating benzene with anhydrous $AlCl_3$ and CH_3COCl.

(3) On treatment with bromine water, compound **P** gives a white precipitate.

(4) Compound **P** is obtained by the hydrogenation of benzoyl chloride with Pd on $BaSO_4$.

87. A 1:3 electrolyte in an aqueous solution is

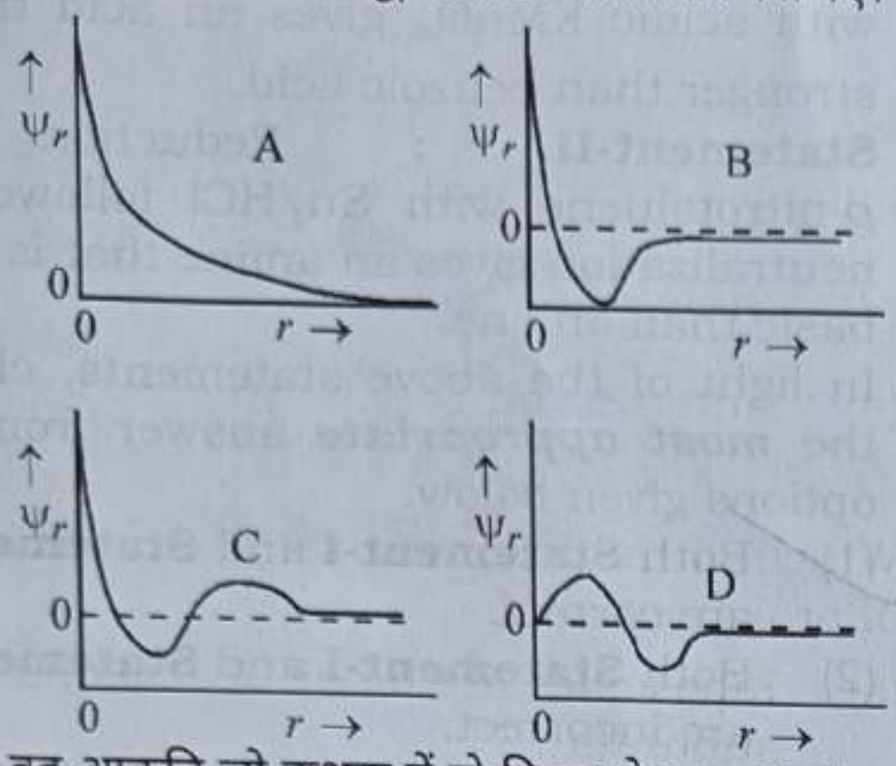
(1) $[CoCl_2(NH_3)_4]Cl$

(2) $[CoCl(NH_3)_5]Cl_2$

(3) $[Co(NH_3)_6]Cl_3$

(4) $[Co(NH_3)_3(NO_2)_3]$

88. नाभिक से दूरी (r) के सापेक्ष कक्षक तरंग फलन (ψ_r) के निम्नलिखित योजनाबद्ध आलेखों पर विचार कीजिए।



वह आकृति जो कक्षक में दो त्रिज्य नोड (radial nodes) को प्रदर्शित करती है, है

(1) A (2) B
(3) C (4) D

89. निम्नलिखित यौगिकों को ध्रुवणता के बढ़ते क्रम में व्यवस्थित कीजिए

A. $CH_3CH_2OCH_2CH_3$
B. CH_3CH_2OH
C. CH_3COCH_3
D. CH_3COOH

नीचे दिए गए विकल्पों में से सही उत्तर चुनिए।

(1) A < B < C < D
(2) C < A < D < B
(3) C < A < B < D
(4) A < C < B < D

90. Ne_2 के लिए उच्चतम अधिग्रहित आणविक कक्षक (highest occupied molecular orbital) है

(1) π_{2p} (2) σ_{2p}
(3) π^*_{2p} (4) σ^*_{2p}

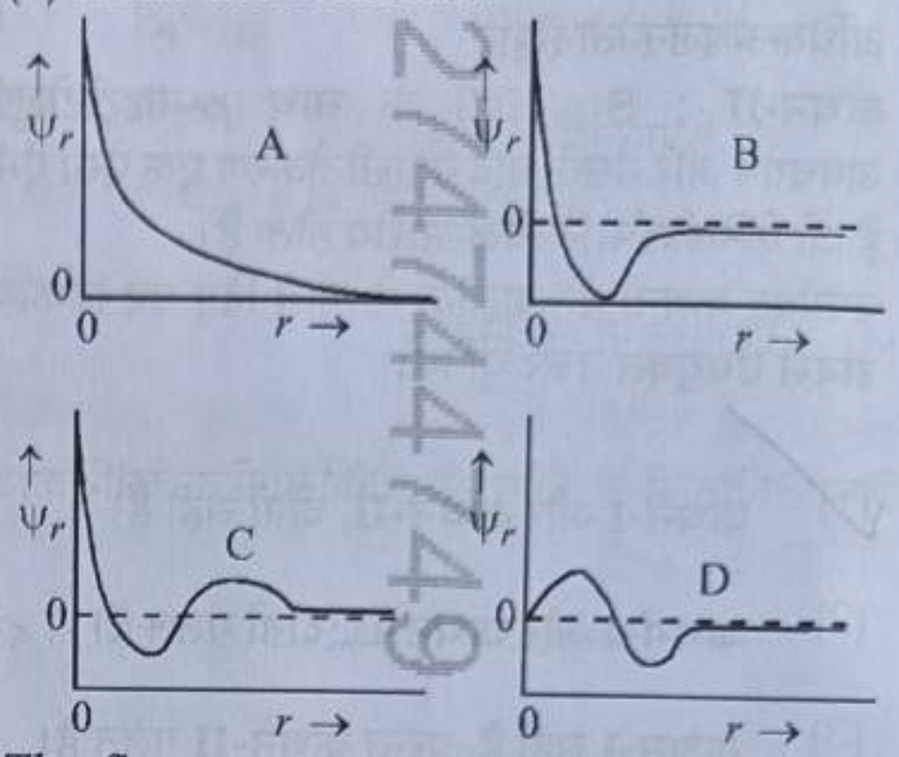
91. मानव में कशेरुकाओं की संख्या _____ है।

(1) 7 (2) 12
(3) 26 (4) 206

92. कवक और शैवाल के बीच सहजीवी सहवास (संबंध) को _____ कहा जाता है।

(1) लाइकेन
(2) स्पंज
(3) माइकोराइजा
(4) क्राइसोफाइट्स

88. Consider the following schematic plots of orbital wavefunction (ψ_r) against distance (r) from the nucleus.



The figure representing two radial nodes in the orbital is

(1) A (2) B
(3) C (4) D

89. Arrange the following compounds in the increasing order of polarity

A. $CH_3CH_2OCH_2CH_3$
B. CH_3CH_2OH
C. CH_3COCH_3
D. CH_3COOH

Choose the correct answer from the options given below.

(1) A < B < C < D
(2) C < A < D < B
(3) C < A < B < D
(4) A < C < B < D

90. The highest occupied molecular orbital for Ne_2 is

(1) π_{2p} (2) σ_{2p}
(3) π^*_{2p} (4) σ^*_{2p}

91. The number of vertebrae in a human is _____.

(1) 7 (2) 12
(3) 26 (4) 206

92. Symbiotic association between fungi and algae are called _____.

(1) lichens
(2) sponges
(3) mycorrhiza
(4) chrysophytes

93. कोशिका सिद्धांत _____ द्वारा प्रतिपादित किया गया था।

(1) श्लाइडेन और श्वान

(2) रॉबर्ट ब्राउन

(3) सिंगर और निकोलसन

(4) एंटोनी वॉन ल्यूवेनहॉक

93. Cell theory was formulated by _____.

(1) Schleiden and Schwann

(2) Robert Brown

(3) Singer and Nicolson

(4) Antonie Von Leeuwenhoek

94. निम्नलिखित में से कौन सी प्रोकैरियोटिक कोशिकाओं की विशेषताएँ हैं ?

(a) राइबोसोम 50S और 30S उप-इकाइयों से बने होते हैं

(b) इनमें प्लास्मिड हो सकते हैं

(c) इनमें मेसोसोम होते हैं

(d) इनमें पेरोक्सिसोम होते हैं

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) केवल (b) और (c)

(2) केवल (a) और (c)

(3) केवल (a), (c) और (d)

(4) केवल (a), (b) और (c)

94. Which of the following are characteristics of prokaryotic cells ?

(a) Ribosomes are made of 50S and 30S subunits

(b) They can have plasmids

(c) They contain mesosome

(d) They have peroxisomes

Choose the **correct** answer from the options given below :

(1) (b) and (c) only

(2) (a) and (c) only

(3) (a), (c) and (d) only

(4) (a), (b) and (c) only

95. निम्नलिखित में से कौन सा मानव केंद्रीय तंत्रिका तंत्र का हिस्सा **नहीं** है ?

(1) एरेक्नॉइड

(2) ड्यूरा मैटर

(3) पाया मैटर

(4) पेरिकार्डियम

95. Which of the following is **not** a part of human central neural system ?

(1) Arachnoid

(2) Dura mater

(3) Pia mater

(4) Pericardium

96. माइटोकॉन्ड्रिया की आंतरिक झिल्ली _____ को घेरे रहती है।

(1) आधात्री (मैट्रिक्स)

(2) कोशिकाद्रव्य (साइटोसोल)

(3) श्लेष्मा (म्यूकस)

(4) जलीय द्रव (एक्वियस ह्यूमर)

96. Mitochondrial inner membrane encloses _____.

(1) matrix

(2) cytosol

(3) mucus

(4) aqueous humor

6747

97. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

	सूची-I		**सूची-II**
A.	क्रिस्टी (Cristae)	I.	क्लोरोप्लास्ट के स्ट्रोमा में चपटी झिल्लीदार थैलियाँ
B.	सिस्टर्नी (Cisternae)	II.	माइटोकॉन्ड्रिया में आंतरिक वलन
C.	थायलाकोइड्स (Thylakoids)	III.	कोशिका झिल्ली
D.	फॉस्फोलिपिड (Phospholipid)	IV.	गॉल्जी उपकरण में डिस्क के आकार की थैलियाँ

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-IV, C-III, D-I

(4) A-IV, B-III, C-I, D-II

98. वह लवक (प्लास्टिड) जो जैन्थोफिल का संचय करता है, _____ के रूप में जाना जाता है।

(1) हरितलवक (क्लोरोप्लास्ट)

(2) वर्णीलवक (क्रोमोप्लास्ट)

(3) प्रोटीन लवक (एल्यूरोप्लास्ट)

(4) मंडलवक (एमाइलोप्लास्ट)

99. पीयूष ग्रंथि (पिट्यूटरी ग्रंथि) से संबंधित निम्नलिखित में से कौन से कथन **सही** हैं ?

(a) यह आन्तरिक रूप से एडिनोहाइपोफिसिस और न्यूरोहाइपोफिसिस में विभाजित है

(b) यह पुटिका प्रेरक हार्मोन (फॉलिकल स्टिमुलेटिंग हार्मोन) स्रावित करता है

(c) यह मेलानोसाइट उत्तेजक हार्मोन स्रावित करता है

(d) यह प्रोलैक्टिन स्रावित नहीं करता है।

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) केवल (a) और (b)

(2) केवल (a), (b) और (c)

(3) केवल (c) और (d)

(4) केवल (b) और (c)

97. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Cristae	I.	Flat membrane sacs in stroma of chloroplast
B.	Cisternae	II.	Infoldings in mitochondria
C.	Thylakoids	III.	Cell membrane
D.	Phospholipid	IV.	Disc shaped sacs in the Golgi apparatus

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-IV, C-III, D-I

(4) A-IV, B-III, C-I, D-II

98. The plastid that stores xanthophyll is known as _____.

(1) chloroplast

(2) chromoplast

(3) aleuroplast

(4) amyloplast

99. Which of the following statements related to pituitary gland are **correct** ?

(a) It is divided anatomically into adenohypophysis and neurohypophysis

(b) It secretes follicle stimulating hormone

(c) It secretes melanocyte stimulating hormone

(d) It does not secrete prolactin

Choose the **correct** answer from the options given below :

(1) (a) and (b) only

(2) (a), (b) and (c) only

(3) (c) and (d) only

(4) (b) and (c) only

100. रुबिस्को (RuBisCo) द्वारा उत्प्रेरित प्रकाश-श्वसन अभिक्रिया नीचे दिखाई गई है :

$RuBP + O_2 \rightarrow 3\text{-Phosphoglycerate} + X$

दिए गए विकल्पों में से "X" की पहचान करें :

(1) फॉस्फोइनोलपायरूवेट
(2) 2-फॉस्फोग्लाइकोलेट
(3) ऑक्सालोएसीटेट
(4) मैलेट

101. मैड काऊ रोग _____ के कारण होता है।

(1) प्रीयॉन्स (प्रोसंक)
(2) वायरोइड्स
(3) *एस्परजिलस स्प.*
(4) *माइकोप्लाज्मा स्प.*

102. प्रकाशसंश्लेषक अभिक्रिया केंद्र PS I (P700) में किस वर्णक का अवशोषण शिखर 700 nm पर होता है ?

(1) क्लोरोफिल b
(2) क्लोरोफिल a
(3) जैन्थोफिल्स
(4) कैरोटीनॉयड्स

103. जल में, मेंढक _____ का उपयोग करके श्वसन करते हैं।

(1) त्वचा
(2) मुख गुहा
(3) फेफड़े
(4) श्वासनली

104. निम्नलिखित में से कौन सा मनुष्यों के पैर (lower limb) में अस्थियों के विन्यास का सही क्रम दर्शाता है ?

(1) फीमर-टिबिया-पटेला-टार्सल
(2) पटेला-फीमर-टिबिया-टार्सल
(3) फीमर-पटेला-टिबिया-टार्सल
(4) फीमर-टार्सल-पटेला-टिबिया

105. पर्णविन्यास (फिलोटैक्सी) _____ के लगने के क्रम को कहते हैं।

(1) पत्तियों
(2) पुष्पों
(3) फलों
(4) बाह्यदलों

100. Photorespiration reaction catalyzed by RuBisCo is shown below :

$RuBP + O_2 \rightarrow 3\text{-Phosphoglycerate} + X$

Identify "X" from the given options :

(1) Phosphoenolpyruvate
(2) 2-Phosphoglycolate
(3) Oxaloacetate
(4) Malate

101. Mad cow disease is caused by _____.

(1) prions
(2) viroids
(3) *Aspergillus sp.*
(4) *Mycoplasma sp.*

102. Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700) ?

(1) Chlorophyll b
(2) Chlorophyll a
(3) Xanthophylls
(4) Carotenoids

103. In water, frogs respire using _____.

(1) skin
(2) buccal cavity
(3) lungs
(4) trachea

104. Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans ?

(1) Femur-tibia-patella-tarsal
(2) Patella-femur-tibia-tarsal
(3) Femur-patella-tibia-tarsal
(4) Femur-tarsal-patella-tibia

105. Phyllotaxy is the pattern of arrangement of _____.

(1) leaves
(2) flowers
(3) fruits
(4) sepals

106. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

सूची-I	**सूची-II**
A. स्टार्च	I. संक्रमण से लड़ता है
B. एंटीबॉडी	II. ऊर्जा संचय
C. कॉनकैनावलिन A	III. ग्लूकोज परिवहन
D. जी.एल.यू.टी -4 (Glut-4)	IV. लेक्टिन

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-I, B-II, C-III, D-IV

107. नीचे दो कथन दिए गए हैं :

कथन I : जब शरीर के केंद्रीय अक्ष से गुजरने वाली कोई भी रेखा जीव को दो समरूप भागों में विभाजित करती है, तो इसे अरीय सममिति (रेडियल सिमिट्री) कहा जाता है।

कथन II : संघ एकाइनोडर्मेटा (शूलयुक्त प्राणी) में, वयस्क और लार्वा दोनों अरीय सममित होते हैं।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं
(2) **कथन I** और **कथन II** दोनों गलत हैं
(3) **कथन I** सही है परन्तु **कथन II** गलत है
(4) **कथन I** गलत है परन्तु **कथन II** सही है

108. अंतःझिल्ली तंत्र (एंडोमेम्ब्रेन सिस्टम) में _____ शामिल हैं।

(1) अंतर्द्रव्यी जालिका, गॉल्जी संकुल, लाइसोसोम और रिक्तिका
(2) अंतर्द्रव्यी जालिका, क्लोरोप्लास्ट, पेरोक्सिसोम और रिक्तिका
(3) माइटोकॉन्ड्रिया, क्लोरोप्लास्ट, पेरोक्सिसोम और रिक्तिका
(4) गॉल्जी संकुल, क्लोरोप्लास्ट, पेरोक्सिसोम और रिक्तिका

106. Match **List-I** with **List-II.**

List-I	**List-II**
A. Starch	I. Fights infection
B. Antibody	II. Energy storage
C. Concanavalin A	III. Glucose transport
D. Glut-4	IV. Lectin

Choose the **correct** answer from the options given below :

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-I, B-II, C-III, D-IV

107. Given below are two statements :

Statement I : When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.

Statement II : In phylum Echinodermata, both adults and larvae are radially symmetrical.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct
(2) Both **Statement I** and **Statement II** are incorrect
(3) **Statement I** is correct but **Statement II** is incorrect
(4) **Statement I** is incorrect but **Statement II** is correct

108. Endomembrane system includes _____.

(1) endoplasmic reticulum, Golgi complex, lysosomes and vacuole
(2) endoplasmic reticulum, chloroplast, peroxisomes and vacuole
(3) mitochondria, chloroplast, peroxisomes and vacuole
(4) Golgi complex, chloroplast, peroxisomes and vacuole

109. ग्लूकोज के 206 अणुओं से ग्लाइकोलिसिस के अंत में पाइरुविक अम्ल के कितने अणु उत्पन्न होते हैं ?

(1) 206 (2) 309
(3) 103 (4) 412

110. निम्नलिखित में से किस पादप वृद्धि नियामक का उपयोग शाकनाशी के रूप में किया जाता है ?

(1) 2,4-डी
(2) काइनेटिन
(3) एब्सिसिक अम्ल
(4) जिब्बेरेलिन

111. नीचे दो कथन दिए गए हैं :

कथन I : जिम्नोस्पर्म (अनावृतबीजी) में, नर और मादा युग्मकोद्भिद बीजाणुधानी के भीतर ही रहते हैं।

कथन II : जिम्नोस्पर्म (अनावृतबीजी) में, बीज ढके हुए नहीं होते हैं।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं
(2) **कथन I** और **कथन II** दोनों गलत हैं
(3) **कथन I** सही है परन्तु **कथन II** गलत है
(4) **कथन I** गलत है परन्तु **कथन II** सही है

112. **सूची-I** का **सूची-II** से मिलान कीजिए।

	सूची-I		**सूची-II**
A.	गोलाकार	I.	विब्रियो
B.	छड़ाकार	II.	कोकाई
C.	अल्पविराम (कॉमा) के आकार का	III.	स्पिरिला
D.	स्पाइरिलम	IV.	बैसिलाई

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-I, B-III, C-II, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-II, B-IV, C-I, D-III

109. How many molecules of pyruvic acid are produced at the end of glycolysis from 206 molecules of glucose ?

(1) 206 (2) 309
(3) 103 (4) 412

110. Which of the following plant growth regulators is used as herbicide ?

(1) 2,4-D
(2) Kinetin
(3) Abscisic acid
(4) Gibberellin

111. Given below are two statements :

Statement I : In gymnosperms, the male and female gametophytes remain within the sporangia.

Statement II : In gymnosperms, seeds are not covered.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct
(2) Both **Statement I** and **Statement II** are incorrect
(3) **Statement I** is correct but **Statement II** is incorrect
(4) **Statement I** is incorrect but **Statement II** is correct

112. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Spherical	I.	Vibrio
B.	Rod	II.	Cocci
C.	Comma	III.	Spirilla
D.	Spirillum	IV.	Bacilli

Choose the **correct** answer from the options given below :

(1) A-I, B-III, C-II, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-II, B-IV, C-I, D-III

113. निम्नलिखित में से कौन सी सोलेनेसी (Solanaceae) कुल की विशेषताएँ हैं ?

(a) पुष्प उभयलिंगी और त्रिज्या सममित (एक्टिनोमोर्फिक) होते हैं

(b) बाह्यदल पुंज में पाँच बाह्यदल होते हैं और वे संयुक्त होते हैं

(c) पुमंग में पाँच पुंकेसर होते हैं और वे दललग्न (एपिपेटल्स) होते हैं

(d) अंडाशय अधोवर्ती होता है

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) केवल (a), (b) और (c)

(2) केवल (d)

(3) केवल (a) और (b)

(4) केवल (b), (c) और (d)

113. Which of the following are characteristic features of Solanaceae family ?

(a) Flowers are bisexual and actinomorphic

(b) Calyx have five sepals and are united

(c) Androecium have five stamens and are epipetalous

(d) Ovary is inferior

Choose the **correct** answer from the options given below :

(1) (a), (b) and (c) only

(2) (d) only

(3) (a) and (b) only

(4) (b), (c) and (d) only

114. उन प्रयोगों के सही क्रम का चयन कीजिए जिनके कारण हरे पौधों में प्रकाशसंश्लेषण की क्रमिक समझ विकसित हुई।

(1) क्लोरोफिल a और b के अवशोषण स्पेक्ट्रा → ग्लूकोज का उत्पादन → ऑक्सीजन का विमोचन → वायु की भूमिका

(2) वायु की भूमिका → ऑक्सीजन का विमोचन → ग्लूकोज का उत्पादन → क्लोरोफिल a और b के अवशोषण स्पेक्ट्रा

(3) ऑक्सीजन का विमोचन → ग्लूकोज का उत्पादन → क्लोरोफिल a और b के अवशोषण स्पेक्ट्रा → वायु की भूमिका

(4) ग्लूकोज का उत्पादन → वायु की भूमिका → ऑक्सीजन का विमोचन → क्लोरोफिल a और b के अवशोषण स्पेक्ट्रा

114. Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.

(1) Absorption spectra of chlorophyll a and b → production of glucose → release of oxygen → role of air

(2) Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b

(3) Release of oxygen → production of glucose → absorption spectra of chlorophyll a and b → role of air

(4) Production of glucose → role of air → release of oxygen → absorption spectra of chlorophyll a and b

115. एक स्वस्थ मनुष्य में साइनो-एट्रियल नोड (SAN) द्वारा उत्पन्न क्रिया विभवों (एक्शन पोटेंशियल) की संख्या _____ प्रति मिनट होती है।

(1) 28 - 30 (2) 70 - 75

(3) 100 - 110 (4) 120 - 140

115. The number of action potentials generated by sino-arterial node (SAN) in a healthy human is _____ per minute.

(1) 28 - 30 (2) 70 - 75

(3) 100 - 110 (4) 120 - 140

116. ग्लूकोज के तीन अणुओं के निर्माण के लिए केल्विन चक्र के कितने फेरों (चक्रों) की आवश्यकता होती है ?

(1) 6 (2) 3

(3) 1 (4) 18

116. How many turns of Calvin cycle are required for the formation of three molecules of glucose ?

(1) 6 (2) 3

(3) 1 (4) 18

117. निम्नलिखित में से कौन सा कथन **गलत** है ?

(1) चोट लगने से रक्त का थक्का जम जाता है।

(2) रक्त का थक्का फाइब्रिन से बना होता है।

(3) फाइब्रिन का निर्माण फाइब्रिनोजेन से होता है।

(4) फाइब्रिनोजेन का निर्माण फाइब्रिन से होता है।

118. **सूची-I** का **सूची-II** के साथ मिलान कीजिए।

	सूची-I		**सूची-II**
A.	कुल (Family)	I.	सैपिन्डेल्स (Sapindales)
B.	वंश (Genus)	II.	डाइकोटीलिडनी (Dicotyledonae)
C.	वर्ग (Class)	III.	एनाकरडिएसी (Anacardiaceae)
D.	संघ (Phylum)	IV.	एंजियोस्पर्मी (Angiospermae)
E.	गण (Order)	V.	*मैंजीफेरा (Mangifera)*

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-I, B-V, C-II, D-IV, E-III

(2) A-II, B-I, C-III, D-IV, E-V

(3) A-II, B-III, C-V, D-I, E-IV

(4) A-III, B-V, C-II, D-IV, E-I

119. निम्नलिखित वर्गीकरण श्रेणियों को आरोही क्रम में व्यवस्थित करें।

(a) वंश (जीनस) (b) वर्ग (क्लास)

(c) गण (ऑर्डर) (d) संघ (फाइलम)

(e) कुल (फैमिली) (f) जगत (किंगडम)

(g) जाति (स्पीशीज)

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) (g), (a), (e), (c), (b), (d), (f)

(2) (a), (c), (d), (g), (f), (b), (e)

(3) (g), (c), (d), (b), (e), (a), (f)

(4) (f), (c), (b), (g), (d), (e), (a)

117. Which of the following statements is **incorrect** ?

(1) Blood coagulates in response to an injury

(2) Blood clot consists of fibrins

(3) Fibrin is produced from fibrinogen

(4) Fibrinogen is produced from fibrin

118. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Family	I.	Sapindales
B.	Genus	II.	Dicotyledonae
C.	Class	III.	Anacardiaceae
D.	Phylum	IV.	Angiospermae
E.	Order	V.	*Mangifera*

Choose the **correct** answer from the options given below :

(1) A-I, B-V, C-II, D-IV, E-III

(2) A-II, B-I, C-III, D-IV, E-V

(3) A-II, B-III, C-V, D-I, E-IV

(4) A-III, B-V, C-II, D-IV, E-I

119. Arrange the following taxonomic categories in ascending order.

(a) Genus (b) Class

(c) Order (d) Phylum

(e) Family (f) Kingdom

(g) Species

Choose the **correct** answer from the options given below :

(1) (g), (a), (e), (c), (b), (d), (f)

(2) (a), (c), (d), (g), (f), (b), (e)

(3) (g), (c), (d), (b), (e), (a), (f)

(4) (f), (c), (b), (g), (d), (e), (a)

120. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

	सूची-I		**सूची-II**
A.	सीमांत बीजाण्डन्यास (Marginal placentation)	I.	*आर्जेमोन* (*Argemone*)
B.	अक्षीय बीजाण्डन्यास (Axile placentation)	II.	टमाटर (Tomato)
C.	भित्ति बीजाण्डन्यास (Parietal placentation)	III.	*प्रिमरोज* (*Primrose*)
D.	मुक्त स्तंभीय बीजाण्डन्यास (Free central placentation)	IV.	मटर (Pea)

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-IV, B-III, C-I, D-II
(4) A-IV, B-II, C-I, D-III

121. स्फीनोप्सिडा (Sphenopsida) वर्ग _____ से संबंधित है।

(1) ब्रायोफाइट्स
(2) एन्जियोस्पर्म्स
(3) जिम्नोस्पर्म्स
(4) टेरिडोफाइट्स

122. प्रकाश-श्वसन (फोटोरेस्पिरेशन) के संबंध में निम्नलिखित में से कौन से कथन सही हैं ?

(a) C3 पौधों में नहीं होता है
(b) CO_2 का उपभोग होता है और O_2 उत्पन्न होता है
(c) फॉस्फोग्लाइकोलेट बनता है
(d) ATP और NADPH का संश्लेषण नहीं होता है।

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) केवल (a) और (d)
(2) केवल (c) और (d)
(3) केवल (b) और (d)
(4) केवल (a) और (b)

123. चिकनी अंतर्द्रव्यी जालिका _____।

(1) इसकी सतह पर राइबोसोम जुड़े होते हैं
(2) लिपिड के संश्लेषण का मुख्य स्थल है
(3) प्रोटीन संश्लेषण में सक्रिय रूप से शामिल होती है
(4) कार्बोहाइड्रेट के संश्लेषण का एक स्थल है

120. Match **List-I** with **List-II**.

	List-I		**List-II**
A.	Marginal placentation	I.	*Argemone*
B.	Axile placentation	II.	Tomato
C.	Parietal placentation	III.	*Primrose*
D.	Free central placentation	IV.	Pea

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-IV, B-III, C-I, D-II
(4) A-IV, B-II, C-I, D-III

121. Sphenopsida class belongs to _____.

(1) bryophytes
(2) angiosperms
(3) gymnosperms
(4) pteridophytes

122. Which of the following statements regarding photorespiration are correct ?

(a) Do not occur in C3 plants
(b) CO_2 is consumed and O_2 is generated
(c) Phosphoglycolate is formed
(d) No synthesis of ATP and NADPH

Choose the **correct** answer from the options given below :

(1) (a) and (d) only
(2) (c) and (d) only
(3) (b) and (d) only
(4) (a) and (b) only

123. Smooth endoplasmic reticulum _____.

(1) has ribosomes attached to its surface
(2) is the major site for the synthesis of lipids
(3) is actively involved in protein synthesis
(4) is a site for the synthesis of carbohydrates

124. निम्नलिखित में से कौन सा कथन **गलत** है ?

(1) अग्न्याशय की α-कोशिकाएं ग्लूकागन का स्राव करती हैं

(2) अग्न्याशय की α- कोशिकाएं इंसुलिन का स्राव करती हैं

(3) ग्लूकागन ग्लाइकोजेनोलिसिस को उत्तेजित करता है

(4) अग्न्याशय की β-कोशिकाएं इंसुलिन का स्राव करती हैं

125. वंश (जीनस) _____ को दर्शाता है।

(1) एक व्यक्तिगत पौधे या जंतु को

(2) पौधों और जंतुओं की एक जनसंख्या को

(3) निकट संबंधी प्रजातियों के एक समूह को

(4) निकट संबंधी कुलों के एक समूह को

126. निम्नलिखित में से कौन सा प्राक्केंद्रक (प्रोकैरियोट) **नहीं** है ?

(1) बैक्टीरिया

(2) नील-हरित शैवाल

(3) माइकोप्लाज्मा

(4) कवक

127. निम्नलिखित में से कौन सा पादप वृद्धि नियामक पत्तागोभी में पुष्पन से अंत:पूर्व पर्व (इंटरनोड) के दीर्घीकरण को बढ़ावा देता है ?

(1) एब्सिसिक अम्ल

(2) जिब्बेरेलिन

(3) इंडोल ब्यूटिरिक अम्ल

(4) एथेफोन

128. वयस्क कोशिका चक्र के चरणों का सही क्रम _____ है।

(1) G1-G2-S-M

(2) G1-M-G2-S

(3) G1-S-G2-M

(4) S-M-G2-G1

124. Which one of the following statements is **incorrect** ?

(1) α-cells of pancreas secrete glucagon

(2) α-cells of pancreas secrete insulin

(3) Glucagon stimulates glycogenolysis

(4) β-cells of pancreas secrete insulin

125. Genus represents _____ .

(1) an individual plant or animal

(2) a population of plants and animals

(3) a group of closely related species

(4) a group of closely related families

126. Which of the following is **not** a prokaryote ?

(1) Bacteria

(2) Blue green algae

(3) Mycoplasma

(4) Fungi

127. Which of the following plant growth regulators promotes internode elongation prior to flowering in cabbage ?

(1) Abscisic acid

(2) Gibberellin

(3) Indole butyric acid

(4) Ethephon

128. The correct sequence of adult cell cycle phases is _____.

(1) G1-G2-S-M

(2) G1-M-G2-S

(3) G1-S-G2-M

(4) S-M-G2-G1

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129. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

	सूची-I		**सूची-II**
A.	युग्मकों के बीच जीवद्रव्य का संलयन	I.	अर्धसूत्री विभाजन
B.	दो केंद्रकों का संलयन	II.	प्लैज्मोगैमी
C.	अगुणित बीजाणुओं का निर्माण	III.	केंद्र संलयन

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-II, B-III, C-I
(2) A-II, B-I, C-III
(3) A-III, B-II, C-I
(4) A-I, B-III, C-II

130. नीचे दो कथन दिए गए हैं :

कथन I : वर्ग सरीसृप (Reptilia) का नाम रेंगने या सरकने की गमन विधि को संदर्भित करता है।

कथन II : सरीसृप वर्ग से संबंधित सभी जीवों का हृदय तीन कक्षीय होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं
(2) **कथन I** और **कथन II** दोनों गलत हैं
(3) **कथन I** सही है लेकिन **कथन II** गलत है
(4) **कथन I** गलत है लेकिन **कथन II** सही है

131. नीचे दो कथन दिए गए हैं :

कथन I : प्रोफेज़ I के अंत में गुणसूत्र पूरी तरह से संघनित हो जाते हैं।

कथन II : अर्धसूत्री विभाजन I (मेयोसिस I), समसूत्री विभाजन (माइटोसिस) के समान होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सत्य हैं
(2) **कथन I** और **कथन II** दोनों असत्य हैं
(3) **कथन I** सही है, लेकिन **कथन II** असत्य है
(4) **कथन I** गलत है, लेकिन **कथन II** सत्य है

129. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Fusion of protoplasms between gametes	I.	Meiosis
B.	Fusion of two nuclei	II.	Plasmogamy
C.	Generation of haploid spores	III.	Karyogamy

Choose the **correct** answer from the options given below :

(1) A-II, B-III, C-I
(2) A-II, B-I, C-III
(3) A-III, B-II, C-I
(4) A-I, B-III, C-II

130. Given below are two statements :

Statement I : The class name Reptilia refers to creeping or crawling mode of locomotion.

Statement II : All organisms belonging to Reptilia have three chambered heart.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct
(2) Both **Statement I** and **Statement II** are incorrect
(3) **Statement I** is correct but **Statement II** is incorrect
(4) **Statement I** is incorrect but **Statement II** is correct

131. Given below are two statements :

Statement I : Chromosomes are fully condensed at the end of prophase I.

Statement II : Meiosis I resembles mitosis.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true
(2) Both **Statement I** and **Statement II** are false
(3) **Statement I** is correct, but **Statement II** is false
(4) **Statement I** is incorrect, but **Statement II** is true

characteristic of chordates

(1) Presence of notochord

(2) Central nervous system is dorsal

(3) Absence of gills

(4) Presence of post anal part (tail)

133. Length of the stem at time 0 is 20 cm. The arithmetic growth rate is 30 cm per day. What is the length of the stem at the end of the 7th day ?

(1) 50 cm

(2) 170 cm

(3) 230 cm

(4) 460 cm

134. Arrange the following elements in descending order of their contribution to percentage weight of the human body.

(a) Oxygen

(b) Carbon

(c) Hydrogen

(d) Nitrogen

Choose the **correct** answer from the options given below :

(1) (a), (b), (c), (d)

(2) (c), (a), (b), (d)

(3) (b), (c), (d), (a)

(4) (b), (a), (c), (d)

135. In frogs, the number of pairs of cranial nerves arising from the brain are _____.

(1) 6 (2) 9

(3) 10 (4) 12

136. Which of the following is used as a clot buster ?

(1) Streptokinase

(2) Penicillin

(3) Cyclosporin A

(4) Statins

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137. बीटी (Bt) विषाक्त पदार्थ का निष्क्रिय रूप कीट की आंत में सक्रिय रूप में परिवर्तित हो जाता है _____।

(1) क्षारीय pH के कारण

(2) अम्लीय pH के कारण

(3) प्रोटीएज द्वारा

(4) न्यूक्लिएज द्वारा

138. नीचे दो कथन दिए गए हैं :

कथन I : डाउन सिंड्रोम एक X-गुणसूत्र की अनुपस्थिति के कारण होता है।

कथन II : टर्नर सिंड्रोम गुणसूत्रों की एक अतिरिक्त प्रति की उपस्थिति के कारण होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं

(2) **कथन I** और **कथन II** दोनों गलत हैं

(3) **कथन I** सही है लेकिन **कथन II** गलत है

(4) **कथन I** गलत है लेकिन **कथन II** सही है

139. निम्नलिखित में से कौन सा रोग यौन संचारित **नहीं** है ?

(1) सिफिलिस

(2) क्षय रोग (ट्यूबरकुलोसिस)

(3) गोनोरिया

(4) जननिक परिसर्प (जेनिटल वार्ट्स)

140. शुक्राणु की गतिशीलता _____ के कारण होती है।

(1) कशाभिकीय गति

(2) पक्ष्माभीय गति

(3) अमीबीय गति

(4) पेशीय गति

137. The inactive form of Bt toxin is converted to the active form in the insect gut _____.

(1) due to alkaline pH

(2) due to acidic pH

(3) by proteases

(4) by nucleases

138. Given below are two statements :

Statement I : Down's syndrome is caused by the absence of one of the X-chromosomes.

Statement II : Turner's syndrome is caused by the presence of an additional copy of the chromosomes.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

139. Which of the following disease is **not** sexually transmitted ?

(1) Syphilis

(2) Tuberculosis

(3) Gonorrhoea

(4) Genital warts

140. Sperm motility is due to _____.

(1) flagellar movement

(2) ciliary movement

(3) amoeboid movement

(4) muscular movement

141. प्राकृतिक वरण _____ प्रदान करता है।
(a) स्थायित्व
(b) आनुवंशिक विचलन
(c) दिशात्मक परिवर्तन
(d) विदारण (डिसरप्शन)
नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :
(1) केवल (a)
(2) केवल (a), (c) और (d)
(3) (a), (b), (c) और (d)
(4) केवल (a) और (c)

142. सहायक जनन प्रौद्योगिकी (assisted reproductive technology) में शुक्राणु को सीधे अंडाणु में इंजेक्ट करने की विधि को क्या कहा जाता है :
(1) युग्मक अंत: फैलोपियन स्थानांतरण (GIFT)
(2) युग्मनज अंत: फैलोपियन स्थानांतरण (ZIFT)
(3) अंत: कोशिकाद्रव्यी शुक्राणु इंजेक्शन (ICSI)
(4) भ्रूण स्थानांतरण (ET)

143. निम्नलिखित में से कौन सी संरचना पुरुष जनन तंत्र का हिस्सा **नहीं** है ?
(1) वृषण जालिकाएँ (रेटे टेस्टिस)
(2) अधिवृषण (ऐपिडिडिमिस)
(3) शुक्र वाहिकाएँ (वास इफेरेंशिया)
(4) इन्फंडिबुलम (कीपक)

144. अमेज़ॅन वर्षावन में प्रजातियों की संख्या के आधार पर निम्नलिखित को अवरोही क्रम (घटते क्रम) में व्यवस्थित करें।
(a) पौधे
(b) पक्षी
(c) मछलियाँ
(d) अकशेरुकी (इनवर्टेब्रेट्स)
(e) स्तनधारी
नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :
(1) (c) > (b) > (d) > (e) > (a)
(2) (d) > (a) > (c) > (b) > (e)
(3) (e) > (b) > (a) > (c) > (d)
(4) (b) > (a) > (d) > (c) > (e)

141. Natural selection can lead to _____.
(a) stabilisation
(b) genetic drift
(c) directional change
(d) disruption
Choose the **correct** answer from the options given below :
(1) (a) only
(2) (a), (c) and (d) only
(3) (a), (b), (c) and (d)
(4) (a) and (c) only

142. The method of directly of injecting a sperm into ovum in assisted reproductive technology is called :
(1) Gamete intra fallopian transfer (GIFT)
(2) Zygote intra fallopian transfer (ZIFT)
(3) Intra cytoplasmic sperm injection (ICSI)
(4) Embryo transfer (ET)

143. Which of the following structure is **not** a part of the male reproductive system ?
(1) Rete testis
(2) Epididymis
(3) Vasa efferentia
(4) Infundibulum

144. Arrange the following in descending order of number of species in the Amazonian rain forest.
(a) Plants
(b) Birds
(c) Fishes
(d) Invertebrates
(e) Mammals
Choose the **correct** answer from the options given below :
(1) (c) > (b) > (d) > (e) > (a)
(2) (d) > (a) > (c) > (b) > (e)
(3) (e) > (b) > (a) > (c) > (d)
(4) (b) > (a) > (d) > (c) > (e)

145. नीचे दो कथन दिए गए हैं :

कथन I : अंडोत्सर्ग (ओव्यूलेशन) LH के तीव्र स्रवण (सर्ज) के कारण होता है जिससे ग्राफी पुटक (ग्रैफ़ियन फॉलिकिल) का विखंडन हो जाता है।

कथन II : अंडोत्सर्ग के बाद शेष बचा ग्राफी पुटक कॉर्पस ल्यूटियम (पीतपिंड) में परिवर्तित हो जाती है और बड़ी मात्रा में एस्ट्रोजन स्रवित करती है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं

(2) **कथन I** और **कथन II** दोनों गलत हैं

(3) **कथन I** सही है परन्तु **कथन II** गलत है

(4) **कथन I** गलत है परन्तु **कथन II** सही है

145. Given below are two statements :

Statement I : Ovulation is caused by LH surge leading to rupture of Graafian follicles.

Statement II : Graafian follicle remaining after ovulation transform into corpus luteum and secretes large amount of estrogen.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

146. खाद्य श्रृंखला में निम्नलिखित में से कौन प्राथमिक उपभोक्ता होते हैं ?

(1) परजीवी

(2) शिकारी

(3) शाकाहारी

(4) मांसाहारी

146. Which of the following are primary consumers in a food chain ?

(1) Parasites

(2) Predators

(3) Herbivores

(4) Carnivores

147. द्विगुणित जीवों की एक जनसंख्या हार्डी-वेनबर्ग साम्यता में है। यदि एलील A की आवृत्ति 0.1 है, तो AA की आवृत्ति होगी

(1) 0.01 (2) 0.02

(3) 0.10 (4) 0.99

147. A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1, the frequency of AA is _____.

(1) 0.01 (2) 0.02

(3) 0.10 (4) 0.99

148. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

सूची-I		सूची-II
A. अत्यधिक वृद्धि हार्मोन	I.	वृक्क में जल और वैद्युत अपघट्य का पुनरावशोषण
B. ल्यूटिनाइजिंग हार्मोन	II.	प्रसव के दौरान गर्भाशय का संकुचन
C. वेसोप्रेसिन	III.	एक्रोमेगाली (अतिकायता)
D. ऑक्सीटोसिन	IV.	अंडोत्सर्ग

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

149. दाहिने आलिंद और दाहिने निलय के रंध्र पर _____ पाया जाता है।

(1) द्विवलनी कपाट (बाइकसपिड वाल्व)
(2) त्रिवलनी कपाट (ट्राइकसपिड वाल्व)
(3) अर्धचंद्र कपाट (सेमील्युनर वाल्व)
(4) शिराअलिंदपर्व (साइनो-आट्रियल नॉड)

150. स्पंज _____ द्वारा O_2 का CO_2 के साथ विनिमय करते हैं।

(1) अपनी पूर्ण शरीर सतह पर सरल विसरण
(2) नम क्यूटिकल
(3) श्वासनलियाँ (ट्रेकियल ट्यूब्स)
(4) गलफड़े (गिल्स)

151. एक प्ररूपी द्विपालित एन्जियोस्पर्म (आवृतबीजी) परागकोष की प्रत्येक पाली में कितने थीका (कोष) उपस्थित होते हैं ?

(1) 2 (2) 6
(3) 8 (4) 12

152. पेशी संकुचन की शुरुआत केंद्रीय तंत्रिका तंत्र द्वारा _____ के विमोचन द्वारा भेजे गए संकेत से होती है।

(1) एसिटिल कोलीन
(2) एसिटिल कोएंजाइम A
(3) चक्रीय ग्वानिन मोनोफॉस्फेट
(4) चक्रीय एडेनिन मोनोफॉस्फेट

148. Match **List-I** with **List-II**.

List-I		List-II
A. Excess growth hormone	I.	Reabsorption of water and electrolytes in kidney
B. Luteinizing hormone	II.	Contraction of uterus during child birth
C. Vasopressin	III.	Acromegaly
D. Oxytocin	IV.	Ovulation

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

149. The opening between the right atrium and the right ventricle is guarded by _____.

(1) bicuspid valve
(2) tricuspid valve
(3) semilunar valve
(4) sino-atrial node

150. Sponges exchange O_2 with CO_2 by _____.

(1) simple diffusion over their entire body surfaces
(2) moist cuticle
(3) tracheal tubes
(4) gills

151. How many theca are present in each lobe of a typical bilobed angiosperm anther ?

(1) 2 (2) 6
(3) 8 (4) 12

152. Muscle contraction is initiated by a signal sent by the central nervous system by the release of _____.

(1) acetyl choline
(2) acetyl coenzyme A
(3) cyclic guanine monophosphate
(4) cyclic adenine monophosphate

153. लैक-ऑपेरॉन (lac-operon) के बारे में निम्नलिखित में से कौन सा कथन **सही** है ?

(1) जीन *i* निरंतर रूप से अभिव्यक्त होता है

(2) लैक्टोज दमनकारी (रिप्रेसर) को ऑपरेटर से जुड़ने के लिए सक्रिय करता है

(3) जीन *i*, *z*, *y* और *a* एक ही सामान्य प्रमोटर साझा करते हैं

(4) गैलेक्टोज लैक ऑपेरॉन के प्रवर्तक (इंड्यूसर) के रूप में कार्य कर सकता है

153. Which of the following statements about lac-operon is **correct** ?

(1) Gene *i* is constitutively expressed

(2) Lactose activates repressor to bind to the operator

(3) Genes *i*, *z*, *y* and *a* share single common promoter

(4) Galactose can act as an inducer of lac operon

154. निम्नलिखित में से कौन एक एन्जियोस्पर्म (आवृतबीजी) के मादा युग्मकोद्भिद (गेमेटोफाइट) में अंडों के निषेचन के लिए परागनलिका का मार्गदर्शन करने में सहायता करता है?

(1) प्रतिव्यासांत कोशिकाएं (एंटीपोडल्स)

(2) सहाय कोशिकाएं (सिनर्जिड्स)

(3) केंद्रीय कोशिकाएं

(4) ध्रुवीय केंद्रक

154. Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the eggs ?

(1) Antipodals

(2) Synergids

(3) Central cells

(4) Polar nucleus

155. निम्नलिखित में से कौन सा पौधा गैर-भ्रूणपोषी (non-albuminous) बीज उत्पन्न करता है?

(1) गेहूँ

(2) मक्का

(3) जौ

(4) मटर

155. Which of the following plant produces non-albuminous seeds ?

(1) Wheat

(2) Maize

(3) Barley

(4) Pea

156. यदि एक प्ररूपी आवृतबीजी (एन्जियोस्पर्म) की द्विगुणित गुणसूत्र संख्या 36 है, तो इसके भ्रूणपोष (एंडोस्पर्म) में गुणसूत्रों की संख्या क्या होगी?

(1) 18 (2) 36

(3) 54 (4) 72

156. If the diploid chromosome number of typical angiosperm is 36, what would be the chromosome number in its endosperm ?

(1) 18 (2) 36

(3) 54 (4) 72

157. हेनले लूप (Henle's loop) में पुनरावशोषण प्रक्रिया के बारे में निम्नलिखित में से कौन से कथन **सही** हैं ?

(a) हेनले लूप की अवरोही भुजा जल के लिए पारगम्य होती है लेकिन वैद्युत अपघट्यों (इलेक्ट्रोलाइट्स) के लिए लगभग अपारगम्य होती है।

(b) हेनले लूप में मूत्र सांद्रित हो जाता है।

(c) हेनले लूप में Na^+ और जल का पुनरावशोषण होता है।

(d) हेनले लूप की आरोही भुजा में विद्युत अपघट्यों का सक्रिय या निष्क्रिय परिवहन होता है।

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) केवल (a) और (b)

(2) केवल (b), (c) और (d)

(3) केवल (a), (b) और (c)

(4) केवल (a), (b) और (d)

157. Which of the following statements about the reabsorption process in Henle's loop are **correct** ?

(a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.

(b) Urine gets concentrated in Henle's loop.

(c) Reabsorption of Na^+ and water takes place in Henle's loop.

(d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop.

Choose the **correct** answer from the options given below :

(1) (a) and (b) only

(2) (b), (c) and (d) only

(3) (a), (b) and (c) only

(4) (a), (b) and (d) only

158. सिर से पैर तक कशेरुकी दंड (वर्टिब्रेट कॉलम) के विन्यास का **सही** क्रम निम्नलिखित में से कौन सा है?

(1) ग्रीवा कशेरुक, वक्षीय कशेरुक, त्रिक (सेक्रम), कटि कशेरुक

(2) त्रिक (सेक्रम), कटि कशेरुक, वक्षीय कशेरुक, ग्रीवा कशेरुक

(3) ग्रीवा कशेरुक, कटि कशेरुक, वक्षीय कशेरुक, त्रिक (सेक्रम)

(4) ग्रीवा कशेरुक, वक्षीय कशेरुक, कटि कशेरुक, त्रिक (सेक्रम)

158. Which of the following is the **correct** order of arrangement of vertebrate column from the head to toe ?

(1) Cervical vertebra, thoracic vertebra, sacrum, lumbar vertebra

(2) Sacrum, lumbar vertebra, thoracic vertebra, cervical vertebra

(3) Cervical vertebra, lumbar vertebra, thoracic vertebra, sacrum

(4) Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

159. निम्नलिखित में से कौन सा जैव विकास का प्रमाण **नहीं है** ?

(1) पक्षियों और तितलियों के पंखों जैसे लक्षणों का अभिसारी विकास (कन्वर्जेंट इवोल्यूशन)

(2) जीवाश्म अभिलेखों से प्राप्त पुराजीवाश्रमिक प्रमाण

(3) अर्न्स्ट हेकेल द्वारा प्रस्तावित जैव विकास के लिए भ्रूणविज्ञानी समर्थन

(4) अग्रपादों जैसी आन्तरिक संरचनाओं का अपसारी विकास (डाइवर्जेंट इवोल्यूशन)

159. Which of the following is **not** evidence for evolution ?

(1) Convergent evolution of traits like wings of birds and butterflies

(2) Paleontological evidence from fossil records

(3) Embryological support for evolution as proposed by Ernst Heckel

(4) Divergent evolution of anatomical structures such as forelimbs

160. नीचे दो कथन दिए गए हैं :

कथन I : आधुनिक *होमो सैपियंस* (*Homo sapiens*) ऑस्ट्रेलिया में उत्पन्न हुए और महाद्वीपों में फैले।

कथन II : *होमो सैपियंस* (*Homo sapiens*) लगभग 75000 से 10000 वर्ष पहले उत्पन्न हुए।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं

(2) **कथन I** और **कथन II** दोनों गलत हैं

(3) **कथन I** सही है लेकिन **कथन II** गलत है

(4) **कथन I** गलत है लेकिन **कथन II** सही है

160. Given below are two statements :

Statement I : Modern *Homo sapiens* arose in Australia and moved across continents.

Statement II : *Homo sapiens* arose around 75000 to 10000 years ago.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

161. 10 मिलियन कोशिकाओं की एक जनसंख्या पर विचार करें। यदि प्रति व्यक्ति जन्म दर 0.002 (प्रति इकाई समय) और प्रति व्यक्ति मृत्यु दर 0.002 (प्रति इकाई समय) दी गई है, तो 10 पीढ़ियों के बाद कोशिकाओं की अपेक्षित संख्या _____ होगी।

(1) 1 मिलियन

(2) 5 मिलियन

(3) 10 मिलियन

(4) 100 मिलियन

161. Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the per-capita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is _____.

(1) 1 million

(2) 5 million

(3) 10 million

(4) 100 million

162. PCR (पीसीआर) के दौरान, प्राइमर _____ चरण में डीएनए स्ट्रैंड्स से जुड़ते हैं।

(1) निष्क्रियकरण (denaturation)

(2) प्रसार (extension)

(3) तापानुशीलन (annealing)

(4) लाइगेशन (ligation)

162. During PCR, primers bind to the DNA strands in the _____ step.

(1) denaturation

(2) extension

(3) annealing

(4) ligation

163. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **तर्क R** के रूप में नामांकित किया गया है।

अभिकथन A : जनसंख्या के संभार तंत्र (लॉजिस्टिक) वृद्धि मॉडल को चरघातांकीय वृद्धि मॉडल की तुलना में अधिक वास्तविक माना जाता है।

तर्क R : संसाधन सीमित होते हैं।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है

(2) **A** और **R** दोनों सही हैं लेकिन **R**, **A** की सही व्याख्या नहीं है

(3) **A** सही है लेकिन **R** सही नहीं है

(4) **A** सही नहीं है लेकिन **R** सही है

164. प्लेसेंटल (अपरा) स्तनधारियों और ऑस्ट्रेलियाई मार्सुपियल्स (शिशुधानी) प्राणियों में अनुकूली विकिरण के परिणामस्वरूप दूर की प्रजातियों के बीच समानता आती है, किसका एक उदाहरण है?

(1) अपसारी विकास

(2) अभिसारी विकास

(3) संस्थापक प्रभाव

(4) आनुवंशिक विचलन

165. निम्नलिखित में से कौन से द्वितीयक लसीका अंग हैं ?

(a) अस्थि मज्जा

(b) टॉन्सिल

(c) प्लीहा

(d) थायमस

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) केवल (a) और (b)

(2) केवल (b) और (c)

(3) केवल (b) और (d)

(4) केवल (a) और (d)

166. निम्नलिखित में से कौन सा हार्मोन मानव अपरा (प्लेसेंटा) द्वारा स्रावित **नहीं** होता है ?

(1) मानव जरायु गोनैडोट्रापिन (hCG)

(2) एस्ट्रोजन

(3) प्रोजेस्टेरोन

(4) ल्यूटिनाइजिंग हार्मोन (LH)

163. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The logistic growth model of populations is considered more realistic than the exponential growth model.

Reason R : Resources are finite.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(2) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(3) **A** is correct but **R** is not correct

(4) **A** is not correct but **R** is correct

164. Adaptive radiation in placental mammals and Australian Marsupials leading to similarity between distant species is an example of _____.

(1) divergent evolution

(2) convergent evolution

(3) founder effect

(4) genetic drift

165. Which of the following are secondary lymphoid organs ?

(a) Bone marrow

(b) Tonsils

(c) Spleen

(d) Thymus

Choose the **correct** answer from the options given below :

(1) (a) and (b) only

(2) (b) and (c) only

(3) (b) and (d) only

(4) (a) and (d) only

166. Which of the following hormone is **not** secreted by human placenta ?

(1) hCG

(2) Estrogen

(3) Progesterone

(4) LH

167. निम्नलिखित में से कौन सा एंजाइम प्रिकर्सर (पूर्वगामी) mRNA का संश्लेषण करता है?

(1) RNA पॉलीमरेज I

(2) RNA पॉलीमरेज II

(3) RNA पॉलीमरेज III

(4) DNA पॉलीमरेज

167. Which of the following enzymes synthesizes precursor mRNA ?

(1) RNA polymerase I

(2) RNA polymerase II

(3) RNA polymerase III

(4) DNA polymerase

168. नीचे दो कथन दिए गए हैं :

कथन I : प्लास्मिड स्वायत्त रूप से प्रतिकृति बनाने वाले डीएनए होते हैं।

कथन II : प्लास्मिड गुणसूत्र-बाह्य डीएनए होते हैं।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **कथन I** और **कथन II** दोनों सही हैं

(2) **कथन I** और **कथन II** दोनों गलत हैं

(3) **कथन I** सही है लेकिन **कथन II** गलत है

(4) **कथन I** गलत है लेकिन **कथन II** सही है

168. Given below are two statements :

Statement I :

Plasmids are autonomously replicating DNA.

Statement II :

Plasmids are extrachromosomal DNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

169. रक्त समूह 'O' वाले व्यक्ति के लिए, निम्नलिखित में से कौन सा माता-पिता के रक्त समूह जीनोटाइप का संभव संयोजन **नहीं है?**

(1) पिता : I^Ai और माता : I^Bi

(2) पिता : I^Ai और माता : I^Ai

(3) पिता : I^Bi और माता : I^Bi

(4) पिता : I^AI^B और माता : I^Ai

169. For a person with blood group 'O', which of the following is **not** a possible combination of parents' blood group genotypes ?

(1) Father : I^Ai and Mother : I^Bi

(2) Father : I^Ai and Mother : I^Ai

(3) Father : I^Bi and Mother : I^Bi

(4) Father : I^AI^B and Mother : I^Ai

170. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **तर्क R** के रूप में नामांकित किया गया है।

अभिकथन A : मनुष्य और चमगादड़ के अग्रपाद समजातीय (होमोलॉगस) होते हैं।

तर्क R : मनुष्य और चमगादड़ के अग्रपादों की शारीरिक संरचना समान होती है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है

(2) **A** और **R** दोनों सत्य हैं, लेकिन **R**, **A** की सही व्याख्या नहीं है

(3) **A** सत्य है लेकिन **R** असत्य है

(4) **A** असत्य है लेकिन **R** सत्य है

171. स्तनपान के शुरुआती दिनों के दौरान माता द्वारा स्रावित कोलोस्ट्रम (पीयूष) में _____ प्रचुर मात्रा में होता है।

(1) IgG (2) IgM

(3) IgA (4) IgD

172. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **तर्क R** के रूप में नामांकित किया गया है।

अभिकथन A : गैलापागोस द्वीपों पर एबिंगडन कछुआ, बकरियों के आने के एक दशक के भीतर विलुप्त हो गया।

तर्क R : बकरियाँ, एबिंगडन कछुए की तुलना में चरने (ब्राउजिंग) में अधिक कुशल थीं।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है

(2) **A** और **R** दोनों सही हैं लेकिन **R**, **A** की सही व्याख्या नहीं है

(3) **A** सही है लेकिन **R** सही नहीं है

(4) **A** सही नहीं है लेकिन **R** सही है

170. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Forelimbs of human and bats are homologous.

Reason R : Forelimbs of humans and bats have similar anatomical structure.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(2) Both **A** and **R** are true, but **R** is not the correct explanation of **A**

(3) **A** is true but **R** is false

(4) **A** is false but **R** is true

171. Colostrum, secreted by mother during initial days of lactation, is abundant in _____.

(1) IgG (2) IgM

(3) IgA (4) IgD

172. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.

Reason R : Goats were more efficient at browsing than Abingdon tortoise.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(2) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(3) **A** is correct but **R** is not correct

(4) **A** is not correct but **R** is correct

173. अंडोत्सर्ग (ओव्यूलेशन) के समय अंडाणु के आवरण को _____ कहते हैं।

(1) एंडोमेट्रियम (गर्भाशय अंतःस्तर)

(2) जोना रेडियाटा

(3) जोना पेलुसिडा

(4) कोरिऑन

174. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

	सूची-I		**सूची-II**
A.	दोनों प्रजातियों को हानि होती है	I.	परभक्षण (Predation)
B.	एक प्रजाति को हानि होती है और दूसरी को लाभ होता है	II.	सहोपकारिता (Mutualism)
C.	दोनों प्रजातियों को लाभ होता है	III.	प्रतिस्पर्धा (Competition)
D.	एक को लाभ होता है जबकि दूसरे पर कोई प्रभाव नहीं पड़ता	IV.	सहभोजिता (Commensalism)

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-III, B-IV, C-II, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-I, C-II, D-IV

173. The covering of ovum at ovulation is _____.

(1) endometrium

(2) zona radiata

(3) zona pellucida

(4) chorion

174. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Both species are harmed	I.	Predation
B.	One species is harmed and the other is benefited	II.	Mutualism
C.	Both species are benefited	III.	Competition
D.	One is benefited while the other has no effect	IV.	Commensalism

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-I, C-II, D-IV

175. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **तर्क R** के रूप में नामांकित किया गया है।

अभिकथन A : एक प्रयोग में, मेंडल ने देखा कि F1 संतति पौधे सभी लंबे थे और कोई भी बौना नहीं था।

तर्क R : तने की ऊँचाई एक विपरीत लक्षण है, जिसमें लंबा प्रभावी और बौना अप्रभावी होता है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है

(2) **A** और **R** दोनों सही हैं, लेकिन **R**, **A** की सही व्याख्या नहीं है

(3) **A** सही है लेकिन **R** सही नहीं है

(4) **A** सही नहीं है लेकिन **R** सही है

175. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarf.

Reason R : Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(2) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(3) **A** is correct but **R** is not correct

(4) **A** is not correct but **R** is correct

176. नीचे दो कथन दिए गए हैं : एक को **अभिकथन A** और दूसरे को **तर्क R** के रूप में नामांकित किया गया है।

अभिकथन A : पुनर्योगज डीएनए तकनीक में, जीवाणु कोशिकाओं को विखंडित करने के लिए लाइसोजाइम का उपयोग किया जाता है जबकि पादप कोशिकाओं के लिए सेलुलेस का उपयोग किया जाता है।

तर्क R : आनुवंशिक पदार्थ के पृथक्करण के लिए कोशिकाओं के विखंडन की आवश्यकता होती है।

उपर्युक्त कथनों के आलोक में, नीचे दिए गए विकल्पों में से **सबसे उपयुक्त** उत्तर चुनें :

(1) **A** और **R** दोनों सही हैं और **R**, **A** की सही व्याख्या है

(2) **A** और **R** दोनों सही हैं लेकिन **R**, **A** की सही व्याख्या नहीं है

(3) **A** सही है लेकिन **R** सही नहीं है

(4) **A** सही नहीं है लेकिन **R** सही है

176. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.

Reason R : Isolation of genetic material needs disruption of cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(2) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(3) **A** is correct but **R** is not correct

(4) **A** is not correct but **R** is correct

177. सर्जरी के बाद के रोगियों के उपचार के लिए निम्नलिखित में से किसका उपयोग एक प्रभावी शामक (sedative) और दर्द निवारक के रूप में किया जाता है?

(1) इंटरफेरॉन

(2) एंटीबायोटिक्स

(3) मॉर्फिन

(4) एंटी-रिट्रोवायरल दवाएं

178. निम्नलिखित में से कौन से कथन **सही** हैं ?

(a) उत्पादकों से उपभोक्ताओं तक ऊर्जा का प्रवाह एकदिशीय होता है

(b) ऊर्जा पिरामिड कभी भी उल्टा नहीं हो सकता

(c) ऊर्जा का स्थानांतरण 1% नियम का पालन करता है

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) (a), (b) और (c)

(2) केवल (a) और (b)

(3) केवल (a) और (c)

(4) केवल (b) और (c)

179. *प्लाज्मोडियम* के विषय में निम्नलिखित में से कौन सा कथन **सही** है?

(1) यकृत कोशिकाओं में लैंगिक जनन करता है

(2) आरबीसी (RBCs) में लैंगिक जनन करता है

(3) गैमेटोसाइट्स मच्छर की आंत में विकसित होते हैं

(4) निषेचन मच्छर की आंत में होता है

177. Which of the following is used as an effective sedative and painkiller for treating post-surgery patients ?

(1) Interferon

(2) Antibiotics

(3) Morphine

(4) Anti-retroviral drugs

178. Which of the following statements are **correct** ?

(a) Energy flow from producers to consumers is unidirectional

(b) Energy pyramid can never be inverted

(c) Transfer of energy follows the 1% law

Choose the **correct** answer from the options given below :

(1) (a), (b) and (c)

(2) (a) and (b) only

(3) (a) and (c) only

(4) (b) and (c) only

179. Which of the following statements is **correct** about *Plasmodium* ?

(1) Reproduces sexually in liver cells

(2) Reproduces sexually in RBCs

(3) Gametocytes develop in mosquito gut

(4) Fertilization takes place in mosquito gut

180. **सूची-I** को **सूची-II** के साथ सुमेलित कीजिए।

सूची-I	**सूची-II**
A. रूपांतरण (Transformation)	I. प्रतिबंध एंजाइम (Restriction enzyme)
B. क्लोनिंग स्थल (Cloning site)	II. मेजबान बैक्टीरिया में डीएनए का स्थानांतरण
C. चयन (Selection)	III. प्रतिकृतियन (Replication)
D. प्रतिकृतियन का उत्पत्ति (Ori)	IV. प्रतिजैविक (Antibiotic)

नीचे दिए गए विकल्पों में से **सही** उत्तर चुनें :

(1) A-II, B-I, C-IV, D-III

(2) A-I, B-II, C-IV, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-I, C-III, D-II

180. Match **List-I** with **List-II.**

List-I	**List-II**
A. Transformation	I. Restriction enzyme
B. Cloning site	II. Transfer DNA to host bacteria
C. Selection	III. Replication
D. Ori	IV. Antibiotic

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-IV, D-III

(2) A-I, B-II, C-IV, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-I, C-III, D-II

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निम्नलिखित निर्देश ध्यान से पढ़ें :	*Read carefully the following instructions :*
9. परीक्षार्थी सुनिश्चित करें कि इस उज़र पत्र को मोड़ा न जाए एवं उस पर कोई अन्य निशान न लगाएं। परीक्षार्थी अपना अनुक्रमांक प्रश्न पुस्तिका/उज़र पत्र में निर्धारित स्थान के अतिरिक्त अन्यत्र ना लिखें।	9. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
10. उज़र पत्र पर किसी प्रकार के संशोधन हेतु व्हाइट ज़्लूइड के प्रयोग की अनुमति **नहीं** है।	10. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
11. पूछे जाने पर प्रत्येक परीक्षार्थी, निरीक्षक को अपना प्रवेश-पत्र दिखाएं।	11. Each candidate must show on-demand his/her Admit Card to the Invigilator.
12. केंद्र अधीक्षक या निरीक्षक की विशेष अनुमति के बिना कोई परीक्षार्थी अपना स्थान न छोड़ें।	12. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
13. कार्यरत निरीक्षक को अपना उज़र पत्र दिए बिना एवं उपस्थिति-पत्रक पर दुबारा हस्ताक्षर (समय के साथ) किए बिना कोई परीक्षार्थी परीक्षा हॉल नहीं छोड़ेंगे। **यदि किसी परीक्षार्थी ने दूसरी बार उपस्थिति-पत्रक पर हस्ताक्षर नहीं किए तो यह माना जाएगा कि उसने उज़र पत्र नहीं लौटाया है और यह अनुचित साधन का मामला माना जाएगा।**	13. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
14. इलेक्ट्रॉनिक/हस्तचालित परिकलक का उपयोग वर्जित है।	14. Use of Electronic/Manual Calculator is prohibited.
15. परीक्षा-कक्ष/हॉल में आचरण के लिए परीक्षार्थी, परीक्षा के नियमों एवं विनियमों द्वारा नियमित हैं। अनुचित साधन के सभी मामलों का फैसला इस परीक्षा के नियमों, विनियमों एवं सार्वजनिक परीक्षा (अनुचित साधनों की रोकथाम अधिनियम - **2024**)के अनुसार होगा।	15. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination along with Public Examinations (Prevention of unfair means act 2024).
16. **किसी हालात में परीक्षा पुस्तिका और उज़र पत्र का कोई भाग अलग न करें।**	16. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
17. परीक्षा पुस्तिका/उज़र पत्र में दिए गए परीक्षा पुस्तिका संकेत को परीक्षार्थी सही तरीके से उपस्थिति-पत्रक में लिखें।	17. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
18. यदि कोई अज़्यर्थी किसी एक प्रश्न पर एक से ज्यादा उज़रों को ओएमआर (OMR) पुस्तिका पर चिह्नित करता है, तो उस प्रश्न को गलत माना जाएगा तथा नकारात्मक अंकन लागू होगा।	18. If a candidate marks more than one answers for a question in the OMR sheet, it will be treated as incorrect and negative marking will be applicable.

SEAL

1. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_p/C_V of the gas is 8/7, then the value of f is :

(1) 3 (2) 2

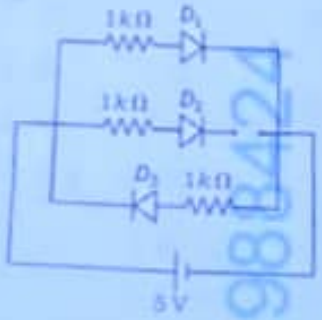
(3) 1 (4) 4

2. One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of a wire to be 1 cm, the actual length of the wire is :

(1) 0.96 cm (2) 1.00 cm

(3) 1.04 cm (4) 0.60 cm

3. Three identical p-n junction diodes D_1, D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is :



(1) $W_3 = W_1 > W_2$

(2) $W_3 > W_2 > W_1$

(3) $W_2 > W_1 = W_3$

(4) $W_1 > W_2 > W_3$

4. Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is k Nm^{-1}. At a given instant, the extension of the spring is x meter and the speed of the particle is v ms^{-1}. On the x - v plane, if the graph of v as a function of x is a circle, then the correct option is :

(1) $k = m$ (2) $k = m^2$

(3) $k = \sqrt{m}$ (4) $k = \frac{1}{m}$

5. An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\epsilon_r = 9$, has the electric field, $E_x = E_0 \sin(kz - 2\pi \times 10^6 t)$ Vm^{-1} where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is :

(1) The wavelength of the electromagnetic wave inside the medium is 300 m

(2) The magnetic field is given by the relation $B_y = \frac{B_0}{v} \sin(kz - 2\pi \times 10^6 t)$ where v is the speed of the electromagnetic wave inside the medium

(3) The direction of propagation of the electromagnetic wave is along +z

(4) The speed of the electromagnetic wave inside the medium is 10^8 ms^{-1}

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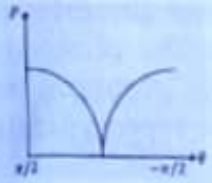
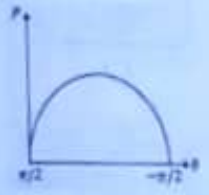
6. A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point $A\left(\theta=\frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point $B\left(\theta=-\frac{\pi}{2}\right)$. During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ ?

(1)



(2)

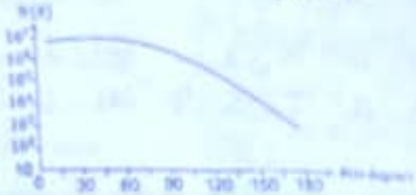


(3)

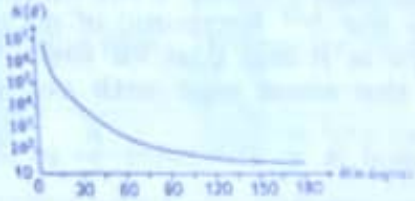
(4)

7. In Geiger-Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the correct plot ?

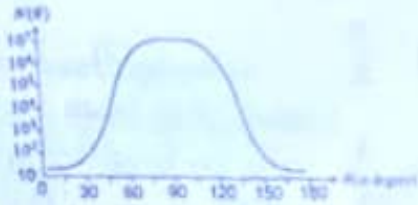
(1)



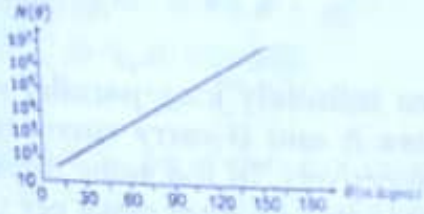
(2)



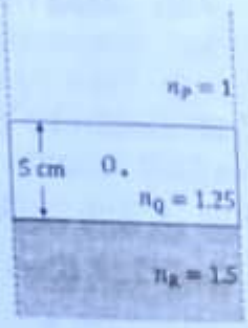
(3)



(4)



8. Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5, respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1-h_2|$, in cm, is :



(1) 1 (2) 2
(3) 3 (4) 0

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9. Consider that σ_s, k_B, b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is :

(1) $[L^{-1}K^{-2}]$ (2) $[L^{-1}T^{-1}K^{-3}]$

(3) $[L^{-1}T^{-1}K^{-4}]$ (4) $[L^{-1}T^{-1}K^{-2}]$

10. For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is :

(1) $\frac{9}{5}$ (2) 1

(3) $\frac{3}{5}$ (4) $\frac{5}{9}$

11. Two infinitely long parallel conducting wires A and B carry currents I and 2I, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is :

[g is the acceleration due to gravity and μ_0 is the permeability of free space.]

(1) $\frac{\mu_0 I^2}{\pi\lambda g}$ (2) $\frac{2\mu_0 I^2}{\pi\lambda g}$

(3) $\frac{4\mu_0 I^2}{\pi\lambda g}$ (4) $\frac{\mu_0 I^2}{2\pi\lambda g}$

12. An ac voltage $V = 220 \sin(2\times10^3 t)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :

(Given : L = 10 mH, C = 25μF, R = 100 Ω)

(1) 5.5 A (2) 11.0 A

(3) 22.0 A (4) 2.2 A

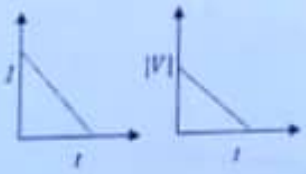
13. Consider a fixed uniformly charged insulating sphere with radius R and total charge +Q. A point charge −q (q<<Q) with mass m is released from rest at a distance of 3R from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is :

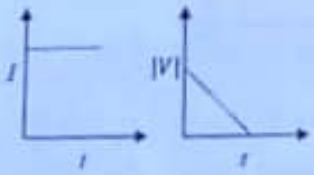
(ϵ_0 is the permittivity of vacuum, neglect gravitational forces).

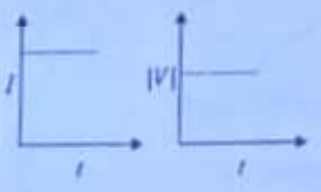
(1) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$ (2) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$

(3) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$ (4) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$

14. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential |V| with time is best represented by :

(1) 

(2) 

(3) 

(4)

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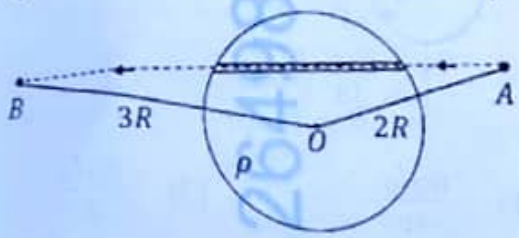
15. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2=\frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is :

(1) 1 (2) $\sqrt{2}$

(3) 2 (4) $\frac{1}{\sqrt{2}}$

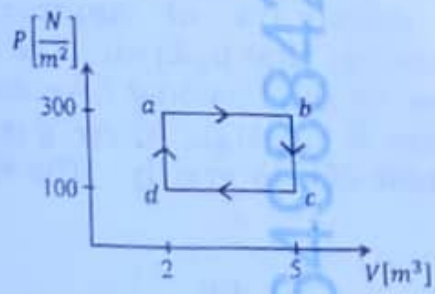
16. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is :
(ϵ_0 is the permittivity of vacuum)



(1) 6 (2) 9

(3) 18 (4) 2

17. One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is :



(1) 500 J (2) 600 J

(3) 800 J (4) 400 J

18. The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the electromagnetic spectrum		Applications	
P	Microwave	I	For purifying the water
Q	UV rays	II	For warming the food
R	Gamma rays	III	For AM and FM communication systems
S	Radio wave	IV	For treating the Cancer cells

The **correct** option is :

(1) P-I, Q-IV, R-II, S-III

(2) P-II, Q-I, R-IV, S-III

(3) P-II, Q-IV, R-III, S-I

(4) P-I, Q-II, R-III, S-IV

19. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed 10 ms^{-1}, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be 10 ms^{-2}, the value of θ is :

(1) $\tan^{-1}\left(\frac{2}{5}\right)$ (2) $\tan^{-1}\left(\sqrt{3}/2\right)$

(3) $\tan^{-1}\left(2\sqrt{3}\right)$ (4) $\tan^{-1}\left(\frac{1}{5}\right)$

20. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t)=\alpha t^2-\beta t+\gamma$, where $\alpha=1$ ms^{-2}, $\beta=6$ ms^{-1} and $\gamma=5$ m. The average speed of the particle, in ms^{-1}, from $t=0$ to $t=6$ s is :

(1) 6 (2) 3

(3) 0 (4) 12

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21. A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B and E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is :

(1) $E_A \neq 0,\ E_B = E_C$

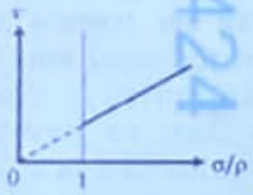
(2) $E_A = 0,\ E_B > E_C$

(3) $E_A \neq 0,\ E_B < E_C$

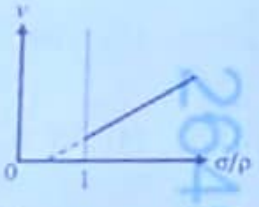
(4) $E_A = 0,\ E_B = E_C$

22. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by :

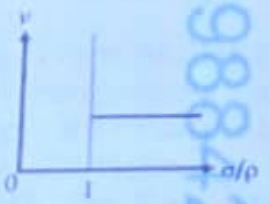
(1)



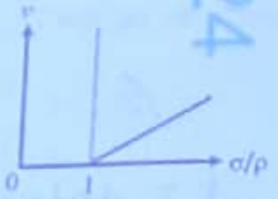
(2)



(3)



(4)



23. A ray of light with wavelength λ is incident on three different photo-electric cells namely 1, 2 and 3. The threshold wavelength of these photo-electric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ and $\lambda_3 >> \lambda$. The correct option is :

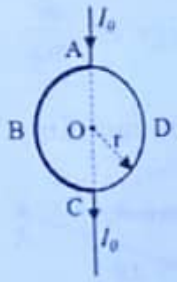
(1) $V_1 = 0,\ V_2 > V_3$

(2) $V_1 > V_2,\ V_3 = 0$

(3) $V_1 < V_2,\ V_3 = 0$

(4) $V_1 = 0,\ V_2 < V_3$

24. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the center O of the loop is :



(1) $\frac{\mu_0 I_0}{4r}$ (2) $\frac{\mu_0 I_0}{2r}$

(3) $\frac{\mu_0 I_0}{2\pi r}$ (4) $\frac{\mu_0 I_0}{12r}$

25. A particle of mass M moves along a horizontal x axis from $x=0$ to $x=L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is :

(1) 1 (2) $\frac{1}{3}$

(3) $\frac{1}{2}$ (4) 3

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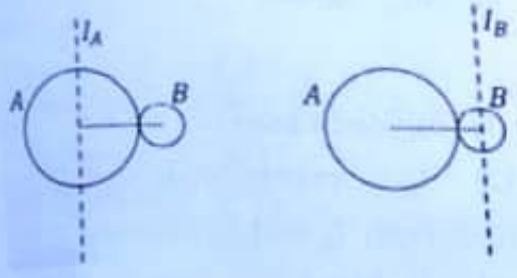
26. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to :

(1) $R^{3/2}$ (2) R^2

(3) R^3 (4) $R^{1/2}$

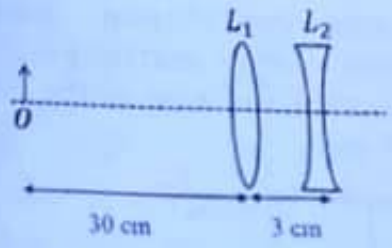
27. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is :



(1) $(m-M)(R+r)^2$ (2) $(m-M)(R-r)^2$

(3) 0 (4) $(M-m)(R+r)^2$

28. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and −10 cm, respectively. The position of the image formed is :



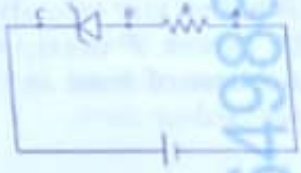
(1) 60 cm to the left of the concave lens

(2) 30 cm to the right of the concave lens

(3) 60 cm to the right of the concave lens

(4) 20 cm to the left of the concave lens

29. An ideal Zener diode with breakdown voltage of −3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between points B and A is :



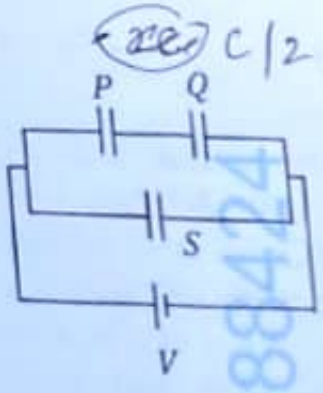
(1) 2 V

(2) 1 V

(3) 0 V

(4) 3 V

30. Three identical capacitors P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is :



(1) 1/3

(2) 1/2

(3) 1/6

(4) 2/3

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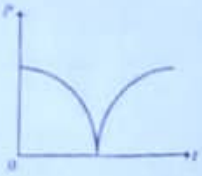
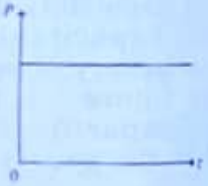
31. A conducting loop of finite resistance lies on the x - y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :

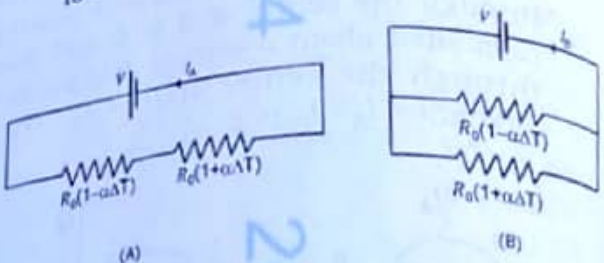
(1) 

(2) 

(3)

(4)

32. Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature of coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is :



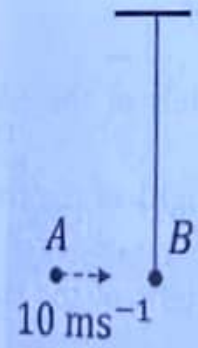
(1) I_A decreases while I_B increases

(2) I_A increases while I_B decreases

(3) both I_A and I_B remain constant

(4) I_A remains constant while I_B increases

33. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed 10 ms^{-1} hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ ms^{-2}$ and neglecting the size of the bob, the value of h is :



(1) 7 (2) 5

(3) 2.5 (4) 8

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A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period 2T. The value of ρ_2/ρ_1 is :

(1) 2 (2) 1/2

(3) 1/4 (4) 4

The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, then the correct option is :

(1) $n_A = 2n_B$ (2) $n_A = \frac{1}{4}n_B$

(3) $n_A = \frac{1}{2}n_B$ (4) $n_A = n_B$

Consider the following nuclear reaction :

$^{238}U \rightarrow {}^{234}Th + {}^{4}He$

Take masses of ^{238}U, ^{234}Th and ^{4}He as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is :

[Given : 1 u = 931.5 MeV c^{-2}]

(1) 3730 (2) 3736

(3) 3740 (4) 3726

Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

(1) $\mu_0 n^2 r^2 l$ (2) $(\mu_0/2\pi) n^2 r^2 l$

(3) $2\mu_0 \pi n^2 r^2 l$ (4) $\mu_0 \pi n^2 r^2 l$

38. Which of the following measurements require 'index correction' ?

(1) Measurement of gravitational acceleration using simple pendulum

(2) Measurement of focal length of lenses using optical bench

(3) Measurement of speed of sound using resonance tube

(4) Measurement of resistance of a wire using meter bridge

39. The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is :

(1) $3\pi R^3 \alpha \Delta T$ (2) $4\pi R^3 \alpha \Delta T$

(3) $6\pi R^3 \alpha \Delta T$ (4) $2\pi R^3 \alpha \Delta T$

40. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is :

(Take speed of light $= 3 \times 10^8$ ms^{-1}, charge of electron $= 1.6 \times 10^{-19}$ C and mass of electron $= 9 \times 10^{-31}$ kg)

(1) $\frac{1}{250}$ (2) 225

(3) 275 (4) $\frac{2}{450}$

41. Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6} \times 10^5$ ms^{-1}. The radius of the orbit is $x \times 10^{-9}$ m. The value of x is :

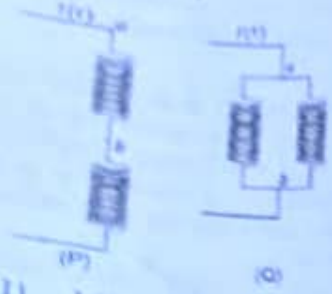
[Take the mass of electron to be 9×10^{-31} kg, charge of electron $= -1.6 \times 10^{-19}$ C and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ N m^2 C^{-2}]

(1) 3 (2) 2

(3) 1 (4) 4

42. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is :

[Neglect the effect of mutual inductance.]



(1) 1/2
(2) 1
(3) 2
(4) 1/4

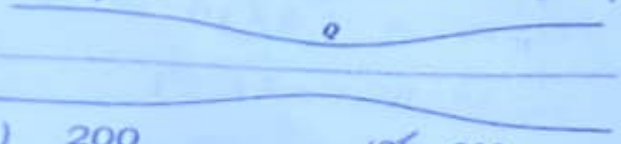
43. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60K to 50K. The work done by the gas in the process is :

[Take the universal gas constant as $R = 8.3\ J\ mol^{-1}\ K^{-1}$]

(1) 83 J
(2) 124.5 J
(3) 166 J
(4) 41.5 J

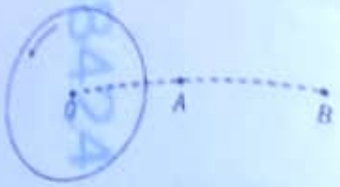
44. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is $15\ Nm^{-2}$. The area of cross-section at P and Q are 40 cm^2 and 20 cm^2, respectively. The rate of flow of water through the pipe, in $cm^3 s^{-1}$, is :

[Take density of water = $1000\ kg\ m^{-3}$]



(1) 200
(2) 300
(3) 400
(4) 100

45. A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B respectively, with $OB = 2 \times OA$. The value of $\frac{L_A}{L_B}$ is :



(1) $\frac{1}{2}$
(2) 1
(3) 2
(4) $\frac{1}{4}$

46. Match the species in **List I** with their geometry in **List II**

List I		**List II**
A.	PCl_5	I. Tetrahedral
B.	BrF_5	II. Square Planar
C.	BF_4^-	III. Trigonal bipyramidal
D.	$[Ni(CN)_4]^{2-}$	IV. Square pyramidal

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-II, D-IV
(3) A-III, B-II, C-I, D-IV
(4) A-IV, B-III, C-I, D-II

47. The amino acid that gives a red-blood colour on treating its sodium fusion extract with sodium nitroprusside is

(1) threonine
(2) methionine
(3) serine
(4) leucine

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Solutions

Physics

Q. 1 A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is :

(Take speed of light $= 3 \times 10^8\ ms^{-1}$, charge of electron $= -1.6 \times 10^{-19} C$ and mass of electron $= 9 \times 10^{-31}\ kg$) me

Option 1:

275

Option 2:

$\frac{2}{450}$

Option 3:

$\frac{1}{250}$

Option 4:

225

Correct Answer:

225

Solution:

For a photon of energy E,

$$p_{ph} = \frac{E}{c}$$

For an electron of kinetic energy E,

$$E = \frac{p_e^2}{2m}$$

$$p_e = \sqrt{2mE}$$

Hence,

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$$\frac{p_e}{p_{ph}} = \frac{\sqrt{2mE}}{E/c}$$

$$= c\sqrt{\frac{2m}{E}}$$

Now,

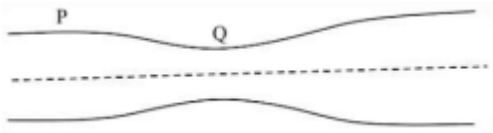
$$\frac{p_e}{p_{ph}} = 3\times10^8\sqrt{\frac{2\times9\times10^{-31}}{3.2\times10^{-18}}}$$

$$= 3\times10^8\sqrt{5.625\times10^{-13}}$$

$$= 3\times10^8\times7.5\times10^{-7} = 225$$

Q. 2 Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is 15Nm^{-2}. The area of cross-section at P and Q are $40\ \text{cm}^2$ and $20\ \text{cm}^2$, respectively. The rate of flow of water through the pipe, in $\text{cm}^3\ \text{s}^{-1}$, is :

[Take density of water $= 1000\ \text{kg m}^{-3}$]



Option 1:

400

Option 2:

100

Option 3:

200

Option 4:

300

Correct Answer:

400

Solution:

Given:

$$P_P - P_Q = 15\text{Nm}^{-2}$$

$$A_P = 40\ \text{cm}^2 = 40\times10^{-4}\ \text{m}^2$$

$$A_Q = 20\ \text{cm}^2 = 20\times10^{-4}\ \text{m}^2$$

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Using equation of continuity,

$$A_P v_P = A_Q v_Q$$
$$40 v_P = 20 v_Q \Rightarrow v_Q = 2 v_P$$

Now using Bernoulli's equation for horizontal flow,

$$P_P + \frac{1}{2}\rho v_P^2 = P_Q + \frac{1}{2}\rho v_Q^2$$
$$P_P - P_Q = \frac{1}{2}\rho\left(v_Q^2 - v_P^2\right)$$
$$15 = \frac{1}{2}(1000)\left[\left(2v_P\right)^2 - v_P^2\right]$$
$$15 = 500\left(4v_P^2 - v_P^2\right) = 500\left(3v_P^2\right)$$
$$15 = 1500 v_P^2 \Rightarrow v_P^2 = 0.01 \Rightarrow v_P = 0.1 \text{ m s}^{-1}$$

Rate of flow:

$$Q = A_P v_P$$
$$Q = 40 \times 10^{-4} \times 0.1 = 4 \times 10^{-4} \text{ m}^3 / \text{s}$$

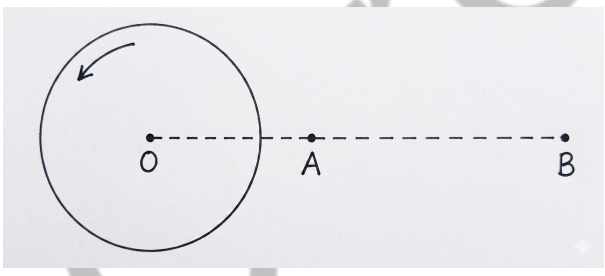
Convert into cm^3 / s :

$$4 \times 10^{-4} \times 10^6 = 400 \text{ cm}^3 / \text{s}$$

Correct Answer: Option (1)

Q. 3

A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2 \times OA$. The value of $\frac{\text{L}_A}{\text{L}_B}$ is :



Option 1:

2

Option 2:

$\frac{1}{4}$

Option 3:

$\frac{1}{2}$

Option 4:

1

Correct Answer:

1

Solution:

A thin disc rotates about the fixed vertical axis through its centre O.
For a rigid body rotating with angular velocity ω, angular momentum about a point on the axis of rotation remains along the axis.
Since points A and B lie on the same line of the axis configuration shown and the disc rotates about the fixed centre, the angular momentum of the rotating disc about points A and B is taken to be the same in this standard result-based question.
Therefore,

$$L_A = L_B$$
$$\frac{L_A}{L_B} = 1$$

Correct Answer: Option (4)

Q. 4 Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

Option 1:

$2\mu_0\pi n^2r^2l$

Option 2:

$\mu_0\pi n^2r^2l$

Option 3:

$\mu_0 n^2r^2l$

Option 4:

$(\mu_0 / 2\pi)\, n^2r^2l$

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Correct Answer:

$\mu_0 \pi n^2 r^2 l$

Solution:

For a long solenoid,

$$L = \mu_0 n^2 A l$$

where
- $n =$ number of turns per unit length
- $A = \pi r^2$

Therefore,

$$L = \mu_0 n^2 \left(\pi r^2 \right) l$$
$$L = \mu_0 \pi n^2 r^2 l$$

Correct Answer: Option (2)

Q. 5 The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is :

Option 1:

$6\pi R^3 \alpha \Delta T$

Option 2:

$2\pi R^3 \alpha \Delta T$

Option 3:

$3\pi R^3 \alpha \Delta T$

Option 4:

$4\pi R^3 \alpha \Delta T$

Correct Answer:

$4\pi R^3 \alpha \Delta T$

Solution:

Volume of a sphere,

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$V = \frac{4}{3}\pi R^3$

For a solid, coefficient of volume expansion is

$\beta = 3\alpha$

Hence increase in volume,

$$\Delta V = \beta V \Delta T$$

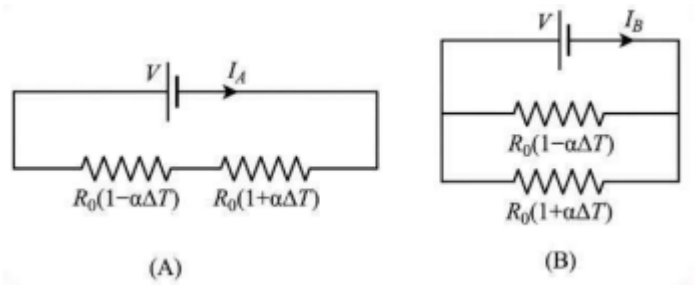
$$\Delta V = 3\alpha\left(\frac{4}{3}\pi R^3\right)\Delta T$$

$$\Delta V = 4\pi R^3 \alpha \Delta T$$

Correct Answer: Option (4)

Q. 6

Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature coefficient, $-\alpha$, as shown in the figure. The currents through these circuits are denoted by I_A and I_B. At the initial temperature, the resistance of the two resistors is R_O. As the temperature is increased, the correct option that describes the variation of current in these circuits is :



Option 1:

both I_A and I_B remain constant

Option 2:

I_A remains constant while I_B increases

Option 3:

I_A decreases while I_B increases

Option 4:

I_A increases while I_B decreases

Correct Answer:

I_A remains constant while I_B increases

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Solution:

In circuit A, the two resistors are in series:

$$R_1 = R_0(1 - \alpha\Delta T), \quad R_2 = R_0(1 + \alpha\Delta T)$$

Total resistance in circuit A:

$$R_A = R_1 + R_2$$
$$R_A = R_0(1 - \alpha\Delta T) + R_0(1 + \alpha\Delta T) = 2R_0$$

So R_A remains constant, hence

$$I_A = \frac{V}{R_A}$$

remains constant.
In circuit B, the two resistors are in parallel:

$$R_B = \frac{R_1 R_2}{R_1 + R_2}$$
$$R_B = \frac{R_0(1 - \alpha\Delta T) \cdot R_0(1 + \alpha\Delta T)}{2R_0}$$
$$R_B = \frac{R_0}{2}\left(1 - \alpha^2\Delta T^2\right)$$

As temperature increases, $\alpha^2\Delta T^2$ increases, so R_B decreases.
Hence current

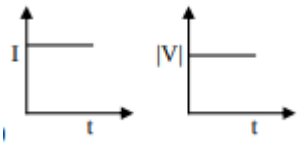
$$I_B = \frac{V}{R_B}$$

increases.
So,
- I_A remains constant
- I_B increases

Correct Answer: Option (2)

Q. 7 A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by :

Option 1:



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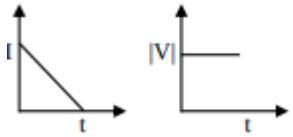
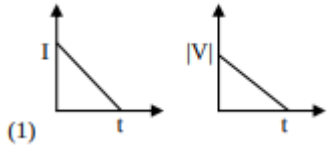
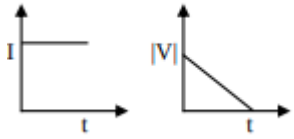
Option 2:



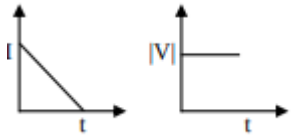
Option 3:



Option 4:



Correct Answer:



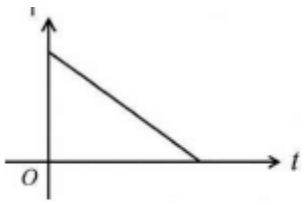
Solution:

$$\text{Intensity} = \frac{\text{Power}}{\text{Area}}$$

∵ Power ↓ se linearly with time, hence intensity ↓ se linearly with time

Also, current $\propto$ intensity

⇒ Current ↓ decreases linearly with time.



→ Stopping potential depends on frequency and not on intensity, so stopping potential remains constant.

Q. 8

An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between points B and A is :

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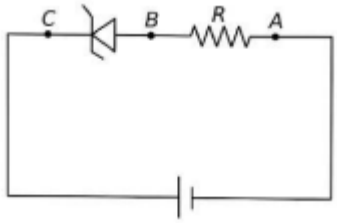


Option 1:

0V

Option 2:

3V

Option 3:

2V

Option 4:

1V

Correct Answer:

2V

Solution:

The Zener diode is reverse biased.
Given:
- Breakdown voltage $= -3\ \text{V}$
- Input voltage $V_i = -5\ \text{V}$

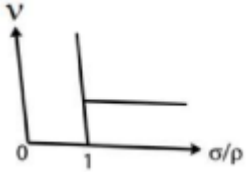
Since the diode is ideal and in reverse breakdown, the potential difference across the Zener diode remains 3 V .
Hence the remaining voltage appears across the resistor segment between A and B.

$$|V_{BA}| = 5 - 3 = 2\ \text{V}$$

Correct Answer: Option (3)

Q. 9 In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by :

Option 1:



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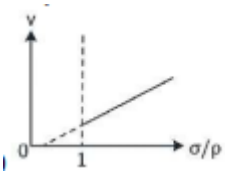
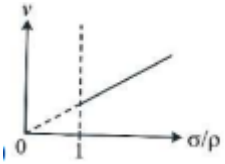
Option 2:

Option 3:



Option 4:



Correct Answer:

Solution:

Terminal velocity of a sphere falling in a viscous liquid is

$$v = \frac{2r^2 g(\sigma - \rho)}{9\eta}$$

where
- $\sigma =$ density of sphere
- $\rho =$ density of liquid

So,

$$v \propto (\sigma - \rho)$$

$$v \propto \rho\left(\frac{\sigma}{\rho} - 1\right)$$

Thus terminal velocity is a linear function of $\frac{\sigma}{\rho}$, and becomes zero when

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$\frac{\sigma}{\rho} = 1$

Therefore the graph is a straight line starting from $\sigma / \rho = 1$.
Correct Answer: Option (2)

Q. 10 Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2 = \frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is :

Option 1:

2

Option 2:

$\frac{1}{\sqrt{2}}$

Option 3:

1

Option 4:

$\sqrt{2}$

Correct Answer:

$\sqrt{2}$

Solution:

Escape speed of a planet is

$$v_e = \sqrt{\frac{2GM}{R}}$$

Given both planets have equal mass, so

$$v \propto \frac{1}{\sqrt{R}}$$

Also,

$$R_2 = \frac{R_1}{2}$$

Hence,

$$\frac{v_2}{v_1} = \sqrt{\frac{R_1}{R_2}}$$

$$= \sqrt{\frac{R_1}{R_1 / 2}} = \sqrt{2}$$

Correct Answer: Option (4)

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Q. 11 An ac voltage $V = 220\sin\left(2\times10^{3}t\right)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :
(Given : $L = 10\text{mH}, C = 25\mu\text{ F}, R = 100\Omega$)

Option 1:

22.0 A

Option 2:

2.2 A

Option 3:

5.5 A

Option 4:

11.0 A

Correct Answer:

2.2 A

Solution:

Given:

$$V = 220\sin\left(2\times10^{3}t\right)$$

So,

$$V_0 = 220\text{ V}, \quad \omega = 2\times10^{3}\text{rad s}^{-1}$$

Also,

$$L = 10\text{mH} = 10\times10^{-3}\text{H}$$
$$C = 25\mu\text{ F} = 25\times10^{-6}\text{ F}$$
$$R = 100\Omega$$

Inductive reactance:

$$X_L = \omega L = \left(2\times10^{3}\right)\left(10\times10^{-3}\right) = 20\Omega$$

Capacitive reactance:

$$X_C = \frac{1}{\omega C} = \frac{1}{\left(2\times10^{3}\right)\left(25\times10^{-6}\right)} = \frac{1}{0.05} = 20\Omega$$

So,

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$X_L = X_C$

Hence the circuit is in resonance and impedance becomes

$Z = R = 100\Omega$

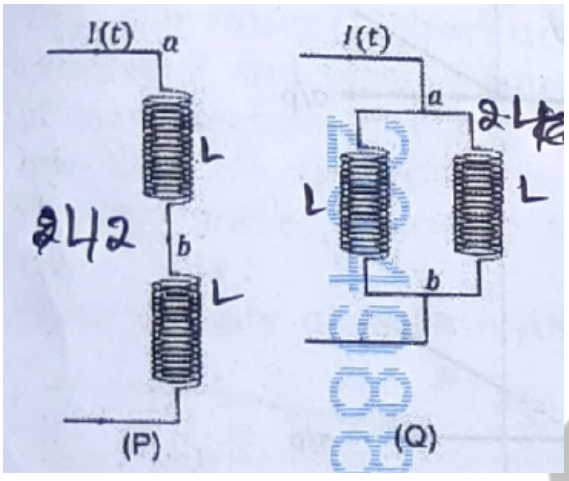
Therefore current amplitude is

$I_0 = \frac{V_0}{R} = \frac{220}{100} = 2.2$ A

Correct Answer: Option (2)

Q. 12

Two identical inductors are connected in two different configurations P and Q , where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P / E_Q is:
Neglect the effect of mutual inductance.]



Option 1:

2

Option 2:

1 / 4

Option 3:

1 / 2

Option 4:

1

Correct Answer:

2

Solution:

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Let self-inductance of each identical inductor be L.
In configuration P :
The two inductors are in series and point a to b covers only one inductor.
Therefore induced emf between a and b is

$$E_P = L\frac{dI}{dt}$$

In configuration Q :
The two identical inductors are connected in parallel between a and b. Equivalent inductance of two equal inductors in parallel is

$$L_{\text{eq}} = \frac{L}{2}$$

Therefore,

$$E_Q = L_{\text{eq}}\frac{dI}{dt}$$
$$E_Q = \frac{L}{2}\frac{dI}{dt}$$

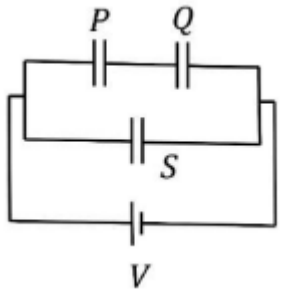
Hence,

$$\frac{E_P}{E_Q} = \frac{L\frac{dI}{dt}}{\frac{L}{2}\frac{dI}{dt}} = 2$$

Correct Answer: Option (1)

Q. 13

Three identical capacitors P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are $\mathrm{U_P}$ and $\mathrm{U_T}$, respectively, then the ratio $\frac{\mathrm{U_P}}{\mathrm{U_T}}$ is :



Option 1:

1 / 6

Option 2:

2 / 3

Option 3:

1 / 3

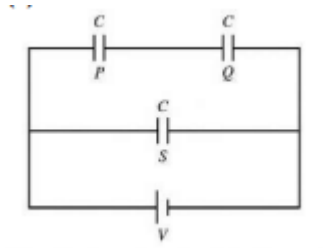
Option 4:

1 / 2

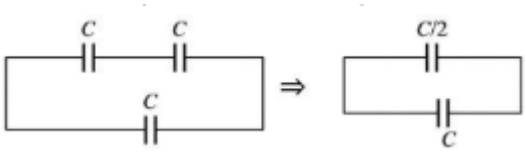
Correct Answer:

1 / 6

Solution:



equal potential difference is there across P and Q = V/2 (since C1 = C2 = C3 = C)



Energy across $(P) = \frac{1}{2}C\left(\frac{V}{2}\right)^2$ ..(i)

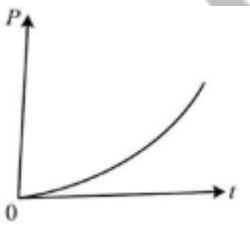
Energy across system $= \frac{1}{2}C_{eq}\left(V^2\right)$

$= \frac{1}{2} \times \frac{3C}{2} \times V^2 = \frac{3C}{4}V^2 \ldots ii$

(i) ÷ (ii) ⇒ $\frac{1}{6}$

Q. 14 A conducting loop of finite resistance lies on the $x - y$ plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :

Option 1:



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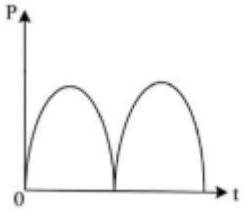
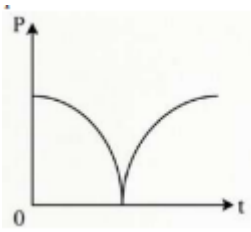
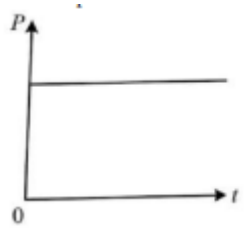
Option 2:



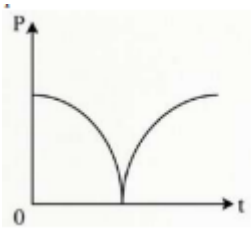
Option 3:



Option 4:



Correct Answer:



Solution:

$$A = A_0(1 + \sin t)$$

$$\phi$$

$$e = \frac{-dt}{dt} = -BA_0\cos t$$

$$|e| = BA_0(1 + \sin t)$$

$$P = \frac{e^2}{R} = B^2\frac{A_0}{R}\cos^2 t$$

$$p \geq 0$$

$$\max t = 0, \pi, 2\pi \ldots \ldots$$

$$\text{zero } t = \frac{\pi}{2}, \frac{3\pi}{2}$$

Q. 15

In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60 K to 50 K . The work done by the gas in the process is :

(Take the universal gas constant as $R = 8.3\ \text{J mol}^{-1}\ \text{K}^{-1}$)

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Option 1:

166 J

Option 2:

41.5 J

Option 3:

83 J

Option 4:

124.5 J

Correct Answer:

124.5 J

Solution:

$$\Delta w = \frac{\mu R\left(T_1 - T_2\right)}{\gamma - 1}$$

$$\Delta w = \frac{1 \times 8.3(60 - 50)}{\frac{5}{3} - 1}$$

$$= \frac{8.3 \times 10}{\frac{2}{3}}$$

$$= 15 \times 8.3$$

$$= 124.5\ \text{J}$$

Q. 16 Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1\ ms^{-2}$, $\beta = 6\ ms^{-1}$ and $\gamma = 5$ m. The average speed of the particle, in ms^{-1}, from $t = 0$ to $t = 6s$ is :

Option 1:

0

Option 2:

12

Option 3:

6

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Option 4:

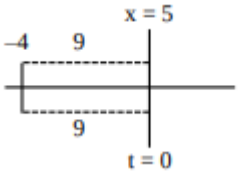
3

Correct Answer:

3

Solution:

$$S = \alpha t^2 - \beta t + C$$
$$S = t^2 - 6t + 5$$
$$V = \frac{ds}{dt} = 2t - 6$$
$$V = 0, 2t - 6 = 0$$
$$t = 3\text{sec}$$
$$att = 0$$
$$S \Rightarrow S = 5$$
$$at = 3\text{sec}$$
$$S = 9 - 18 + 5 = -4$$
$$at = t = 6\text{sec}$$
$$S = 36 - 36 + 3 = 5$$



18/6= 3m/s

Q. 17

The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the electromagnetic spectrum		Applications	
P	Microwave	I	For purifying the water
Q	UV rays	II	For warming the food
R	Gamma rays	III	For AM and FM communication systems
S	Radio wave	IV	For treating the Cancer cells

The correct option is :

Option 1:

P-II, Q-IV, R-III, S-I

Option 2:

P-I, Q-II, R-III, S-IV †

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Option 3:

P-I, Q-IV, R-II, S-III

Option 4:

P-II, Q-I, R-IV, S-III

Correct Answer:

P-II, Q-I, R-IV, S-III

Solution:

We match each part of electromagnetic spectrum with its correct application:
- Microwaves → warming food
- UV rays → purifying water
- Gamma rays → treating cancer cells
- Radio waves → AM and FM communication systems

So the correct matching is:

$P \rightarrow II, \quad Q \rightarrow I, \quad R \rightarrow IV, \quad S \rightarrow III$

According to the answer key provided, the marked option is (4).
Correct Answer: Option (4)

Q. 18 An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_P / C_V of the gas is $8 / 7$, then the value of f is :

Option 1:

1

Option 2:

4

Option 3:

3

Option 4:

2

Correct Answer:

4

Solution:

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$$r = 1 + \frac{2}{(3+3+2f)}$$

$$\frac{8}{7} = 1 + \frac{2}{6+2f}$$

$$\left(\frac{2}{6+2f}\right) = \frac{1}{7}$$

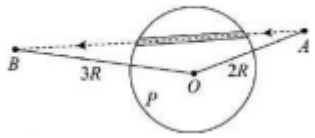
$$14 = 6 + 2f$$

$$2f = 8$$

$$f = 4$$

A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this processs, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is:

Q. 19 (ϵ_0 is the permittivity of vacuum)



Option 1:

18

Option 2:

2

Option 3:

3

Option 4:

9

Correct Answer:

18

Solution:

$$\because V_{\text{out}} = \frac{\rho R^3}{3\varepsilon_0 r}$$

At point $A, V_A = \frac{\rho R^2}{6\varepsilon_0}$, when $(r = 2R)$ At point $B, V_B = \frac{\rho R^2}{9\varepsilon_0}$, when $(r = 3R)$

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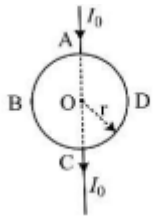
$$\Delta W = q\left|V_B - V_A\right| = 1 \times \frac{\rho R^2}{18\varepsilon_0}$$

$$\therefore n = 18$$

A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the center O of the loop is :

Q. 20



Option 1:

$\frac{\mu_0 I_0}{2\pi r}$

Option 2:

$\frac{\mu_0 I_0}{12r}$

Option 3:

$\frac{\mu_0 I_0}{4r}$

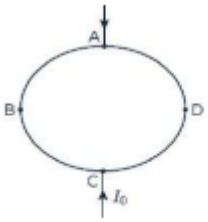
Option 4:

$\frac{\mu_0 I_0}{2r}$

Correct Answer:

$\frac{\mu_0 I_0}{12r}$

Solution:



$B_{ADC} = \frac{\mu_0 I_0}{3(4r)}$

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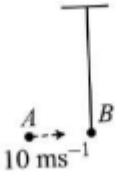
$$B_{ABC} = \frac{\mu_0 (2I_0)}{3(4r)}$$

$$B_{ABC} > B_{ADC}$$

$$B_{ABDC} - B_{ADC} = \frac{\mu_0 I_0}{12r}$$

Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m , as shown in the figure. Point mass A of mass m travelling horizontally with speed $10\ \text{ms}^{-1}$ hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ \text{ms}^{-2}$ and neglecting the size of the bob, the value of h is :

Q. 21



Option 1:

2.5

Option 2:

8

Option 3:

7

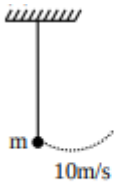
Option 4:

5

Correct Answer:

5

Solution:



Due to elastic collision

$$\text{K.E.} = \frac{1}{2}mv^2 = mgH$$

$$\frac{1}{2} \times (10)^2 = 10H$$

$$H = 5\ \text{m}$$

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Q. 22 An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant, $\epsilon_r = 9$, has the electric field, $E_x = E_0 \sin\left(kz - 2\pi \times 10^6 t\right) Vm^{-1}$ where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is :

Option 1:

The direction of propagation of the electromagnetic wave is along $+z$

Option 2:

The speed of the electromagnetic wave inside the medium is $10^8\ ms^{-1}$

Option 3:

The wavelength of the electromagnetic wave inside the medium is 300 m

Option 4:

The magnetic field is given by the relation $B_y = \frac{B_0}{v} \sin\left(kz - 2\pi \times 10^6 t\right)$ where v is the speed of the electromagnetic wave inside the medium

Correct Answer:

The direction of propagation of the electromagnetic wave is along $+z$

Solution:

For the wave

$$E_x = E_0 \sin\left(kz - 2\pi \times 10^6 t\right)$$

this is of the form

$$E = E_0 \sin(kz - \omega t)$$

Hence the wave propagates along $+z$ direction Given dielectric constant,

$$\varepsilon_r = 9$$

Speed of electromagnetic wave in the medium is

$$v = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{3} = 10^8\ m\ s^{-1}$$

Also,

$$\omega = 2\pi \times 10^6$$

So frequency is

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$f = 10^6$ Hz

Therefore wavelength is

$$\lambda = \frac{v}{f} = \frac{10^8}{10^6} = 100 \text{ m}$$

So the statement "wavelength is 300 m " is incorrect.
Hence this question is treated as Bonus in the given answer key.
Correct Answer: Bonus

Q. 23 A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = \mathrm{L}$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(\mathrm{L}) = 0$. The total work done by the frictional force during the motion is $n\mu_0 \mathrm{MgL}$, where g is the acceleration due to gravity. The value of n is :

Option 1:

$\frac{1}{2}$

Option 2:

3

Option 3:

1

Option 4:

$\frac{1}{3}$

Correct Answer:

$\frac{1}{2}$

Solution:

$$\mu_k = \mu_0 - \alpha x$$

$$W \cdot D = m\mu_0 MgL$$

$$n = ?$$

$$W \cdot D = \int_0 F_r \cdot ds$$

$$W \cdot D = \int_0^1 (\mu_0 - \alpha_n)\, mg \cdot dx$$

$$W \cdot D = \mu_0 mgL - \alpha \frac{L^2}{2}$$

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According to relation

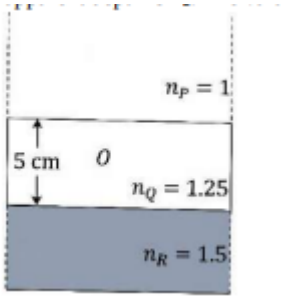
$\mu_k = \mu_0 - \alpha x$

But at $x = L$

$\mu_x = 0$

Consider three media P, Q and R with refractive indices $1, 1.25$, and 1.5 , respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm , is :

Q. 24



Option 1:

3

Option 2:

0

Option 3:

1

Option 4:

2

Correct Answer:

1

Solution:

Refractive indices are:

$n_P = 1, \quad n_Q = 1.25, \quad n_R = 1.5$

Thickness of medium Q is 5 cm .
Object is at the centre of medium Q, so real depth from either face is

$\frac{5}{2} = 2.5 \text{ cm}$

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Apparent depth when viewed from medium P :

$$h_1 = \frac{n_P}{n_Q} \times 2.5$$

$$h_1 = \frac{1}{1.25} \times 2.5 = 2 \text{ cm}$$

Apparent depth when viewed from medium R :

$$h_2 = \frac{n_R}{n_Q} \times 2.5$$

$$h_2 = \frac{1.5}{1.25} \times 2.5 = 3 \text{ cm}$$

Hence,

$$|h_1 - h_2| = |2 - 3| = 1 \text{ cm}$$

Correct Answer: Option (3)

Q. 25 Consider a fixed uniformly charged insulating sphere with radius R and total charge $+Q$. A point charge $-q(q \ll Q)$ with mass m is released from rest at a distance of $3R$ from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is :
(ϵ_0 is the permittivity of vacuum, neglect gravitational forces).

Option 1:

$\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$

Option 2:

$\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$

Option 3:

$\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$

Option 4:

$\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$

Correct Answer:

$\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$

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Solution:

$$U_i + x_i = U_t + k_t$$

$$\frac{-kQq}{3R} + 0 = \frac{-kQq}{R} + \frac{1}{2}mv^2$$

$$\frac{2kQq}{3R} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{Qq}{3\pi\varepsilon_0 mR}}$$

Q. 26 A car travels on a circular racetrack of radius 50 m , which is banked at an angle θ. If the car travels at a speed $10\ ms^{-1}$, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be $10\ ms^{-2}$, the value of θ is :

Option 1:

$\tan ^{-1}(2 \sqrt{3})

Option 2:

$\tan^{-1}\left(\frac{1}{5}\right)$.

Option 3:

$\tan^{-1}\left(\frac{2}{5}\right)$

Option 4:

$\tan^{-1}(\sqrt{3}\ /\ 2)$

Correct Answer:

$\tan^{-1}\left(\frac{1}{5}\right)$.

Solution:

For a vehicle moving on a banked road with minimum wear and tear, friction is not required. Therefore,

$$\tan\theta = \frac{v^2}{rg}$$

Given,

$$v = 10\ m\ s^{-1}, \quad r = 50\ m, \quad g = 10\ m\ s^{-2}$$

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So,

$$\tan\theta = \frac{10^2}{50 \times 10} = \frac{100}{500} = \frac{1}{5}$$

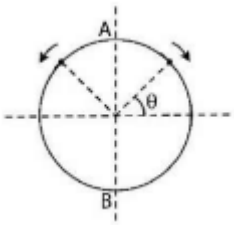
Hence,

$$\theta = \tan^{-1}\left(\frac{1}{5}\right)$$

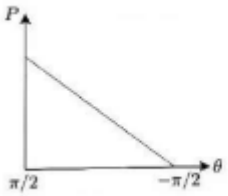
Correct Answer: Option (2)

Q. 27

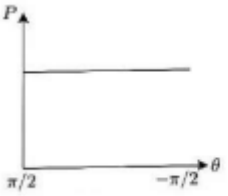
A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point $A\left(\theta = \frac{\pi}{2}\right)$ with identical uniform angular speeds in opposite directions, and meet again at point $B\left(\theta = -\frac{\pi}{2}\right)$. During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ ?



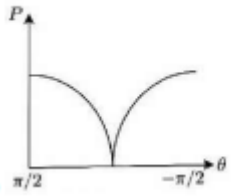
Option 1:



Option 2:



Option 3:



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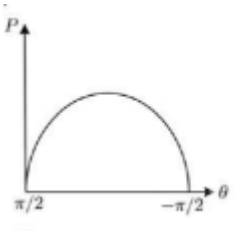
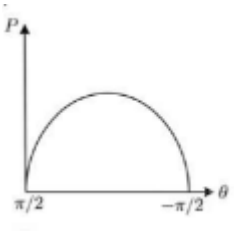
Option 4:



Correct Answer:

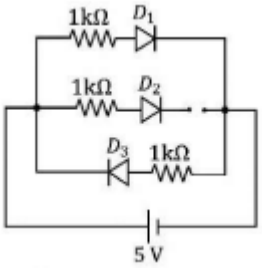


Solution:

$\because$ Initial momentum at $\theta = \frac{\pi}{2}$ is zero.
Initial momentum at $\theta = \frac{-\pi}{2}$ is also zero.

Q. 28

Three identical $p - n$ junction diodes D_1, D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is :



Option 1:

$W_2 > W_1 = W_3$

Option 2:

$W_1 > W_2 > W_3$

Option 3:

$W_3 = W_1 > W_2$

Option 4:

$W_3 > W_2 > W_1$

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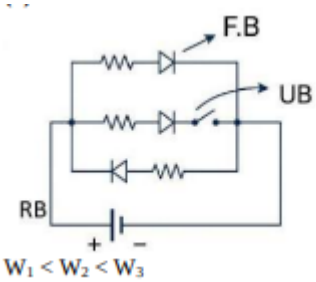
Correct Answer:

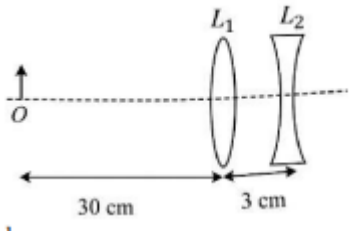
$W_2 > W_1 = W_3$

Solution:



in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and -10 cm, respectively. The position of the image formed is :

Q. 29



Option 1:

60 cm to the right of the concave lens

Option 2:

20 cm to the left of the concave lens

Option 3:

60 cm to the left of the concave lens

Option 4:

30 cm to the right of the concave lens

Correct Answer:

60 cm to the left of the concave lens

Solution:

For lens L_1,

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$f_1 = +10\ \text{cm}, \quad u_1 = -30\ \text{cm}$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{10} = \frac{1}{v_1} - \left(-\frac{1}{30}\right)$$

$$\frac{1}{10} = \frac{1}{v_1} + \frac{1}{30}$$

$$\frac{1}{v_1} = \frac{1}{10} - \frac{1}{30} = \frac{2}{30} = \frac{1}{15}$$

$$v_1 = 15\ \text{cm}$$

So the image formed by L_1 is 15 cm to the right of L_1.
Distance between the two lenses is 3 cm , therefore this image lies

$$15 - 3 = 12\ \text{cm}$$

to the right of lens L_2.
Thus for lens L_2, object distance is

$$u_2 = +12\ \text{cm}$$

Now for concave lens L_2,

$$f_2 = -10\ \text{cm}$$

Using lens formula,

$$\frac{1}{f_2} = \frac{1}{v_2} - \frac{1}{u_2}$$

$$\frac{1}{-10} = \frac{1}{v_2} - \frac{1}{12}$$

$$\frac{1}{v_2} = -\frac{1}{10} + \frac{1}{12}$$

$$\frac{1}{v_2} = \frac{-6+5}{60} = -\frac{1}{60}$$

$$\frac{1}{v_2} = \frac{-6+5}{60} = -\frac{1}{60}$$

$$v_2 = -60\ \text{cm}$$

Negative sign shows that the final image is formed 60 cm to the left of the concave lens.
Correct Answer: Option (3)

Q. 30

A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is :

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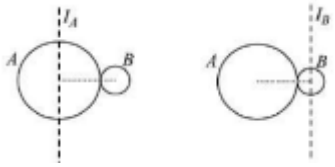


Option 1:

0

Option 2:

$(M-m)(R+r)^2$

Option 3:

$(\mathrm{m}-\mathrm{M})(\mathrm{R}+\mathrm{r})^2$

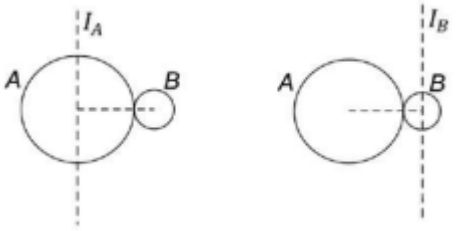
Option 4:

$(\mathrm{m}-\mathrm{M})(\mathrm{R}-\mathrm{r})^2$

Correct Answer:

$(\mathrm{m}-\mathrm{M})(\mathrm{R}+\mathrm{r})^2$

Solution:



$$I_A = \frac{2}{5}MR^2 + \frac{2}{5}mr^2 + m(R+r)^2$$

$$I_B = \frac{2}{5}mr^2 + \frac{2}{5}MR^2 + M(r+R)^2$$

$$I_A - I_B = m(R+r)^2 - M\left(r+R^2\right)$$

$$== (R+r)^2(m-M)$$

Q. 31

Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6} \times 10^5\ \mathrm{ms}^{-1}$. The radius of the orbit is $x \times 10^{-9}$ m. The value of x is :

[Take the mass of electron to be 9×10^{-31} kg, charge of electron

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$= -1.6 \times 10^{-19}\text{C}$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \text{N m}^2\text{C}^{-2}$]

Option 1:

1

Option 2:

4

Option 3:

3

Option 4:

2

Correct Answer:

1

Solution:

$$V = \sqrt{25.6} \times 10^5$$
$$V^2 = 25.6 \times 10^{10}$$
$$V^2 = 2.56 \times 10^{11}$$
$$\Rightarrow \frac{mv^2}{r} = \frac{Ke^2}{r^2}$$
$$r = \frac{Ke^2}{mv^2} = \frac{9 \times 10^9 \times \left(1.6 \times 10^{-19}\right)^2}{9 \times 10^{-31} \times 2.56 \times 10^{11}}$$
$$r = 10^{-9}$$
$$r = 1 \times 10^{-9}$$
$$\therefore x = 1$$

Q. 32 The mean free path of molecules in an ideal gas A is half that of another ideal gas B The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, then the correct option is :

Option 1:

$n_\text{A} = \frac{1}{2} n_\text{B}$

Option 2:

$n_A = n_B$

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Option 3:

$n_A = 2n_B$

Option 4:

$n_A = \frac{1}{4}n_B$

Correct Answer:

$n_A = \frac{1}{2}n_B$

Solution:

$$(\lambda_m)_A = \frac{1}{2}(\lambda_m)_B$$

$$d_A = 2\,d_B$$

$$\lambda_m = \frac{1}{\sqrt{2}\pi\, d^2 n}$$

$$n \propto \frac{1}{\lambda_m d^2}$$

$$\frac{n_A}{n_B} = \frac{\lambda m_B}{\lambda m_A} \times \left(\frac{d_B}{d_A}\right)^2$$

$$= \left(\frac{2}{1}\right) \times \left(\frac{1}{2}\right)^2$$

$$\frac{n_A}{n_B} = \frac{1}{2} \text{ so } n_A = \frac{1}{2}n_B$$

Q. 33 A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period 2 T . The value of ρ_2 / ρ_1 is :

Option 1:

1 / 4

Option 2:

4

Option 3:

2

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Option 4:

1 / 2

Correct Answer:

1 / 4

Solution:

$$T = 2\pi\sqrt{\frac{m}{A\rho \cdot g}}$$

$$T \propto \frac{1}{\sqrt{\rho_1}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\sqrt{\frac{\rho_2}{\rho_1}}\right)^2$$

$$\frac{\rho_2}{\rho_1} = \left(\frac{T}{2}\right)^2$$

$$\frac{\rho_2}{\rho_1} = \frac{1}{4}$$

Q. 34 For sound waves, if the number of nodes for the 5^{th} harmonic of an open-ended pipe is n and that for the 9^{th} harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is :

Option 1:

$\frac{3}{5}$

Option 2:

$\frac{5}{9}$

Option 3:

$\frac{9}{5}$

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Option 4:

1

Correct Answer:

1

Solution:

1. opens end (5^{th} hormonic)

Harmonic : $\mathrm{p} = 5$
Nodes (n) = 5

2. One end closed (9^{th} harmonic)

Harmonic: $\mathrm{p} = 1$
Nodes (m): $m = \frac{p+1}{2}$

$$\mathrm{m} = \frac{\mathrm{a}+1}{2} = 5$$

$$\frac{\mathrm{n}}{\mathrm{m}} = \frac{5}{5} = 1$$

Consider the following nuclear reaction :

$^{238}\mathrm{U} \rightarrow {}^{234}\mathrm{Th} + {}^{4}\mathrm{He}$

Q. 35 Take masses of $^{238}\mathrm{U}$, $^{234}\mathrm{Th}$ and $^{4}\mathrm{He}$ as $238.050u$, $234.043u$ and $4.003u$, respectively. The Q value for the reaction, in keV , is :

[Given : $1\mathrm{u} = 931.5\mathrm{MeVc}^{-2}$]

Option 1:

3740

Option 2:

3726

Option 3:

3730

Option 4:

3736

Correct Answer:

3726

Solution:

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Nuclear reaction:

$$^{238}\mathrm{U} \rightarrow {}^{234}\mathrm{Th} + {}^{4}\mathrm{He}$$

Given masses:

$$m\left({}^{238}\mathrm{U}\right) = 238.050u$$
$$m\left({}^{234}\mathrm{Th}\right) = 234.043u$$
$$m\left({}^{4}\mathrm{He}\right) = 4.003u$$

Mass defect:

$$\Delta m = 238.050 - (234.043 + 4.003)$$
$$\Delta m = 238.050 - 238.046 = 0.004u$$

Q-value:

$$Q = \Delta m \times 931.5\mathrm{MeV}$$
$$Q = 0.004 \times 931.5 = 3.726\mathrm{MeV}$$

Converting to keV :

$$Q = 3726\mathrm{keV}$$

Correct Answer: Option (2)

Q. 36 Which of the following measurements require 'index correction'?

Option 1:

Measurement of speed of sound using resonance tube

Option 2:

Measurement of resistance of a wire using meter bridge

Option 3:

Measurement of gravitational acceleration using simple pendulum

Option 4:

Measurement of focal length of lenses using optical bench

Correct Answer:

Measurement of focal length of lenses using optical bench

Solution:

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index correction is and adjustment applied to optical bench reading to account for the physical discrepancy between on the measured distance on the bench scale and the true optical distance from the centre pole of a lens or mirror

Q. 37 In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to :

Option 1:

R^3

Option 2:

$R^{1/2}$

Option 3:

$R^{3/2}$

Option 4:

R^2

Correct Answer:

$R^{3/2}$

Solution:

For a planet revolving around the sun,

$$\frac{GMm}{R^2} = \frac{mv^2}{R}$$

$$v = \sqrt{\frac{GM}{R}}$$

Time period,

$$T = \frac{2\pi R}{v}$$

$$T = 2\pi R\sqrt{\frac{R}{GM}}$$

$$T \propto R^{3/2}$$

Correct Answer: Option (3)

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Consider that σ_s, k_B, b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1}$ b is :

Q. 38

Option 1:

$\left[L^{-1} T^{-1} K^{-4}\right]$

Option 2:

$\left[L^{-1} T^{-1} K^{-2}\right]$

Option 3:

$\left[L^{-1} K^{-2}\right]$

Option 4:

$\left[L^{-1} T^{-1} K^{-3}\right]$

Correct Answer:

$\left[L^{-1} T^{-1} K^{-2}\right]$

Solution:

We need the dimensions of

$\sigma_s k_B^{-1} b$

where
- $\sigma_s =$ Stefan constant
- $k_B =$ Boltzmann constant
- $b =$ Wien's displacement constant

From Stefan's law,

$j = \sigma_s T^4$

where j is power per unit area.

$$[j] = \frac{\text{energy}}{\text{time} \times \text{area}} = \frac{ML^2T^{-2}}{T \cdot L^2} = MT^{-3}$$

So,

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$$[\sigma_s] = \frac{MT^{-3}}{K^4} = MT^{-3}K^{-4}$$

$$[\sigma_s] = \frac{MT^{-3}}{K^4} = MT^{-3}K^{-4}$$

$$k_B = \frac{\text{energy}}{\text{temperature}}$$
$$[k_B] = ML^2T^{-2}K^{-1}$$

So,

$$\left[k_B^{-1}\right] = M^{-1}L^{-2}T^2K$$

Dimensions of Wien's constant From Wien's law,

$$\lambda_m T = b$$
$$[b] = LK$$

$$\left[\sigma_s k_B^{-1} b\right] = \left(MT^{-3}K^{-4}\right)\left(M^{-1}L^{-2}T^2K\right)(LK)$$
$$= L^{-1}T^{-1}K^{-2}$$

Correct Answer: Option (2)

Q. 39 A ray of light with wavelength λ is incident on three different photoelectric cells namely 1,2 and 3 . The threshold wavelength of these photoelectric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1,V_2 andV_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda, \lambda_2 > \lambda$ and $\lambda_3 \gg \lambda$. The correct option is :

Option 1:

$V_1 < V_2, V_3 = 0$

Option 2:

$V_1 = 0,\ V_2 < V_3$

Option 3:

$V_1 = 0,\ V_2 > V_3$

Option 4:

$V_1 > V_2, V_3 = 0$

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Correct Answer:

$V_1 = 0,\ V_2 < V_3$

Solution:

Q39 Solution
In photoelectric effect, emission takes place only when incident wavelength is less than threshold wavelength.
Given:

$$\lambda_1 < \lambda, \quad \lambda_2 > \lambda, \quad \lambda_3 \gg \lambda_1$$

Now compare:
- For cell 1: $\lambda > \lambda_1$, so incident wavelength is greater than threshold wavelength → no photoelectric emission

$V_1 = 0$

- For cell 2 and 3 , photoelectric emission occurs and stopping potential is proportional to maximum kinetic energy:

$eV_s = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$

Smaller threshold wavelength means larger threshold frequency and hence smaller kinetic energy for the same incident light.
Thus,

$\boldsymbol{V_2} < \boldsymbol{V_3}$

Correct Answer: Option (2)

Q. 40 One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10 . When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4^{th} Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of a wire to be 1 cm , the actual length of the wire is :

Option 1:

1.04 cm

Option 2:

0.60 cm

Option 3:

0.96 cm

Option 4:

1.00 cm

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Correct Answer:

1.04 cm

Solution:

Given:
- 1 main scale division $= 1\ \text{mm}$
- Number of Vernier divisions = 10

Least count:

$$LC = \frac{1\text{MSD}}{10} = 0.1\ \text{mm} = 0.01\ \text{cm}$$

When jaws touch, Vernier zero is to the left of main scale zero. So the instrument has negative zero error.
Given 4th Vernier division coincides with a main scale division.
Hence zero error

$$= -4 \times LC = -4(0.01) = -0.04\ \text{cm}$$

Measured length of wire $= 1.00\ \text{cm}$
True length

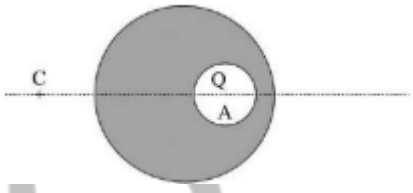
$$= \text{measured value} - (\text{zero error})$$

$$= 1.00 - (-0.04) = 1.04\ \text{cm}$$

Correct Answer: Option (1)

Q. 41

A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are $\mathrm{E_A}$, E_B and E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is :



Option 1:

$\mathrm{E_A} \neq 0, \mathrm{E_B} \leqslant \mathrm{E_C}$

Option 2:

$\mathrm{E_A} = 0, \mathrm{E_B} = \mathrm{E_C}$

Option 3:

$\mathrm{E_A} \neq 0, \mathrm{E_B} = \mathrm{E_C}$

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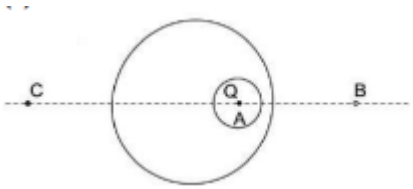
Option 4:

$E_A = 0, E_B > E_C$.

Correct Answer:

$E_A \neq 0, E_B = E_C$

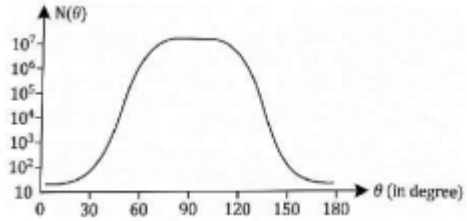
Solution:



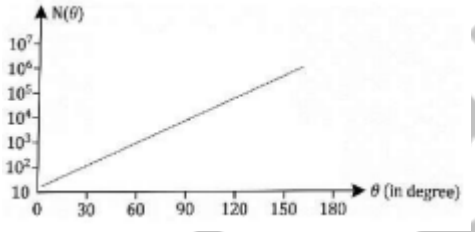
Change is present in cavity, so EA □ 0 B and C are equidistant from centre So, EB = EC

Q. 42 In Geiger-Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the correct plot?

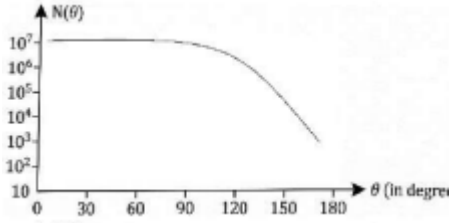
Option 1:



Option 2:



Option 3:



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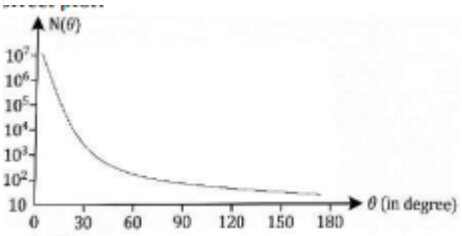
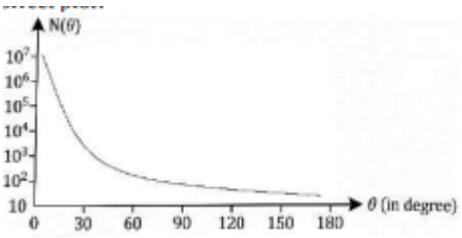
Option 4:



Correct Answer:



Solution:

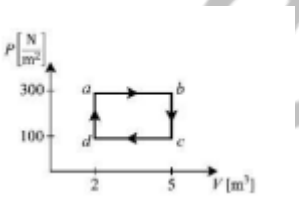
In Geiger-Marsden scattering experiment, Rutherford scattering formula gives

$$N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$$

Thus:
- $N(\theta)$ is very large for small θ
- it decreases sharply as θ increases
- it becomes very small for large scattering angles

Therefore the correct graph is the one which falls continuously from a large value at small angle to a very small value at large angle.
Correct Answer: Option (4)

Q. 43 One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is :



Option 1:

800 J

Option 2:

400 J

Option 3:

500 J

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Option 4:

600 J

Correct Answer:

600 J

Solution:

For a cyclic process,

$\Delta U = 0$

Hence total heat supplied to the gas is equal to net work done by the gas:

$Q = W$

From the $P - V$ diagram, the process forms a rectangle with:

- Pressure change: $300 - 100 = 200\ \text{N m}^{-2}$
- Volume change: $5 - 2 = 3\ \text{m}^3$

Area enclosed = net work done

$$W = (300 - 100)(5 - 2)$$
$$W = 200 \times 3 = 600\ \text{J}$$

Hence total heat supplied,

$Q = 600\ \text{J}$

Correct Answer: Option (4)

Q. 44 Two infinitely long parallel conducting wires A and B carry currents I and 2I, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is
[g is the acceleration due to gravity and μ_0 is the permeability of free space.]

Option 1:

$\frac{4\mu_0 I^2}{\pi\lambda g}$

Option 2:

$\frac{\mu_0 I^2}{2\pi\lambda g}$

Option 3:

$\frac{\mu_0 \text{I}^2}{\pi\lambda\ \text{g}}$

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Option 4:

$\frac{2\mu_0 I^2}{\pi\lambda\, g}$

Correct Answer:

$\frac{\mu_0 I^2}{\pi\lambda\, g}$

Solution:

Two long parallel wires carry currents:
- wire A: I
- wire B: $2I$

Distance between them $= h$
Force per unit length between parallel wires is

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi h}$$

So on wire A,

$$\frac{F}{L} = \frac{\mu_0 (I)(2I)}{2\pi h} = \frac{\mu_0 I^2}{\pi h}$$

For wire A to just not rise, magnetic attraction must balance its weight per unit length:

$$\frac{\mu_0 I^2}{\pi h} = \lambda g$$

$$h = \frac{\mu_0 I^2}{\pi\lambda g}$$

This is the maximum allowed value of h. Thus the required option is

$$\frac{\mu_0 I^2}{\pi\lambda g}$$

Correct Answer: Option (3)

Q. 45 Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is $k\text{Nm}^{-1}$. At a given instant, the extension of the spring is x meter and the speed of the particle is v ms^{-1}. On the $x - v$ plane, if the graph of v as a function of x is a circle, then the correct option is :

Option 1:

$k = \sqrt{m}$

Option 2:

$k = \frac{1}{m}$

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Option 3:

$k = m$

Option 4:

$k = m^2$

Correct Answer:

$k = m$

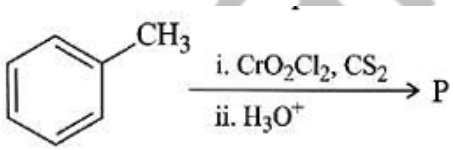
Solution:

$$K = m\omega^2$$
$$\omega^2 = \frac{K}{m}$$
$$v = \omega\sqrt{A^2 - x^2}$$
$$v^2 = \omega^2\left(A^2 - x^2\right)$$
$$v^2 = \left(\frac{K}{\omega}\right)\left(A^2 - x^2\right)$$
$$v^2 + x^2 = (\text{radius})^2$$
$$\text{Let } K = m$$
$$v^2 + x^2 = A^2$$
$$\text{circle}$$

Chemistry

Consider the following reaction, and choose the correct option.

Q. 1



Option 1:

Compound P is obtained by the hydrogenation of benzoyl chloride with Pd on $BaSO_4$.

Option 2:

On treating compound P with saturated $NaHCO_3$ solution, brisk effervescence is observed.

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Option 3:

Compound P can be prepared by treating benzene with anhydrous $AlCl_3$ and CH_3COCl.

Option 4:

On treatment with bromine water, compound P gives a white precipitate.

Correct Answer:

Compound P is obtained by the hydrogenation of benzoyl chloride with Pd on $BaSO_4$.

Solution:

Toluene reacts with CrO_2Cl_2 in CS_2 followed by hydrolysis (Etard reaction) to form Benzaldehyde (Compound P).

Rosenmund reduction i.e, hydrogenation of benzoyl chloride with Pd on $BaSO_4$) also produces benzaldehyde.

Hence, the correct answer is option (1)

Q. 2 The formula of tetraammineaquachloridocobalt(III) chloride is

Option 1:

$\left[Co\left(NH_3\right)_4\left(H_2O\right)Cl\right]Cl_2$

Option 2:

$\left[Co\left(NH_3\right)_4Cl_2\right]\times H_2O$

Option 3:

$\left[Co\left(NH_3\right)_4\right]Cl_3\times H_2O$

Option 4:

$\left[Co\left(NH_3\right)_4\left(H_2O\right)Cl\right]Cl$

Correct Answer:

$\left[Co\left(NH_3\right)_4\left(H_2O\right)Cl\right]Cl_2$

Solution:

Coordination Name: tetraammoniaaquachloridocobalt(III) chloride

Central metal: Co^{3+}

Ligands inside coordination sphere: $4\times NH_3, 1\times H_2O, 1\times Cl^-$

Sphere charge calculation: $+3+0+0-1=+2$.

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To balance a +2 charge, $2 \times Cl^-$ ions are needed outside the sphere: $\left[Co\left(NH_3\right)_4\left(H_2O\right)Cl\right]Cl_2$.

Hence the correct answer is option (1)

Q. 3 The lanthanide ion having four unpaired electrons is
(Given : Atomic numbers of $Ce = 58, Nd = 60, Tb = 65$ and $(Hg = 67)$

Option 1:

Ho^{3+}

Option 2:

Nd^{3+}

Option 3:

Ce^{3+}

Option 4:

Tb^{3+}

Correct Answer:

Ho^{3+}

Solution:

Electronic Configurations (Ln^{3+}):

Nd^{3+} $(Z = 60)$: $[Xe]4f^3 \rightarrow$ 3 unpaired electrons

Tb^{3+} $(Z = 65)$: $[Xe]4f^8 \rightarrow$ 6 unpaired electrons

Ho^{3+} $(Z = 67)$: $[Xe]4f^{10} \rightarrow$ 4 unpaired electrons

Ce^{3+} $(Z = 58)$: $[Xe]4f^1 \rightarrow$ 1 unpaired electron

Ho^{3+} has 4 unpaired electrons

Hence, the correct answer is option (1)

Q. 4 For an elementary chemical reaction, the Arrhenius plot is given below

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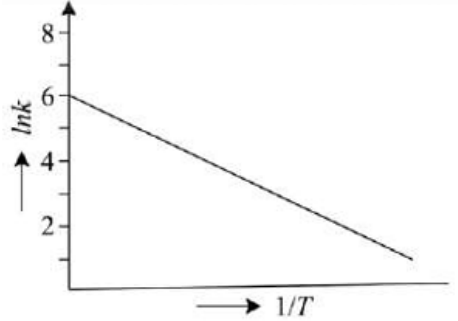

If the energy of activation is 6.64 kJ mol^{-1} and $\text{R} = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1}$, the temperature at which the rate constant becomes $e^2\ \text{min}^{-1}$, is

Option 1:

250 K

Option 2:

125 K

Option 3:

150 K

Option 4:

200 K

Correct Answer:

200 K

Solution:

Arrhenius Equation: $\ln k = \ln A - \frac{E_a}{RT}$

The y-intercept ($\ln A$) is 6 . Therefore, $A = e^6$.

Given: $k = e^2, E_a = 6.64\ \text{kJ mol}^{-1} = 6640\ \text{J mol}^{-1}, R = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1}$.

$$\ln\left(\frac{A}{k}\right) = \frac{E_a}{RT}$$

$$\ln\left(\frac{e^6}{e^2}\right) = \frac{6640}{8.3 \times T} \Longrightarrow 4 = \frac{800}{T} \Longrightarrow T = 200\ \text{K}$$

Hence the correct answer is option (4)

Q. 5 The green paramagnetic species formed by heating KMnO_4 at 513 K is

Option 1:

KO_2

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Option 2:

K_2MnO_4

Option 3:

Mn_3O_4

Option 4:

MnO

Correct Answer:

K_2MnO_4

Solution:

Heating potassium permanganate $\left(KMnO_4\right)$ at 513 K decomposes it into potassium manganate, manganese dioxide, and oxygen gas:

$$2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 \text{ (green, paramagnetic) } + MnO_2 + O_2$$

K_2MnO_4 formed from this reaction is green and paramagnetic.

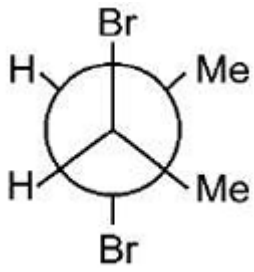
Hence, the correct answer is option (2)

Q. 6

Given below are two statements:

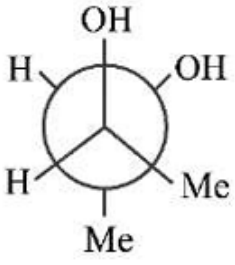
Statement I:

trans-But-2-ene uponf treatment with Br_2 in CCl_4 gives the following product



Statement II:

cis-But-2-ene upon treatment with alkaline $KMnO_4$ gives the following product



In the light of the above statements, choose the ***most appropriate*** answer from the questions given below.

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Option 1:

Statement I is incorrect but **Statement II** is correct

Option 2:

Both Statement I and Statement II are correct

Option 3:

Both Statement I and Statement II are incorrect

Option 4:

Statement I is correct but Statement II is incorrect

Correct Answer:

Statement I is incorrect but **Statement II** is correct

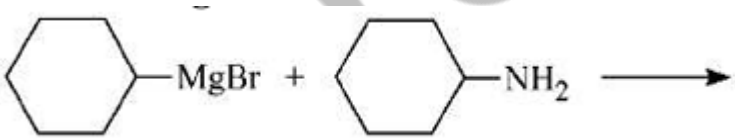
Solution:

Statement I: trans-But-2-ene undergoes anti-addition with Br_2 / CCl_4 to form a meso compound. The Newman projection shown represents a meso form. It has a center of inversion upon rotation.

Statement II: cis-But-2-ene undergoes syn-addition with cold alkaline $KMnO_4$ (syn-dihydroxylation) to yield a meso compound. The Newman projection shown shows a molecule lacking symmetry and represents a chiral threo pair/enantiomer instead of the meso form.

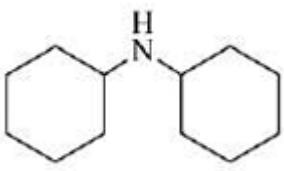
Hence, the correct answer is option (1)

Q. 7 One of the products formed in the following reaction is



Option 1:

Option 2:



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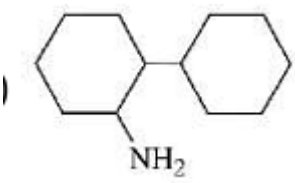
Option 3:



Option 4:

Correct Answer:

Solution:

Cyclohexylmagnesium bromide ($R-MgBr$, a strong base/nucleophile) reacts with cyclohexylamine.

The Grignard reagent acts as a strong base and abstracts the highly acidic proton from the amine group (NH_2), producing cyclohexane:

$$C_6H_{11}MgBr + C_6H_{11}NH_2 \rightarrow C_6H_{12} + C_6H_{11}NHMgBr$$

Hence, the correct answer is option (1)

Q. 8

Given below are two statements:
Statement-I : Heating NaCl with concentrated H_2SO_4 and MnO_2 results in oxidation of Mn .
Statement-II : Heating NaI with concentrated H_2SO_4 and MnO_2 results in reduction of Mn .
In light of the above statements, choose the ***most appropriate*** answer from the options given below:

Option 1:

Statement-I is incorrect but Statement-II is correct.

Option 2:

Both Statement-I and Statement II are correct.

Option 3:

Both Statement-I and Statement II are incorrect.

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Option 4:

Statement-I is correct but Statement-II is incorrect.

Correct Answer:

Statement-I is incorrect but Statement-II is correct.

Solution:

Statement I: Heating NaCl with conc. H_2SO_4 and MnO_2 generates Cl_2 gas. Here, Mn gets reduced from $+4$ (in MnO_2) to $+2$ (in $MnSO_4$). Oxidation of Mn does not happen.

Thus, Statement I is incorrect.

Statement II: Similarly, heating NaI with conc. H_2SO_4 and MnO_2 reduces Mn from +4 to +2 .

Thus, Statement II is correct.

Statement-I is incorrect, but Statement-II is correct.

Hence, the correct answer is option (1)

Q. 9 Among the following options, the correct trend in the electron gain enthalpy is

Option 1:

$I > Br > Cl > F$

Option 2:

$F > Cl > Br > I$

Option 3:

$Br > Cl > F > I$

Option 4:

$Cl > F > Br > I$

Correct Answer:

$Cl > F > Br > I$

Solution:

Electron gain enthalpy values generally become less negative down a group. However, due to the exceptionally small size of the Fluorine atom, strong inter-electronic repulsions in its compact $2p$ subshell make its electron gain enthalpy less negative than Chlorine.

Order of negative electron gain enthalpy: $Cl > F > Br > I$

Hence the correct answer is option (4)

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Q. 10 Given below are two statements:

Statement-I : $\left[Fe(ox)_3\right]^{3-}$ is chiral.

Statement-II : trans $= \left[Cr\left(H_2O\right)_2(ox)_2\right]^{-}$ is chiral.

(Given : $oxH_2 = HOOC - COOH$)

In light of the above statements, choose the ***most appropriate*** answer from the options given below:

Option 1:

Statement-I is incorrect but Statement-II is correct.

Option 2:

Both Statement-I and Statement II are correct.

Option 3:

Both Statement-I and Statement II are incorrect.

Option 4:

Statement-I is correct but Statement-II is incorrect.

Correct Answer:

Statement-I is correct but Statement-II is incorrect.

Solution:

Statement I: $\left[Fe(ox)_3\right]^{3-}$ is a homoleptic tris-chelated complex. It exists as non-superimposable d- and lenantiomers, meaning it lacks a plane or center of symmetry. Thus, it is chiral.

Statement I is correct

Statement II: trans- $\left[Cr\left(H_2O\right)_2(ox)_2\right]^{-}$ possesses a center of inversion and planes of symmetry due to the transposition of the unidentate water ligands across the central metal. Thus, it is achiral

Statement II is incorrect

Statement-I is correct but Statement-II is incorrect.

Hence, the correct answer is option (4)

Q. 11 The correct statement about peptides and proteins is

Option 1:

In α-helices, the polypeptide chain is twisted into a left-handed screw (helix) through intramolecular hydrogen bonds.

Option 2:

Tertiary structure of proteins has two or more polypeptide subunits.

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Option 3:

Only the proteins having a quaternary structure are biologically active.

Option 4:

In β-pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

Correct Answer:

In β-pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

Solution:

α-helices are predominantly right-handed screws. (Incorrect)

Tertiary structure relates to the overall folding of a single polypeptide chain; quaternary structure involves two or more subunits. (Incorrect)

Many proteins with primary, secondary, or tertiary structures are fully biologically active. (Incorrect)
In β-pleated sheets, separate adjacent peptide chains are held together side-by-side via intermolecular hydrogen bonds. (Correct)

Hence, the correct answer is option (4)

Q. 12

Given below are two statements:
Statement-I : Oxidation of p nitrotoluene with acidic $KMnO_4$ gives an acid that is stronger than benzoic acid.

Statement-II : Reduction of p nitrotoluene with Sn / HCl followed by neutralization gives an amine that is more basic than aniline.

In light of the above statements, choose the **most appropriate** answer from the Options

Option 1:

Statement-I is incorrect but Statement-II is correct.

Option 2:

Both Statement-I and Statement-II are correct.

Option 3:

Both Statement-I and Statement-II are incorrect.

Option 4:

Statement-I is correct but Statement-II is incorrect.

Correct Answer:

Both Statement-I and Statement-II are correct.

Solution:

Oxidation of p-nitrotoluene with acidic $KMnO_4$ yields p-nitrobenzoic acid. Due to the electron-withdrawing I and M effects of the NO_2 group, it is a stronger acid than benzoic acid.

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Thus, Statement I is correct.

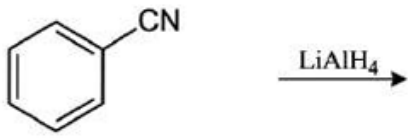
Reduction of p-nitrotoluene with Sn / HCl followed by neutralization yields p-toluidine (an aniline ring with an electron-donating CH_3 group). The $+I$ and hyperconjugation effects of CH_3 increase electron density on nitrogen, making it more basic than aniline.

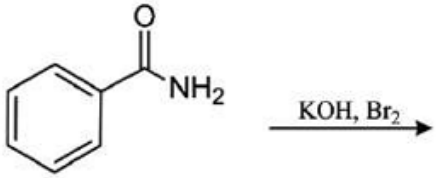
Thus, Statement II is correct.

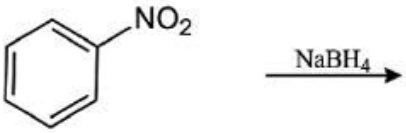
Hence, the correct answer is option (2)

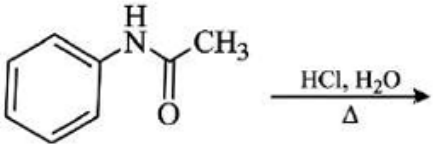
Q. 13

Identify the reactions which give aniline as the major product.

A. 

B. 

C. 

D. 

Choose the correct answer from the options given below.

Option 1:

C and D only

Option 2:

A and B only

Option 3:

B and D only

Option 4:

A and C only

Correct Answer:

B and D only

Solution:

Reaction A: Reduction of Benzonitrile with $LiAlH_4$ gives benzylamine $\left(C_6H_5CH_2NH_2\right)$.

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Reaction B: Hoffmann bromamide degradation of Benzamide with KOH / Br_2 yields Aniline $\left(C_6H_5NH_2\right)$.

Reaction C: $NaBH_4$ is a mild reducing agent and generally cannot reduce the aromatic nitro group of Nitrobenzene.

Reaction D: Acid-catalyzed hydrolysis $\left(HCl, H_2O, \Delta\right)$ of Acetanilide cleaves the amide bond to yield Aniline and acetic acid.

B and D give aniline as the major product.

Hence, the correct answer is option (3)

Q. 14 Two moles of an ideal gas undergo free expansion from 10 L to 100 L at 300 K . The values of ΔS_{system} and $\Delta S_{surroundings}$ are
(R is universal gas constant)

Option 1:

$\Delta S_{system} = 4.606R; \Delta S_{surroundings} = 0$

Option 2:

$\Delta S_{system} = 0; \Delta S_{surroundings} = 0$

Option 3:

$\Delta S_{system} = 4.606R; \Delta S_{surroundings} =$ -4.606 R

Option 4:

$\Delta S_{system} = 0;$ $\Delta S_{surroundings} =$ 4.606 R

Correct Answer:

$\Delta S_{system} = 4.606R; \Delta S_{surroundings} = 0$

Solution:

Entropy of System (ΔS_{system}): For the isothermal expansion of an ideal gas:

$$\Delta S_{system} = nR\ln\left(\frac{V_2}{V_1}\right)$$

$$= 2 \times R \times \ln\left(\frac{100}{10}\right)$$

$$= 2 \times R \times 2.303\log_{10}(10)$$

$$= 4.606R$$

Entropy of Surroundings ($\Delta S_{surroundings}$): During "free expansion" into a vacuum, the external pressure $P_{ext} = 0$, meaning work done $W = 0$. Since it is an ideal gas, $\Delta U = 0$, so heat exchanged $q = 0$. Because no heat is exchanged with the surroundings ($q_{surr} = 0$):

$\Delta S_{surroundings} = 0$

Hence, the correct answer is option (1)

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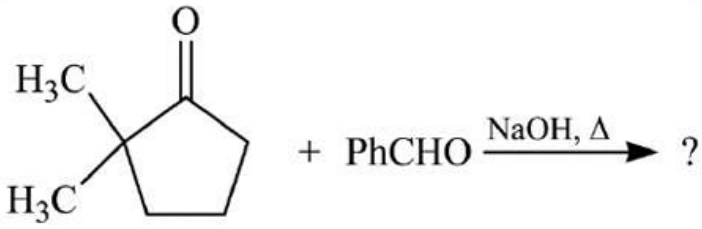
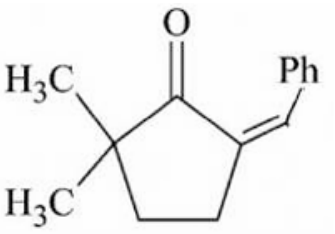
The compound that CANNOT be obtained from the aldol condeasation reaction shown below, is

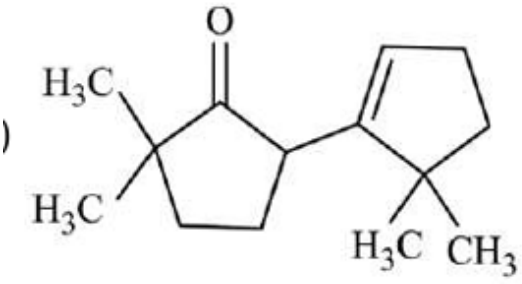
Q. 15



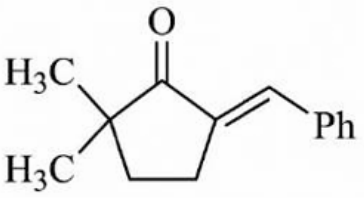
Option 1:



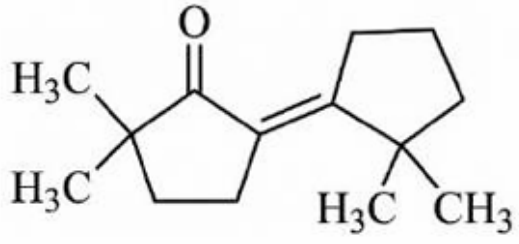
Option 2:



Option 3:



Option 4:



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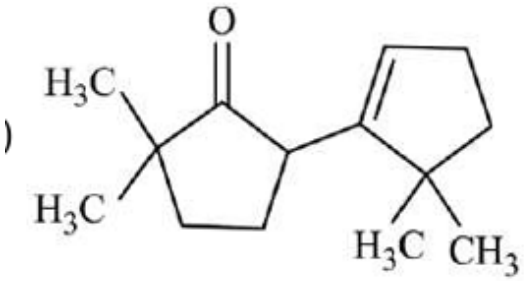
Correct Answer:



Solution:

The reaction shows 2,2-dimethylcyclopentanone reacting with benzaldehyde $(PhCHO)$ in the presence of NaOH and heat.

The ketone has only one position with α-hydrogens (the $-CH_2-$ group at position 5) because position 2 is fully substituted with two methyl groups. Therefore, aldol condensation can only occur at that single open α-carbon.

Options (1) and (3) show the correct cross-aldol condensation products with benzaldehyde.
Option (4) shows a self-condensation product of two ketone molecules using the correct α-position.
Option (2) shows a structure where condensation has taken place at the fully substituted carbon carrying the two methyl groups, which is impossible due to the complete lack of α-hydrogens there.

Hence the correct answer is option (2)

Q. 16 The complex which has facial and meridional isomers is
(Given : py $=$ pyridine and en $= H_2N-CH_2-CH_2-NH_2$)

Option 1:

$\left[Ni(en)_2(H_2O)_2\right]^{2+}$

Option 2:

$\left[Cr(py)_3(Cl)_3\right]$

Option 3:

$\left[Cr(H_2O)_6\right]^{3+}$

Option 4:

$\left[Co(NH_3)_4(H_2O)_2\right]^{3+}$

Correct Answer:

$\left[Cr(py)_3(Cl)_3\right]$

Solution:

Geometric facial (fac) and meridional (mer) isomerism is specifically observed in octahedral complexes of the general formula $\left[MA_3B_3\right]$.

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$\left[Cr(py)_3(Cl)_3\right]$ perfectly fits the $\left[MA_3\ B_3\right]$ template, where py and Cl^- act as the unidentate ligands.

Hence, the correct answer is option (2)

Q. 17 The numbers 17.0145 and 21.0235 were rounded to three figures after the decimal point. The resulting numbers, respectively, are

Option 1:

17.015 and 21.024

Option 2:

17.014 and 21.023

Option 3:

17.015 and 21.023

Option 4:

17.014 and 21.024

Correct Answer:

17.014 and 21.024

Solution:

17.0145 : The digit at the fourth decimal place is 5, preceded by an even number(4).

According to the round - to - nearest - even rule, it rounds downt o17.014.

21.0235: The digit at the fourth decimal place is 5, preceded by an odd number (3). According to the rounding rule, it rounds up to the nearest even digit, giving 21.024.

Hence, the correct answer is option (4)

Q. 18 The amount of carbon dioxide evolved upon complete combustion of 116 g of n-butane is
(Given: atomic mass ini annu $n = 1$, $C = 12$ and $O = 16$)

Option 1:

362 g

Option 2:

352 g

Option 3:

322 g

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Option 4:

176 g

Correct Answer:

352 g

Solution:

$2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$

Molar mass of n-butane $\left(C_4H_{10}\right) = (4 \times 12) + (10 \times 1) = 58\ g / mol$

Moles of n-butane given $= \frac{116\ g}{58\ g / mol} = 2 moles$

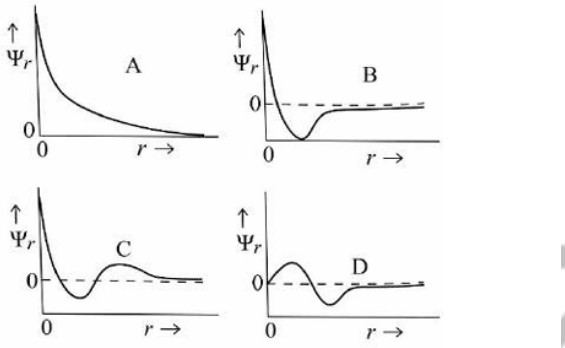
From the balanced equation, 2 moles of C_4H_{10} produce 8 moles of CO_2.

Mass of $CO_2 = 8$ moles $\times 44\ g / mol = 352\ g$

Hence, the correct answer is option (2)

Q. 19

Consider the following schematic plots of orbital wavefunction $\left(\Psi_r\right)$ against distance (r) from the nucleus.



The figure representing two radial nodes in the orbital is

Option 1:

D

Option 2:

A

Option 3:

B

Option 4:

C

Correct Answer:

C

Solution:

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Radial nodes occur where the radial wavefunction Ψ_r passes through zero (crosses the r-axis, excluding $r = 0$).

A: 0 crossings (0 radial nodes)

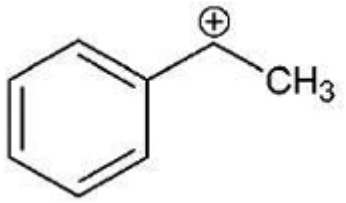
B: 1 crossing (1 radial node)

C: 2 crossings (2 radial nodes)

D: 1 crossing (1 radial node)

Hence, the correct answer is option (4)

Q. 20 The following carbocation is stabilised by the interaction of the empty p orbital with



Option 1:

empty σ^* and empty π^* orbitals

Option 2:

filled σ and filled π orbitals

Option 3:

empty σ and empty π^* orbitals

Option 4:

empty σ^* and filled π orbitals

Correct Answer:

filled σ and filled π orbitals

Solution:

The given carbocation is a benzylic carbocation. The positively charged carbon has an empty p orbital.

This empty orbital is stabilized via resonance delocalization by accepting electron density from the adjacent aromatic ring's pi system. Thus, it interacts directly with the filled π orbitals of the benzene ring.

The empty p-orbital of the carbocation is stabilized by interacting with filled σ and filled π orbitals.

Hence, the correct answer is option (2)

Q. 21 A $1:3$ electrolyte in an aqueous solution is

Option 1:

$\left[Co\left(NH_3\right)_3\left(NO_2\right)_3\right]$

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Option 2:

$\left[CoCl_2\left(NH_3\right)_4\right]Cl$

Option 3:

$\left[CoCl\left(NH_3\right)_5\right]Cl_2$

Option 4:

$\left[Co\left(NH_3\right)_6\right]Cl_3$

Correct Answer:

$\left[Co\left(NH_3\right)_6\right]Cl_3$

Solution:

A $1:3$ electrolyte in an aqueous solution is:

A 1:3 electrolyte dissociates to give one cation and three anions (or vice versa).

$\left[Co\left(NH_3\right)_6\right]Cl_3 \rightarrow \left[Co\left(NH_3\right)_6\right]^{3+} + 3Cl^-$

This yields a total ratio of $1:3$ ions.
Hence, the correct answer is option (4)

Q. 22 The standard electrode potential (E°) for the half-cell reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$ at 298 K is

(Given: $E^\circ\left(Fe^{3+}/Fe\right) = -0.04$ V and $E^\circ\left(Fe^{2+}/Fe\right) = -0.44$ V at 298 K)

Option 1:

+0.92 V

Option 2:

+0.40 V

Option 3:

+0.76 V

Option 4:

-0.48 V 0.40

Correct Answer:

+0.76 V

Solution:

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The standard electrode potential (E°) for the half-cell reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$ at 298 K is:

$\Delta G^\circ = -nFE^\circ$

For the target reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$:

$$\Delta G_3^\circ = \Delta G_1^\circ - \Delta G_2^\circ = 0.12F - 0.88F = -0.76F$$
$$-1 \cdot F \cdot E^\circ = -0.76F \Longrightarrow E^\circ = +0.76\ \text{V}$$

1. $Fe^{3+} + 3e^- \rightarrow Fe \quad \left(\Delta G_1^\circ = -3 \cdot F \cdot (-0.04) = +0.12F\right)$
2. $Fe^{2+} + 2e^- \rightarrow Fe \quad \left(\Delta G_2^\circ = -2 \cdot F \cdot (-0.44) = +0.88F\right)$

Hence, the correct answer is option (3)

Q. 23 In potash alum, the ratio of K^+ and SO_4^{2-} ions is

Option 1:

3: 2

Option 2:

1: 2

Option 3:

2: 1

Option 4:

2: 3

Correct Answer:

1: 2

Solution:

The formula for potash alum is $K_2SO_4 \cdot Al_2\left(SO_4\right)_3 \cdot 24H_2O$.

Total K^+ ions $= 2$. Total SO_4^{2-} ions $= 1 + 3 = 4$.

Ratio $= \frac{K^+}{SO_4^{2-}} = \frac{2}{4} = 1:2$

Hence, the correct answer is option (2)

Q. 24 Consider the following statements about the solutions formed by mixing two liquids.
A. An ideal solution thus formed obeys Raoult's law throughout the composition range.
B. Mixture of chloroform and acetone shows negative deviation from Raoult's law.
C. Mixture of aniline and phenol shows positive deviation from Raoult's law.

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Option 1:

A and C only

Option 2:

A and B only

Option 3:

B and C only

Option 4:

A only

Correct Answer:

A and B only

Solution:

Ideal solutions obey Raoult's law over the entire composition range.

Therefore, statements A and B are correct.
Chloroform and acetone form hydrogen bonds, resulting in a negative deviation.
Aniline and phenol form intermolecular hydrogen bonds, showing a negative deviation (not positive).
Hence, the correct answer is option (2)

Q. 25

For a salt **XY**, which is a strong electrolyte, the plot of Λ_m versus $\sqrt{c}$ has a slope of $-90.0\ \text{S cm}^2\ \text{mol}^{-3/2}\ \text{L}^{1/2}$ at 298 K . At 0.01 M concentration of XY, the value of Λ_m is $145.0\ \text{S cm}^2\ \text{mol}^{-1}$. The limiting molar conductivity of Y^- ion $\left(\lambda^0_{Y^-}\right.$, in $\left.\text{Scm}^2\ \text{mol}^{-1}\right)$ at 298 K will be (Given: $\lambda^0_{X^+} = 74.0\ \text{S cm}^2\ \text{mol}^{-1}$)

Option 1:

76.0

Option 2:

80.0

Option 3:

100.0

Option 4:

90.0

Correct Answer:

80.0

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Solution:

70. According to Kohlrausch's law and Debye-Hückel-Onsager equation:

$$\Lambda_m = \Lambda_m^\circ - b\sqrt{c}$$

$$145.0 = \Lambda_m^\circ - (-90.0) \cdot \sqrt{0.01}$$

$$145.0 = \Lambda_m^\circ - 90.0 \cdot 0.1 \Longrightarrow 145.0 = \Lambda_m^\circ - 9.0 \Longrightarrow \Lambda_m^\circ = 154.0 \text{ S cm}^2 \text{ mol}^{-1}$$

By Kohlrausch's law: $\Lambda_m^\circ(\text{XY}) = \lambda_{\text{X}^+}^\circ + \lambda_{\text{Y}^-}^\circ$

$$154.0 = 74.0 + \lambda_{\text{Y}^-}^\circ \Longrightarrow \lambda_{\text{Y}^-}^\circ = 154.0 - 74.0 = 80.0 \text{ S cm}^2 \text{ mol}^{-1}$$

Hence, the correct answer is option (2)

Q. 26

Arrange the following compounds in the increasing order of polarity

A. $CH_3CH_2OCH_2CH_3$
B. CH_3CH_2OH
C. CH_3COCH_3
D. CH_3COOH

Choose the correct answer from the options given below.

Option 1:

$\text{A} < \text{C} < \text{B} < \text{D}$

Option 2:

$\text{A} < \text{B} < \text{C} < \text{D}$

Option 3:

$\text{C} < \text{A} < \text{D} < \text{B}$

Option 4:

$\text{C} < \text{A} < \text{B} < \text{D}$

Correct Answer:

$\text{A} < \text{C} < \text{B} < \text{D}$

Solution:

Polarity depends on the dipole moment and functional groups.

Order: $\text{A} < \text{C} < \text{B} < \text{D}$
Diethyl ether $\left(CH_3CH_2OCH_2CH_3,\ \text{A}\right)$ has the lowest dipole moment.
Acetone $\left(CH_3COCH_3, \text{C}\right)$ has a highly polar $\text{C} = \text{O}$ bond.

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Ethanol (CH_3CH_2OH, B) has strong hydrogen bonding capabilities.
Acetic acid (CH_3COOH, D) is the most polar due to resonance stability and extreme hydrogen bonding.

Hence, the correct answer is option (1)

Q. 27 According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is

Option 1:

$Cl^- > NH_3 > H_2O > CO$

Option 2:

$CO > NH_3 > H_2O > Cl^-$

Option 3:

$CO > H_2O > NH_3 > Cl^-$

Option 4:

$Cl^- > H_2O > NH_3 > CO$

Correct Answer:

$CO > NH_3 > H_2O > Cl^-$

Solution:

According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is:

$CO > NH_3 > H_2O > Cl^-$

Referring to the Spectrochemical Series, strong field ligands come first:

$CO > NH_3 > H_2O > Cl^-$

Hence, the correct answer is option (2)

Q. 28 The amino acid that gives a red-blood colour on treating its sodium fusion extract with sodium nitroprusside is

Option 1:

serine

Option 2:

leucine

Option 3:

threonine

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Option 4:

methionine

Correct Answer:

methionine

Solution:

A blood-red coloration $\left([Fe(SCN)]^{2+} \right)$ in Lassaigne's test indicates the simultaneous presence of both Nitrogen (N) and Sulfur (S). Among the given options, only methionine contains both N and S in its chemical structure.

Hence the correct answer is option (4)

Q. 29 In an acidic medium, 10 mL of 0.25 M oxalic acid is titrated with $KMnO_4$ solution. If the volume of $KMnO_4$ solution required to reach end point is 10 mL , the strength of the $KMnO_4$ solution is

Option 1:

0.15 M

Option 2:

0.10 M

Option 3:

0.20 M

Option 4:

0.25 M

Correct Answer:

0.10 M

Solution:

$N_1V_1(\text{ oxalic acid}) = N_2V_2\left(KMnO_4\right)$

Law of Equivalence

$M_1 \times n_1 \times V_1 = M_2 \times n_2 \times V_2$

Where

For $KMnO_4$ (acidic medium): $n_1 = 5,\ V_1 = 10$ mL
For Oxalic Acid: $n_2 = 2,\ M_2 = 0.25$ M, $V_2 = 10$ mL

$M_1 \times 5 \times 10 = 0.25 \times 2 \times 10$

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$5 \times M_1 = 0.50$

$M_1 = \frac{0.50}{5} = 0.10\ \text{M}$

Hence, the correct answer is option (2)

Q. 30 The correct statement is

Option 1:

Aluminium has five valence orbitals.

Option 2:

Boron has a maximum covalency of four.

Option 3:

Beryllium has three valence orbitals.

Option 4:

Magnesium has a maximum covalency of four.

Correct Answer:

Boron has a maximum covalency of four.

Solution:

Boron belongs to the second period and lacks d-orbitals. It only has one $2s$ and three $2p$ orbitals available for bonding, limiting its maximum covalency strictly to 4 (e.g., in BF_4^-). Statement 2 is correct.

Hence, the correct answer is option (2)

Q. 31 Among the following, the compound having conjugated double bonds is

Option 1:

hepta-1,6-diene

Option 2:

hepta-1,3-diene

Option 3:

hepta-1,4-diene

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Option 4:

hepta-1,5-diene

Correct Answer:

hepta-1,3-diene

Solution:

Double bonds separated by exactly one single bond $(-\mathrm{CH}=\mathrm{CH}-\mathrm{CH}=\mathrm{CH}-)$.

In hepta-1,3-diene, the double bonds are at positions 1 and 3, meaning they are alternate and perfectly conjugated.

Hence, the correct answer is option (2).

Q. 32 $2\ \mathrm{A} \xrightarrow{k} \mathrm{B}$ is a zero-order reaction, whe: $k = 1.0\ \mathrm{mol\ L^{-1}\ min^{-1}}$. If the initi concentration of A is 2 M , then th time taken to complete 75% of th reaction will be

Option 1:

2.0 min

Option 2:

1.5 min

Option 3:

0.75 min

Option 4:

1.0 min

Correct Answer:

0.75 min

Solution:

For a zero-order reaction with a stoichiometric coefficient of $2(2\ \mathrm{A} \rightarrow\ \mathrm{B})$, the integrated rate law is:

$$\Delta[\mathrm{A}] = 2kt$$

For 75% completion, the change in concentration $\Delta[A] = 0.75 \times 2\mathrm{M} = 1.5\mathrm{M}$.
Substituting the values:

$$1.5 = 2 \times 1.0 \times t \Longrightarrow t = 0.75\ \mathrm{min}$$

Hence, the correct answer is option (3).

Q. 33 The correct order of solubility of the given salts in water at 298 K is

Salt	$\mathrm{K_{sn}}$ **at 298 K**

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AgBr	5.0×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

Option 1:

$Zn(OH)_2 > AgBr > Hg_2Cl_2$

Option 2:

$Hg_2Cl_2 > Zn(OH)_2 > AgBr$

Option 3:

$AgBr > Zn(OH)_2 > Hg_2Cl_2$

Option 4:

$Hg_2Cl_2 > AgBr > Zn(OH)_2$

Correct Answer:

$Zn(OH)_2 > AgBr > Hg_2Cl_2$

Solution:

$AgBr(AB \text{ type}): K_{sp} = S^2 \Longrightarrow S = \sqrt{5.0 \times 10^{-13}} \approx 7 \times 10^{-7} M$

$Zn(OH)_2\ (AB_2 \text{ type}): K_{sp} = 4S^3 \Longrightarrow S = \sqrt[3]{\frac{1.0 \times 10^{-15}}{4}} \approx 6.3 \times 10^{-6} M$

$Hg_2Cl_2\ (A_2B_2 \text{ type, dissociates as } Hg_2^{2+} + 2Cl^-): K_{sp} = 4S^3 \Longrightarrow S = \sqrt[3]{\frac{1.3 \times 10^{-18}}{4}} \approx 6.8 \times 10^{-7} M$

Comparing solubilities (S): $Zn(OH)_2 > AgBr > Hg_2Cl_2$

Hence, the correct answer is option (1).

Q. 34 The correct decreasing order of oxidation state of the underlined atom in each molecule is

Option 1:

$\underline{P}_4O_6 > \underline{Cl}_2O_7 > \underline{Al}H_3$

Option 2:

$\underline{P}_4O_{10} > \underline{S}O_3 > H_2\underline{O}$

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Option 3:

$\underline{N_2O_5} > \underline{Al_2O_3} > \underline{H_2S}$

Option 4:

$\underline{PbO_2} > \underline{N_2O_3} > \underline{SO_3}$

Correct Answer:

$\underline{N_2O_5} > \underline{Al_2O_3} > \underline{H_2S}$

Solution:

$\underline{N}_2O_5: 2x + 5(-2) = 0 \Longrightarrow x = +5$

$Al_2O_3: 2x + 3(-2) = 0 \Longrightarrow x = +3$

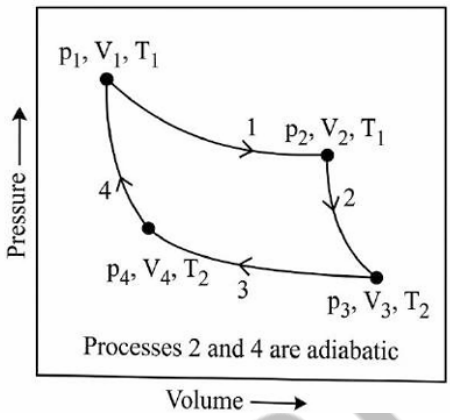
$H_2\underline{S}: 2(+1) + x = 0 \Longrightarrow x = -2$

Decreasing order: $+5 > +3 > -2$

Hence, the correct answer is option (3).

Q. 35

Consider the reversible processes for 1.0 mol of an ideal gas as shown in the figure.



w_1, w_2, w_3 and w_4 represent work done (in calories) in the processes $1, 2, 3$ and 4 , respectively; ΔU_2 and ΔU_4 are changes in the internal energy for the processes 2 and 4 ,respectively.

[use $R = 2cal^{-1}\ mol^{-1}$]

The correct option is

Option 1:

$w_1 + w_2 + w_3 + w_4 = 0$

Option 2:

$w_1 + w_3 = -2\ T_1 ln\frac{V_2}{V_1} - 2\ T_2 ln\frac{V_4}{V_3}$

Option 3:

$w_2 + w_4 = \Delta U_2 - \Delta U_4$

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Option 4:

$w_1 + w_2 = 2\ T_1 \ln\frac{V_2}{V_1}$

Correct Answer:

$w_1 + w_3 = -2\ T_1 \ln\frac{V_2}{V_1} - 2\ T_2 \ln\frac{V_4}{V_3}$

Solution:

Processes 2 and 4 are adiabatic ($\Delta Q = 0$).

Processes 1 and 3 are isothermal at temperatures T_1 and T_2 respectively.

Work done in isothermal process 1 $w_1 = -RT_1 \ln\left(\frac{V_2}{V_1}\right)$

Work done in isothermal process 3 $w_3 = -RT_2 \ln\left(\frac{V_4}{V_3}\right) = 2T_2 \ln\left(\frac{V_3}{V_4}\right)$ (Since $R = 2$). w_1 and w_3 gives:

$w_1 + w_3 = -2T_1 \ln\frac{V_2}{V_1} - 2T_2 \ln\frac{V_4}{V_3}$

Hence, the correct answer is option (2).

Q. 36

Assertion A :
For an ideal solution formed by mixing liquids P and Q, $\Delta_{mix} H = 0$ and $\Delta_{mix} V = 0$

Reason R :
No interactions occur between P and Q In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:

A is not correct but R is correct

Option 2:

Both A and R are correct and R is the correct explanation of A

Option 3:

Both A and R are correct but R is NOT the correct explanation of A

Option 4:

A is correct but R is not correct

Correct Answer:

A is correct but R is not correct

Solution:

Assertion: For an ideal solution formed by mixing two liquids P and Q, there is no change in enthalpy ($\Delta_{mix}H = 0$) and no change in volume ($\Delta_{mix}V = 0$) upon mixing.

Reason R is false: It is incorrect to say that no interactions occur between P and Q. In an ideal solution, intermolecular interactions do exist, but the strength of the new solute-solvent interactions (P − Q) is exactly equal to the strength of the original pure component interactions (P − P and Q − Q).

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Interaction Energy: $(P-Q)=(P-P)=(Q-Q)$

A is correct but R is not correct

Hence, the correct answer is option (4).

Q. 37 Among the species spin-only magnetic moment is highest for
Given : Atomic number of Ti = 22, $Mn=25, Fe=26$ and $Co=27$)

Option 1:

$\left[Ti\left(H_2O\right)_6\right]^{3+}$

Option 2:

$\left[Mn(CN)_6\right]^{3-}$

Option 3:

$\left[Fe(CN)_6\right]^{3-}$

Option 4:

$\left[Co\left(NH_3\right)_6\right]^{3+}$

Correct Answer:

$\left[Mn(CN)_6\right]^{3-}$

Solution:

The spin-only magnetic moment (μ) depends directly on the number of unpaired electrons (n).
$\left[Ti\left(H_2O\right)_6\right]^{3+}: Ti^{3+}\left(3d^1\right) \rightarrow 1$ unpaired electron.
$\left[Mn(CN)_6\right]^{3-}: Mn^{3+}\left(3d^4\right)$, strong-field ligand $\left(CN^-\right)$ pairs electrons $\rightarrow t_{2g}^4 e_g^0 \rightarrow 2$ unpaired electrons.

$\left[Fe(CN)_6\right]^{3-}: Fe^{3+}\left(3d^5\right)$, strong-field ligand pairs electrons $\rightarrow t_{2g}^5 e_g^0 \rightarrow 1$ unpaired electron.
$\left[Co\left(NH_3\right)_6\right]^{3+}: Co^{3+}\left(3d^6\right)$, strong-field ligand pairs electrons $\rightarrow t_{2g}^6 e_g^0 \rightarrow 0$ unpaired electrons.

Hence, the correct answer is option (2).

Q. 38 A protein undergoes reversible thermal denaturation from its initial state N to denatured state D according to $N \rightleftharpoons D$. At $60^{\circ}C$, the concentrations of both N and D are equal at equilibrium, and the standard enthalpy change of denaturation is $666\ kJ\ mol^{-1}$. The standard entropy change (ΔS° in $kJKK^{-1}\ mol^{-1}$) of the protein upon denaturation at $60^{\circ}C$ is closest to

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Option 1:

11.1

Option 2:

2.0

Option 3:

2000.0

Option 4:

333.0

Correct Answer:

2.0

Solution:

At equilibrium, $[\mathrm{N}] = [\mathrm{D}]$, so the equilibrium constant $K_{eq} = 1$.

Since $K_{eq} = 1$, the standard Gibbs free energy change is $\Delta G^\circ = -RT\ln\left(K_{eq}\right) = 0$.

Using the relation $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$:

$$0 = \Delta H^\circ - T\Delta S^\circ \Longrightarrow \Delta S^\circ = \frac{\Delta H^\circ}{T}$$

Given $T = 60^\circ\mathrm{C} = 333\ \mathrm{K}$ and $\Delta H^\circ = 666\ \mathrm{kJ\ mol^{-1}}$:

$$\Delta S^\circ = \frac{666}{333} = 2.0\ \mathrm{kJ\ K^{-1}\ mol^{-1}}$$

Q. 39

Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R.**
Assertion A : Generally, $3d$ transition metals have high melting points.
Reason R : Involvement of $3d$-electrons in addition to 4 s-electrons in the interatomic metallic bonding.
In light of the above statements, choose the most appropriate answer from the Options given below:

Option 1:

A is not correct but R is correct.

Option 2:

Both A and R are correct and R is the correct explanation of A.

Option 3:

Both A and R are correct and R is NOT the correct explanation of A.

Option 4:

A is correct but R is not correct.

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Correct Answer:

Both A and R are correct and R is the correct explanation of A.

Solution:

$3d$ transition elements possess high melting and boiling points due to strong interatomic metallic bonding.

This strong bonding is a direct result of the involvement of both outer $4s$ electrons and unpaired inner $3d$ electrons in forming metallic bonds.

Both A and R are correct and R is the correct explanation of A.

Hence, the correct answer is option (2).

Q. 40 Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R.**
Assertion A: The first ionization enthalpy of O is lower than that of N and F.
Reason R: The loss of an electron from O leads to a stable half-filled p orbital.
In light of the above statements, choose the most appropriate answer from the options given below:

Option 1:

A is not correct but R is correct.

Option 2:

Both A and R are correct and R is the correct explanation of A.

Option 3:

Both A and R are correct and R is NOT the correct explanation of A.

Option 4:

A is correct but R is not correct.

Correct Answer:

Both A and R are correct and R is NOT the correct explanation of A.

Solution:

Nitrogen $\left(2s^2 2p^3\right)$ has a stable half-filled electronic configuration, and Fluorine has a higher nuclear charge, making Oxygen's ($2s^2 2p^4$) first ionization enthalpy lower than both.

Removing one electron from Oxygen $\left(2p^4 \rightarrow 2p^3\right)$ generates a stable half-filled configuration.

Comparison with N: The lower ionization energy of O compared to N is due to the high stability of Nitrogen's already existing half-filled $2p^3$ configuration and the inter-electronic repulsion between the paired electrons in Oxygen's $2p^4$ orbital.

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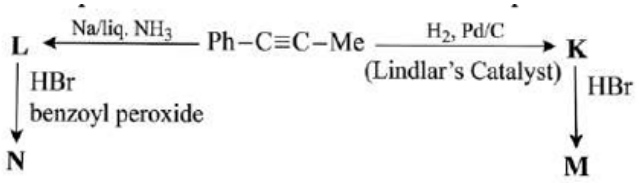
Comparison with F: The lower ionization energy of O compared to F is due to an increase in effective nuclear charge (Z_{eff}) across the period, which Reason R fails to address.

Both A and R are correct and R is NOT the correct explanation of A.

Hence, the correct answer is option (3).

Q. 41 Consider the following reaction sequences and choose the correct option.



Option 1:

M and N are stereoisomers

Option 2:

K and L are geometrical isomers

Option 3:

K and L are enantiomers

Option 4:

M and N are geometrical isomers

Correct Answer:

K and L are geometrical isomers

Solution:

Reduction of an alkyne ($Ph - C \equiv C - Me$) using sodium in liquid ammonia (Na / liq. NH_3) yields a transalkene (L).

Reduction of the same alkyne using hydrogen over Lindlar's catalyst ($H_2, Pd / C$) yields a cis-alkene (K).
Since K and L share the same molecular formula but differ in spatial geometry around the double bond, they are geometrical isomers.

Hence, the correct answer is option (2).

Q. 42 The highest occupied molecular orbital for Ne_2 is

Option 1:

σ^*_{2p}

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Option 2:

π_{2p}

Option 3:

σ_{2p}

Option 4:

π^*_{2p}

Correct Answer:

σ^*_{2p}

Solution:

Neon (Ne) has an atomic number of 10 , so a hypothetical Ne_2 molecule contains 20 electrons.

The molecular orbital electronic configuration for 20 electrons is:

$\sigma^2_{1s}\sigma^{*2}_{1s}\sigma^2_{2s}\sigma^{*2}_{2s}\sigma^2_{2p_z}\pi^2_{2p_x}\pi^2_{2p_y}\pi^{*2}_{2p_x}\pi^{*2}_{2p_y}\sigma^{*2}_{2p_z}$

The last and highest energy molecular orbital filled with electrons is σ^*_{2p}.

Hence, the correct answer is option (1).

Q. 43

Match the species in **List I** with their geometry in **List II**

	List 1		List II
A	PCl_5	I.	Tetrahedral
B	BrF_5	II.	Square Planar
C	BF_4^-	III.	Trigonal bipyramidal
D	$\left[Ni(CN)_4\right]^{2-}$	IV.	Square pyramidal

Choose the **correct** answer from the options given

Option 1:

A-III, B-II, C-I, D-IV

Option 2:

A-IV, B-III, C-I, D-II

Option 3:

A-III, B-IV, C-I, D-II

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Option 4:

A-III, B-I, C-II, D-IV

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

A. PCl_5: 5σ-bonds +0 lone pairs $\Longrightarrow sp^3d$ hybridization → Trigonal bipyramidal (III)

B. BrF_5: 5σ-bonds +1 lone pair $\Longrightarrow sp^3d^2$ hybridization → Square pyramidal (IV)

C. BF_4^-: 4σ-bonds +0 lone pairs $\Longrightarrow sp^3$ hybridization → Tetrahedral (I)

D. $\left[Ni(CN)_4\right]^{2-}$: Ni^{2+} is $3d^8$, strong field ligand (CN^-) causes pairing $\Longrightarrow dsp^2$ hybridization → Square Planar (II)

A-III, B-IV, C-I, D-II

Hence, the correct answer is option (3).

Q. 44

Match the species in **List I** with their geometry in **List II**

	List I		List II
A.	vitamin A	I.	meat
B.	vitamin B_{12}	II.	Sunflower oil
C.	vitamin E	III.	green leafy vegetables
D.	vitamin K	IV.	carrots

Choose the **correct** answer from the options given

Option 1:

A-III, B-I, C-IV, D-II

Option 2:

A-II, B-III, C-IV, D-I

Option 3:

A-IV, B-I, C-II, D-III

Option 4:

A-IV, B-II, C-I, D-III

Correct Answer:

A-IV, B-I, C-II, D-III

Solution:

A. Vitamin A: Richly found in carrots (IV).
B. Vitamin B_{12} : Primarily found in animal products like meat (I).

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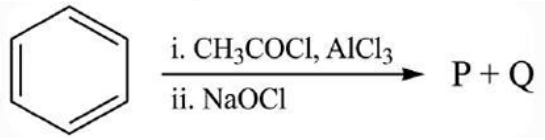
C. Vitamin E: Commonly sourced from vegetable oils like sunflower oil (II).

D. Vitamin K: Plentiful in green leafy vegetables (III).

Hence, the correct answer is option (3).

Q. 45 For the following reaction sequence, choose the correct option



Option 1:

Both P and Q are carbonyl compounds.

Option 2:

If P is the sodium salt of a carboxylic acid, Q is a primary alcohol.

Option 3:

P and Q are aromatic compounds.

Option 4:

If P gives a carboxylic acid on acidification, Q gives a poisonous gas on exposure to air and light.

Correct Answer:

If P gives a carboxylic acid on acidification, Q gives a poisonous gas on exposure to air and light.

Solution:

Benzene reacts with CH_3COCl in the presence of anhydrous $AlCl_3$ to give acetophenone $\left(C_6H_5COCH_3\right)$.

Acetophenone contains a methyl ketone ($-COCH_3$) group. Treatment with sodium hypochlorite (NaOCl) converts it into the sodium salt of benzoic acid $\left(C_6H_5COONa\right.$, let's call it product P) and chloroform $\left(CHCl_3\right.$, product Q).

Acidification of P $\left(C_6H_5COONa\right)$ gives benzoic acid (a carboxylic acid).
Chloroform ($Q = CHCl_3$) oxidation by air and light yields phosgene ($COCl_2$), which is a highly poisonous gas.
Hence, the correct answer is option (4).

Biology

Q. 1 Given below are two statements :

Statement I : The class name Reptilia refers to creeping or crawling mode of locomotion.

Statement II : All organisms belonging to Reptilia have three chambered heart. In the light of the above statements, choose the most appropriate answer from the options given below :

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Option 1:

Statement I is incorrect but Statement II is correct

Option 2:

Both Statement I and Statement II are correct

Option 3:

Both Statement I and Statement II are incorrect

Option 4:

Statement I is correct but Statement II is incorrect

Correct Answer:

Statement I is correct but Statement II is incorrect

Solution:

Statement I is correct — the word Reptilia comes from Latin, meaning to creep or crawl.

Statement II is incorrect — most reptiles have three-chambered hearts; only crocodiles have four-chambered hearts. So both statements cannot be correct together.

Hence, the correct answer is option (4) Statement I is correct but Statement II is incorrect.

Q. 2 How many turns of Calvin cycle are required for the formation of three molecules of glucose?

Option 1:

18

Option 2:

6

Option 3:

3

Option 4:

1

Correct Answer:

18

Solution:

In the Calvin cycle, one turn fixes one molecule of CO_2. To synthesise one glucose molecule, 6 turns are needed. For 3 glucose molecules, $6 \times 3 = 18$ turns are required.

Hence, the correct answer is option 1) 18.

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Q. 3

Photorespiration reaction catalyzed by RuBisCo is shown below :

$RuBP + O_2 \rightarrow$ 3-Phosphoglycerate +X Identify " X " from the given options :

Option 1:

Malate

Option 2:

Phosphoenolpyruvate

Option 3:

2-Phosphoglycolate

Option 4:

Oxaloacetate

Correct Answer:

2-Phosphoglycolate

Solution:

During photorespiration, RuBisCO enzyme binds with O_2 instead of CO_2. This splits RuBP into one molecule of 3-phosphoglycerate and one molecule of 2-phosphoglycolate. So ' x ' is 2-phosphoglycolate.

Hence, the correct answer is option 2)2-Phosphoglycolate

Q. 4

Given below are two statements :

Statement I: In gymnosperms, the male and female gametophytes remain within the sporangia.

Statement II :In gymnosperms, seeds are not covered.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:

Statement I is incorrect but Statement II is correct

Option 2:

Both Statement I and Statement II are correct

Option 3:

Both Statement I and Statement II are incorrect

Option 4:

Statement I is correct but Statement II is incorrect

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Correct Answer:

Both Statement I and Statement II are correct

Solution:

Both statements are correct. In gymnosperms, male and female gametophytes are not independent - they remain within the sporangia itself. Also, gymnosperm seeds are naked meaning they are not enclosed within a fruit.

Hence, the correct answer is option 2)Both Statement I and Statement II are correct.

Q. 5 How many molecules of pyruvic acid are produced at the end of glycolysis from 206 molecules of glucose?

Option 1:

412

Option 2:

206

Option 3:

309

Option 4:

103

Correct Answer:

412

Solution:

Glycolysis breaks down one glucose molecule into two pyruvate molecules. So 206 glucose molecules will produce $206 \times 2 = 412$ pyruvate molecules.

Hence, the correct answer is option 1) 412.

Q. 6

Match LIST -I with LIST -II

List I	List II
A. Fusion of protoplasms between gametes	I. Meiosis
B. Fusion of two nuclei	II. Plasmogamy
C. Generation of haploid spores	III. Karyogamy

Choose the correct answer from the options given below :

Option 1:

A-I, B-III, C-II

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Option 2:

A-II, B-III, C-I

Option 3:

A-II, B-I, C-III

Option 4:

A-III, B-II, C-I

Correct Answer:

A-II, B-III, C-I

Solution:

A. Fusion of protoplasms between gametes → II. Plasmogamy

Plasmogamy means the fusion of the cytoplasm (protoplasm) of two gametes. Nuclei do not fuse yet.
B. Fusion of two nuclei → III. Karyogamy

Karyogamy means the fusion of two nuclei. "Karyo" means nucleus. This results in a diploid nucleus.
C. Generation of haploid spores → I. Meiosis

Meiosis is the reductive division that produces haploid spores from diploid cells.

Hence, the correct answer is option 2) A-II, B-III, C-I

Q. 7 Mitochondrial inner membrane encloses _____ .

Option 1:

aqueous humor

Option 2:

matrix

Option 3:

cytosol

Option 4:

mucus

Correct Answer:

matrix

Solution:

The inner mitochondrial membrane is folded into cristae and encloses a fluid-filled space called the matrix. The matrix contains enzymes for Krebs cycle, mitochondrial DNA, and ribosomes.

Hence, the correct answer is option 2) Matrix.

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Q. 8 Phyllotaxy is the pattern of arrangement of ____ .

Option 1:

sepals

Option 2:

leaves

Option 3:

flowers

Option 4:

fruits

Correct Answer:

leaves

Solution:

Phyllotaxy refers to the pattern of arrangement of leaves on the stem or branch. It ensures maximum light absorption by leaves without overlapping each other.

Hence, the correct answer is option 2) Leaves.

Q. 9 Mad cow disease is caused by ____ .

Option 1:

Mycoplasma

Option 2:

prions

Option 3:

viroids

Option 4:

Aspergillus

Correct Answer:

prions

Solution:

Mad cow disease or Bovine Spongiform Encephalopathy is caused by prions. Prions are misfolded infectious proteins that contain no nucleic acid. They were first described by Stanley Prusiner.

Hence, the correct answer is option 3)prions.

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Q. 10 Cell theory was formulated by ____ .

Option 1:

Antonie Von Leeuwenhoek

Option 2:

Schleiden and Schwann

Option 3:

Robert Brown

Option 4:

Singer and Nicolson

Correct Answer:

Schleiden and Schwann

Solution:

Cell theory was proposed by Matthias Schleiden in 1838 for plants and Theodor Schwann in 1839 for animals. Together they stated that all living organisms are composed of cells and the cell is the basic unit of life.

Hence, the correct answer is option 2) Schleiden and Schwann.

Q. 11 Which of the following plant growth regulators promotes internode elongation prior to flowering in cabbage?

Option 1:

Ethephon

Option 2:

Abscisic acid

Option 3:

Gibberellin

Option 4:

Indole butyric acid

Correct Answer:

Gibberellin

Solution:

Gibberellins promote a phenomenon called bolting in rosette plants like cabbage. Bolting refers to rapid elongation of internodes just before flowering.

Hence, the correct answer is option 3)Gibberellin.

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Q. 12 Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700) ?

Option 1:

Carotenoids

Option 2:

Chlorophyll b

Option 3:

Chlorophylla

Option 4:

Xanthophylls

Correct Answer:

Chlorophylla

Solution:

Photosystem I has its reaction centre called P700. The P700 name is given because it contains a special form of Chlorophyll a that absorbs light maximally at 700 nm wavelength.

Hence, the correct answer is option 3)Chlorophylla.

Q. 13 Sphenopsida class belongs to ____ .

Option 1:

pteridophytes

Option 2:

bryophytes

Option 3:

angiosperms

Option 4:

gymnosperms

Correct Answer:

pteridophytes

Solution:

Sphenopsida is a class that includes plants like Equisetum (horsetail). These are vascular plants without seeds and are classified under Pteridophyta

Hence, the correct answer is option 1)pteridophytes.

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Q. 14 Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans?

Option 1:

Femur-tarsal-patella-tibia

Option 2:

Femur-tibia-patella-tarsal

Option 3:

Patella-femur-tibia-tarsal

Option 4:

Femur-patella-tibia-tarsal

Correct Answer:

Femur-patella-tibia-tarsal

Solution:

The correct sequence of bones in the lower limb from top to bottom is: Femur (thigh) → Patella (knee cap) → Tibia and Fibula (leg) → Tarsals → Metatarsals → Phalanges. Option (4) correctly represents this sequence.

Hence, the correct answer is option 4)Femur-patella-tibia-tarsal.

Q. 15 Which of the following plant growth regulators is used as herbicide ?

Option 1:

Gibberellin

Option 2:

$2,4-\mathrm{D}$

Option 3:

Kinetin

Option 4:

Abscisic acid

Correct Answer:

$2,4-\mathrm{D}$

Solution:

2,4-Dichlorophenoxyacetic acid (2,4-D) is a synthetic auxin. When applied in high concentrations, it selectively kills broad-leaved dicot weeds without harming monocot crops. It is the most commonly used herbicide.

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Hence, the correct answer is option 2) 2,4-D

Q. 16 Genus represents ____

Option 1:

a group of closely related families

Option 2:

an individual plant or animal

Option 3:

a population of plants and animals

Option 4:

a group of closely related species

Correct Answer:

a group of closely related species

Solution:

A genus is a taxonomic rank that groups together closely related species sharing common evolutionary characteristics. For example, the genus Panthera includes lion (P. leo), tiger (P. tigris), and leopard (P. pardus).

Hence, the correct answer is option 4) a group of closely related species.

Q. 17 The plastid that stores xanthophyll is known as_____

Option 1:

amyloplast

Option 2:

chloroplast

Option 3:

chromoplast

Option 4:

aleuroplast

Correct Answer:

chromoplast

Solution:

Chromoplasts are coloured plastids that store fat-soluble pigments. They contain xanthophyll which gives yellow colour and carotene which gives orange-red colour to flowers and fruits. They are different from chloroplasts which store chlorophyll.

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Hence, the correct answer is option 3) chromoplast.

Q. 18 In water, frogs respire using_____

Option 1:

trache

Option 2:

skin

Option 3:

buccal cavity

Option 4:

lungs

Correct Answer:

skin

Solution:

Frogs are amphibians that can live both on land and in water. When in water, frogs breathe through their skin - a process called cutaneous respiration. Their skin is thin, moist, and richly supplied with blood vessels to allow gas exchange.

Hence, the correct answer is option 2) skin.

Q. 19 Which of the following is not a characteristic of chordates ?

Option 1:

Presence of post anal part (tail)

Option 2:

Presence of notochord -

Option 3:

Central nervous system is dorsal

Option 4:

Absence of gills

Correct Answer:

Absence of gills

Solution:

The fundamental characteristics of all chordates include a notochord, a dorsal hollow nerve cord, pharyngeal gill slits, and a post-anal tail at least at some stage of their life. Therefore, "absence of gills" is not a chordate characteristic; pharyngeal gill slits are

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actually present.

Hence, the correct answer is option 4)Absence of gills.

Q. 20 Smooth endoplasmic reticulum____ .

Option 1:

is a site for the synthesis of carbohydrates

Option 2:

has ribosomes attached to its surface

Option 3:

is the major site for the synthesis of lipids

Option 4:

is actively involved in protein synthesis

Correct Answer:

is the major site for the synthesis of lipids

Solution:

Smooth endoplasmic reticulum does not have ribosomes on its surface. Its primary function is the synthesis of lipids, phospholipids, and steroids. It also plays a role in the detoxification of drugs and poisons in liver cells.

Hence, the correct answer is option 3)is the major site for the synthesis of lipids.

Q. 21

Which of the following are characteristics of prokaryotic cells ?

(a) Ribosomes are made of 50 S and 30 S subunits
(b) They can have plasmids
(c) They contain mesosome
(d) They have peroxisomes

Choose the correct answer from the options given below :

Option 1:

(a), (b) and (c) only

Option 2:

(b) and (c) only

Option 3:

(a) and (c) only

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Option 4:

(a), (c) and (d) only

Correct Answer:

(a), (b) and (c) only

Solution:

Prokaryotes possess 70 S ribosomes $(50\ S + 30\ S)$, may contain plasmids and mesosomes, but lack membrane-bound organelles such as peroxisomes.

However, peroxisomes are membrane-bound organelles found in eukaryotic cells. Since prokaryotes lack membrane-bound organelles, they do not possess peroxisomes.
Therefore, statements (a), (b), and (c) are correct, while statement (d) is incorrect.

Hence, the correct answer is option 1)1) (a), (b) and (c) only.

Q. 22

Match List - I with LIst -II

List - I	List - II
A. Cristae	I. Flat membrane sacs in stroma of chloroplast
B. Cisternae	II. Infoldings in mitochondria
C. Thylakoids	III. Cell membrane
D. Phospholipid	IV.Disc shaped sacs in the Golgi apparatus

Choose the correct answer from the options given below :

Option 1:

A-IV, B-III, C-I, D-II

Option 2:

A-III, B-IV, C-I, D-II

Option 3:

A-II, B-IV, C-I, D-III

Option 4:

A-II, B-IV, C-III, D-I

Correct Answer:

A-II, B-IV, C-I, D-III

Solution:

Cristae are infoldings of the inner mitochondrial membrane, cisternae are disc-shaped sacs of the Golgi apparatus, thylakoids are flat membrane sacs in the chloroplast, and phospholipids are major constituents of the cell membrane.

Hence, the correct answer is option 3)A-II, B-IV, C-I, D-III.

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Q. 23

Which of the following statements related to pituitary gland are correct ?

(a) It is divided anatomically into adenohypophysis neurohypophysis
(b) It secretes follicle stimulating hormone
(c) It secretes melanocyte stimulating hormone
(d) It does not secrete prolactin

Choose the correct answer from the options given below

Option 1:

(b) and (c) only

Option 2:

(a) and (b) only

Option 3:

(a), (b) and (c) only

Option 4:

(c) and (d) only

Correct Answer:

(a), (b) and (c) only

Solution:

The pituitary gland is anatomically divided into adenohypophysis and neurohypophysis (a). It secretes Melanocyte Stimulating Hormone or MSH (d). It does secrete prolactin, making statement (b) wrong. FSH is secreted by adenohypophysis, making (c) a correct fact but wrong as a statement in this context.

Hence, the correct answer is option 3)(a), (b) and (c) only.

Q. 24

Which of the following statements regarding photorespiration are correct?
(a) Do not occur in C 3 plants
(b) CO_2 is consumed and O_2 is generated
(c) Phosphoglycolate is formed
(d) No synthesis of ATP and NADPH

Choose the correct answer from the options given below :

Option 1:

(a) and (b) only

Option 2:

(a) and (d) only

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Option 3:

(c) and (d) only

Option 4:

(b) and (d) only

Correct Answer:

(c) and (d) only

Solution:

In photorespiration, O_2 is consumed, and CO_2 is released - opposite of photosynthesis (a). Phosphoglycolate is formed when RuBisCO acts on O_2 (c). Photorespiration does occur in C3 plants, so (b) is wrong. Option (d) about gametocytes is completely unrelated.

Hence, the correct answer is option 3)(c) and (d) only.

Q. 25 Which of the following statements is incorrect?

Option 1:

Fibrinogen is produced from fibrin

Option 2:

Blood coagulates in response to an injury

Option 3:

Blood clots consist of fibrins

Option 4:

Fibrin is produced from fibrinogen

Correct Answer:

Fibrinogen is produced from fibrin

Solution:

Statement (1) is incorrect and therefore the answer. Fibrinogen is NOT produced from fibrin - it is the other way around. During blood clotting, the enzyme thrombin converts soluble fibrinogen into insoluble fibrin threads that form the clot.

Hence, the correct answer is option 1)Fibrinogen is produced from fibrin.

Q. 26

Arange the following taxonomic categories in ascending order.

(a) Genus
(d) Phylum
(c) Order
(f) Kingdom

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(e) Family
(g) Spécies

Choose the correct answer from the opions given below :

Option 1:

(f), (c), (b), (g), (d), (e), (a)

Option 2:

(g), (a), (e), (c), (b), (d), (f)

Option 3:

(a), (c), (d), (g), (f), (b), (e)

Option 4:

(g), (c), (d), (b), (e), (a), (f)

Correct Answer:

(g), (a), (e), (c), (b), (d), (f)

Solution:

The correct ascending order of taxonomic hierarchy is: Species (g) → Genus (a) → Family (e) → Order (c) → Class (b) → Phylum (d) → Kingdom (f). This goes from the most specific to the most general category.

Hence, the correct answer is option 2)2) (g), (a), (e), (c), (b), (d), (f).

Q. 27 Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.

Option 1:

Production of glucose → role of air → release of oxygen → absorption spectra of chlorophyll a and b .

Option 2:

Absorption spectra of chlorophyll a and b → production of glucose → release of oxygen → role of air

Option 3:

Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b .

Option 4:

Release of oxygen → production of glucose → absorption spectra of chlorophyll a and b → role of air

Correct Answer:

Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b .

Solution:

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The historical sequence of photosynthesis discovery: Priestley (1770) demonstrated the role of air → Ingenhousz showed oxygen is released only in sunlight → Calvin and others established that glucose is produced → Emerson studied the absorption spectra of chlorophyll a and b .

Hence, the correct answer is option 3)Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b.

Q. 28 Match List-I with List-II.

List - I	List -II
A. Starch	I. Fights infection
B. Antibody	II. Energy storage
C. Concanavalin	III. Glucose transport
D.Glut-4	IV. Lectin

Choose the correct answer from the options given below

Option 1:

A-I, B-II, C-III, D-IV

Option 2:

A-I, B-II, C-IV, D-III

Option 3:

A-II, B-I, C-IV, D-III

Option 4:

A-II, B-I, C-III, D-IV

Correct Answer:

A-II, B-I, C-IV, D-III

Solution:

Starch (A) → Energy storage (II) - polysaccharide storing energy in plants
Antibody (B) → Fights infection (I) - immunoglobulins that neutralise pathogens
Concanavalin A (C) → Lectin (IV) - a carbohydrate-binding plant protein
Glut-4 (D) → Glucose transport (III) - transporter protein in muscle and fat cells

Hence, the correct answer is option 3)A-II, B-I, C-IV, D-III.

Q. 29 The number of vertebrae in a human is ____ .

Option 1:

206

Option 2:

7

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Option 3:

12

Option 4:

26

Correct Answer:

26

Solution:

In an adult human, the vertebral column consists of 26 bones (vertebrae). Although there are initially 33 vertebrae, the 5 sacral vertebrae fuse to form the sacrum and the 4 coccygeal vertebrae fuse to form the coccyx. Therefore, the adult vertebral column contains a total of 26 vertebrae/bones.

Hence, the correct answer is option 4) 26

Q. 30 Endomembrane system includes ____ .

Option 1:

Golgi complex, chloroplast, peroxisomes and vacuole

Option 2:

endoplasmic reticulum, Golgi complex, lysosomes and vacuole

Option 3:

endoplasmic reticulum, chloroplast, peroxisomes and vacuole

Option 4:

mitochondria, chloroplast, peroxisomes and vacuole

Correct Answer:

endoplasmic reticulum, Golgi complex, lysosomes and vacuole

Solution:

The endomembrane system comprises the endoplasmic reticulum, Golgi apparatus, lysosomes, and vacuoles, while mitochondria, chloroplasts, and peroxisomes are not included because they function independently.

Hence, the correct answer is option 2) Endoplasmic reticulum, Golgi complex, lysosomes and vacuole.

Q. 31 Length of the stem at time 0 is 20 cm . The arithmetic growth rate is 30 cm per day. What is the length of the stem at the end of the 7^{th} day ?

Option 1:

460 cm

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Option 2:

50 cm

Option 3:

170 cm

Option 4:

230 cm

Correct Answer:

230 cm

Solution:

Correct answer - (4) 230 cm

In arithmetic growth, the organism increases by a constant amount in each unit of time.

Length after 7 days $= 20 + (30 \times 7) = 230$ cm.

Hence, the correct answer is option 4) 230 cm

Q. 32

Match List-I with List-II.

List-I	List -II
A. Spherical,	I. Vibrio
B. Rod	II. Cocci
C. Comma	III. Spirilla
D. Spirillum	IV. Bacilli

Choose the correct answer from the options given below :

Option 1:

A-II, B-IV, C-I, D-III .

Option 2:

A-I, B-III, C-II, D-IV

Option 3:

A-III, B-II, C-I, D-IV

Option 4:

A-II, B-I, C-IV, D-III

Correct Answer:

A-II, B-IV, C-I, D-III .

Solution:

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Spherical bacteria are cocci, rod-shaped bacteria are bacilli, comma-shaped bacteria are vibrio, and spiral-shaped bacteria are spirilla.

Hence, the correct answer is option 1) A-II, B-IV, C-I, D-III

Q. 33 The number of action potentials generated by sino-arterial node (SAN) in a healthy human is ____ per minute.

Option 1:

$120 - 140$

Option 2:

$28 - 30$

Option 3:

$70 - 75$

Option 4:

$100 - 110$

Correct Answer:

$70 - 75$

Solution:

The SAN generates about $70 - 75$ action potentials per minute and acts as the natural pacemaker of the heart. These electrical impulses regulate the normal rhythm of cardiac contractions.

Hence, the correct answer is option 3) 70-75

Q. 34

Match List-I with List-II.

List-I	List -II
A. Family	I. Sapindales
B. Genus	II. Dicotyledonae
C. Class	III. Anacardiaceae
D. Phylum	IV. Angiospermae
E. Order	V. Mangifera

Choose the correct answer from the options given below

Option 1:

A-III, B-V, C-II, D-IV, E-I

Option 2:

A-I, B-V, C-II, D-IV, E-III

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Option 3:

A-II, B-I, C-III, D-IV, E-V

Option 4:

A-II, B-III, C-V, D-I, E-IV

Correct Answer:

A-III, B-V, C-II, D-IV, E-I

Solution:

Anacardiaceae (Family), Mangifera (Genus), Dicotyledonae (Class), Angiospermae (Phylum/Division), Sapindales (Order).

Mangifera belongs to the family Anacardiaceae, placed under the order Sapindales. This order belongs to the class Dicotyledonae, included in flowering plants(Angiosperms)

Hence, the correct answer is option 1) A-III, B-V, C-II, D-IV, E-I

Q. 35 Which of the following is not a part of human central neural system ?

Option 1:

Pericardium

Option 2:

Arachnoid

Option 3:

Dura mater

Option 4:

Pia mater

Correct Answer:

Pericardium

Solution:

Pericardium surrounds the heart, whereas the arachnoid mater, dura mater, and pia mater are the three protective membranes that surround the brain and spinal cord of the CNS. Therefore, pericardium is not the part of CNS.

Hence, the correct answer is option 1) Pericardium

Q. 36 Given below are two statements :

Statement I : Chromosomes are fully condensed at the end of prophase I.-
Statement II : Meiosis I resembles mitosis. ↑

In the light of the above statements, choose the most appropriate answer from the options given below :

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Option 1:

Statement I is incorrect, but Statement II is true

Option 2:

Both Statement I and Statement II are true

Option 3:

Both Statement I and Statement II are false

Option 4:

Statement I is correct, but Statement II is false

Correct Answer:

Statement I is correct, but Statement II is false

Solution:

Chromosomes become fully condensed during the diakinesis stage; however, Meiosis I is a reductional division involving pairing and separation of the homologous chromosomes. Meiosis II, on the other hand, resembles mitosis because sister chromatids separate during this division.

Hence, the correct answer is option 4) Statement I is correct, but Statement II is false

Q. 37

Match List-I with List-II. List-I

List - I	List -II
A. Marginal placentation	I. Argemone
B. Axile placentation	II. Tomato
C. Parietal placentation	III. Primrose
D. Free central placentation	IV. Pea

Choose the correct answer from the options given below :

Option 1:

A-IV, B-II, C-I, D-III

Option 2:

A-II, B-IV, C-I, D-III

Option 3:

A-IV, B-II, C-III, D-I

Option 4:

A-IV, B-III, C-I, D-II

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Correct Answer:

A-IV, B-II, C-I, D-III

Solution:

Pea shows marginal placentation, where ovules are attached along the ventral suture of a single carpel. Tomato shows axile placentation, Argemone shows parietal placentation, while Primrose has free central placentation.

Hence, the correct answer is option 1) A-IV, B-II, C-I, D-III

Q. 38 Symbiotic association between fungi and algae are called ____ .

Option 1:

chrysophyte

Option 2:

lichen

Option 3:

sponges

Option 4:

mycorrhiza

Correct Answer:

lichen

Solution:

A symbiotic association is a close relationship between two different organisms living together. In this relationship, at least one member benefits. Lichens are a symbiotic association between algae (or cyanobacteria) and fungi

Hence, the correct answer is option 2) Lichens

Q. 39 Which of the following is not a prokaryote?

Option 1:

Fungi

Option 2:

Bacteria

Option 3:

Blue green algae

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Option 4:

Mycoplasma

Correct Answer:

Fungi

Solution:

Fungi are eukaryotic organisms with a well-defined nucleus and membrane-bound organelles, whereas bacteria, blue-green algae (cyanobacteria), and mycoplasma are prokaryotes.

Hence, the correct answer is option 1) Fungi

Q. 40 Arrange the following elements in descending order of their contribution to percentage weight of the human body.

(a) Oxygen
(b) Carbon
(c) Hydrogen
(d) Nitrogen

Choose the correct answer from the options given below

Option 1:

(b), (a), (c), (d)

Option 2:

(a), (b), (c), (d)

Option 3:

(c), (a), (b), (d)

Option 4:

(b), (c), (d), (a)

Correct Answer:

(a), (b), (c), (d)

Solution:

The descending order of elemental abundance in the human body is Oxygen > Carbon > Hydrogen > Nitrogen. These four elements constitute the majority of the body's mass.

Hence, the correct answer is option 2) (a), (b), (c), (d)

Q. 41 Which one of the following statements is incorrect?

Option 1:

β-cells of pancreas secrete insulin

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Option 2:

α-cells of pancreas secrete glucagon

Option 3:

α-cells of pancreas secrete insulin

Option 4:

Glucagon stimulates glycogenolysis

Correct Answer:

α-cells of pancreas secrete insulin

Solution:

α-cells secrete glucagon, while β-cells secrete insulin.Glucagon increases blood glucose levels, whereas insulin helps lower blood glucose levels by promoting the uptake and breakdown of glucose.

Hence, the correct answer is option3) α-cells of pancreas secrete insulin

Q. 42

Which of the following are characteristic features of Samily?
(a) Flowers are bisexual and actinomorphic
(b) Calyx have five sepals and are united
(c) Androecium have five stamens and are epipetalous
(d) Ovary is inferiory

Choose the correct answer from the options given below :

Option 1:

(b), (c) and (d) only

Option 2:

(a), (b) and (c) only

Option 3:

(d) only

Option 4:

(a) and (b) only

Correct Answer:

(a), (b) and (c) only

Solution:

Flowers of the family Solanaceae are bisexual and actinomorphic with united sepals and epipetalous stamens. The gynoecium consists of a superior ovary, which distinguishes it from an inferior ovary.

Hence, the correct answer is option 2) (a), (b), and (c) only

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Given below are two statements :

Q. 43 **Statement I:** When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.
Statement II : In phylum Echinodermata, both adults and larvae are radially symmetrical.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:

Statement I is incorrect but Statement II is correct

Option 2:

Both Statement I and Statement II are correct

Option 3:

Both Statement I and Statement II are incorrect

Option 4:

Statement I is correct but Statement II is incorrect

Correct Answer:

Statement I is correct but Statement II is incorrect

Solution:

Radial symmetry means any plane through the central axis divides the body into equal halves; in Echinodermata, adults are radial, but larvae are bilateral.

Hence, the correct answer is option 4) Statement I is correct but Statement II is incorrect

Q. 44 The correct sequence of adult cell cycle phases is ____ .

Option 1:

S-M-G2-G1

Option 2:

G1-G2-S-M

Option 3:

G1-M-G2-S

Option 4:

G1-S-G2-M

Correct Answer:

G1-S-G2-M

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Solution:

The normal sequence of the cell cycle is G1 phase → S phase → G2 phase → M phase. Cell cycle proceeds in an orderly sequence of growth, DNA replication, preparation for division, and finally cell division.

Hence, the correct answer is option 4) G1-S-G2-M

Q. 45 In frogs, the number of pairs of cranial nerves arising from the brain are ____

Option 1:

12

Option 2:

6

Option 3:

9

Option 4:

10

Correct Answer:

10

Solution:

Frogs have 10 pairs of cranial nerves arising from the brain. These nerves are responsible for transmitting sensory information to the brain and carrying motor impulses to different parts of the body.

Hence, the correct answer is option 4) 10

Q. 46 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.
Reason R : Isolation of genetic material needs disruption of cells.
In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:

A is not correct but *R* is correct

Option 2:

Both *A* and *R* are correct and *R* is the correct explanation of *A*

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Option 3:

Both *A* and *R* are correct but *R* is not the correct explanation of *A*

Option 4:

A is correct but *R* is not correct

Correct Answer:

Both *A* and *R* are correct but *R* is not the correct explanation of *A*

Solution:

Lysozyme and cellulase are used because bacterial and plant cell walls are different from each other in their composition. Lysozyme digests the peptidoglycan cell wall of bacteria, whereas cellulase helps break down the cellulose cell wall of plants.

Hence, the correct answer is option 3) Both *A* and *R* are correct but *R* is not the correct explanation of *A*

Q. 47 The method of directly of injecting a sperm into ovum in assisted reproductive technology is called :

Option 1:

Embryo transfer (ET)

Option 2:

Gamete intra fallopian transfer (GIFT)

Option 3:

Zygote intra fallopian transfer (ZIFT)

Option 4:

Intra cytoplasmic sperm injection (ICSI)

Correct Answer:

Intra cytoplasmic sperm injection (ICSI)

Solution:

Intracytoplasmic Sperm Injection (ICSI) is an assisted reproductive technology in which a single sperm is injected directly into the cytoplasm of an ovum. This is commonly used in the case of severe male infertility and facilitates fertilisation.

Hence, the correct answer is option 4) Intra cytoplasmic sperm injection (ICSI)

Q. 48 Adaptive radiation in placental mammals and Australian Marsupials leading to similarity between distant species is an example of _____.

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Option 1:

genetic drift

Option 2:

divergent evolution

Option 3:

convergent evolution

Option 4:

founder effect

Correct Answer:

convergent evolution

Solution:

Placental mammals and Australian marsupials evolved separately but developed similar traits due to similar environmental pressures. This is convergent evolution, unrelated species developing similar characteristics independently. A classic example is the placental wolf and Tasmanian wolf (marsupial), looking similar despite being unrelated.

Hence, the correct answer is option 3) Convergent evolution

Q. 49

Given below are two statements: one is labelled as Assertion A, and the other is labelled as Reason R.
Assertion A: In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarfs.
Reason R: Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

In light of the above statements, choose the most appropriate answer from the options given below :

Option 1:

A is not correct but R is correct

Option 2:

Both A and R are correct and R is the correct explanation of A

Option 3:

Both A and R are correct but R is not the correct explanation of A

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Option 4:

A is correct but R is not correct

Correct Answer:

Both A and R are correct and R is the correct explanation of A

Solution:

Tallness is dominant over dwarfness; therefore, all F_1 progeny obtained from a cross between pure tall and pure dwarf plants are tall.

Hence, the correct answer is option 2) Both A and R are correct and R is the correct explanation of A

Q. 50

Arrange the following in descending order of number of species in the Amazonian rain forest.
(a) Plants
(b) Birds
(c) Fishes
(d) Invertebrates
(e) Mammals

Choose the **correct** answer from the options given below -

Option 1:

(b) > (a) > (d) > (c) > (e) .

Option 2:

(c) > (b) > (d) > (e) > (a)

Option 3:

(d) > (a) > (c) > (b) > (e)

Option 4:

(e) > (b) > (a) > (c) > (d)

Correct Answer:

(d) > (a) > (c) > (b) > (e)

Solution:

In the Amazonian rainforest, the correct descending order of species number is:

(d) Invertebrates — highest diversity, millions of species, especially insects
(a) Plants — second highest, thousands of plant species

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(c) Fishes — The Amazon River has an enormous fish diversity
(b) Birds — less diverse than fishes
(e) Mammals — the least number of species among all

Hence, the correct answer is option 3)(d) > (a) > (c) > (b) > (e)

Q. 51 Sponges exchange O_2 with CO_2 by _____.

Option 1:

gills

Option 2:

simple diffusion over their entire body surfaces

Option 3:

moist cuticle

Option 4:

tracheal tubes

Correct Answer:

simple diffusion over their entire body surfaces

Solution:

Sponges are the most primitive animals and lack any specialised respiratory organs. Oxygen and CO2 are exchanged directly by simple diffusion across the entire body surface. Water continuously flows through their pores (ostia), bringing fresh oxygen to all cells. This is sufficient for sponges as their body wall is very thin and all cells are close to water.

Hence, the correct answer is option 2) Simple diffusion over their entire body surface

Q. 52 For a person with blood group 'O', which of the following is not a possible combination of parents' blood group genotypes?

Option 1:

Father : P^AP^B and Mother : P^Ai

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Option 2:

Father : $P^{A}i$ and Mother :P_{i}

Option 3:

Father : $F^{A}i$ and Mother : $F^{A}i$

Option 4:

Father : $I^{B_{i}}$ and Mother :P_{i}

Correct Answer:

Father : $P^{A}P^{B}$ and Mother : $P^{A}i$

Solution:

For blood group O, a child must have genotype "ii" — getting one "i" allele from each parent. Father $I^{A}I^{B}$ does not carry any "i" allele at all, so he can never pass "i" to his child. Without "i" from the father, the child cannot get the "ii" genotype. Hence, a blood group O child is completely impossible from this combination. All other options have at least one "i" allele in both parents, making blood group O possible.

Hence, the correct answer is option 1)Father: $P^{A}P^{B}$ and Mother: $P^{A}i$

Q. 53

Given below are two statements :

Statement I : Modern Homo sapiens arose in Australia and moved across continents.
Statement II : Homo sapiens arose around 75000 to 10000 years ago. In the light of the above statements, choose the **most appropriate answer** from the options given below :

Option 1:

Statement I is incorrect but Statement II is correct

Option 2:

Both **Statement I** and **Statement II** are correct

Option 3:

Both **Statement I** and **Statement II** are incorrect

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Option 4:

Statement I is correct but **Statement II** is incorrect

Correct Answer:

Statement I is incorrect but Statement II is correct

Solution:

Modern *Homo sapiens* originated in Africa, not Australia, whereas NCERT states that modern *Homo sapiens* arose during the period 75,000–10,000 years ago.

Hence, the correct answer is option 1)Statement I is incorrect but Statement II is correct

Q. 54 Which of the following is used as an effective sedative and painkiller for treating post-surgery patients?

Option 1:

Anti-retroviral drugs

Option 2:

Interferon

Option 3:

Antibiotics

Option 4:

Morphine

Correct Answer:

Morphine

Solution:

Morphine is a potent opioid analgesic and sedative used for relieving severe pain, including post-surgical pain.

Hence, the correct answer is option 4)Morphine

Q. 55 Which of the following plant produces non-albuminous seeds ?

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Option 1:

Pea

Option 2:

Wheat

Option 3:

Maize

Option 4:

Barley

Correct Answer:

Pea

Solution:

Pea seeds are non-albuminous because the endosperm is consumed during seed development, whereas maize, wheat, and barley are albuminous seeds.

Hence, the correct answer is option 1) Pea

Q. 56 Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.
Reason R: Goats were more efficient at browsing than Abingdon tortoise.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:

A is not correct but R is correct

Option 2:

Both A and R are correct and R is the correct explanation of A

Option 3:

Both A and R are correct but R is not the correct explanation of A

Option 4:

A is correct but R is not correct

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Correct Answer:

Both *A* and *R* are correct and *R* is the correct explanation of *A*

Solution:

The Abingdon tortoise became extinct within a decade after goats were introduced in the Galapagos Islands. Goats were more efficient browsers than tortoises and consumed all vegetation. This is a classic example of competitive exclusion where goats outcompeted tortoises for food resources. R correctly explains A.

Hence, the correct answer is option 2) Both A and R are correct and R is the correct explanation of A

Q. 57 The covering of ovum at ovulation is ____.

Option 1:

chorion

Option 2:

endometrium

Option 3:

zona radiata

Option 4:

zona pellucida

Correct Answer:

zona pellucida

Solution:

At the time of ovulation, the ovum is released from the Graafian follicle. The ovum is covered by a thin transparent membrane called Zona Pellucida. It is a glycoprotein layer surrounding the egg cell.

Hence, the correct answer is option 4) Zona pellucida.

Q. 58 Which of the following is used as a clot buster?

Option 1:

Statins

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Option 2:

Streptokinase

Option 3:

Penicillin

Option 4:

Cyclosporin A

Correct Answer:

Streptokinase

Solution:

Streptokinase acts as a clot buster by dissolving blood clots in affected vessels. It is produced from bacteria Streptococcus. It dissolves blood clots in patients with heart attacks by activating plasminogen to plasmin.

Hence, the correct answer is option 2) Streptokinase

Q. 59 Which of the following structure is **not** a part of the male reproductive system ?

Option 1:

Infundibulum

Option 2:

Rete testis

Option 3:

Epididymis

Option 4:

Vasa efferntia

Correct Answer:

Infundibulum

Solution:

The infundibulum is a part of the female oviduct, whereas the rete testis, epididymis, and vasa efferentia are components of the male reproductive system.

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Hence, the correct answer is option 1) Infundibulum.

Q. 60

Given below are two statements :

Statement I: Ovulation is caused by LH surge leading to rupture of Graafian follicles.
Statement II: Graafian follicle remaining after ovulation transform into corpus luteum and secretes large amount of estrogen
In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:

Statement I is incorrect but **Statement II** is correct

Option 2:

Both **Statement I** and **Statement II** are correct

Option 3:

Both **Statement I** and **Statement II** are incorrect

Option 4:

Statement I is correct but **Statement II** is incorrect

Correct Answer:

Statement I is correct but **Statement II** is incorrect

Solution:

LH surge causes ovulation, but the corpus luteum mainly secretes progesterone, not a large amount of estrogen.

Hence, the correct answer is option 4)Statement I is correct, but Statement II is incorrect

Q. 61

The opening between the right atrium and the right ventricle is guarded by _____.

Option 1:

sino-atrial node

Option 2:

bicuspid valve

Option 3:

tricuspid valve

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Option 4:

semilunar valve

Correct Answer:

tricuspid valve

Solution:

The tricuspid valve (right atrioventricular valve) guards the opening between the right atrium and right ventricle. It has three cusps/flaps and prevents backflow of blood from the right ventricle into the right atrium during ventricular contraction.

Hence, the correct answer is option 3) Tricuspid valve.

Q. 62 Which of the following is **not** evidence for evolution?

Option 1:

Divergent evolution of anatomical structures such as forelimbs

Option 2:

Convergent evolution of traits like wings of birds and butterflies

Option 3:

Paleontological evidence from fossil records

Option 4:

Embryological support for evolution as proposed by Ernst Heckel

Correct Answer:

Embryological support for evolution as proposed by Ernst Heckel

Solution:

Embryological evidence (Haeckel's biogenetic law — "ontogeny recapitulates phylogeny") is now considered largely inaccurate and not reliable evidence for evolution. Divergent evolution, convergent evolution, and paleontological fossil records are all accepted as genuine evidence for evolution.

Hence, the correct answer is option 4)Embryological support for evolution as proposed by Ernst Heckel.

Q. 63 The inactive form of Bt toxin is converted to the active form in the insect gut _____.

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Option 1:

by nucleases

Option 2:

due to alkaline pH

Option 3:

due to acidic pH

Option 4:

by proteases

Correct Answer:

due to alkaline pH

Solution:

The inactive protoxin (Bt toxin) produced by Bacillus thuringiensis gets solubilised in the alkaline pH of the insect's gut. This alkaline environment converts the inactive protoxin into its active toxic form, which then binds to the midgut epithelial cells and kills the insect larva.

Hence, the correct answer is option 2) due to alkaline pH

Q. 64 Colostrum, secreted by mother during initial days of lactation, is abundant in _____.

Option 1:

IgD

Option 2:

IgG

Option 3:

IgM

Option 4:

IgA

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Correct Answer:

IgA

Solution:

Colostrum (the first milk secreted by the mother after childbirth) is particularly rich in IgA antibodies. These antibodies provide passive immunity to the newborn, protecting against infections, especially in the gut, as IgA is the primary immunoglobulin found in secretions.

Hence, the correct answer is option 4) IgA

Q. 65 Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the eggs ?

Option 1:

Polar nucleus

Option 2:

Antipodals

Option 3:

Synergids.

Option 4:

Central cells

Correct Answer:

Synergids.

Solution:

Synergids are two special cells present at the micropylar end of the embryo sac, flanking the egg cell. They have special finger-like projections called the filiform apparatus that help in guiding and directing the pollen tube towards the egg cell for fertilisation.

Hence, the correct answer is option 3) Synergids.

Q. 66 Which of the following disease is **not** sexually transmitted?

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Option 1:

Genital warts

Option 2:

Syphilis

Option 3:

Tuberculosis

Option 4:

Gonorrhoea

Correct Answer:

Tuberculosis

Solution:

Tuberculosis is not a sexually transmitted disease, unlike Gonorrhoea, Syphilis, and Genital warts.

Hence, the correct answer is Option 2)Tuberculosis.

Q. 67 Which of the following statements about lac-operon is **correct**?

Option 1:

Galactose can act as an inducer of lac operon

Option 2:

Gene i is constitutively expressed

Option 3:

Lactose activates repressor to bind to the operator

Option 4:

Genes i, z, y and a share single common promoter

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Correct Answer:

Gene i is constitutively expressed

Solution:

In lac operon, gene i codes for the repressor protein and is always active, independent of inducer presence.

Hence, the correct answer is option 2)Gene i is constitutively expressed

Q. 68

Match **List-I** with **List-II**.

	List-I		**List-II**
A	Transformation	I	Restriction enzyme
B	Cloning site	II	Transfer DNA to host bacteria
C	Selection	III	Replication
D	Ori	IV	Antibiotic

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-I, C-III, D-II

Option 2:

A-II, B-I, C-IV, D-III

Option 3:

A-I, B-II, C-IV, D-III

Option 4:

A-III, B-IV, C-II, D-I

Correct Answer:

A-II, B-I, C-IV, D-III

Solution:

Transformation involves DNA transfer, the cloning site is a restriction enzyme site, the selection marker is antibiotic resistance, and Ori is the replication origin.

Hence, the correct answer is option 2)A-II, B-I, C-IV, D-III

Q. 69

A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1 , the frequency of AA is _____.

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Option 1:

0.99

Option 2:

0.01

Option 3:

0.02

Option 4:

0.10

Correct Answer:

0.01

Solution:

Hardy-Weinberg law: frequency of $AA = p^2$. Given allele $A = 0.1 \rightarrow (0.1)^2 = 0.01$.

Hence, the correct answer is option 2) 0.01

Q. 70 Sperm motility is due to _____.

Option 1:

muscular movement

Option 2:

flagellar movement

Option 3:

ciliary movement

Option 4:

amoeboid movement

Correct Answer:

flagellar movement

Solution:

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Sperm cells possess a flagellum — a long, whip-like tail structure. The rhythmic beating of this flagellum propels the sperm forward, enabling motility. This is distinct from ciliary, muscular, or amoeboid movement, none of which apply to sperm locomotion.

Hence, the correct answer is option 2)flagellar movement.

Q. 71 Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the per-capita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is _____.

Option 1:

100 million

Option 2:

1 million

Option 3:

5 million

Option 4:

10 million

Correct Answer:

10 million

Solution:

When birth rate equals death rate (both 0.002 per unit time), the net growth rate (r) = 0. With zero net growth, the population size remains constant regardless of generations. So, 10 million cells after 10 generations stays 10 million.

Hence, the correct answer is option 4) 10 million.

Q. 72 Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason *R*.
Assertion *A* **:** Forelimbs of human and bats are homologous.
Reason *R* **:** Forelimbs of humans and bats have similar anatomical structure.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:

A is false but *R* is true

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Option 2:

Both A and R are correct and R is the correct explanation of A

Option 3:

Both A and R are true, but R is not the correct explanation of A

Option 4:

A is true but R is false

Correct Answer:

Both A and R are correct and R is the correct explanation of A

Solution:

Forelimbs of humans and bats are indeed homologous structures — they share the same basic skeletal framework (humerus, radius, ulna, carpals) despite different functions. The reason is correct because this similar anatomical structure is precisely what makes them homologous.

Hence, the correct answer option 2)Both A and R are correct and R is the correct explanation of A

Q. 73 Muscle contraction is initiated by a signal sent by the central nervous system by the release of _____.

Option 1:

cyclic adenine monophosphate

Option 2:

acetyl choline

Option 3:

acetyl coenzyme A

Option 4:

cyclic guanine monophosphate

Correct Answer:

acetyl choline

Solution:

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At the neuromuscular junction, motor neurons release the neurotransmitter acetylcholine (ACh). ACh binds to receptors on the muscle fibre membrane, triggering depolarisation and ultimately initiating the sliding filament mechanism of muscle contraction.

Hence, the correct answer is option 2)acetyl choline.

Q. 74 Which of the following hormone is **not** secreted by human placenta?

Option 1:

LH

Option 2:

hCG

Option 3:

Estrogen

Option 4:

Progesterone

Correct Answer:

LH

Solution:

The human placenta secretes several hormones, including hCG (human chorionic gonadotropin), estrogen, and progesterone. LH (Luteinizing Hormone) is secreted by the anterior pituitary gland, not by the placenta, making it the correct answer here.

Hence, the correct answer is option 1)LH.

Q. 75 Which of the following statements is **correct** about *Plasmodium*?

Option 1:

Fertilization takes place in mosquito gut

Option 2:

Reproduces sexually in liver cells

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Option 3:

Reproduces sexually in RBCs

Option 4:

Gametocytes develop in mosquito gut

Correct Answer:

Fertilization takes place in mosquito gut

Solution:

In Plasmodium, sexual reproduction (fertilisation) occurs in the stomach/gut of the female Anopheles mosquito, where male and female gametocytes fuse to form a zygote (ookinete). The asexual stages occur in the human liver and RBCs.

Hence, the correct answer is option 1)Fertilisation takes place in the mosquito gut.

Q. 76 Which of the following are primary consumers in a food chain ?

Option 1:

Carnivores

Option 2:

Parasites

Option 3:

Predators

Option 4:

Herbivores

Correct Answer:

Herbivores

Solution:

Primary consumers are those that directly feed on producers (green plants/autotrophs). Herbivores eat plants directly, placing them at the second trophic level as primary consumers. Carnivores feed on herbivores, making them secondary consumers. Hence, herbivores are the correct answer.

Hence, the correct answer is option 4)Herbivores.

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Q. 77

Which of the following statements about the reabsorption process in Henle's loop are **correct** ?
(a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.
(b) Urine gets concentrated in Henle's loop.
(c) Reabsorption of Na^+ and water takes place in Henle's loop.
(d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop.

Choose the **correct** answer from the options given below :

Option 1:

(a), (b) and (d) only

Option 2:

(a) and (b) only o

Option 3:

(b), (c) and (d) only

Option 4:

(a), (b) and (c) only

Correct Answer:

(a), (b) and (d) only

Solution:

Descending limb is permeable to water but impermeable to electrolytes (a). Urine gets concentrated in Henle's loop due to the countercurrent mechanism (b). Statement (c) is WRONG because the ascending limb is impermeable to water; only electrolytes are reabsorbed there. Active and passive electrolyte transport occur in the ascending limb (d).

Hence, the correct answer is option 1)(a), (b) and (d) only

Q. 78

Given below are two statements : one is labelled as **Assertion A** and the other is labelled as Reason R.
Assertion A : The logistic growth model of populations is considered more realistic than the exponential growth model.
Reason R: Resources are finite. In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:

A is not correct but R is correct

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Option 2:

Both A and R are correct and R is the correct explanation of A

Option 3:

Both A and R are correct but R is not the correct explanation of A

Option 4:

A is correct but R is not correct

Correct Answer:

Both A and R are correct and R is the correct explanation of A

Solution:

The logistic growth model is more realistic than the exponential growth model (A correct) because natural resources are always finite and limited (R correct). Limited resources cause the population to reach the carrying capacity (K) and stabilise, which is exactly what the logistic model shows. R correctly explains A.

Hence, the correct answer is option 2)(2) Both A and R are correct and R is the correct explanation of A

Q. 79 Which of the following is the **correct** order of arrangement of vertebrate column from the head to toe ?

Option 1:

Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

Option 2:

Cervical vertebra, thoracic vertebra, sacrum, lumbar vertebral

Option 3:

Sacrum, lumbar vertebra, thoracic vertebra, cervical vertebra

Option 4:

Cervical vertebra, lumbar vertebra, thoracic vertebra, sacrum

Correct Answer:

Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

Solution:

The correct anatomical order of the vertebral column from head to toe is:

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- Cervical → 7 vertebrae
- Thoracic → 12 vertebrae
- Lumbar → 5 vertebrae
- Sacrum → 5 fused vertebrae
- Coccyx → 4 fused vertebrae

This is the standard arrangement in humans.

Hence, the correct answer is option 1)Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum.

Q. 80

Match **List-I** with **List-II.**

List-I		**List-II**	
A	Both species are harmed	I	Predation
B	One species is harmed other is benefied	II	Mutualism
C	Both species are benefied	III	compitition
D	One species is while the other has no effect	IV	Commenselism

Choose the **correct** answer from the options given below :

Option 1:

A-III, B-I, C-II, D-IV

Option 2:

A-III, B-IV, C-II, D-I

Option 3:

A-I, B-II, C-III, D-IV

Option 4:

A-II, B-I, C-IV, D-III

Correct Answer:

A-III, B-I, C-II, D-IV

Solution:

Correct matching of ecological interactions:

- A. Both species harmed → III. Competition
- B. One harmed, other benefited → I. Predation
- C. Both species benefited → II. Mutualism
- D. One benefited, other unaffected → IV. Commensalism

Hence, the correct answer is option 1)A-III, B-I, C-II, D-IV.

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Q. 81 If the diploid chromosome number of typical angiosperm is 36 , what would be the chromosome number in its endosperm ?

Option 1:

72

Option 2:

18

Option 3:

36

Option 4:

54

Correct Answer:

54

Solution:

Endosperm is triploid (3n) in angiosperms. If the diploid chromosome number (2n) is 36, then n = 18. Endosperm chromosome number = 3n = 3 × 18 = 54. Endosperm forms by the fusion of two polar nuclei with one male gamete.

Hence, the correct answer is option 4) 54.

Q. 82 Which of the following enzymes synthesizes precursor mRNA ?

Option 1:

DNA polymerase

Option 2:

RNA polymerase I

Option 3:

RNA polymerase II

Option 4:

RNA polymerase III

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Correct Answer:

RNA polymerase II

Solution:

In eukaryotes, RNA polymerase II synthesises precursor mRNA (hnRNA), which is later processed into mature mRNA. RNA polymerase I synthesises rRNA, while RNA polymerase III synthesises tRNA and 5S rRNA.

Hence, the correct answer is option 3)RNA polymerase II.

Q. 83 Given below are two statements :

Statement I : Plasmids are autonomously replicating DNA.

Statement II : Plasmids are extrachromosomal DNA.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:

Statement I is incorrect but Statement II is correct

Option 2:

Both Statement I and Statement II are correct

Option 3:

Both **Statement I** and **Statement II** are incorrect

Option 4:

Statement I is correct but **Statement II** is incorrect

Correct Answer:

Both Statement I and Statement II are correct

Solution:

Plasmids are indeed autonomously replicating circular DNA molecules (Statement I correct). They are also extrachromosomal DNA, existing independently of the main bacterial chromosome (Statement II correct). Both properties make plasmids valuable vectors in genetic engineering.

Hence, the correct answer is option 2) Both Statement I and Statement II are correct.

Q. 84 How many theca are present in each lobe of a typical bilobed angiosperm anther?

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Option 1:

12

Option 2:

2

Option 3:

6

Option 4:

8

Correct Answer:

2

Solution:

Each lobe of a typical bilobed angiosperm anther contains 2 thecae (singular: theca). Each theca contains 2 pollen sacs (microsporangia). So each anther lobe has 2 thecae, and the complete anther has 4 microsporangia in total.

Hence, the correct answer is option 2)2

Q. 85

Natural selection can lead to _____.
(a) stabilisation
(b) genetic drift
(c) directional change
(d) disruption

Choose the **correct** answer from the options given below :

Option 1:

(a) and (c) only

Option 2:

(a) only

Option 3:

(a), (c) and (d) only

Option 4:

(a), (b), (c) and (d)

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Correct Answer:

(a), (c) and (d) only

Solution:

Natural selection leads to stabilisation (a), directional change (c), and disruption (d) — these are the three recognised types of natural selection. Genetic drift (b) is a separate evolutionary mechanism that occurs due to random chance, not natural selection, so it is excluded.

Hence, the correct answer is option 3) (a), (c) and (d) only.

Q. 86

Which of the following statements are **correct**?
(a) Energy flow from producers to consumers is unidirectional
(b) Energy pyramid can never be inverted
(c) Transfer of energy follows the 1% law .
Choose the **correct** answer from the options given below :

Option 1:

(b) and (c) only

Option 2:

(a), (b) and (c)

Option 3:

(a) and (b) only

Option 4:

(a) and (c) only

Correct Answer:

(a) and (b) only

Solution:

- Energy flows from producers to consumers is strictly unidirectional (a) — correct.
- Energy pyramids can never be inverted (b) — correct, unlike biomass or number pyramids.
- The 10% law (not 1%) governs energy transfer between trophic levels, making statement (c) incorrect.

Hence, the correct answer is option 3)(a) and (b) only

Q. 87

Match **List-I** with **List-II.**

List-I	List-II

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A	Excess growth hormone	I	Reabsorption of water and electrolytes in kidney
B	Luteinising Hormone	II	Contraction of uterus during child birth
C	Vasopressin	III	Acromegaly
D	Oxytocin	IV	Ovulation

Choose the **correct** answer from the options given below :

Option 1:

A-IV, B-III, C-I, D-II

Option 2:

A-III, B-IV, C-II, D-I

Option 3:

A-III, B-IV, C-I, D-II

Option 4:

A-II, B-IV, C-I, D-III

Correct Answer:

A-III, B-IV, C-I, D-II

Solution:

- Excess growth hormone → Acromegaly (III).
- Luteinizing hormone → Ovulation (IV).
- Vasopressin → Reabsorption of water and electrolytes in the kidney (I).
- Oxytocin → Contraction of the uterus during childbirth (II).

This is the correct hormonal matching.

Hence, the correc tnaswer is option 3) A-III, B-IV, C-I, D-II.

Q. 88

Which of the following are secondary lymphoid organs ?
(a) Bone marrow
(b) Tonsils
(c) Spleen
(d) Thymus

Choose the **correct** answer from the options given below :

Option 1:

(a) and (d) only

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Option 2:

(a) and (b) only

Option 3:

(b) and (c) only

Option 4:

(b) and (d) only

Correct Answer:

(b) and (c) only

Solution:

- Primary lymphoid organs = Bone marrow + Thymus
- Secondary lymphoid organs = Spleen, Tonsils, Lymph nodes, MALT, Peyer's patches

Hence, the correct answer is option 3) (b) and (c) only

Q. 89 During PCR, primers bind to the DNA strands in the _____ step.

Option 1:

ligation

Option 2:

denaturation

Option 3:

extension

Option 4:

annealing

Correct Answer:

annealing

Solution:

During PCR, primers bind to the denatured single-stranded DNA template during the annealing step (typically at 50–65°C). Denaturation separates strands, annealing allows primer binding, and extension is where DNA polymerase synthesises the new strand.

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Hence, the correct answer is option 4) annealing

Q. 90

Given below are two statements :

Statement I: Down's syndrome is caused by the absence of one of the X -chromosomes.
Statement II : Turner's syndrome is caused by the presence of an additional copy of the chromosomes.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:

Statement I is incorrect but Statement II is correct

Option 2:

Both Statement I and Statement II are correct

Option 3:

Both Statement I and Statement II are incorrect ,

Option 4:

Statement I is correct but Statement II is incorrect

Correct Answer:

Both Statement I and Statement II are incorrect ,

Solution:

Down syndrome is caused by trisomy of chromosome 21 (extra chromosome), NOT absence of the X chromosome. Turner's syndrome is caused by the absence of one X chromosome (45, XO), NOT by an additional chromosome. Both statements are therefore incorrect.

Hence, the correct answer is option 3) Both Statement I and Statement II are incorrect

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